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[https://platformwiki.o9solutions.com/index.php/Category:Supply Chain Solver \(SCS\)](https://platformwiki.o9solutions.com/index.php/Category:Supply_Chain_Solver_(SCS))

The premier AI platform for
**Integrated Planning and
Operations Management**



Supply Chain Solver

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About Supply Chain Solver

A "supply chain" encompasses mainly the following three functions:

- i. Supply of raw materials from the supplier to the manufacturer, through the procurement process;

- ii. Conversion from raw materials and components to finished goods through manufacturing process
- iii. Fulfilment of finished goods through distributors and retailers via distribution process

To facilitate the flow of products, information is shared across the entire supply chain, i.e. both upstream and downstream. It enables all parties to plan appropriately to meet current and future needs.

The primary goal of the supply chain solver is to make a demand supply match by adhering to the business priorities/rules.

Others are secondary goals, some of which are

- Excess inventory to be minimized
- Costs to be reduced
- Product time to market to be improved
- Agility and resilience to be enhanced

o9 is capable of fulfilling these goals using supply chain solver, which is a plug-in and runs within the GraphCube Server process.

Supply Chain Solver Algorithm

Supply Chain Solver (SCS) follows an order by order heuristic, where it takes one order at a time and checks for material and capacity availability upstream. SCS supports segregation of material on the basis of Lot as well. However, segregation on the basis of lots or document IDs is not a hard constraint and can be relaxed as per the convenience of the planners. If capacity/material is not available in JIT time, it will then check material and capacity in earlier buckets, bucket by bucket till it reaches the build ahead limit. This is all accomplished in a single pass. If the demand quantity is not fulfilled on primary path, it will explore other paths in the order of priority.

When checking for build late, it will check for all activities providing that material and for each such activity it will go bucket by bucket till build late limit is reached. It will also check for capacity during this process. Least late supplies among all late supplies are selected. The primary objective here is to minimize the delay. If supply is not available in these periods, then that portion of the demand will be shorted.

To summarize, solver does one pass with the option fulfilling the demand JIT and another pass for the rest of the quantity where it can late the demand within the build late limit.

Planning algorithm (Basic Material-Capacity coordination)

1. Sort the orders as per demand priority
2. Plan order by order by traversing the supply chain from downstream to upstream checking material (and their corresponding lots), and capacity and also adhering to the supply chain network and business priorities as per the below sequence

- a. Ask immediate connected upstream activity (in order of priority) for the quantity requested.
- b. Check for current capacity availability in that time bucket for that activity node
 - i. If available, ask upstream node for the material {minimum [demand request, capacity]} in the time bucket with lead time offset
 - ii. If cumulative material is available (Look for existing surplus / inventory / expected receipt supplies in JIT & early buckets), provide material and update the inventory profile of the node
 - iii. Else recurse upstream asking activity (capacity) and node (material) in similar way till the most upstream node is reached
- c. If not, check for capacity in earlier buckets one at a time (till the build ahead limit or plan start is reached) and go to upstream nodes in similar fashion as earlier
- d. If demand is not met fully, then build late, by looking ahead one bucket at a time, till the build late limit is reached (or till plan end).
 - i. For each bucket check alternate activities so that the lateness is reduced

Additional Notes

1. If it is BOM, then all components are needed to satisfy the ask on that node at that time
2. Look for existing surplus / inventory / expected receipt supplies in bucket, before asking for new supplies from upstream
3. When build ahead the default option is to build ahead on primary, bucket by bucket till build ahead limit is reached and then go to alternate activities. Based on a flag, one can minimize this earliness. In that case solver checks all activities in the same bucket and then proceed to the immediate preceding bucket and so on. In each bucket it checks for all activities.
4. This should work in below planning categories as well
 - unconstrained planning - Both material and capacity unconstrained
 - capacity unconstrained – material constrained planning
 - material unconstrained – capacity constrained planning
 - Constrained Planning - Both material and capacity constrained

Pegging

Planning works out of a profile (which is the net of all productions and consumptions in a time bucket). It does not know which flow plan is producing and at what time. Peggings work out of actual productions and consumptions.

Example:

- a) Producer P1 of 100 exists on Jan 1.

- b) Consumer/Demand C1 for 100 comes on Jan 10. Got satisfied using the production from existing inventory from P1
- c) Another Consumer C2 of 100 arrives at Jan 5. Optimal planning logic would eat up/steal existing production (meant for C1) and make another production P2 later on Jan 10. Call this P2
- d) So we cannot use planning logic in pegging here i.e. P1 cannot be pegged to C1. Else we need to recompute peggings whenever required, which is quite expensive (especially when we have safety stock across levels)

The proposal is to have the producer and consumer at each item-location node during planning. At the end of the planning, follow FIFO(first in first out) to peg producers to consumers. The required peggings are

1. Order Pegging: Given a demand, it will tell which material and capacity are consumed in the supply chain from downstream to upstream. It helps create the entire supply path through which it was planned along with time dimension
2. Material Pegging: Given an item-location node and time bucket, it should point out all the orders which utilized the material in that particular time bucket
3. Capacity Pegging: Given a capacity node and time bucket, it should point out all the orders which utilize the capacity in that time bucket.
4. Lot Pegging: Given the material node and the lots available at that material node in a specified time bucket, SCS should point out all Lots at any material node that were utilized for demand fulfillment.

Note: The peggings need to have lot-level information (just the flow in a bucket won't suffice) and we will need to write it to some measure group, if we want to re-instantiate the supply chain along with plans at a later point and do incremental solves.

SCS Configuration

o9 Platform is a multi tenant SaaS offering. To the benefit of our consultants we provide a model in the sandbox environment, which users can copy and start playing around to understand the solver better. The details of the model are provided below.

The tenant and the login details are provided below

Tenant Name: SCS-o9-2019_0.com
 User ID: etluser@scs-o9-2019_0.com
 Password: Welcome123
 Environment : [Sandbox](#)

Note: Please do not make changes directly to this model on sandbox. Create a copy for yourself before you play around the same. Also uninstall and delete when you are done.

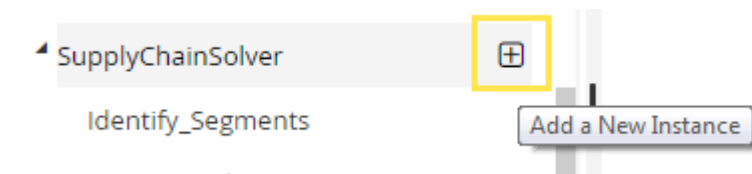
Configuration Wizard

Goto Workspace→Designer→Edit

model→Rule→Plugin→Global→o9.GraphCubePlugins→SupplyChainSolver→Click the



sign as shown below



A form shows towards the right as follows where in the desired instance name can be typed in the text box

SupplyChainSolver Adding a new instance for Class 'SupplyChainSolver'.

OLDER

Instance Options Parameters Node Demand Inventory Manufacturing Distribution Procurement

Plugin Name o9.GraphCube.Plugins.SupplyChainSolver.SupplyChainSolver

Instance Name

Invocation Type Active

The form has the following tabs

1. Instance Options
2. Node
3. Demand
4. Inventory
5. Parameters
6. Manufacturing
7. Distribution
8. Procurement

Let us go through the contents of each tab.

Instance Options

This tab contains information with respect to the instance of the supply chain Plugin SupplyChainSolver

SupplyChainSolver

Instance Options Node Demand Inventory Parameters Manufacturing Distribution Procurement

Plugin Name o9.GraphCube.Plugins.SupplyChainSolver.SupplyChainSolver

Instance Name SupplyChainSolver

Invocation Type Active

Update Cancel

Invocation Type:

Choose Active from the drop-down Menu. However, It is an On-Demand call. All On-Demand calls are initiated by the user, hence the name. User can trigger the plugin through action buttons.

Note: we have only one option by default in the drop-down menu.

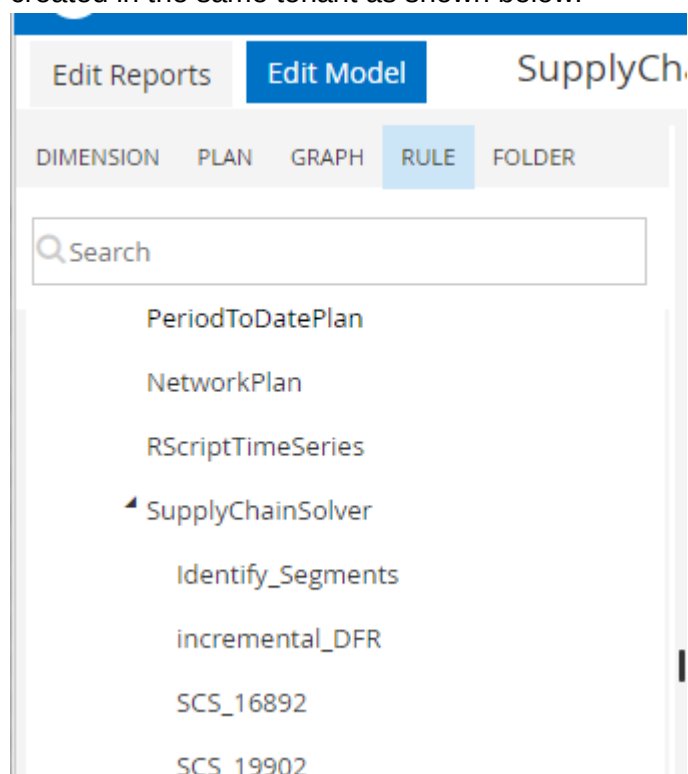
Plugin Name:

It is a drop down menu. The user selects the name of the plugin that (s)he wants to invoke. SupplyChainSolver is a global plugin which appears in the drop down menu as `o9.GraphCube.Plugins.SupplyChainSolver.SupplyChainSolver`.

Instance Name:

This is considered as an alias name given to the plugin that one wants to invoke. Alias name should be kept as readable and understandable so that it can be used across the tenant in IBPL or MERU interface. It is advised that spaces should not be there in the Instance Name. One can use "_" but not "-" as separators.

SupplyChainSolver is a global plugin in Platform instance(s) of which can be created in the same tenant as shown below.



The above instances can be invoked as follows:

```
exec plugin instance SCS_test for Measures {[Demand Quantity]} using scope
([Version].[Version Name].[CurrentWorkingView] * Time.[Fiscal Week] * Order.[Order
Name]) ;
```

```
exec plugin instance SupplyChainSolver for Measures {[Demand Quantity]} using
scope ([Version].[Version Name].[CurrentWorkingView] * Time.[Fiscal Week] *
Order.[Order Name]) ;
```

```
exec plugin instance SupplyChainSolver1 for Measures {[Demand Quantity], [BOH],
[Expected Receipts]} using scope ([Version].[Version Name].[CurrentWorkingView] *
Time.[Fiscal Week] * Order.[Order Name]);
```

Parameters

SupplyChainSolver [Successfully loaded instance 'SupplyChainSolver'.](#)

Instance Options **Parameters** Node Demand Inventory Manufacturing Distribution Procurement

This tab has the following fields:

Parameter	Default Value	Dependencies	Category	Details
Time Attribute Name			Basic	Here
Telescopic Attribute Name			Advanced	Here
Start Time Bucket Name			Basic	Here
Current Time Bucket Name			Basic	Here
End Time Bucket Name			Basic	Here
SCS Lot Name			Basic	Name of the Lot member, reserved for SCS generated plans.
Material Constraint	True		Basic	Here
Capacity Constraint	True		Basic	Here
Storage Constraint	True		Advanced	Here
Constrained Unconstrained	False		Basic	Here
Allow NOH	False		Advanced	Here
Respect Freeze Window For WIPs	False	SKIP WIP PLAN	Advanced	Here
Respect Activity Holiday For WIPs	True	SKIP WIP PLAN	Advanced	Creates production and consumption WIPs in a holiday

				bucket for activity when this flag is set to False.
Minimize Inventory Build Ahead	False	Ignore BAL For NoCarry Buffer	Basic	Here
No Write Execution	False		Performance	Here
No Peggings	False		Performance	Here
No Constraints Report	False		Performance	Here
SS Build Ahead Limit	NULL	Inventory Plan Type	Advanced	Here
Try Material Reallocation In Inventory Plan	False	Inventory Plan Type	Advanced	Allows material reallocation during safety stock planning.
Load Downstream Supply Chain	True		Performance	Here
Is SupplyCode Plan	False		Advanced	Here
Advanced SupplyCode Plan	False		Advanced	Here
Optimize Demand Supportability	None		Performance	Multi select picklist: When set to ReplanShortLateDemands , it will replan short/late demands for late+short quantity. When set to ReplanAtDeliveryNode , it will unplan all delivery flows and replan demands in the order of their priority. ReplanForOboRca , it will generate OBO RCA after unplanning and replanning short/late demands Functionality
Plan Date Hard Peg	True		Advanced	Here
Min Time Hard Peg	True		Advanced	Here

Concurrent Planning	False		Performance	Here
Number of Planning Threads	1	Concurrent Planning	Performance	Here
Number of Export Threads Per Segment	1	Concurrent Planning	Performance	Here
Number of Import Threads	3	Concurrent Planning	Performance	Number of parallel threads used while loading data from LS to SCS data structures.
Excess Clean Up Frequency	SS		Advanced	Here
Segment Split Min Node Count Limit	500		Performance	Here
Segment Split Max Node Count Limit	5000		Performance	Here
Delete SCS Plan Data For Version	False		Basic	Here
Respect Max Stock Constraint In JIT Planning	False		Advanced	Here
Avoid Pegging Through Max Stock Node	False	Respect Max Stock Constraint In JIT Planning	Performance	Here
Varying Capacity Buckets	False		Advanced	Here
Use Bands To Minimize Lateness	False		Performance	Here
Use Bands To Minimize Build Ahead	False		Performance	Here
Activity Marker Exclusion List			Basic	Here
Intra Bucket Slicing Only	False	Demand Slice Factor	Advanced	Here

Demand Slice Factor	1		Basic	Here
Build Ahead Limit		Current Time Bucket Name	Basic	Here
Build Late Limit		End Time Bucket Name	Basic	Here
Inventory Plan Type	NONE		Basic	Here
Inventory Plan Policy	TARGET_LEVEL	Inventory Plan Type	Basic	Here
Inventory Plan Bands		Inventory Plan Type Inventory Plan Policy	Advanced	Here
Excess Clean Up Strategy	RELEASE	Excess Clean Up Frequency	Performance	Three options supported RELEASE, MOVE_OUT and None
Late Plan Strategy	LINEAR	Build Late Limit	Performance	Here
Maximum Fractional Precision	4		Advanced	This specifies the maximum fractional precision for decimals, float and double values that are exported
Plan Type		Segment Split Min Edge Count Limit Segment Split Max Node Count Limit	Basic	Takes values as FULL_PLAN , INCREMENTAL_PLAN , IDENTIFY_SEGMENTS
Band Priority	MIN_MAX_BAND	Use Bands To Minimize Lateness Material Constrained Capacity Constrained Storage Constrained Unconstrained Build Late Limit	Advanced	Here
Objective Rank List	Band, Lot, Activity		Basic	Here

	Marker, Late			
Lot Value Delimiter	~	Is supply code plan	Advanced	Here
Demand Slice Threshold		Demand Slice Factor Intra Bucket Slicing Only	Advanced	Here
Split Slice Qty As Integer	False	Demand Slice Factor Intra Bucket Slicing Only	Advanced	Here
Demand RCA	False	Material Constrained Capacity Constrained Storage Constrained Build Late Limit	Basic	Here
Intermediate Material Pegging Type	None		Advanced	Here
Export Zero Inventory Material Nodes	True		Advanced	When set to False, solver skips inventory plan reporting for the material nodes with 0 as BOH & EOH throughout horizon (for all buckets). Inventory plan reported for a material node if it has at least one non zero value for BOH or EOH for a bucket.

These parameters are not exposed in Parameter UI

Parameter	Default Value	Dependencies	Category	Details
Avoid Material Reallocation	False		Advanced	When set to True, material reallocation will be turned off during demand planning
Expedite WIP	False		Advanced	Here
Ignore BAL For NoCarry Buffer	False		Advanced	When set to true, the demand build ahead limit is violated to

				minimize the no carry violation by moving in consumption earlier in time during demand planning
Inventory Plan Strategy		Inventory Plan Type	Advanced	Allowable value is MOVE_IN. When this not used, code follows default SS plan i.e. Release and Produce. In case of MOVE_IN, solver tries to do level by level pull-in of flow plans. However in default SS plan it releases the flow plan (producer) completely till upstream and then try to produce in the earlier bucket to meet the SS.
Minimize NoCarry Violations		Excess Cleanup Strategy	Advanced	When set to true, during final excess cleanup, solver tries to push the material at no carry node to downstream node by pulling in the consumer ahead in time subject to feasible constraints. This way it tries to minimize the no carry violation.
Skip All Exports	False		Debug	Used for performance testing. When set to true, no output will be exported.
Skip Delivery Plan Export	False		Debug	When set to true, delivery plan will not be exported.
Skip Inventory Plan Export	False		Debug	When set to true inventory plan will not be exported.
Skip JIT Plan	False		Debug	When set to true Solver will not do demand planning
Try Material Reallocation In RCA	False		Advanced	When set to true, material reallocation will be allowed during demand RCA phase similar to demand planning.
Write SCS Arguments	False		Debug	When Set to True it prints all the SCS Arguments in the Logs whenever solver is invoked

Write SCS Measures	False		Debug	When Set to True it prints all the SCS input and output measures in the Logs whenever solver is invoked

Time Attribute Name: SCS has the capability of planning demands on a daily, weekly, monthly, quarterly and yearly basis. However, time component of planning totally lies on the planners' discretion. Planners can thus set the unit of measure of the time bucket in *Time Attribute Name*.

If the planning to be done on a weekly basis, then, Time Attribute Name = [time].[Fiscal Week]. Other examples are [time].[Fiscal Month] and [time].[Fiscal Year].

Start Time Bucket Name: Once the value of the Time attribute name is set, be it [time].[Fiscal Week] or [time].[Fiscal Month] or [time].[Fiscal Year], it is required that the solver is provided with the time frame between which it should run and solve demands. If Time Attribute Name = [time].[Fiscal Week] then Start Time Bucket Name = <any Week value mentioned in time dimension that one wants to start with>.

If Time Attribute Name = [time].[Fiscal Month] then Start Time Bucket Name = <any Month value mentioned in time dimension that one wants to start with>

If Time Attribute Name = [time].[Fiscal Year] then Start Time Bucket Name = <any Year value mentioned in time dimension that one wants to start with>.

Current Time Bucket Name: By default, Current time bucket is the current time specified in LiveServer. It takes after the format which is in sync with the Time Attribute Name. However, while invoking the solver, user can override the default time bucket by entering any other value.

If Time Attribute Name = [time].[Fiscal Week] then Current Time Bucket Name = <any Week value mentioned in time dimension that user wants as Current Time>.

If Time Attribute Name = [time].[Fiscal Month] then Start Time Bucket Name = <any Month value mentioned in time dimension that user wants as Current Time>

If Time Attribute Name = [time].[Fiscal Year] then Start Time Bucket Name = <any Year value mentioned in time dimension that user wants as Current Time>.

End Time Bucket Name: After the start time bucket name is set, one can either complete the time frame by setting an end time bucket name to a value that is in sync with that in Start Time Bucket name or keep it as Null. In case of Null, the solver will run until the last time bucket that exists in the model

If Time Attribute Name = [time].[Fiscal Week] then End Time Bucket Name = <any Week value mentioned in time dimension that one wants to start with>.

If Time Attribute Name = [time].[Fiscal Month] then End Time Bucket Name = <any Month value mentioned in time dimension that one wants to start with>

If Time Attribute Name = [time].[Fiscal Year] then end Time Bucket Name = <any Year value mentioned in time dimension that one wants to start with>.

Note: The values mentioned in Start Time Bucket Name and end Time Bucket Name should be in Sync with the Time Attribute name that we set in the configuration. It's a mistake to set Start Time Bucket Name = W01-2015 and End Time Bucket Name = M01-2015 if the Time Attribute Name is

explicitly set to [Time].[Fiscal Month]. Start Time Bucket Name and end Time Bucket name should contain month values.

Split Slice Qty As Integer: Demand quantity will be split as whole number to the extent possible. Default is False.

Build Ahead Limit: The unit of measure is the same as that mentioned in Time Attribute Name. It means that the solver is allowed to make goods to supply for a demand upto “n” time buckets in advance (earlier than JIT bucket) where n = Build Ahead Limit.

If Time Attribute Name = [Time].[Fiscal Week] and Build Ahead Limit = 5, this means that it can make supply available for a demand upto 5 weeks in advance. Similarly it can keep the unit of measure of Build Ahead Limit as Month or Year.

A disadvantage of stating Build Ahead limit is High Inventory costs. Therefore Build Ahead Limit should be kept as small as possible unless really necessary. By Default Build Ahead limit is NULL, which means the solver can build material for supply in advance in case of material or capacity constraint.

Build Late Limit: By stating Build Late limit, the solver is allowed to delay the fulfilment of demand by the specified limit. Build Late limit is used when there is material constraints or capacity constraint (which in turn restricts Just-In-Time Production)and there is flexibility of satisfying a demand at a later date than which is originally mentioned.

If Time Attribute Name = [Time].[Fiscal Week] and Build Late Limit = 5, it means that solver can make supply available for a demand upto 5 weeks Late. Similarly the unit of measure of Build Ahead Limit can be set as Month or Year.

Inventory Plan Type: This is used to specify how much inventory is needed at unique material Node. Inventory Plan type can be specified in terms of

- a) Quantity
- b) Weeks of Supply (WOS)
- c) Quantity and Weeks of Supply (WOS)

By Default, Inventory Plan Type = NONE i.e. inventory planning is not done at material node level.

Load Downstream Supply Chain: Businesses require the SCS runs to be fast. In many cases, supply chains are structured such that only the upstream nodes can be loaded without requiring to load the downstream nodes for each of the upstream nodes. This is now possible depending on this flag. The default is true i.e. the checkbox is ticked (loads both upstream and downstream and hence the fully connected supply chain network). Set it to false to load only upstream network.

With the flag set to false

- a) Only upstream of demand node gets loaded
- b) Alternate upstream will be loaded
- c) In case of co prod, only the co prod node (1 level) is loaded
- d) In case of shared capacity, the other part is not loaded (only the path upstream of demand node loaded)
- e) The downstream of demand nodes will not be loaded

Limitation: Since the other part of the network is not loaded, WIP, SS etc specified on those nodes are not honoured

SS Build Ahead Limit: This flag determines upto how many planning buckets earlier i.e. build ahead is allowed during inventory planning (i.e. safety stock planning).

SupplyChainSolver Successfully loaded Instance 'SupplyChainSolver'.

Instance Options Parameters Node Demand Inventory Manufacturing Distribution Procurement

SS Build Ahead Limit ?

SS Build Ahead Limit	Description
NULL [Default Value]	Build ahead limit during Safety stock Planning stretches till start of Time Horizon
0	No Build Ahead During Safety Stock Planning
n	Build ahead by n buckets during Safety Stock planning. Here n stands for any positive integer like 1,2 etc

[SS Build Ahead Limit] could be used at material node grain for setting SS build ahead uniquely for a SS material node, thus having more flexibility.

Configuration

We need to create a measure at Material Node grain

Example

Material Node Parameters ? ? Cancelled edit settings for Measure 'SS BAL'.

Granularity Item Location Version Name

SS BAL

Add the measure in the Inventory Tab of SCS Config UI as follows

SupplyChainSolver Successfully loaded instance 'SupplyChainSolver'.

Instance Options Parameters Node Demand Inventory Manufacturing Distribution Procurement

SS Build Ahead Limit Measure Name ?

No Write Execution: This is used for performance evaluation to see the runtime of solver without writing the output to LiveServer. The SCS export time is consisting of following 3 steps

- Pegging creation
- Flow plan export
- Pegging export

The default of this flag is False. When this is set to True, flow plan and pegging will not be exported to LiveServer. However solver would generate the delivery plans, peggings in memory. The export time will contain only the pegging creation duration.

Excess Clean Up Frequency:

In order to provide the planner with the handle to control the stage at which the excess clean up is triggered, a new parameter known as the Excess Clean Up Frequency is now introduced.

Functionality Description

The Excess Clean Up frequency, a new parameter is now added based on its value excess clean up which is triggered at various stages during the planning. The table below displays how this parameter functions:

Excess Clean Up Frequency	What does this mean?
NONE	Excess clean up is not triggered. 
WIP	Excess clean up is triggered post the WIP Planning. 
JIT	Excess clean up is triggered post the Demand Planning. 
SS	Excess clean up is triggered post the SS Planning i.e., in the end. This is the Default Value . 
WIP,JIT	Excess clean up is triggered post the WIP Planning and the Demand Planning. 
WIP,SS	Excess clean up is triggered post the WIP Planning and the SS Planning. 
JIT,SS	Excess clean up is triggered post the JIT Planning and the SS Planning. 
WIP,JIT,SS	Excess clean up is triggered post the WIP Planning, the Demand Planning and the SS Planning. 



Below Parameters are not recommended and have been added only for Testing purpose

Excess Clean Up Frequency	What does this mean?
Demand	Excess clean up is triggered after planning each demand during demand planning
Demand_WIP	Excess clean up is triggered post the WIP Planning and after planning each demand during demand planning
Demand_SS	Excess clean up is triggered after planning each demand during demand planning and after Safety Stock Planning
Demand_WIP_SS	Excess clean up is triggered post the WIP Planning, after planning each demand during demand planning and after Safety Stock Planning

The following parameter is used to select one of the excess cleanup strategy types which is followed during the excess cleanup post SS (i.e. at the end):

- Parameter: Excess Clean Up Strategy
- Supported values: MOVE_OUT, RELEASE, NONE

At other places (post WIP, JIT) it will be RELEASE type irrespective of the above parameter setting.

The no carry violation reduction workflow gets triggered during the excess cleanup when the **Minimize No Carry Violation** is **True**. However, this is done only during the final excess cleanup i.e. post SS.

Configuration

“SS” excess cleanup frequency is the default value.

This parameter is exposed in the UI as a Multi-select drop down:

Parameter

Goto Workspace → Designer → Edit model → Rule → Plugin → Global o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameter → Excess Clean Up Frequency → select corresponding measure name(s) from drop-down menu

Excess Clean Up Frequency ⓘ

There are 3 options available on the parameter (WIP, JIT and SS).

Excess Clean Up Frequency ⓘ

8 Different values (as explained above in the table) are possible from different permutations and combinations of these 3 options. For eg:

- If the user selects JIT, then “JIT” excess cleanup frequency would be activated
- If the user selects JIT and SS, then “JIT,SS” excess cleanup frequency would be activated
- If the user selects WIP and JIT and SS, then “WIP,JIT,SS” excess cleanup frequency would be activated
- If the user deselects and doesn’t select any of the options, then “NONE” excess cleanup frequency would be activated

The plugin command example can be used to invoke the parameter.

```
exec plugin instance SupplyChainSolver for Measures {[Demand Quantity]} using scope ([Version].[Version Name].[CurrentWorkingView] * Time.[Fiscal Week] * Order.[Order Name]) arguments {[Material Constrained], "True"}, ([Capacity Constrained] , "False"), ([Delete SCS Plan Data For Version], "True"),([Excess Clean Up Frequency], "WIP")};
```

```
exec plugin instance SupplyChainSolver for Measures {[Demand Quantity]} using scope ([Version].[Version Name].[CurrentWorkingView] * Time.[Fiscal Week] * Order.[Order Name]) arguments {[Material Constrained], "True"}, ([Capacity Constrained] , "False"), ([Delete SCS Plan Data For Version], "True"),([Excess Clean Up Frequency], "WIP,SS")};
```

Node

A supply chain network is built of nodes and edges. The node tab in this form consists of all the node formats that take part in building the supply chain network in o9 Platform. The node formats are the following:

Node tab has the following fields:

Node Components	Details
Material Node Components	Can be a composite node of different dimensions. Mostly represented by Item, Location. Eg: [Item].[Item Name], [Location].[Location Name]
Activity Node Components	Can be a single (Eg: [Activity].[Activity]) or composite node. For manufacturing operations usually represented by BOM,Routing (Eg: [BOM].[BOM ID], [Routing].[Routing ID])
Capacity Node Components	Can be a composite node, but usually denoted by the resource dimension. Eg: [Resource].[Resource Name]
Storage Node Components	Can be a composite node, but usually denoted by the Storage dimension. Eg: [Storage].[Storage Name]
Demand Node Components	A composite node, which represents the demand signal.

	Usually represented by [Demand][Demand ID], [Time].[Fiscal Week], [Customer].[Customer ID]
Distribution Activity Node Components	Composite Activity Node. can be [BOD],[BOD ID], [TransMode].[ID] or just [BOD].[BOD ID].
Supplier Activity Node Components	Composite Activity Node. can be [BOD],[BOD ID], [TransMode].[ID] or just [BOD].[BOD ID].
Group Components	Composite node used for grouping. Used primarily for simultaneous capacity with [Group].[Group ID]

Demand

Demand Tab in SCS UI has the following fields:

Measure Name	Details
Inventory Demand Name	Here
Demand Qty Measure Name	Measure which specified the requested quantity of demand
Demand Fulfillment Report Name	Delivery Graph Name
Demand Fulfillment Property	Edge representing the quantity fulfilled for a particular demand
Demand Layer Measure	Measure representing priority of demand
Demand Build Ahead Limit Measure Name	Here
Demand Build Late Limit Measure Name	Here
Plan Date Offset Measure Name	Here
Demand Trace Measure Name	Here
Demand Split Measure Name	Here
Demand Target Date Measure Name	Here
Is Eligible For Incremental Plan Measure Name	Here
Total MetOnTime Quantity Measure Name	Delivery plan measure that represents the total quantity met on time for the demand
Total Short Quantity Measure Name	Delivery plan measure that represents the short quantity for a demand

Total Late Quantity Measure Name	Delivery plan measure that represents the short quantity for a demand
Weighted Late Quantity Measure Name	Delivery plan measure that represents the short quantity for a demand
Maximum Lateness Buckets Measure Name	Delivery plan measure determine the number of buckets the order is late
Commit Date Measure Name	Here
Total Time Taken By Demand Measure Name	Here
Total Tries Measure Name	Here
Total Time Taken By Delay Demand Measure Name	Here
Re-Planned By Interactive Plan Measure Name	Here
Cascading RCA Limit Measure Name	Here
Acceptable Lot Sequence Measure Name	Here
Demand Segment Measure Name	Here
Plan Min Time During Demand Run Measure Name	Here
Fulfilment Policy Measure Name	Here
Demand Lot Size Multiple Measure Name	Here

Total Time Taken By Demand Measure Name:Delivery Plan Measure determine the total time taken by a demand to get Planned. The unit of measure is Milliseconds by default.

Total Tries Measure Name:Delivery Plan Measure determine the total number of tries during delay demand planning. The unit of measure is Milliseconds by default.

Total Time Taken By Delay Demand Measure Name:Delivery Plan Measure determine the total time taken delay demand planning for a demand. The unit of measure is Milliseconds by default.

Build Ahead Limit: The unit of measure is the same as that mentioned in Time Attribute Name. It means that the solver is allowed to make goods to supply for a demand upto “n” time buckets in advance (earlier than JIT bucket) where n = Build Ahead Limit. If Time Attribute Name = [Time].[Fiscal Week] and Build Ahead Limit = 5, this means that it can make supply available for a demand upto 5 weeks in advance. Similarly it can keep the unit of measure of Build Ahead Limit as Month or Year.

A disadvantage of stating Build Ahead limit is High Inventory costs. Therefore Build Ahead Limit should be kept as small as possible unless really necessary. By Default Build Ahead limit is NULL, which means the solver can build material for supply in

advance in case it cannot build an order Just-In-Time due to material and/or capacity constraint.

Build Late Limit: By stating Build Late limit, the solver is allowed to delay the fulfilment of demand by the specified limit. Build Late limit is used when there is material constraint or capacity constraint (which in turn restricts Just-In-Time Production) and there is flexibility of satisfying a demand at a later date than which is originally mentioned.

If Time Attribute Name = [Time].[Fiscal Week] and Build Late Limit = 5, it means that solver can make supply available for a demand upto 5 weeks Late. Similarly the unit of measure of Build Ahead Limit can be set as Month or Year.

Default value of Build Late Limit is NULL. This means that if an order can't be built in JIT bucket, then SCS will try to fulfil the same demand by JIT+1 bucket. If that is not possible, then SCS tries to build the same order by JIT+2 buckets and so on until it reaches the end of the planning horizon. Therefore when Build Late Limit is Null, planners can assume that SCS will build an order n buckets late, where n tends to be infinite/very large number.

Local setting for build late limit takes precedence over the global setting for build late limit. This means if a demand has its own build late limit specified by the planner, the global setting of the same order is waived by SCS.

Inventory

The screenshot shows the 'Inventory' tab of the 'SupplyChainSolver' application. The tab is selected among 'Instance Options', 'Node', 'Demand', 'Parameters', 'Manufacturing', 'Distribution', and 'Procurement'. The 'Inventory' tab contains several fields, each with an information icon (i) to its left:

- BOH Activity**: A text input field containing 'BOH Activity'.
- Expected Receipt Activity Name**: A text input field containing 'Expected Receipt Activity'.
- Expected Receipt Measure Name**: A dropdown menu with 'Expected Receipts' selected.
- BOH Measure Name**: A dropdown menu with 'BOH' selected.
- Constant WOS Target Measure Name**: A dropdown menu with 'Constant WOS Targ' selected.
- WOS Target Measure Name**: A dropdown menu with 'IL WOS Target' selected.
- WOS Reorder Point Measure Name**: A dropdown menu with 'IL WOS Reorder Point' selected.
- Constant Target Inventory Measure**: A dropdown menu with 'Constant Target Inventory' selected.

At the bottom right of the form, there are two buttons: 'Update' (with a checkmark icon) and 'Cancel' (with a circle and slash icon).

It has the following fields:

Argument Name	Default Name	Input type
1. BOH Activity *	BOH Activity	TextBox
2. Expected Receipt Activity Name *	Expected Receipt Activity	TextBox

3. Expected Receipt Measure Name *	Expected Receipts	DropDown Menu
4. BOH Measure Name *	BOH	DropDown Menu
5. Constant WOS Target Measure Name	Constant WOS Target	DropDown Menu
6. WOS Target Measure Name	IL WOS Target	DropDown Menu
7. Constant Target Inventory Measure Name	Constant Target Inventory	DropDown Menu
8. Target Inventory Measure Name	IL Target Inventory	DropDown Menu
9. Has Infinite Inventory Measure Name	Infinite Inventory	DropDown Menu
11. Beginning On Hand Material Quantity Measure Name	Beginning On Hand Material Quantity	DropDown Menu
12. Ending On Hand Material Quantity Measure Name	Ending On Hand Material Quantity	DropDown Menu
13. Lot Attribute Name	[Lot].[Lot ID]	Text Box

BOH Activity: It is a production activity. This activity produces inventory at a material node in the form of Beginning on Hand inventory. By default it is named as “BOH activity”. BOH activity is a required member in the Activity Dimension.

Expected Receipts Activity Name: This is a production activity. This means that this produces inventory at a material node in the form of Expected Receipts Inventory. By default it is named as “Expected Receipt Activity”. Expected Receipt Activity is a required member in the Activity Dimension.

Expected Receipt Measure Name: This measure name is defined for the Inventory Graph table. By default it is named as “Expected Receipts”. It stores the quantity of material that is expected at a unique [Item].[Location] in a specific Time Bucket.

BOH Measure Name: BOH stands for beginning on hand. This measure name is also defined in the Inventory Graph table. By default it is named as BOH. It stores the quantity of material a unique [Item].[Location] in Current Time bucket that is physically ready to be shipped to interested customers.

Constant WOS Target Measure Name: WOS is an acronym for Weeks of Supply. This measure refers to the WOS target Measure which is constant over time. This measure will not be grained by time member.

WOS Target Measure Name: This measure refers to the WOS target Measure which varies over time for same/different material nodes.

Constant Target Inventory Measure Name: It is the Inventory Quantity target Measure name which is constant over time. This measure won't be grained by time member.

Target Inventory Measure Name: The measure that stores the amount of safety stock a unique [Item].[Location] MUST HAVE at the specified time bucket even after the solver is done planning for the demands. For example, If planners want 100 units of FG_1 at Plant_1 on W04, in the form of stock, they can set the target inventory measure name corresponding to this item , location and time bucket to 100.

Manufacturing

SupplyChainSolver ×

Instance Options
Node
Demand
Inventory
Parameters
Manufacturing
Distribution
Procurement

Lead Time Measure Name ⓘ

Material Production Plan Measure Name ⓘ

Material Consumption Plan Measure Name ⓘ

Capacity Consumption Plan Measure Name ⓘ

Storage Plan Measure Name ⓘ

Material Production Pegging Graph Name ⓘ

Material Production Pegging Property ⓘ

Activity Lead Time ▼

Material Quantity ▼

Material Consumption Quantity ▼

Capacity Consumption Plan ▼

Storage Plan ▼

Production_Pegging ▼

Material Production Pegged Quantity ▼

✓ Update
⊗ Cancel

This tab has the following fields:

The default name here in the table are the relevant measures for the corresponding argument. These measures need to be created before so that you can see those available in the drop down menu to select appropriately.

Argument Name	Default Name	Input type
1. Lead Time Measure Name*	Activity Lead Time	DropDown Menu
2. Material Production Plan Measure Name*	Material Quantity	DropDown Menu
3. Material Consumption Plan Measure Name*	Material Consumption Quantity	DropDown Menu

4. Capacity Consumption Plan Measure Name	Capacity Consumption Plan	DropDown Menu
5. Material Production Pegging Graph Name	Production_Pegging	DropDown Menu
6. Material Production Pegging Property	Material Production Pegging Property	DropDown Menu
7. Material Production Constraint Type Property	MaterialConstraintType	DropDown Menu
8. Material Production Constraint Qty Property	Material Production Constraint Qty Property	DropDown Menu
9. Production Graph Measure Name	Material Production Graph Association	DropDown Menu
10. Prod Graph Yield Measure Name	Material Production Qty Per	DropDown Menu
11. Prod Graph Priority Measure Name	Material Production Priority	DropDown Menu
12. Prod Graph Min Qty Measure Name	Material Production Min Qty	DropDown Menu
13. Prod Graph Mul Qty Measure Name	Material Production Multiple Qty	DropDown Menu
14. Prod Graph Planned WIP Measure Name	Material Production Planned WIP	DropDown Menu
15. Prod Graph Locked WIP Measure Name	Material Production Locked WIP	DropDown Menu
16. Prod Graph No Build Measure Name	No Build	DropDown Menu
17. Prod Graph Time Yield Measure Name	Time Varying Material Production	DropDown Menu
18. Prod Graph Holiday Measure Name	Production Holiday	DropDown Menu
19. Material Consumption Pegging Graph Name	Consumption_Pegging	DropDown Menu
20. Material Consumption Pegging Property	Material Consumption Pegging Property	DropDown Menu
21. Consumption Graph Measure Name	Material Consumption	DropDown Menu

	Graph Association	
22. Consumption Graph Qty Per Measure Name	Material Consumption Qty Per	DropDown Menu
23. Consumption Graph Planned WIP Measure Name	Material Consumption Planned WIP	DropDown Menu
24. Consumption Graph Locked WIP Measure Name	Material Consumption Locked WIP	DropDown Menu
25. Consumption Graph Time Qty Per Measure Name	Time Varying Material Consumption	DropDown Menu
26. Consumption Graph Holiday Measure Name	Consumption Holiday	DropDown Menu
27. Capacity Consumption Pegging Graph Name	Capacity_Pegging	DropDown Menu
28. Capacity Consumption Pegging Property	Capacity Consumption Pegging Property	DropDown Menu
29. Capacity Consumption Constraint Type Property	Capacity Consumption Constraint Type Property	DropDown Menu
30. Capacity Consumption Constraint Qty Property	Capacity Consumption Constraint Qty Property	DropDown Menu
31. Capacity Graph Measure Name	Capacity Consumption Graph Association	DropDown Menu
32. Capacity Graph Qty Per Measure Name	Capacity Consumption Qty Per	DropDown Menu
33. Capacity Graph Priority Measure Name	Capacity Graph Priority	DropDown Menu
34. Capacity Graph Planned WIP Measure Name	Capacity Consumption Planned WIP	DropDown Menu DropDown Menu
35. Capacity Graph Locked WIP Measure Name	Capacity Consumption Locked WIP	DropDown Menu

36. Capacity Graph Time Qty Per Measure Name	Time Varying Capacity Consumption	DropDown Menu
37. Simultaneous Capacity Consumption Graph Association Measure Name	Simultaneous Cap Cons Association	DropDown Menu
38. Start Date Measure Name		DropDown Menu
39. End Date Measure Name		DropDown Menu
40. Activity Marker Measure Name		DropDown Menu
41. Activity Freeze Time Bucket Offset Measure Name	Activity Freeze Time Bucket Offset	DropDown Menu
42. Is Holiday Measure Name	Is Holiday	DropDown Menu
43. Demand Activity Priority Measure Name	RevisedPriority	DropDown Menu
44. Aval Cap Measure Name	Capacity Availability	DropDown Menu
45. Time Dep Aval Cap Measure Name		DropDown Menu
46. Activity Proportion Measure Name	Split Ratio	DropDown Menu

Distribution

SupplyChainSolver ×

Instance Options
Node
Demand
Inventory
Parameters
Manufacturing
Distribution
Procurement

Distribution Lead Time Measure Name ?

▼

Distribution Material Production Plan Measure Name ?

▼

Distribution Material Consumption Plan Measure Name ?

▼

Distribution Capacity Consumption Plan Measure Name ?

▼

Distribution Material Production Pegging Graph Name ?

▼

Distribution Material Production Pegging Property ?

▼

Distribution Material Production Constraint Type Property

▼

✓ Update
⌂ Cancel

This tab has the following fields:

Argument Name	Default Name	Input type
1. Distribution Lead Time Measure Name	Distribution LeadTime	DropDown Menu
2. Distribution Material Production Plan Measure Name	Distribution Production Quantity	DropDown Menu
3. Distribution Material Consumption Plan Measure Name	Distribution Consumption Quantity	DropDown Menu
4. Distribution Capacity Consumption Plan Measure Name	Dist Cap Cons Qty	DropDown Menu
5. Distribution Material Production Pegging Graph Name	Dist_Production_Pegging	DropDown Menu
6. Distribution Material Production Pegging Property	Distribution Production Pegged Quantity	DropDown Menu
7. Distribution Material Production Constraint Type Property		DropDown Menu

8. Distribution Material Production Constraint Qty Property		DropDown Menu
9. Distribution Production Graph Measure Name	Dist Prod Graph Association	DropDown Menu
10. Distribution Prod Graph Yield Measure Name		DropDown Menu
11. Distribution Prod Graph Min Qty Measure Name		DropDown Menu
12. Distribution Prod Graph Mul Qty Measure Name		DropDown Menu
13. Distribution Prod Graph Priority Measure Name	Distribution Priority	DropDown Menu
14. Distribution Prod Graph Planned WIP Measure Name		DropDown Menu
15. Distribution Prod Graph Locked WIP Measure Name		DropDown Menu
16. Distribution Prod Graph Holiday Measure Name		DropDown Menu
17. Distribution Material Consumption Pegging Graph Name	Dist_Consumption_Pegging	DropDown Menu
18. Distribution Material Consumption Pegging Property	Distribution Consumption Pegged Quantity	DropDown Menu
19. Distribution Consumption Graph Measure Name	Dist Cons Graph Association	DropDown Menu
20. Distribution Consumption Graph Qty Per Measure Name		DropDown Menu
21. Distribution Consumption Graph Planned WIP Measure Name		DropDown Menu
22. Distribution Consumption Graph Locked WIP Measure Name		DropDown Menu
23. Distribution Consumption Graph Holiday Measure Name		DropDown Menu

24. Distribution Capacity Consumption Pegging Graph Name	Dist_Capacity_Pegging	DropDown Menu
25. Distribution Capacity Consumption Pegging Property	DCC Pegged Qty	DropDown Menu
26. Distribution Overtime1 Capacity Consumption Pegging Property		DropDown Menu
27. Distribution Overtime2 Capacity Consumption Pegging Property		DropDown Menu
28. Distribution Capacity Consumption Constraint Type Property		DropDown Menu
29. Distribution Capacity Consumption Constraint Qty Property		DropDown Menu
30. Distribution Capacity Graph Measure Name	DistCapCons Association	DropDown Menu
31. Distribution Capacity Graph Qty Per Measure Name	DistCapCons Qty Per	DropDown Menu
32. Distribution Capacity Graph Priority Measure Name		DropDown Menu
33. Distribution Resource Category Measure Name		DropDown Menu
34. Distribution Capacity Graph Planned WIP Measure Name		DropDown Menu DropDown Menu
35. Distribution Capacity Graph Locked WIP Measure Name		DropDown Menu
36. Simultaneous Distribution Capacity Consumption Graph Association Measure Name		DropDown Menu
37. Distribution Start Date Measure Name		DropDown Menu

38. Distribution End Date Measure Name		DropDown Menu
39. Distribution Activity Marker Measure Name		DropDown Menu
40. Distribution Activity Freeze Time Bucket Offset Measure Name		
41. Is Distribution Holiday Measure Name		
42. Distribution Activity Segment Measure Name		
43. Demand Distribution Activity Priority Measure Name		
44. Demand Distribution Activity Start Date Measure Name		
45. Demand Distribution Activity End Date Measure Name		
46. Distribution Activity Proportion Measure Name		

Procurement

SupplyChainSolver ×

Instance Options
Node
Demand
Inventory
Parameters
Manufacturing
Distribution
Procurement

Supplier Lead Time Measure Name ⓘ

Supplier Material Production Plan Measure Name ⓘ

Supplier Material Consumption Plan Measure Name ⓘ

Supplier Capacity Consumption Plan Measure Name ⓘ

Supplier Material Production Pegging Graph Name ⓘ

Supplier Material Production Pegging Property ⓘ

Supplier Material Production Constraint Type Property ⓘ

✓ Update
⊘ Cancel

This tab has the following fields:

Argument Name	Default Name	Input type
1. Supplier Lead Time Measure Name	Supplier LeadTime	DropDown Menu
2. Supplier Material Production Plan Measure Name	Supply Produced	DropDown Menu
3. Supplier Material Consumption Plan Measure Name	Supply Consumed	DropDown Menu
4. Supplier Capacity Consumption Plan Measure Name	Supplier Capacity Consumed	DropDown Menu
5. Supplier Material Production Pegging Graph Name	Supplier_Production_Pegging	DropDown Menu
6. Supplier Material Production Pegging Property	Supplier Production Pegged Quantity	DropDown Menu
7. Supplier Material Production Constraint Type Property		DropDown Menu

8. Supplier Material Production Constraint Qty Property		DropDown Menu
9. Supplier Production Graph Measure Name	Supp Prod Association	DropDown Menu
10. Supplier Prod Graph Yield Measure Name		DropDown Menu
11. Supplier Prod Graph Min Qty Measure Name		DropDown Menu
12. Supplier Prod Graph Mul Qty Measure Name		DropDown Menu
13. Supplier Prod Graph Priority Measure Name	Supp Prod Priority	DropDown Menu
14. Supplier Prod Graph Planned WIP Measure Name		DropDown Menu
15. Supplier Prod Graph WIP Change Measure Name		DropDown Menu
16. Supplier Activity Mandatory Prebuild Offset Measure Name		DropDown Menu
17. Supplier Consume Outbound Capacity for WIP Measure Name		DropDown Menu
18. Supplier Prod Graph Locked WIP Measure Name		DropDown Menu
19. Supplier Prod Graph Holiday Measure Name		DropDown Menu
20. Supplier Material Consumption Pegging Graph Name	Supplier_Consumption_Pegging	DropDown Menu
21. Supplier Material Consumption Pegging Property	Supplier Consumption Pegged Quantity	DropDown Menu
22. Supplier Consumption Graph Measure Name	Supp Cons Association	DropDown Menu
23. Supplier Consumption Graph Qty Per Measure Name		DropDown Menu
24. Supplier Consumption Graph Planned WIP Measure Name		DropDown Menu

25. Supplier Consumption Graph WIP Change Measure Name		DropDown Menu
26. Supplier Consumption Graph Locked WIP Measure Name		DropDown Menu
27. Supplier Consumption Graph Holiday Measure Name		DropDown Menu
28. Supplier Capacity Consumption Pegging Graph Name	Supplier_Capacity_Consumption_Pegging	DropDown Menu
29. Supplier Capacity Consumption Pegging Property	SCC Pegged Quantity	DropDown Menu
30. Supplier Overtime1 Capacity Consumption Pegging Property		DropDown Menu
31. Supplier Overtime2 Capacity Consumption Pegging Property		DropDown Menu
32. Supplier Capacity Consumption Constraint Type Property		DropDown Menu
33. Supplier Capacity Consumption Constraint Qty Property		DropDown Menu
34. Supplier Capacity Graph Measure Name	Supp Cap Cons Association	DropDown Menu
35. Supplier Capacity Graph Qty Per Measure Name		DropDown Menu
36. Supplier Capacity Graph Priority Measure Name		DropDown Menu
37. Supplier Resource Category Measure Name		DropDown Menu
38. Supplier Capacity Graph Planned WIP Measure Name		DropDown Menu
39. Supplier Capacity Graph WIP Change Measure Name		DropDown Menu

40. Supplier Capacity Graph Locked WIP Measure Name		DropDown Menu
41. Simultaneous Supplier Capacity Consumption Graph Association Measure Name		DropDown Menu
42. Supplier Start Date Measure Name		DropDown Menu
43. Supplier End Date Measure Name		DropDown Menu
44. Supplier Activity Marker Measure Name		DropDown Menu
45. Supplier Activity Freeze Time Bucket Offset Measure Name		DropDown Menu
46. Is Supplier Holiday Measure Name		DropDown Menu
47. Supplier Activity Segment Measure Name		DropDown Menu
48. Demand Supplier Activity Priority Measure Name		DropDown Menu
49. Demand Supplier Activity Start Date Measure Name		DropDown Menu
50. Demand Supplier Activity End Date Measure Name		DropDown Menu
51. Supplier Activity Proportion Measure Name		DropDown Menu
52. Supplier Load Plan Graph Name		DropDown Menu
53. Supplier Load Plan Capacity		DropDown Menu

Consumption Property Name		
54. Supplier Load Plan Material Production Property Name		DropDown Menu

Supply Chain Data Model

The tenant and the login details are provided below

Tenant Name: SCS-o9-2019_0.com

User ID: etluser@scs-o9-2019_0.com

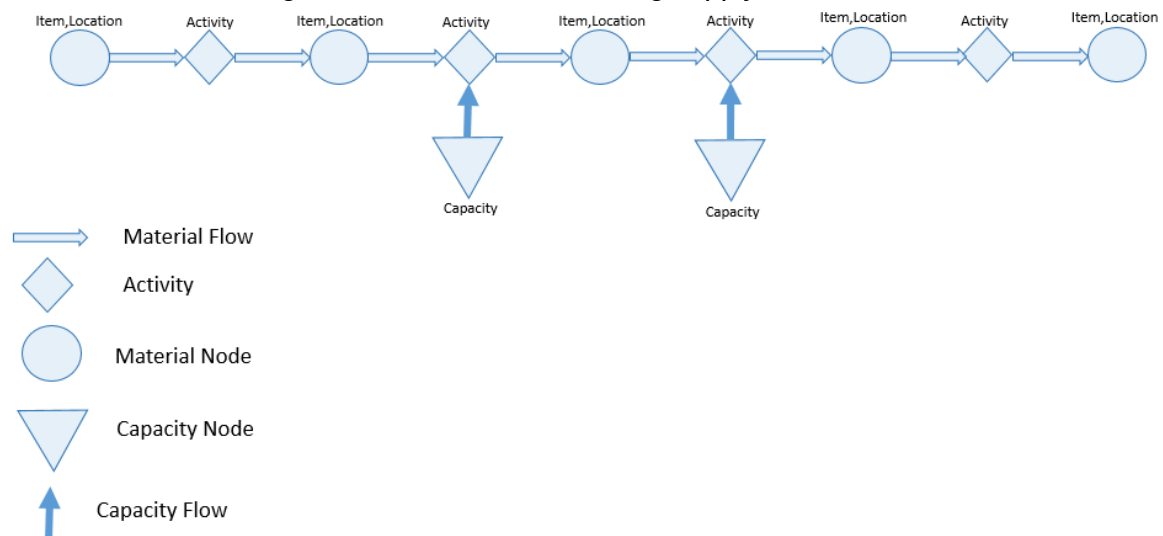
Password: Welcome123

Environment : [Sandbox](#)

Assumption: User knows basic tenant terminologies like Plan, Models, layout, deployment etc.

Link to [Platform Guide](#)

For our understanding, let us assume the following supply chain network.



Details on node definitions and flow of material and capacity can be found [here](#)

Dimensions needed for SCS-Plugin

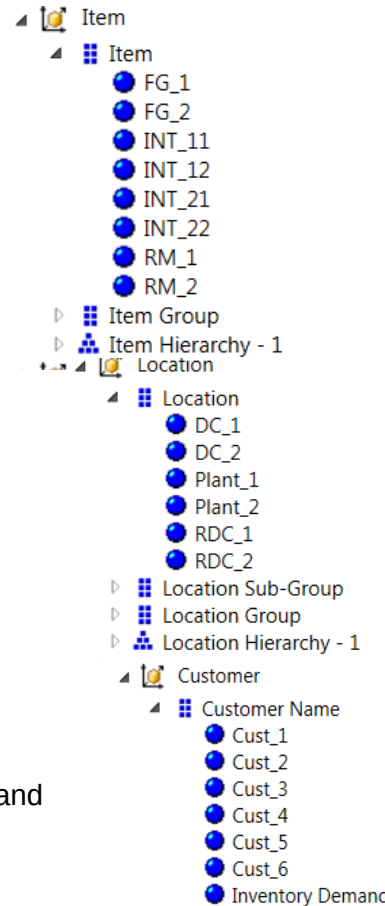
Let us now understand the supply chain model maintained in scs-o9-2019_0.com tenant. There are two raw materials required at the plant which are procured from suppliers. It is processed in the plant and converted to intermediate parts. These parts are then converted to finished goods, using the manufacturing process. After the finished goods are produced inside the plant, they are shipped from plant to Distribution centres via Air or Ocean modes of transport. After they reach Distribution centres, they get shipped to Regional Distribution Centres again via Air or Ocean. Finally from Regional Distribution

Centres these finished goods get to the Customer. Demand can be placed at any material node (Item, Location composite) in the supply chain.

The Supply chain network master data for this tenant is as follows

1. Item

- FG_1
- FG_2
- INT_11
- INT_12
- INT_21
- INT_22
- RM_1
- RM_2



2. Location

- DC_1
- DC_2
- Plant_1
- Plant_2
- RDC_1
- RDC_2

3. Customer

- Cust_1
- Cust_2
- Inventory Demand

4. Time

- Has time data that spans for from 2012 to 2015

Dimension.DimTime.c
SV

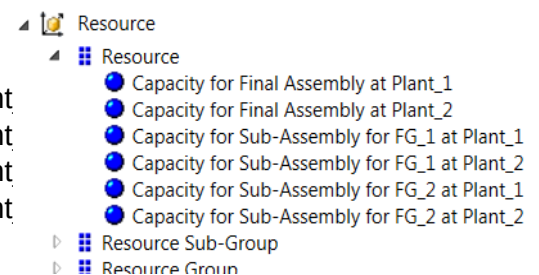
5. Activity

- 54 activities that cover the whole supply chain network

Dimension.DimActivit
y.csv

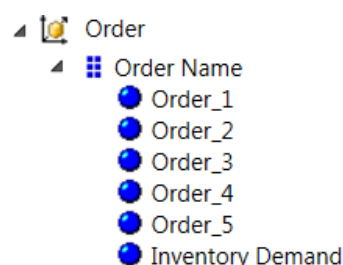
6. Resource

- Capacity for Final Assembly of FG_1
- Capacity for Final Assembly of FG_2
- Capacity for Sub-Assembly for FG_1 at Plant
- Capacity for Sub-Assembly for FG_1 at Plant
- Capacity for Sub-Assembly for FG_2 at Plant
- Capacity for Sub-Assembly for FG_2 at Plant



7. Order

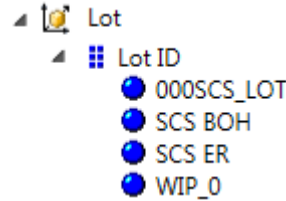
- Order_1



- Order_2
- Order_3
- Order_4
- Order_5
- Inventory Demand

8. Lot

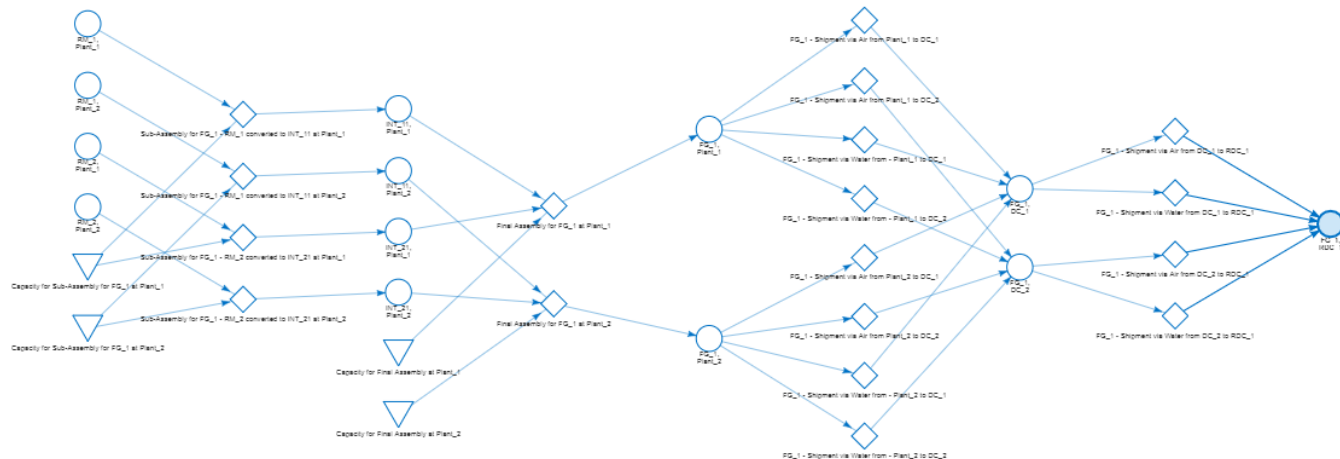
- 000SCS_LOT
- SCS BOH
- SCS ER
- SCS WIP_0



Note: The above screen shots give a minimalistic view of the member data of tenant scs-o9-2019_0. To accommodate other features of SCS, corresponding dimensions and their member data have also been added, all of which can be seen in sandbox and sandbox2.

Note: Inventory Demand is a MUST HAVE member of all the dimensions present in the demand signal. In the above tenant, it is a must have in Order and Customer dimension. Its value is internal to the working of the plugin, so that remaining inventory can be pegged to.

How to Design SCS-o9 Supply Chain Network



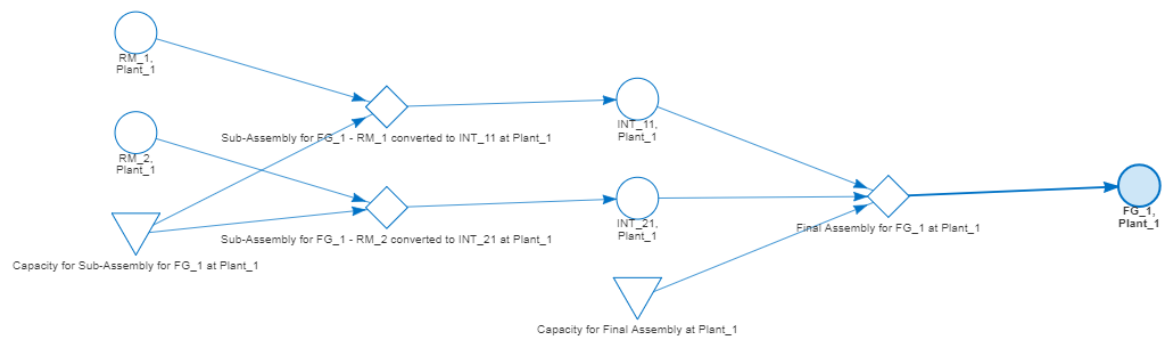
It is in the homogenous supply chain workspace. Please click the [link](#) to see the network (Login required)

The above shown supply chain can be broken down into two parts

- Manufacturing
- Distribution

Note: In general, apart from manufacturing and distribution, there will be procurement, to indicate the supply of raw material or components into the factory

Manufacturing



Description:

In the above diagram,

{RM_1} is converted to {INT_11} at {Plant_1} via Activity {Sub-Assembly for FG_1 – RM_1 converted to INT_11 at Plant_1}

{RM_2} is converted to {INT_21} at {Plant_1} via Activity {Sub-Assembly for FG_1 – RM_2 converted to INT_21 at Plant_1}

The activities:

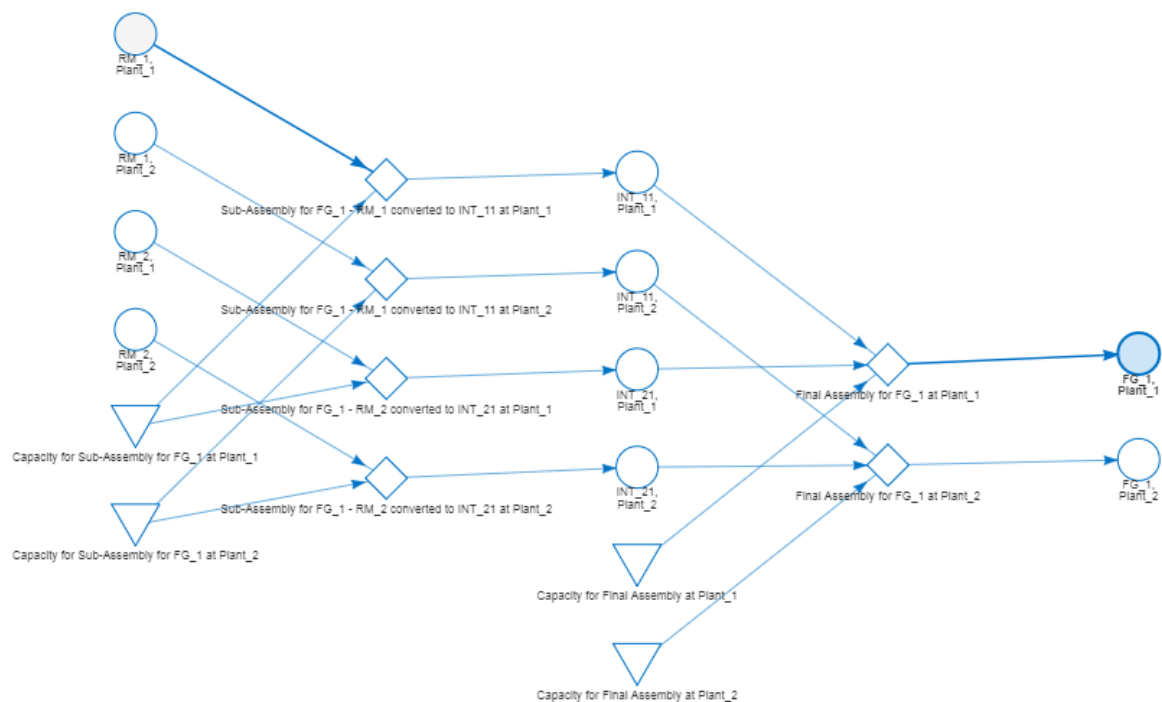
- {Sub-Assembly for FG_1 – RM_1 converted to INT_11 at Plant_1}
- And {Sub-Assembly for FG_1 – RM_2 converted to INT_21 at Plant_1}

Share the following capacity

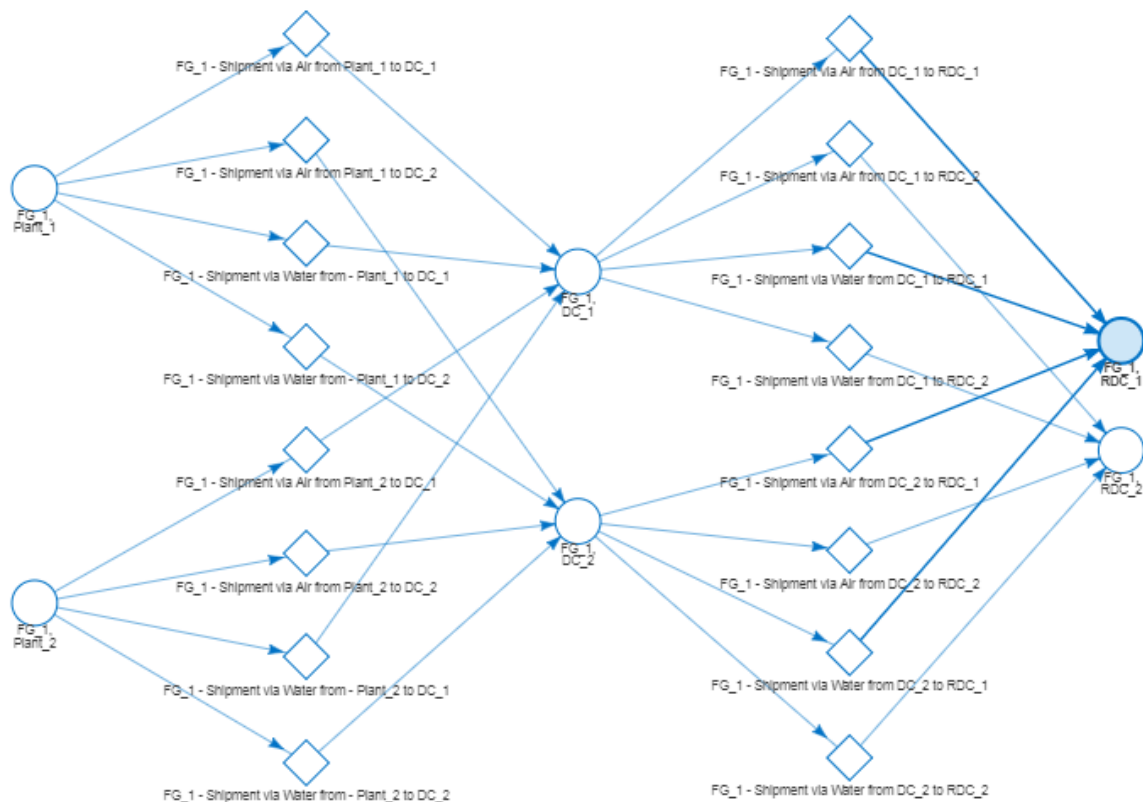
- Capacity for Sub-Assembly for FG_1 at Plant_1

Intermediate products INT_11 and INT_21 are converted to FG_1 via Activity { Final Assembly for FG_1 at Plant_1}

Note: All this manufacturing takes place in Pant_1. Similarly we have a manufacturing process for FG_1 at Plant_2. A consolidated Manufacturing Diagram for FG_1 in both Plant_1 and Plant_2 is provided below.



Distribution



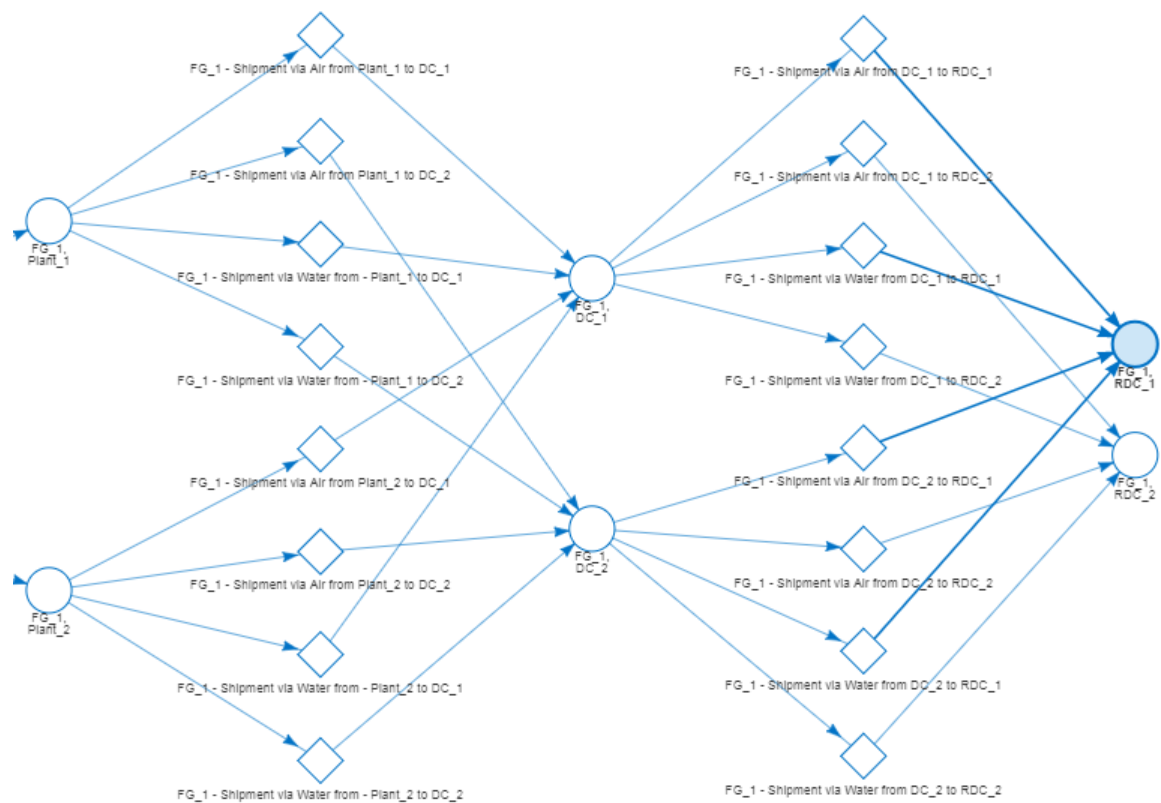
The flow of Finished Goods takes place across different locations through various activities before it can reach the final Customer as desired. FG_1 can be shipped from Plant_1 to DC_1 via Air. The activity performed to do so is named as {FG_1 - Shipment via Air from Plant_1 to DC_1}

Note: Similarly FG_1 can be shipped from Plant_1 to DC_2 via Air. The activity performed to do so will be named as {FG_1 - Shipment via Air from Plant_1 to DC_2}

FG_1 can be shipped from DC_1 to RDC_1 via Air. The activity performed to do so is named as {FG_1 - Shipment via Air from DC_1 to RDC_1}

Similarly we can connect all item locations with each other. We can have various modes of shipment i.e AIR, WATER and/or LAND. Accordingly, we need to maintain the activities.

A consolidated distribution unit that can enable shipment of FG_1 to RDC_1 through various Plants and DCs looks as follows:



Note : Unique Plants, DCs , RDCs are connected to each other via Unique Activities.

Nodes required in SCS-o9 Supply Chain Network

There are following node types

1. Mandatory
 - a. Material node
 - b. Demand node
 - c. Activity node
2. Optional
 - a. Capacity node
 - b. Storage node

1) Material Node



A Material Node is an Inventory carrying node in the supply chain, and can be represented by composite (many) dimensions. Often, Material Node is represented by the combination of two dimension. [Item].[Item Name] and [Location].[Location Name]. It is represented in the Supply Chain Configuration in the same manner i.e [Item].[Item Name], [Location].[Location Name]

Example: [Item].[FG_1], [Location].[RDC_1]

2) Demand Node



A demand Node is the requesting entity for the demand. Node is a misnomer here. Demand node is defined as a demand signal incident on a material node
Demand Node is a composite Node consisting of the following dimensions

- Mandatory(in that order)
 - Order
 - Time

Demand Node Components ⓘ

[Order].[Order Name], [Time].[Fiscal Week]

- Optional
 - Customer

Demand Node Components ⓘ

[Order].[Order Name], [Customer].[Customer Name], [Time].[Fiscal Week]

{Order} is a participating dimension in the representation of the demand Node. It can either represent a specific Order (e.g have a Order-ID obtained from transactional systems), or it could be logical demand type, such as "Forecast", or multiple such logical types, such as "Committed Forecast" vs "Upside Forecast". Having this dimension present as a participating dimension to the demand allows us to represent multiple demand types for the same demand node, and for the same time bucket.

Note: The name of order dimension may vary from one tenant to another. In case of SCS-o9-hm ,the name of the order dimension is kept as Order.

Representation of Demand

A given demand can be represented through a combination of the following 2 nodes

- Demand Node
 - Demand node consists of dimension Order, Time and Customer(optional)
- Material Node
 - Material node consists of dimension Item and Location

Example :

[Order].[Order_1], [Time].[W09-2015], [Item].[FG_1], [Location].[RDC_1] demand quantity = 100, Demand Priority = 1

could be read as

There is a demand with demand ID = Order_1, which is due on Week 9 2015 . This demand is for 100 units of item FG_1 at location RDC_1. This demand is of the highest priority

[Order].[Order_1], [Time].[W09-2015], [Customer].[Cust_1], [Item].[FG_1], [Location].[RDC_1] demand quantity = 100, Demand Priority = 1.

could be read as

There is a demand with demand ID = Order_1, which is due on Week 9 2015 for Customer named Cust_1. This demand is for 100 units of item FG_1 at location RDC_1. This demand is of the highest priority

Note : the order dimension has to have a mandatory entry of Order Name = Inventory Demand, irrespective of the number of orders any business model has.

3) Activity Node



An Activity Node represents the required activity to either transform or transport material between two material nodes. This can also be a composite node. The Activity node(s) can also be connected to Capacity node(s). Activity node is represented as [Activity].[Activity Name].

[Activity].[FG_1 - Shipment via Air from DC_1 to RDC_1]

[Activity].[Final Assembly of FG_1 at Plant_1]

Note: Activity node can also be a composite of 2 or more dimensions. Eg: BOM, Routing

4) Capacity Node



A capacity node represents the resource required to carry out any supply chain activity. Capacity Node can be represented by composite dimensions. Here we just represent it by [Resource].[Resource] for simplicity. An example is subassembly of components for a laptop might need a machine.

Example : [Resource].[Capacity for Sub-Assembly for FG_1 at Plant_1]

[Resource].[Capacity for Sub-Assembly for FG_1 at Plant_2]

Therefore the Master data Configuration of the Supply Chain solver requires the following dimensions at the very least:

- 1) Material node
 - a) Item
 - b) Location
- 2) Activity node
 - a) Activity
- 3) Capacity node
 - a) Resource
- 4) Storage Node
 - a) Storage
- 5) Demand Node
 - a) Order
 - b) Time
 - c) Customer (optional)

Note: For the Capacity to be considered during planning, Capacity node components should be explicitly defined in the SCS config. If not defined, SCS will simply ignore the capacity during planning. If capacity node is configured in SCS and corresponding plan and peggings are not specified in the same config then SCS gives an error.

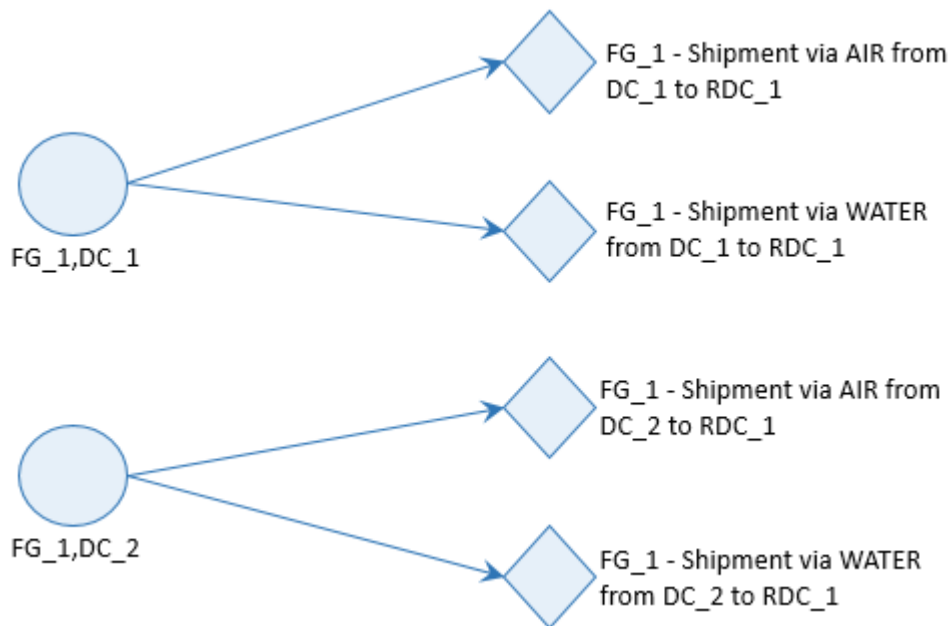
Data for Dimension, Fact and Graphs are available on [google drive](#)

Designing SCS-o9 Supply Chain Network through Graphs

After the master data is created, one can move to building the supply chain network. In order to do so, the above dimensions need to be linked with each other. This can be done through Graphs. Graphs can also be represented by association tables. One can create association across various nodes in the supply chain network through graph tables. Following are the 3 graph tables that can be used as input to the solver.

- Material Consumption Graph
- Material Production Graph
- Capacity Consumption Graph

1. Material Consumption Graph



Material consumption graph denotes if an activity is consuming any product at any Location.

It consists of 4 dimensions:

1. Activity Node
 - a. Activity.[Activity]
2. Material Node
 - a. Item.[Item]
 - b. Location.[Location]
3. Version.[Version Name]

and 2 measures:

a) Material Consumption Graph Association: Material Consumption Graph Association takes Boolean values 1/0.

1 means that the corresponding edge(in this case represented as a measure, which connects the activity node and material node) is active in the supply chain network.

0 means that the corresponding edge(in this case represented as a measure, which connects the activity node and material node) is inactive in the supply chain network.

b) Material Consumption Qty Per: It is a multiplier and takes numeric values. It denotes the number of items a particular activity consumes at that location.

2. Material Production Graph

Final Assembly for
FG_1 at Plant_1



Material Production Graph helps us know if an activity is producing any product at a Location.

It consists of 4 dimensions

1. Activity Node
 - a. Activity.[Activity]
2. Material Node
 - a. Item.[Item]
 - b. Location.[Location]
3. Version.[Version Name]

and 3 measures:

a) Material Production Graph Association: Material Production Graph Association takes Boolean values 1/0.

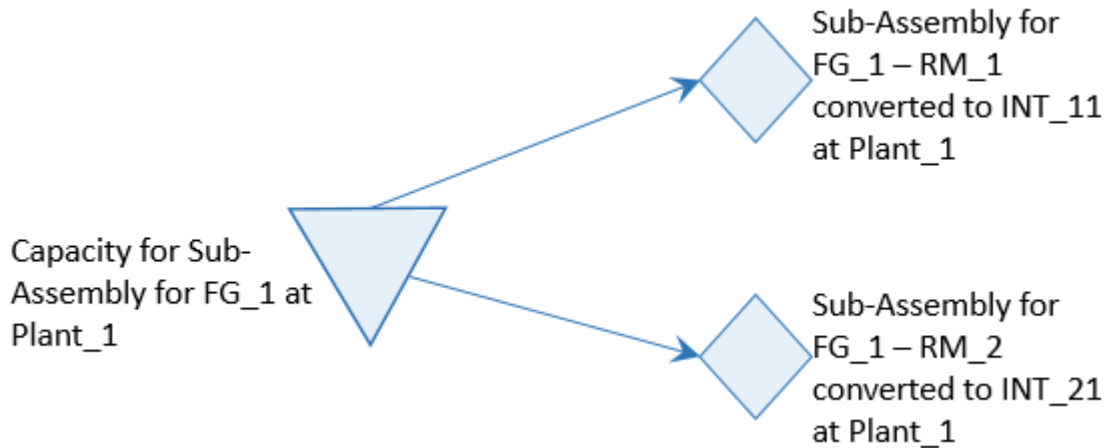
1 means that the corresponding edge(in this case represented as a measure, which connects the activity node and material node) is active in the supply chain network.

0 means that the corresponding edge(in this case represented as a measure, which connects the activity node and material node) is inactive in the supply chain network.

b) Material Production Qty Per: It is a multiplier and takes numeric values. It represents the number of items an activity produces at that location.

c) Material Production Priority: There may be various activities producing onto a material Node. For Eg. Plant_1 and Plant_2 both can ship FG_1 to DC_1 and DC_2. In that case, through Material Production Priority we specify to the solver which activity takes precedence over others. Material Production Priority takes numbers as input. Lower number denotes higher priority.

3. Capacity Consumption Graph



Sometimes certain activities require resources to complement them. Eg. As sub-assembly activity, which assembles various components into one finished product might require usage of certain machines to finish the task. That machine is known as a resource and configured in the Resource dimension. A resource has some capacity, which implies the quantity of material it can produce for the activity it is linked to.

Capacity Consumption Graph has 3 Dimensions

1. Activity Node
 - a. Activity.[Activity]
2. Capacity Node
 - a. Resource.[Resource]
3. Version.[Version Name]

And 2 Measures

a) Capacity Consumption Graph Association: Capacity Consumption Graph Association takes Boolean values 1/0.

1 means that the corresponding activity node is constrained by the capacity available from the resource that it is attached to.

0 means that the corresponding Activity can complete itself without using any capacity of that Resource. It is equivalent to having an infinite supply.

b) Capacity Consumption Qty Per: It is a multiplier and takes numeric values. It represents the quantity of resources required to produce one unit of output.

Association Graph Measure

Along with below measure groups which are required by SCS for planning, one needs to define corresponding mPower Graph tables to see the supply chain network through graph widget.

1. Capacity Consumption Graph
2. Material Consumption Graph
3. Material Production Graph

However, there is a feature called association graph duality through which one can make a measure group behave as regular graph. In those cases, there is no need to create or define the actual graph tables for SCS to view the supply chain network.

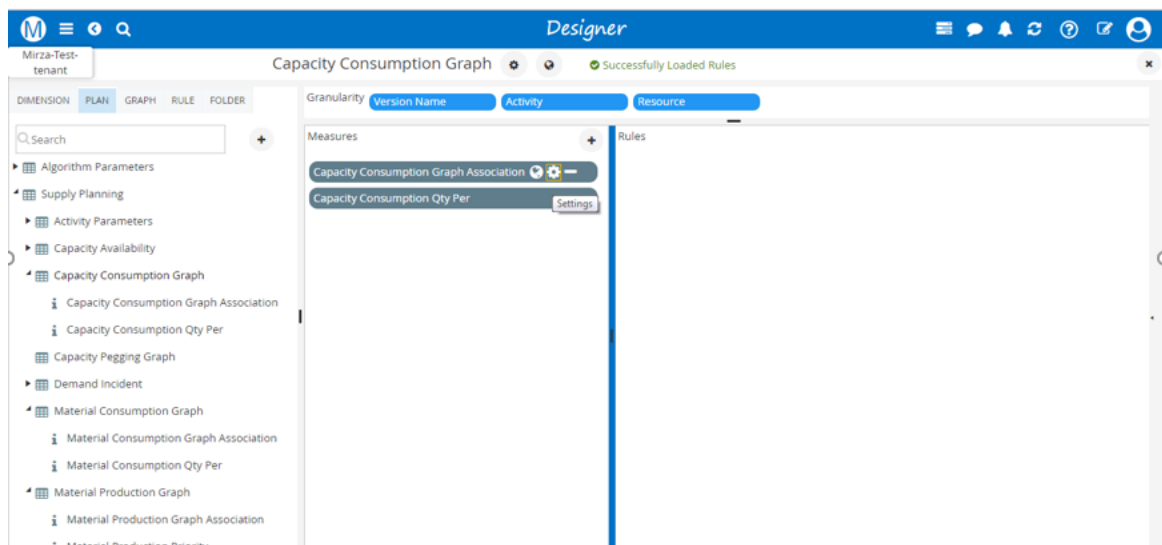
Let's go through the steps to see how to make the above 3 measure groups play duality (measure group as well as graph) in mPower through model designer through an example. This is available only in Kibo 2.0.

1. Mark one of the measures of these below measure group as “association graph measure” in model designer

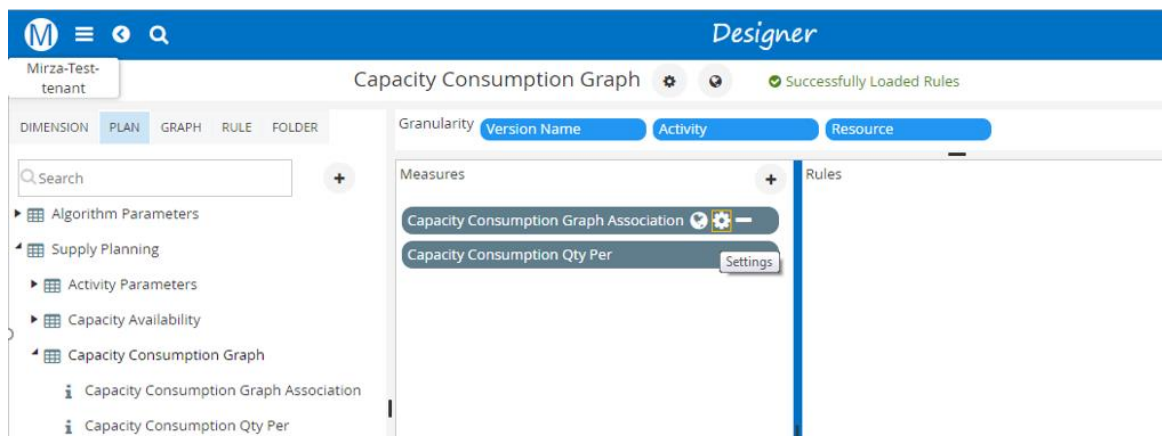
Capacity Consumption Graph

Material Consumption Graph

Material Production Graph



Click on Settings



Capacity Consumption Gra.. ✓ ✕

Granularity ← Resource

Measures +

Capacity Category

Capacity Consumption.. ⚙️ -

Capacity Consumption Qty Per

Capacity Graph Priority

Name *

Capacity Consumption Graph Assoc

Description *

Capacity Consumption Graph Assoc

ToolTip

Capacity Consumption Graph Assoc

Data Type

Number ▼

Type

Regular ▼

☐ Reporting Measure

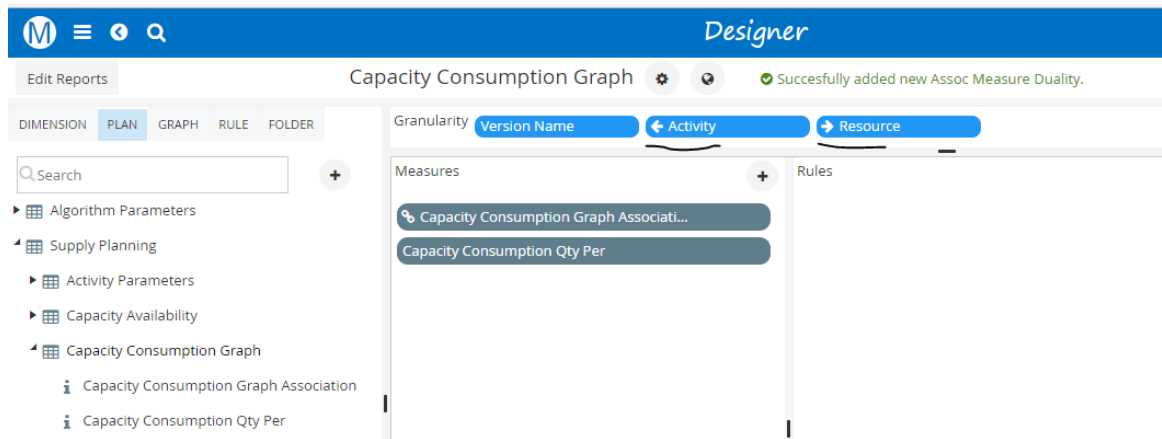
☒ Editable

☒ Association Graph Measure

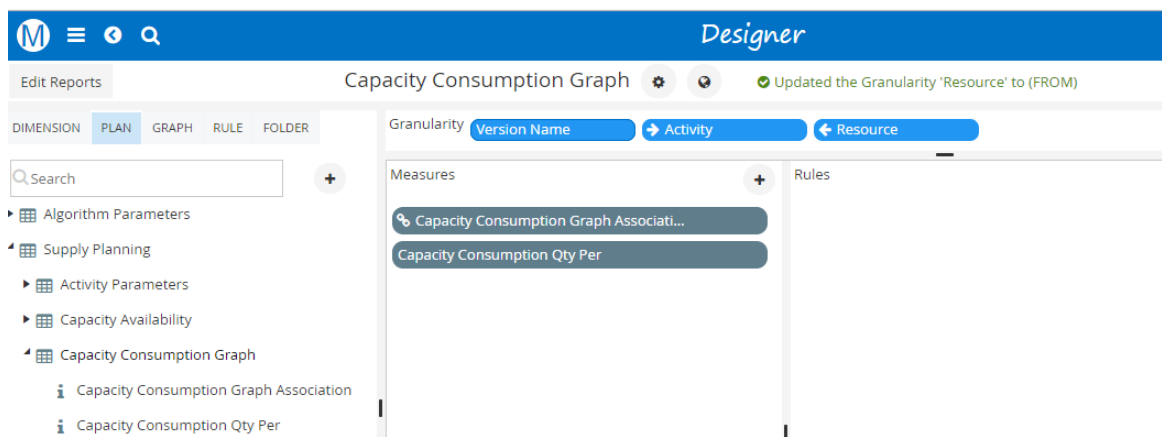
☐ Apply Conversion

Conversion Formula

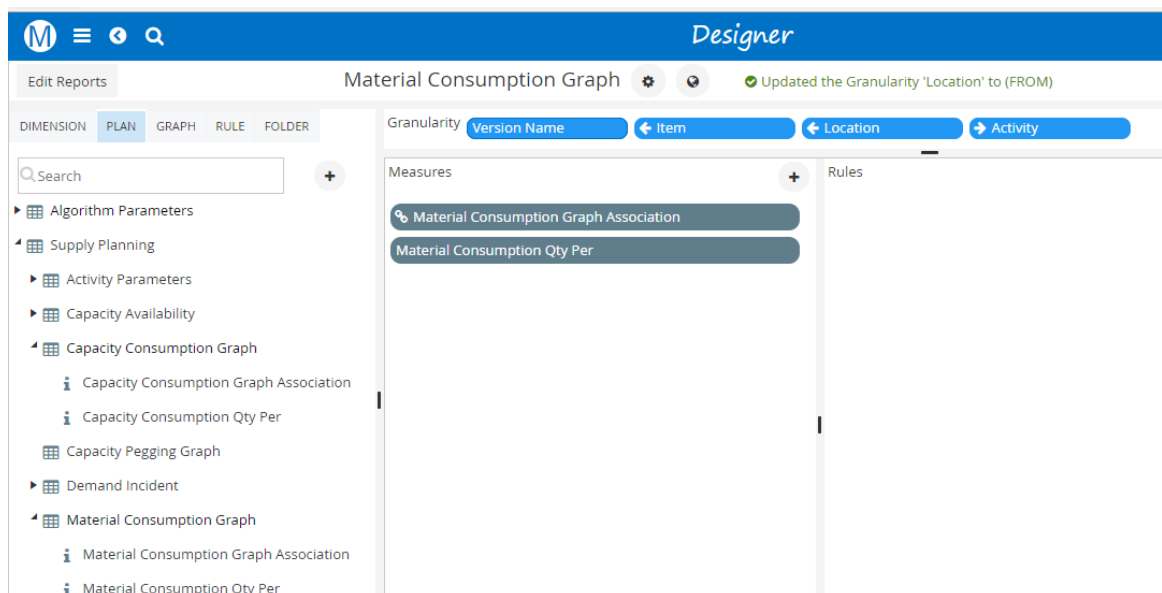
Now you see the arrow signs (showing these are from & to nodes of the graph) in the granularity tab. Make Sure you correct the From and To nodes here. Just click on the arrow to switch it from “From” node to “To” node and vice-versa. Also, see an icon preceding the measure that is marked as association graph measure.

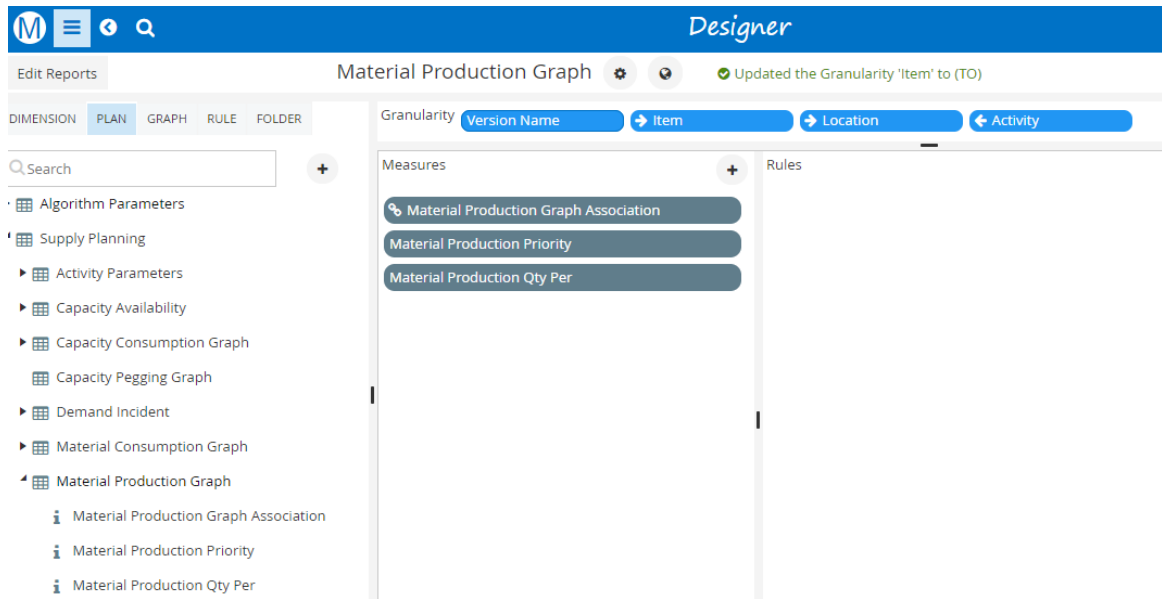


In the above corrected the from and to nodes



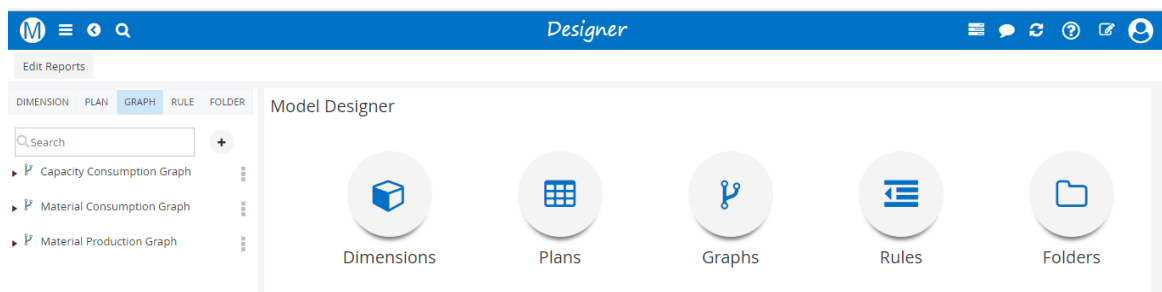
Similarly do it for material consumption and production graphs



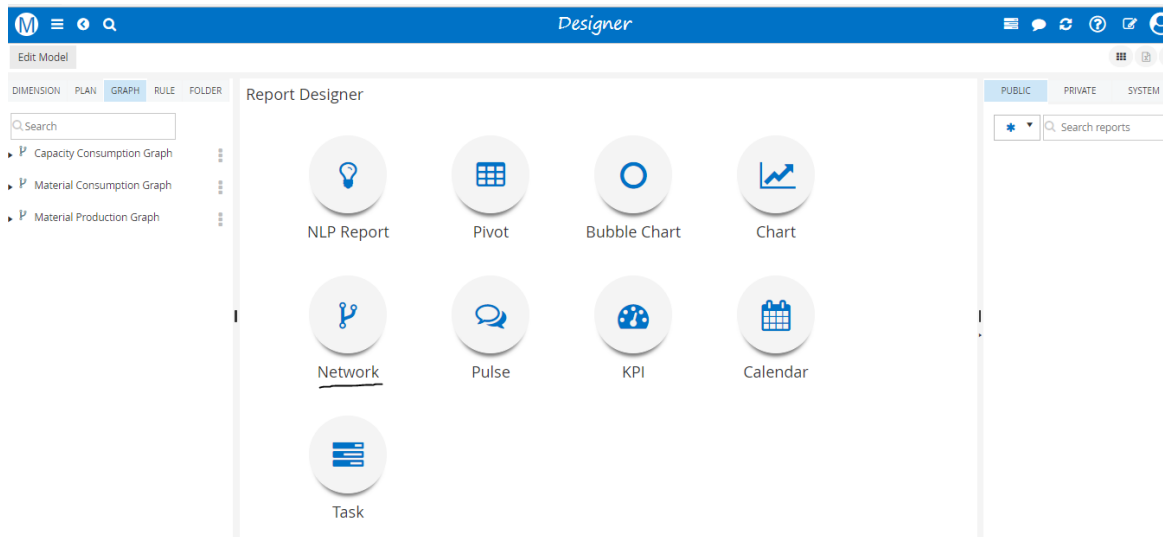


2. Now go to GRAPH tab and you should see these 3 measure groups converted graphs.

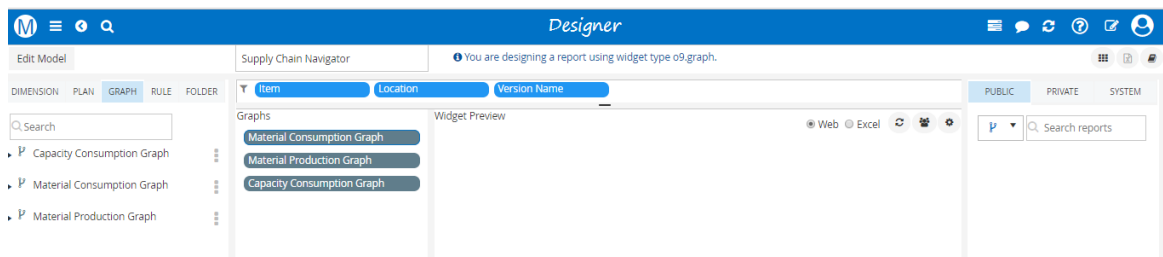
[For now we need to save and restart the LS to see these in GRAPH tab, finally one can see these in the GRAPH tab as soon as they get converted to association graph measures.]



3. Now turn on to Report Designer mode by clicking on the toggle switch “Edit Reports” on the left hand side of the page. Click on the “Network” type widget.



4. Give a name of the report, drag and drop attribute from Dimension tab to the report filter, drag and drop graphs to the Graphs pane on the report designer.



5. Click on the gear icon (just do mouse hover at the end of the graphs added to Graphs pane) and edit the necessary properties (like the images of the nodes to be displayed, mark one of the graphs as start graph).

Designer

Edit Model

SupplyChainNavigator

Opened report SupplyChainNavigator.

Material Production Graph

Show Node As Image

Show Shape

From Node Shape

diamond

☒ Show From Node Label

To Node Shape

dot

☒ Show To Node Label

☒ Is Start Graph

Start Node

TO

☐ Has Pegging Graph

Item

Location

Version Name

Graphs

Material Production Graph

Material Consumption Graph

Capacity Consumption Graph

Widget Preview

Designer

Edit Model

SupplyChainNavigator

Opened report SupplyChainNavigator.

Material Consumption Graph

Show Node As Image

Show Shape

From Node Shape

dot

☒ Show From Node Label

To Node Shape

diamond

☒ Show To Node Label

☐ Is Start Graph

☐ Has Pegging Graph

Item

Location

Version Name

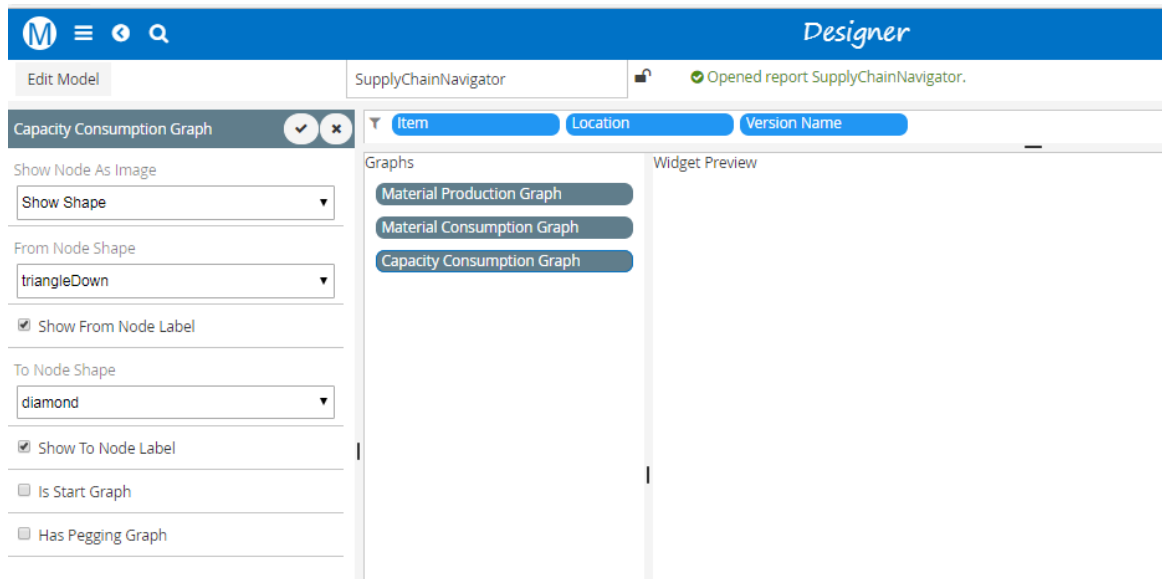
Graphs

Material Production Graph

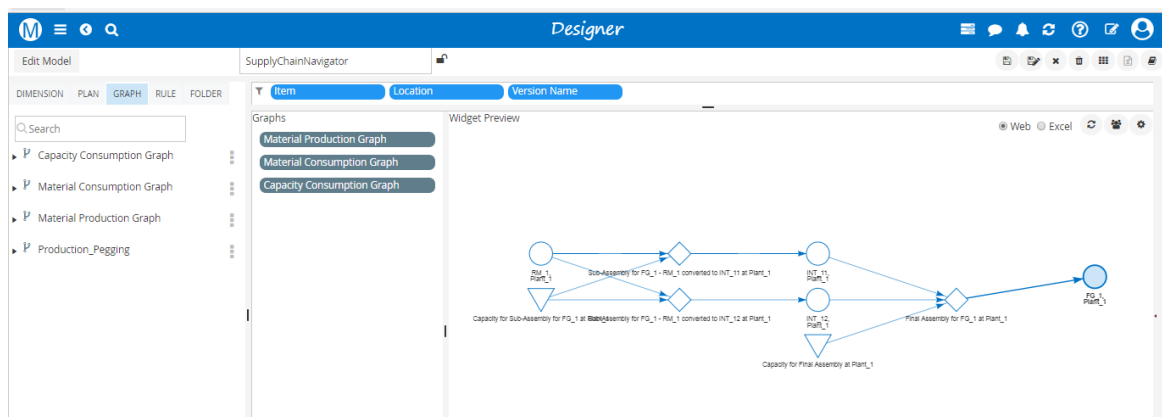
Material Consumption Graph

Capacity Consumption Graph

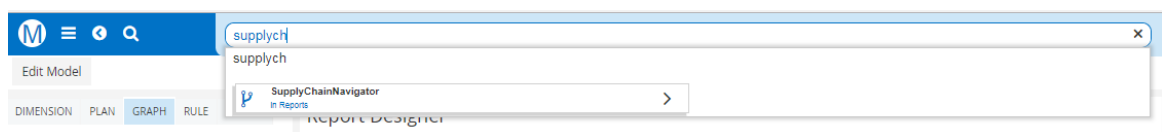
Widget Preview

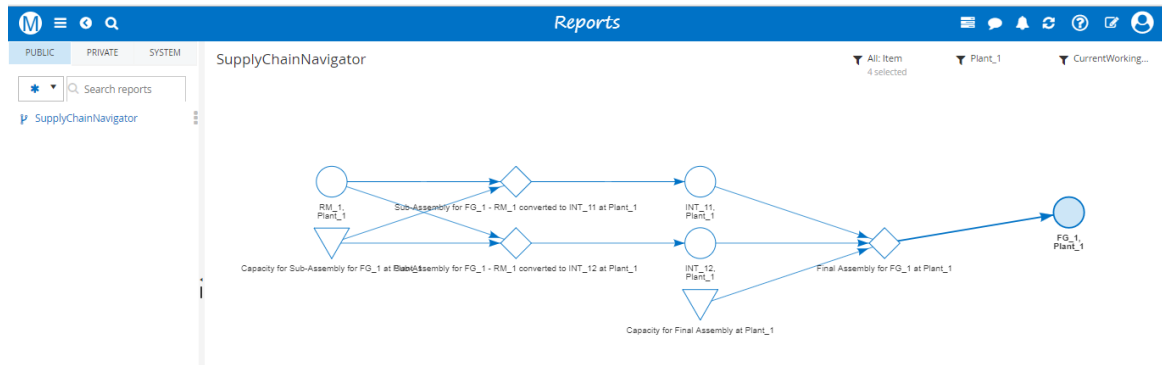


6. After this save the report. You can see the preview in the “Widget Preview” pane.



7. Now you can access the report from report repository. For quick access you can search the report name in the NLP search box and click on report name.





Similarly one can do for Storage Graph.

Transactional Data

After creating master data in the 4 Graphs we move on to creating transactional data to the solver. They are as follows:

1. Demand Quantity

Order Name	Customer Name	Item	Location	Data	W30-2015	W31-2015
Order_1	Cust_1	FG_1	RDC_1	Demand Quantity	100	
				Demand Priority	1	
Order_2	Cust_1	FG_2	RDC_2	Demand Quantity		100
				Demand Priority		1

Demand Quantity helps us identify the order Incident on any Material node in a particular time bucket.

It consists of 5 dimensions

1. Demand Node
 - a. Order.[Order Name]
 - b. Time.[Fiscal Week]
 - c. Customer.[Customer Name] (This is optional dimension)
2. Material Node
 - a. Item.[Item]
 - b. Location.[Location]
3. Version.[Version Name]

and 3 measures

- a) Demand Quantity: This measure represents the requested quantity of finished goods for the corresponding demand.
- b) Demand Priority: This measure sets the priority of the order. It helps the solver to identify the sequence in which the orders are picked to plan. Lower the number the higher the priority
- c) Quantity Fulfilled: This Measure stores the quantity of goods allocated by the solver for the corresponding demand. It is the output of the Solver. It is left blank by the user while uploading.

2. Inventory

Location	Item	Lot ID	Data	W01-2015	W02-2015
Plant_1	FG_1	SCS BOH	BOH	90	
			Expected Receipts		
		SCS ER	BOH		
			Expected Receipts	10	
	FG_2	SCS ER	BOH		
			Expected Receipts		100

SCS needs to know the starting state and prior commitments. This is represented in the form of beginning on hand and expected receipts. For this one needs to maintain a Measure Group called Inventory. It has 4 Dimensions, namely

1. Material Node
 - a. Item.[Item]
 - b. Location.[Location]
2. Lot.[Lot ID]
3. Time.[Fiscal Week]
4. Version.[Version Name]

And 2 measures:

- a) BOH: It is the Beginning On Hand. It stores the Inventory at the current time bucket.
- b) Expected Receipts: This measure stores the inventory accruals expected in time buckets other than the first time bucket.

3. Activity Parameters

Activity	Activity Lead Time
FG_9 - Shipment via Water from Plant_1 to DC_1	2
FG_9 - Shipment via Air from Plant_1 to DC_1	1
Final Assembly of FG_9 at Plant_1	1
Sub-Assembly for FG_9 - RM_1 converted to INT_19 at Plant_1	1
Sub-Assembly for FG_9 - RM_1 converted to INT_29 at Plant_1	1

Activity parameters stores the various measuring parameters of any activity under consideration. It has 2 dimensions, namely

1. Activity Node
 - a. Activity.[Activity]
2. Version.[Version Name]

and 5 measures

Mandatory:

- a) Activity Lead Time:

This measure denotes the lead time of the activity. i.e. the time an activity takes to complete itself. It is given in the form of number of buckets.

Optional:

b) Solver Activity Start Date:

This measure denotes the start date after which an activity is active. The supply chain solver will not consider that activity as valid before the mentioned date.

c) Solver Activity End Date:

The end date measure signifies the date after which an activity is inactive. The solver will not consider the activity valid after the mentioned date.

d) Activity Marker Measure Name:

List of arbitrary names that can be assigned to an activity or as the name suggests, mark an activity. It is a string or a list of strings

e) Activity freeze time bucket offset:

This measure takes numeric values. The offset denotes number of buckets upto which the solver will not consider producing fresh starts. The solver does not instantiate an activity in the time buckets starting from current bucket based on the offset value provided. Hence, new activity plans will not be created for the mentioned time buckets.

Datasets

Steps to restore the tenant through UI.

1. Login as super-admin in [sandbox](#)
2. Go to My profile -> select "o9solutions" tenant in the drop down.
3. Goto → Workspace → Deployment → Manage Backups
4. Filter backups by tenant name → scs-o9-2019_0

The screenshot shows the 'Deployment' section of a web application. The 'Manage Backups' tab is active. A table displays backup information for tenant 'scs-o9-hm'. A modal window titled 'Tenant' is open, showing a list of backup files for 'scs-o9'.

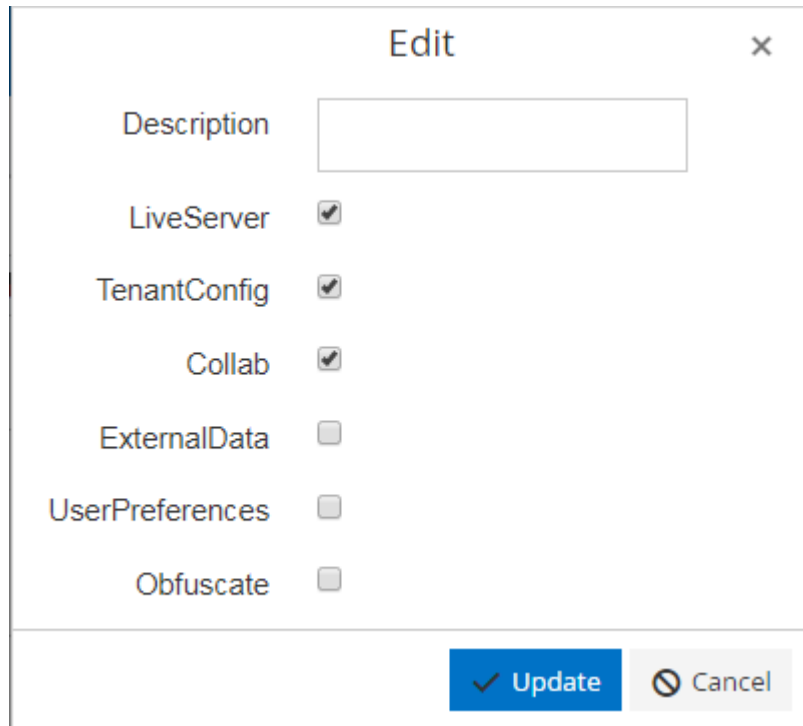
TenantName	Status	Description	CreatedAt	Components	Environme...	MachineNa...
scs-o9-hm	COMPLETED	For Testing	04-Sep-2017 12:10:37 PM	LiveServer,T...	sandbox	
scs-o9-hm	COMPLETED	dabde	01-Sep-2017 21:37:32 PM	LiveServer,T...	genie	
scs-o9-hm	COMPLETED	31st Aug	31-Aug-2017 08:17:15 AM	LiveServer,T...	genie	

The modal window 'Tenant' shows a list of backup files for 'scs-o9':

- Mark-SCS-o9
- Mark-SCS-o9-composite
- Mark-SCS-o9-lot
- SCS-o9-hm
- SCS-o9-hm_Mirza

5. Take Backup

Click on "+Backup".



The 'Edit' dialog box contains the following fields and controls:

- Description:** A text input field.
- LiveServer:** A checkbox that is checked.
- TenantConfig:** A checkbox that is checked.
- Collab:** A checkbox that is checked.
- ExternalData:** An unchecked checkbox.
- UserPreferences:** An unchecked checkbox.
- Obfuscate:** An unchecked checkbox.
- Buttons:** 'Update' (blue) and 'Cancel' (grey).

In the popup Edit dialog box give some description and click on "update".
Keep checking the backup status. There are different steps all need to be completed.

6. Restore Backup

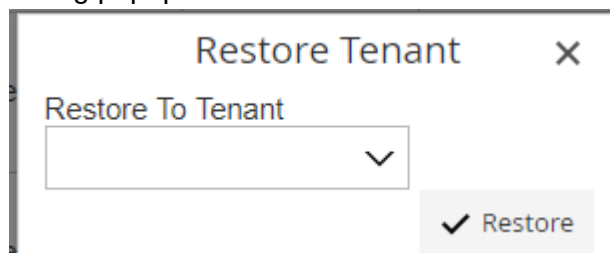
Goto → Workspace → Deployment → Manage Backups

Find the backup you created.

Click on the link "Restore Backup" on the right hand side.

^	scs-o9-hm	COMPLETED	For Testing	04-Sep-2017 12:10:37 PM	LiveServer,T...	sandbox		60987	1705d590- d93a-4976- 9c4f- df01a0c2d0c4	false	Restore Backup Delete
---	-----------	-----------	-------------	----------------------------	-----------------	---------	--	-------	--	-------	--------------------------

You will get following popup box.



The 'Restore Tenant' dialog box contains the following elements:

- Title:** Restore Tenant
- Label:** Restore To Tenant
- Field:** A dropdown menu for selecting a tenant.
- Button:** 'Restore' (grey).

In the box, give a name for the new tenant.
As soon as you give a name, you can see few other properties like below.

Restore Tenant

Restore To Tenant

SCS-o9-hm-Testing

LiveServer Machine IP

10.0.1.151

LiveServer Data Folder

D:\mPowerStore

Organisation Name

o9solutions

Restore

Provide the details as shown above and click “Restore”.

Restore Tenant

Restore request submitted successfully.
You may monitor the progress from the
restore tab view

7. Click <Restore Status> to view the status of the ongoing restore

TenantName ▼	Status ▼	Description ▼	CreatedAt ▼	Components ▼	Environme... ▼	MachineNa... ▼	Created By ▼	WorkflowId ▼	Obfuscated ▼	
▼ scs-o9-hm	COMPLETED	fresh backup for sdbx1 and sdbx 2	16-Oct-2017 12:34:25 PM	LiveServer,T...	genie		192184	bce59595-07ee-4586-b7c2-2e4b6c38a3d9	false	Restore Backup Delete

BACKUP STATUS

RESTORE STATUS

Refresh

ActivityType	CreatedAt	Environment	UserId	Status ▲	Message
▼ Restore Attempt: 15163feb-f684-4fe5-b4d0-391efa3b4cff					
RestoreLiveServerDataAc...	16-Oct-2017 12:43:32 PM	sandbox2	54485	COMPLETED	Successfully replaced live server data folder at D:\mPowerStore\SCS-o9-hm\data
RestoreUserConfigActivity	16-Oct-2017 12:42:32 PM	sandbox2	54485	COMPLETED	Successfully restored user config with message -
RestoreTenantConfigActivity	16-Oct-2017 12:44:10 PM	sandbox2	54485	COMPLETED	Successfully restore config Json with message -
RestoreDownloadAttachm	16-Oct-2017 12:43:31 PM	sandbox2	54485	COMPLETED	Successfully downloaded

8. Goto→Workspace→Deployment View the newly created tenant. On the Tenant filter, give a string to find the tenant.

Show items with value that:

Contains

SCS-o9-hm-T

And

Contains

Clear

Filter

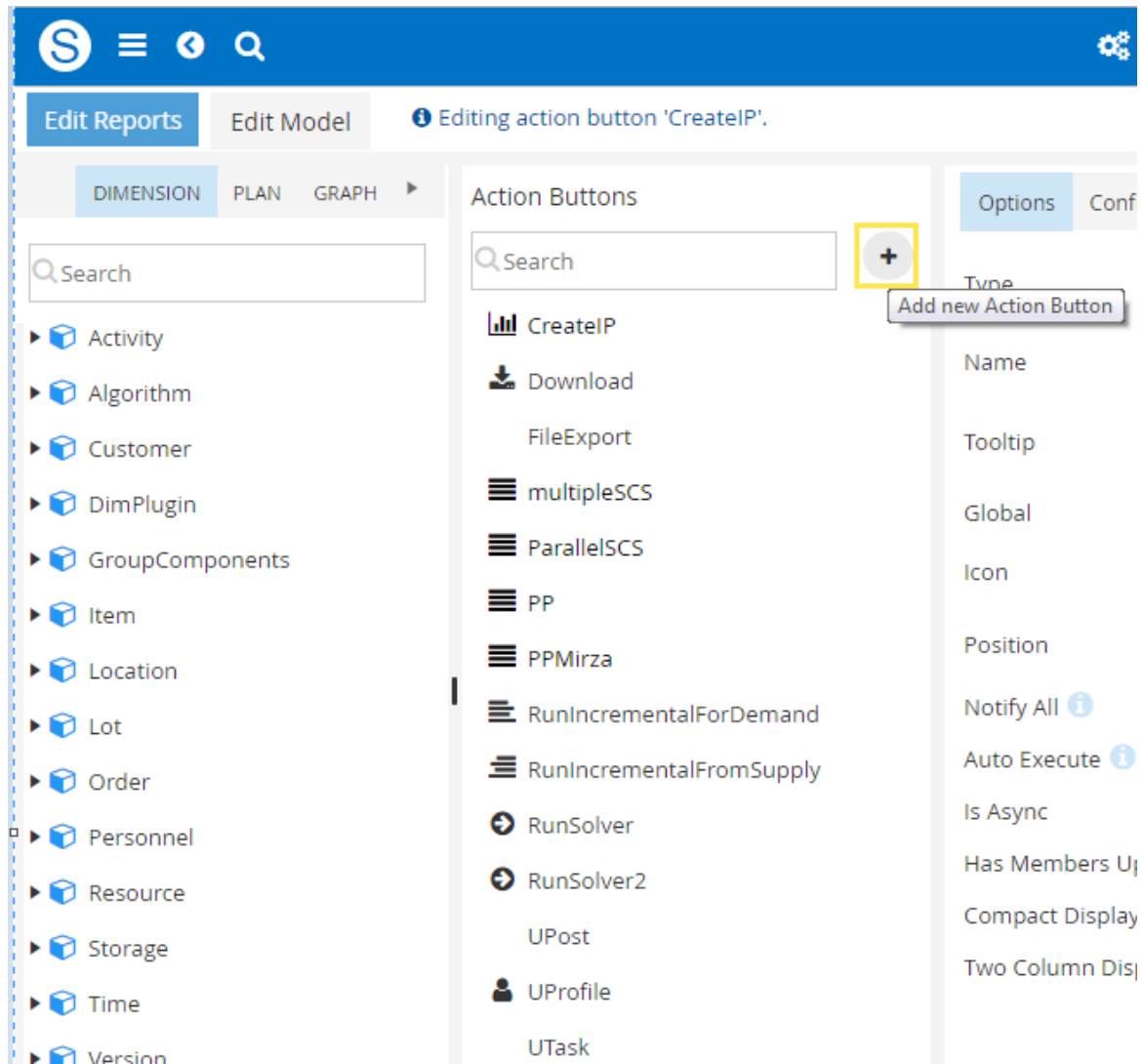
Deployment										
Live Servers		Manage Collab	Manage Backups	SSO Config	System Settings					
Id	Tenant *	Server Name	Server Port	PID	Restore Fr...	Data Folder	Temp Folder	Last Start T...	Status	
8762	SCS-o9-hm-Testing	10.0.1.151	27254	33128	true	D:\mPowerSt...o9-hm-Testing	D:\mPowerSt...o9-hm-Testing\Clou...	Mon Sep 04 2017 06:53:41 GMT+0530 (India Standard Time)	Running	Edit

Invoking the solver in Meru

exec plugin instance SupplyChainSolver **for** Measures {[Demand Quantity]} **using scope** ([Version].[Version Name].[CurrentWorkingView] * Time.[Fiscal Week] * Order.[Order Name]);

The above query can be a part of the action button

Goto Workspace→Designer→Edit Reports→Action Button →Add New Action Button

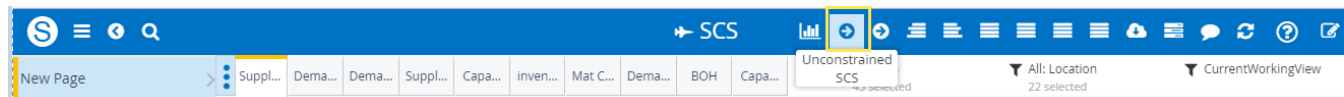


A configuration wizard for action button appears to the right as shown below. User can navigate through various tabs to configure the created action button

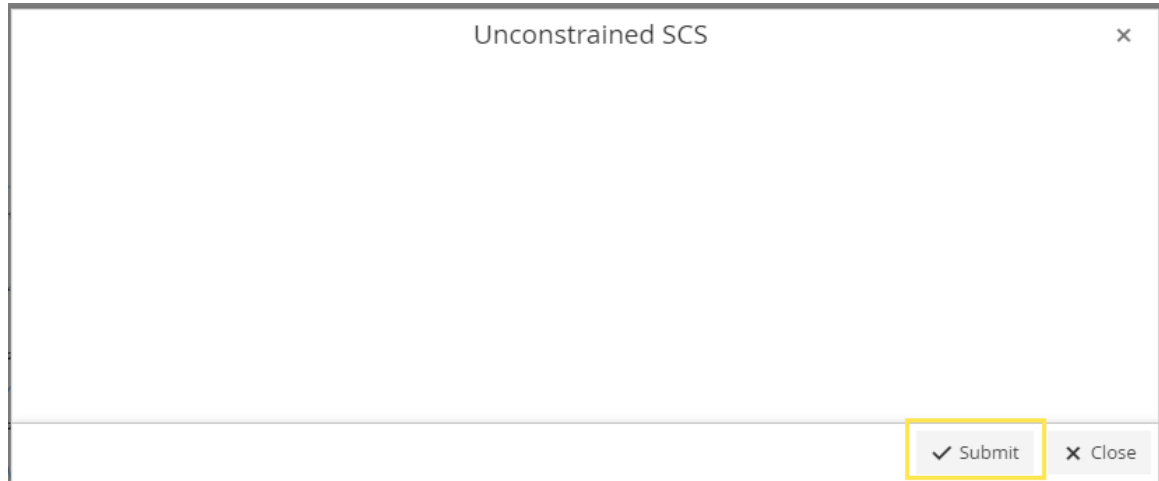
Options	Config	RuleOptions	FieldBindings	DataSources
Type	Rule			
Name	RunSolver			
Tooltip	Unconstrained SCS			
Global	☒			
Icon	arrow-circle-right			
Position	2			
Notify All <i>i</i>	☐			
Auto Execute <i>i</i>	☐			
Is Async	☒			
Has Members Updates <i>i</i>	☐			
Compact Display <i>i</i>	☐			
Two Column Display <i>i</i>	☐			

Options	Config	RuleOptions	FieldBindings	DataSources
☐ Validate				
SubmitButton				
Label	Submit			
ProgressMessage				
SuccessMessage				
ErrorMessage				
ConfirmMessage				
ResetButton				
Label	Reset			
CloseButton				
Label	Close			
ConfirmMessage	Unsaved changes will be lost. Continue?			

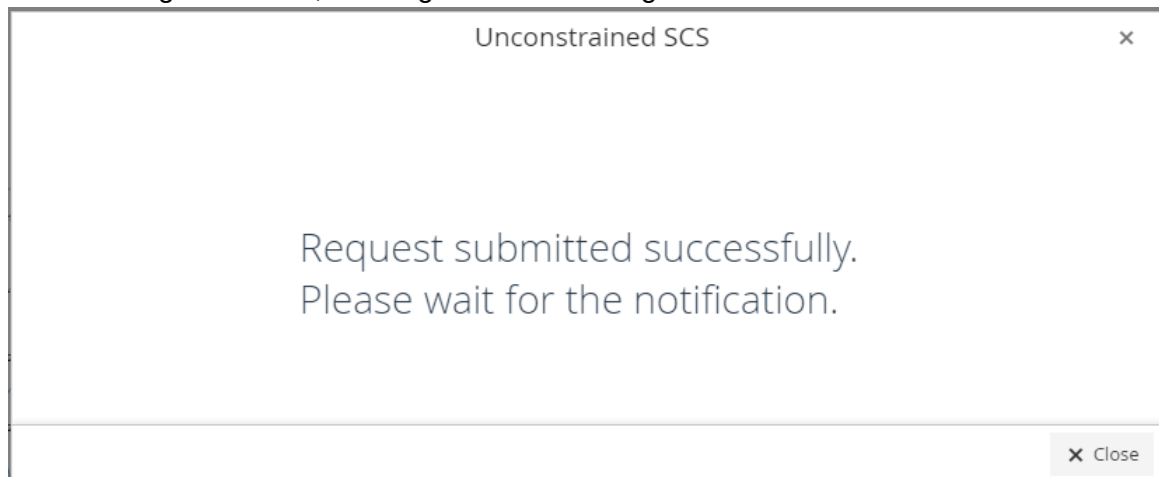
After the action button has been configured, SCS plugin can be executed as follows:
Goto <any> WorkSpace→the above configured action button appears on the top right of the workspace



Click on the action button and on the screen that appears next click on “Submit”

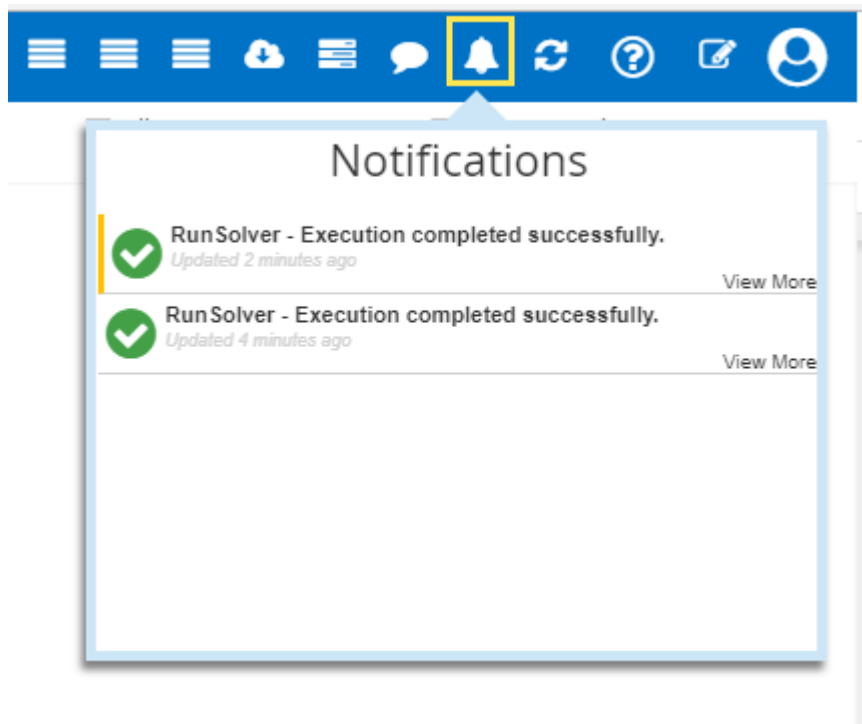


After clicking on Submit, the UI gives the following information.



Click on Close.

To confirm if the solver ran successfully click on the bell icon on the top right corner of the Workspace



exec plugin instance **SupplyChainSolver** for Measures {[Demand Quantity]} using scope ([Version].[Version Name].[CurrentWorkingView] * Time.[Fiscal Week] * Order.[Order Name]) ;

“SupplyChainSolver” is the name provided by the user in the settings tab. This is a variable name that is given to the plugin. Therefore instead of just typing `<o9.GraphCube.Plugins.SupplyChainSolver.SupplyChainSolver >`, a user can give an alias name as “SupplyChainSolver” and invoke the same plugin.

Do not use a hyphen (“-”) in the name while creating an instance on UI for SCS. You can use underscore (“_”) though. Thus “Test_SCS” will work fine, but “Test-SCS” will throw error while executing the plugin.

[Demand Quantity]: It is a measure that is defined in the Demand Quantity Measure Group. Please refer the Measure group again. As mentioned, this measure sets the quantity of finished goods required by the corresponding order. It is a user input. The solver is running with primary objective of solving this demand. We can therefore run the solver for different Demands (Orders or Forecasts). We simply populate the forecast values under Demand Quantity Measure group.

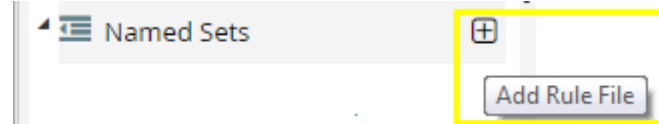
In the arguments you can mention the parameters (which you find in Designer→ Edit Model → Rule → Plugins → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver→ Parameters) like below
([Parameter Name], "Value")

You can define multiple parameters like above using comma (“,”) as separator. Note that the parameter name is case sensitive.

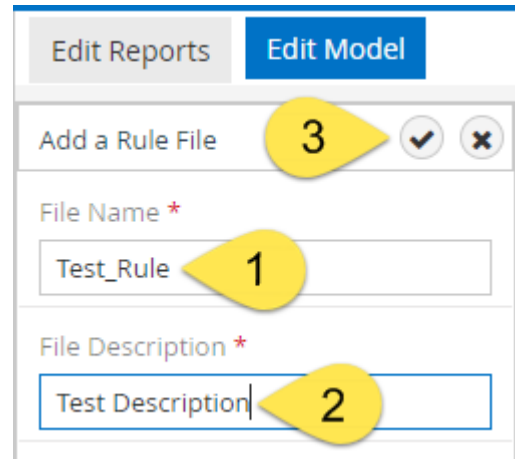
Planners can add Namesets to the command line signature configurations of which are as follows:.

Goto Workspace → Designer → Edit Model → Rule → Named Sets → Add rule by clicking

the  present on the right as shown below



New Rule File can be added as follows



1. Type File Name
2. Type Description
3. Click on  to save

After the Rule File Test_Rule is created , namesets can be added as follows:



1. Click on the  to enter details in the form
2. Type the comments in the comment section
3. Type the name of the nameset
4. Type the rule that it represents
5. Click on Save to save the nameset

Namesets can be used in the solver query as follows:

```
exec plugin instance SupplyChainSolver for Measures {[Demand Quantity]} using
scope (&CWV *
&PlanningHorizon *
Order.[Order Name])
arguments {[Material Constrained], "False"},
[Capacity Constrained] , "False",
[Delete SCS Plan Data For Version], "True",
[Current Time Bucket Name],&CurrentWeek.element(0).Name),
```

`(([Start Time Bucket Name],&StartWeek.element(0).Name),
([End Time Bucket Name],&StartWeek.element(0).leadoffset(51).Name));`

Deleting Graph data using IBPL commands:

```
DELETE DATA FOR RELATIONSHIPTYPE [TraversalGraph A]  
WHERE {Version.[Version Name].[scenarioDGE], ([TraversalGraph A],  
From.Product.[BDC Name].Filter(#.Key <= 20))};
```

Example:

```
DELETE DATA FOR RELATIONSHIPTYPE [Mat_Production_Graph]  
WHERE {Version.[Version Name].[CurrentWorkingView]};
```

```
DELETE DATA FOR RELATIONSHIPTYPE [Mat_Consumption_Graph]  
WHERE {Version.[Version Name].[CurrentWorkingView]};
```

```
DELETE DATA FOR RELATIONSHIPTYPE [GraphCapacityConsumption]  
WHERE {Version.[Version Name].[CurrentWorkingView]};
```

The above queries can be a part of action button as described [here](#)

Output of the Solver

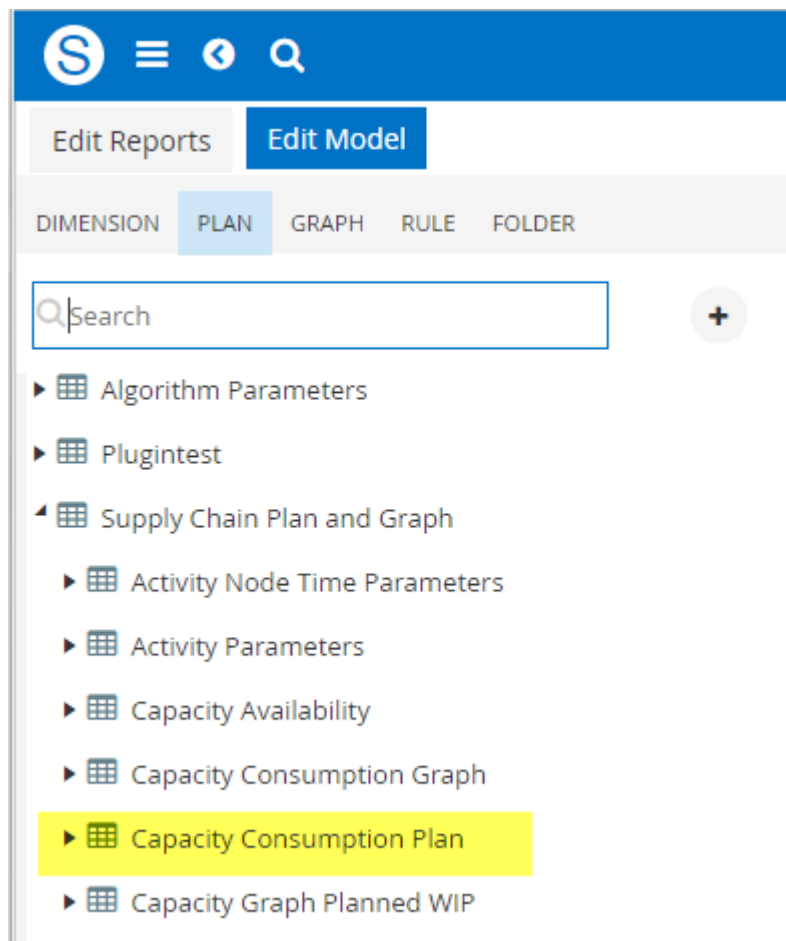
Flow Plans

Based on what quantities we maintain in the Demand Quantity Measure and what inventory we maintain at each material node in the Inventory Measure Group the Solver generates the following Flow plans.

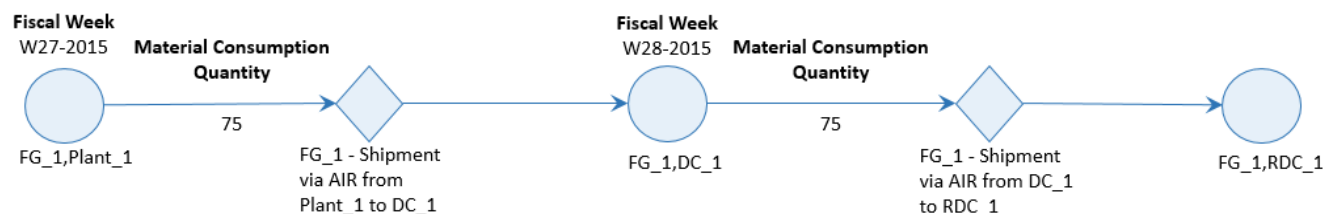
1. Material Consumption plan
2. Material Production plan
3. Capacity Consumption plan

The Above 3 plans can be configured in the tenant as follows:

Goto Designer → Edit Models → Plans → Supply Chain Plan and Graph → Add measure Group



Material Consumption Plan



Material consumption plan caters to all the material nodes in the supply chain network. A row in Material consumption plan simply means in time bucket T an activity A will consume N quantities of Item I having Lot Id x at Location L.

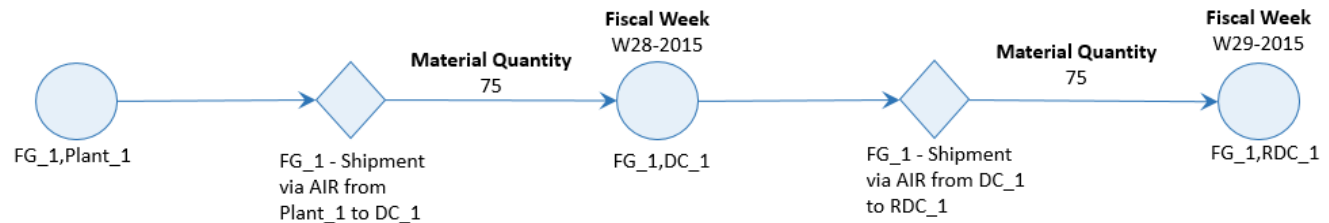
It has dimension

1. Activity Node
 - a. [Activity].[Activity] = A
2. Material Node
 - a. [Item].[Item] = I
 - b. [Location].[Location] = L
3. Lot.[Lot ID] = x
4. [Time].[Fiscal Week] = T
5. [Version].[Version Name] = CurrentWorkingView

And 1 measures

- a) Material Consumption Quantity = N

Material Production Plan



Material production plan caters to all the material nodes in the supply chain network. A row in Material production plan simply means in time bucket T an activity A will produce N quantities of Item I having Lot Id x at Location L.

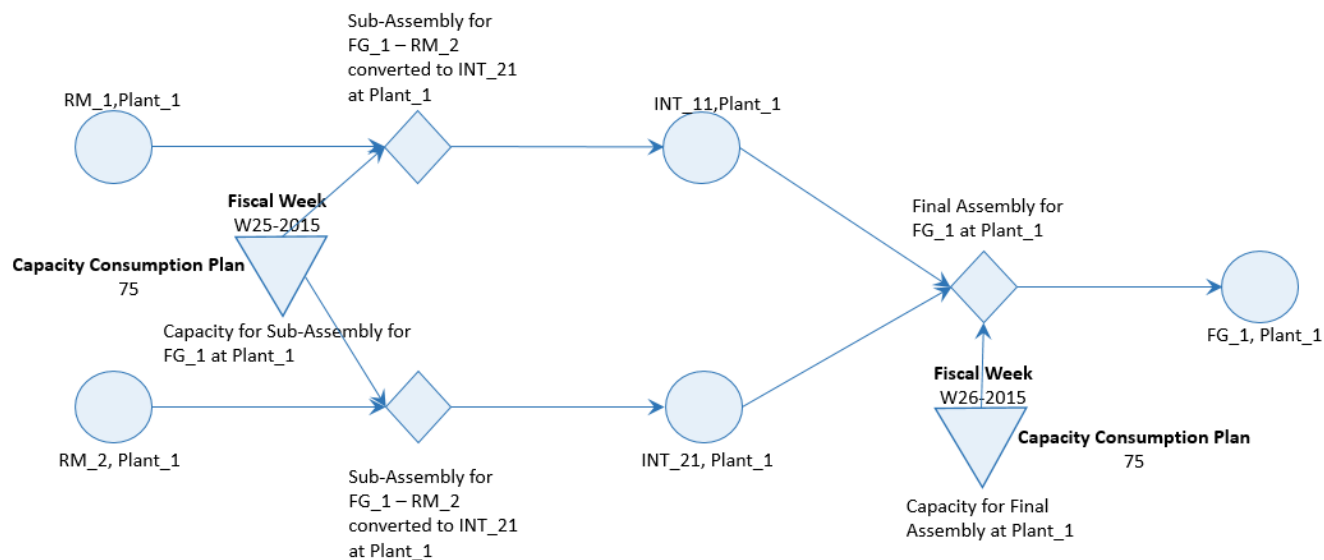
It has dimension

1. Activity Node
 - a. [Activity].[Activity] = A
2. Material Node
 - a. [Item].[Item] = I
 - b. [Location].[Location] = L
3. Lot.[Lot ID] = x
4. [Time].[Fiscal Week] = T
5. [Version].[Version Name] = CurrentWorkingView

And 1 measures

- a) Material Production Quantity = N.

Capacity Consumption Plan



Capacity consumption plan caters to all the Capacity nodes i.e. [Resource].[Resource]. A row in Capacity consumption plan means in time bucket T an activity A will consume C capacity of Resource R to process items having Lot ID x.

It has dimensions

1. Activity Node
 - a. [Activity].[Activity] = A
2. Capacity Node
 - a. [Resource].[Resource] = R

3. Lot.[Lot ID] = x
4. [Time].[Fiscal Week] = T
5. [Version].[Version Name] = CurrentWorkingView

And 1 measure

- a) Capacity consumption plan = C.

Pegging Tables

In addition to flow plans, solver also does the pegging. Pegging is nothing but linking the quantities of item/resources that are produced/consumed at various material/capacity nodes across time buckets towards fulfilment of a particular order.

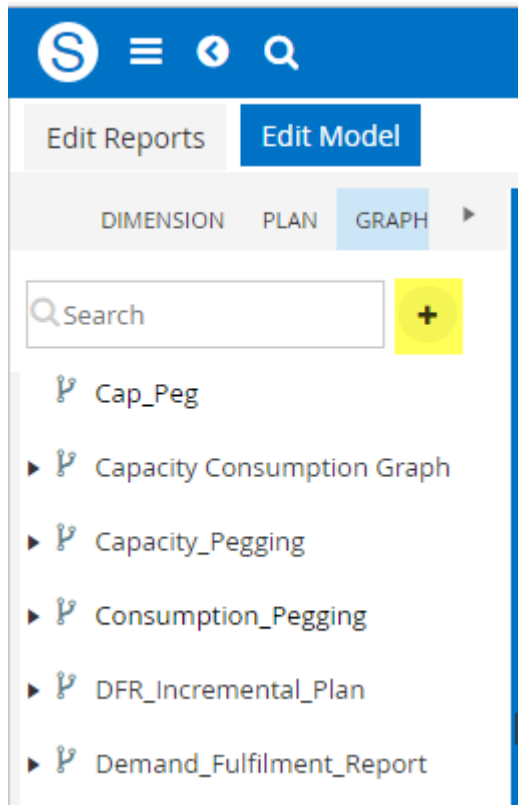
The pegging tables stores information of the quantity of flow of goods from source to destination across the levels of supply chain. From the pegging tables one can infer that for a given order how much quantity of goods is consumed/produced at intermediate material nodes and in which time bucket before it reaches the end item Material Node (on which demand is incident). Pegging tables also stores information on the lot ids that participated in the demand fulfillment.

Note: orders have different / same priority. SCS sorts the order in 2 phases. 1st the sorting is done on the basis of initially set priorities. Orders having same/no priorities set to them are then sorted on the basis of their respective due date. This means demands on earlier time buckets are relatively of higher priority than demands of later due date(assuming both are assigned same / no priorities)

Sometimes demand fair share with slice factor is used and still demands are not satisfied. In such a scenario whatever part of the demand is satisfied is reported to the planners, and whatever part of the demand is not satisfied is reported to the planners via constraint report.[\[Functional test 45\]](#)

Pegging tables can be configured as follows:

Goto Designer → Edit Model → Graph → Click on  to add new relationship type or graph



Add new Records:

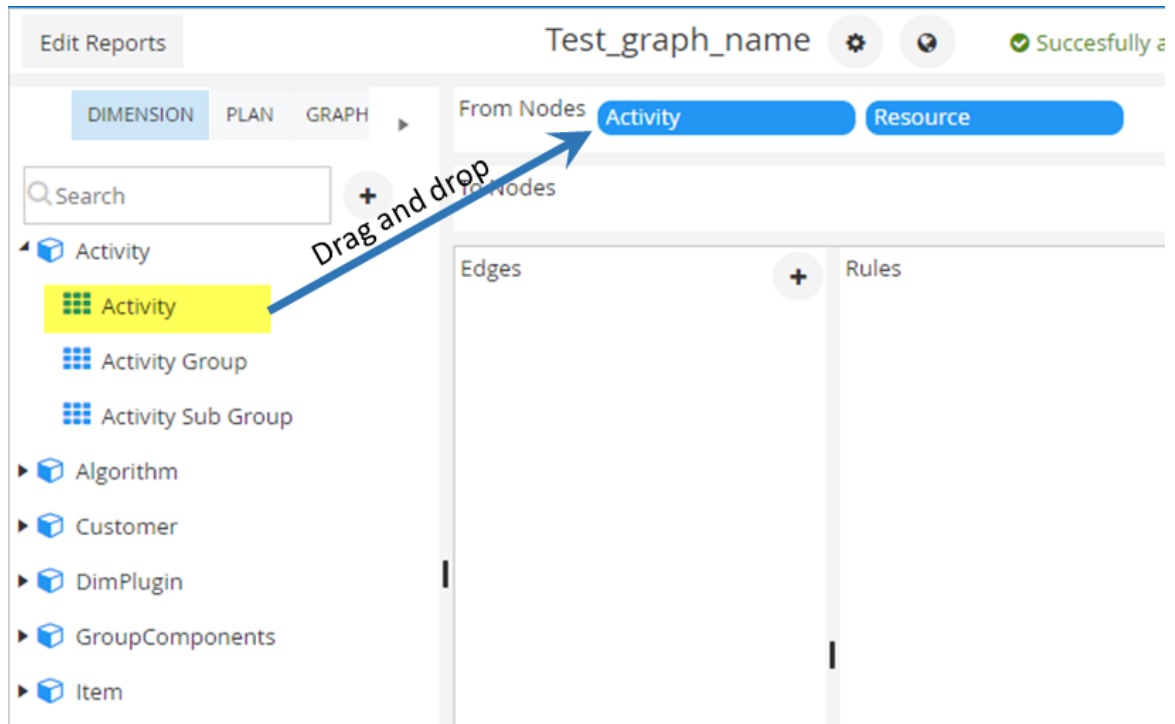
Relationship Type Name: Name of the pegging table you want to define.

Description: Brief Description of that the defined pegging table does.

Click on  to proceed.

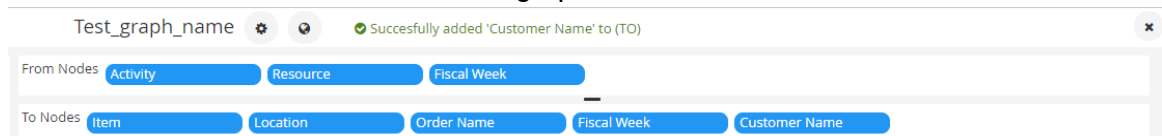
On clicking  we see the following screen to add from_node to_node and edge properties

Using Designer, drag and drop dimensions onto the From Nodes



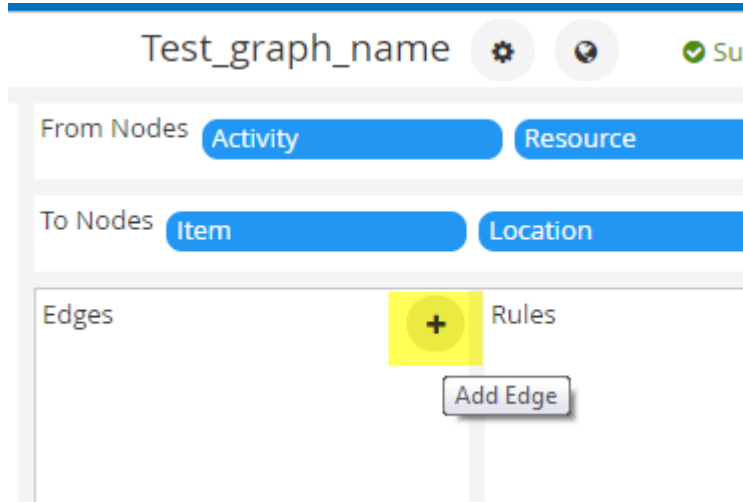
Repeat the same with To Node

Thus from Nodes and To Nodes of the graph looks as follows:



Edges

Click on the highlighted “+” sign to add an edge property



As soon as one clicks the “+” and pane appears to the right as follows:

Add a Measure

✓

✕

Name *

Description *

Data Type

Number

Aggregation

Sum

☐ Editable

Format

(0,000 a)

From Nodes

Activity

To Nodes

Item

Edges

One may edit the relevant fields as follows:

Click on  to create edge property.

We configure the following pegging tables in our Supply Chain Model-

1. Demand_Fulfilment_Report
2. Consumption_Pegging
3. Production_Pegging
4. Capacity_Pegging

1. Demand Fulfilment Report

from.[Time]. [Fiscal Week]	to.[Order]. [Order Name]	to.[Time]. [Fiscal Week]	to.[Item]. [Item]	to.[Location]. [Location]	to.[Customer]. [Customer]	Data	Value
W31-2015	Order_1	W31-2015	FG_1	RDC_1	Cust_3	Demand Fulfilment Quantity	100.00

This report outlines the quantity of demand satisfied by the solver for an Order based on the available Inventory.

<from_members> is a combination of following Dimensions.

- [Time].[Fiscal Week]

<to_members> is a combination of following Dimensions.

- Demand Node
 - [Order].[Order Name]

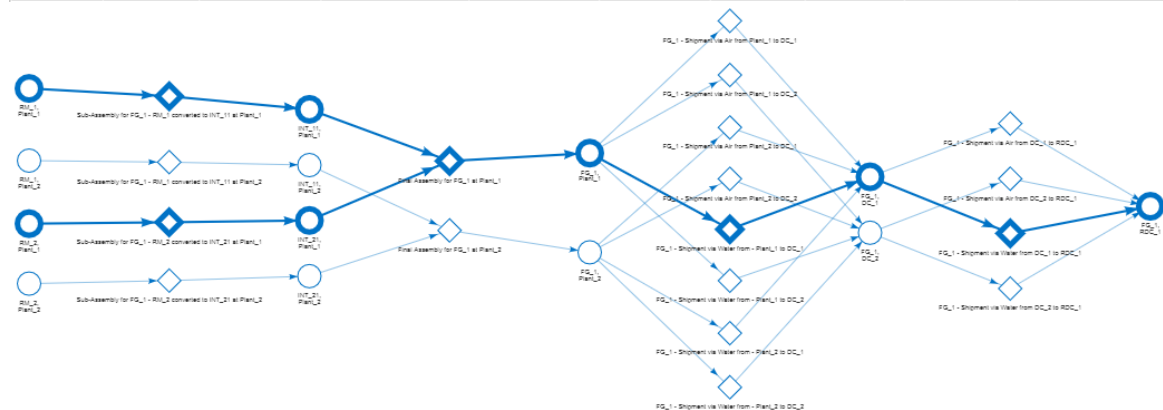
- [Time].[Fiscal Week]
- [Customer].[Customer Name]
- Material Node
 - [Item].[Item]
 - [Location].[Location]

Properties:

Demand Fulfilment Quantity: The data type of this edge property is number and the aggregation level is sum. This property stores the amount of demand that is satisfied for an Order after the solver runs successfully. We give the Demand Quantity as input and get the Demand Fulfilment Quantity as corresponding output.

2. Material Consumption Pegging

from. [Item]. [Item]	from. [Location]. [Location]	from.[Time]. [Fiscal Week]	from.[Activity]. [Activity]	from.[Lot].[Lot ID]	to.[Order]. [Order Name]	to.[Item]. [Item]	to. [Location]. [Location]	to.[Time]. [Fiscal Week]	Material Consumption Pegged Quantity
FG_1	DC_1	W29-2015	FG_1 - Shipme...	000SCS_LOT	Order_1	FG_1	RDC_1	W31-2015	100.00
FG_1	Plant_1	W27-2015	FG_1 - Shipme...	000SCS_LOT	Order_1	FG_1	RDC_1	W31-2015	100.00
INT_11	Plant_1	W26-2015	Final Assembl...	000SCS_LOT	Order_1	FG_1	RDC_1	W31-2015	100.00
INT_21	Plant_1	W26-2015	Final Assembl...	000SCS_LOT	Order_1	FG_1	RDC_1	W31-2015	100.00
RM_1	Plant_1	W25-2015	Sub-Assembly...	000SCS_LOT	Order_1	FG_1	RDC_1	W31-2015	100.00
RM_2	Plant_1	W25-2015	Sub-Assembly...	000SCS_LOT	Order_1	FG_1	RDC_1	W31-2015	100.00



<from_members> is a combination of following Dimensions

- Material Node
 - [Item].[Item]
 - [Location].[Location]
- Activity Node
 - [Activity].[Activity]
- Lot.[Lot ID]
- [Time].[Fiscal Week]

<to_members> is a combination of following Dimensions

- Demand Node
 - [Order].[Order Name]
 - [Time].[Fiscal Week]
 - [Customer].[Customer]
- Material Node
 - [Item].[Item]
 - [Location].[Location]

Properties:

Material consumption pegged quantity: The data type of this edge property is number and the aggregation level is sum. It stores the quantity of material consumed corresponding to each activity, all of which are incident on a unique order.

Holistically Material consumption pegging says that to fulfil a particular order, the path that it takes through the supply chain providing visibility into the consumption flows from the intermediate material nodes in different time buckets.

3. Material Production Pegging

Material production pegging is similar to material consumption pegging. The only difference is that instead of letting us know consumption activities, this pegging table lists all production activities at intermediate material nodes in different time buckets to fulfil the chosen order (Demand Signal incident on material node).

SKU	from. [Location]. [Location]	from.[Activity].[Activity]	Supply Timebucket	from.[Lot].[Lot ID]	to.[Location]. [Location]	to.[Item]. [Item]	to.[Order]. [Order Name]	Order Timebucket	Material Production Pegged Quantity
RM_1	Plant_1	BOH Activity	W01-2015	SCS BOH	RDC_1	FG_1	Order_1	W31-2015	90.00
RM_1	Plant_1	Expected Receipt Activity(1)	W01-2015	SCS ER	RDC_1	FG_1	Order_1	W31-2015	10.00
RM_2	Plant_1	BOH Activity	W01-2015	SCS BOH	RDC_1	FG_1	Order_1	W31-2015	90.00
RM_2	Plant_1	Expected Receipt Activity(1)	W01-2015	SCS ER	RDC_1	FG_1	Order_1	W31-2015	10.00
INT_11	Plant_1	Sub-Assembly for FG_1 - RM_...	W26-2015	000SCS_LOT	RDC_1	FG_1	Order_1	W31-2015	100.00
INT_21	Plant_1	Sub-Assembly for FG_1 - RM_...	W26-2015	000SCS_LOT	RDC_1	FG_1	Order_1	W31-2015	100.00
FG_1	Plant_1	Final Assembly for FG_1 at Pl...	W27-2015	000SCS_LOT	RDC_1	FG_1	Order_1	W31-2015	100.00
FG_1	DC_1	FG_1 - Shipment via Water fr...	W29-2015	000SCS_LOT	RDC_1	FG_1	Order_1	W31-2015	100.00
FG_1	RDC_1	FG_1 - Shipment via Water fr...	W31-2015	000SCS_LOT	RDC_1	FG_1	Order_1	W31-2015	100.00

<from_members> is a combination of following Dimensions.

- Material Node
 - [Item].[Item]
 - [Location].[Location]
- Activity Node
 - [Activity].[Activity]
- Lot.[Lot ID]
- [Time].[Fiscal Week]

<to_members> is a combination of following Dimensions.

- Demand Node
 - [Order].[Order Name]
 - [Time].[Fiscal Week]
 - [Customer].[Customer]
- Material Node
 - [Item].[Item]
 - [Location].[Location]

Properties:

Material production pegged quantity: The data type of this edge property is number and the aggregation level is sum. It stores information about the quantity of goods produced by each activity at each material node to satisfy a unique demand at Demand node.

Note: The BOH and Expected Receipts that are used to fulfill an order (during either constrained or unconstrained run) are pegged to that order in Production Pegging table. When BOH is pegged to an order, it means that order is getting fulfilled through the existing stock. Similarly, if Expected receipts is pegged to an order, it means that order is getting fulfilled from the projected supply.

4. Capacity Consumption Pegging

Capacity pegging tables gives us information of the amount of capacity consumed at various material nodes in various time buckets to fulfil a unique order.

from.[Resource].[Resource]	from.[Lot].[Lot ID]	from.[Activity].[Activity]	Supply Bucket US	to.[Item].[Item]	to.[Location].[Location]	to.[Order].[Order Name]	to.[Time].[Fiscal Week]	Capacity Pegged Quantity
Capacity for Final Assembly at Plant_1	000SCS_LOT	Final Assembly ...	W26-2015	FG_1	RDC_1	Order_1	W31-2015	100.00
Capacity for Sub-Assembly for FG_1 at Plant_1	000SCS_LOT	Sub-Assembly f...	W25-2015	FG_1	RDC_1	Order_1	W31-2015	100.00
Capacity for Sub-Assembly for FG_1 at Plant_1	000SCS_LOT	Sub-Assembly f...	W25-2015	FG_1	RDC_1	Order_1	W31-2015	100.00

<from_members> is a combination of following Dimensions.

- Capacity Node
 - [Resource].[Resource]
- Activity Node
 - [Activity].[Activity]
- Lot.[Lot ID]
- [Time].[Fiscal Week]

<to_members> is a combination of following Dimensions.

- Demand Node
 - [Order].[Order Name]
 - [Time].[Fiscal Week]
 - [Customer].[Customer]
- Material Node
 - [Item].[Item]
 - [Location].[Location]

Properties:

Capacity Pegged Quantity: The data type of this edge property is number and the aggregation level is sum. It links a unique order (demand signal incident on Material node) to all the resources (connected via activity) that are required for the fulfilment of that particular order in various time buckets. In simple terms, if an order requires some capacities of certain resources to process itself, the information of the capacity pegged to that order is stored in Capacity Pegged Quantity.

Measure Groups

Apart from the graphs, flow plans and peggings, here are the other measure groups that can be configured based on the functionality desired.

1. Delivery Plan

Measures	Granularity
1. Total Met OnTime Quantity	1. Demand Signal

2. Total Late Quantity 3. Maximum Lateness Buckets 4. Weighted Late Quantity 5. Total Short Quantity	2. Material Node 3. Version
---	--------------------------------

2. Activity Node Time Parameter

Measures	Granularity
1. Is Holiday	1. Activity Node 2. Time 3. Version

3. Capacity Availability

Measures	Granularity
1. Capacity Availability	1. Capacity Node 2. Time 3. Version

4. Material Production Graph Planned WIP

Measures	Granularity
1. Material Consumption Planned WIP	1. Activity Node 2. Material Node 3. Lot 4. Time 5. Version

5. Material Consumption Graph Planned WIP

Measures	Granularity
2. Material Consumption Planned WIP	1. Activity Node 2. Material Node 3. Lot 4. Time 5. Version

6. Capacity Graph Planned WIP

Measures	Granularity
Capacity Consumption Planned WIP	- Activity Node - Capacity Node - Lot - Time - Version

7. Inventory Plan

Measures	Granularity
- Beginning On Hand Material Quantity - Ending On Hand Material Quantity	- Material Node - Time - Version

8. Material Node Parameters

Measures	Granularity
- Constant Target Inventory - Constant WOS Target - Infinite Inventory - Segment	- Material Node - Version

9. Material Node Time Parameters

Measures	Granularity
- IL Target Inventory - IL WOS Target	- Material Node Components - Time - Version

Model Validations - Node properties and measure groups

One of the prerequisites to successfully run the supply chain solver is to model and configure it appropriately. Logging of errors also need to be appropriate for the configurators to know what is not modeled correctly. SCS now has these second level of validations in place. If a measure which needs to be grained at composite dimensions as defined in node parameters is not correct, then the following warnings/errors are thrown in the GraphCubeServer log.

Example to be checked in the GraphCubeServer log (Can be seen in the debugging workspace)

```
TenantId:<TenantID> UserId:<UserId> (<Server IP>) ERROR LiveServer (<Thread>)
[Capacity Consumption Plan Measure Name] = Measure.[Ending On Hand Material Quantity]
has invalid granularity.

TenantId:<TenantID> UserId:<UserId> (<Server IP>) WARN LiveServer (<Thread>)
Optional config value [Constant WOS Target Measure Name] has been assigned invalid
Measure name [Constant WOS Targ].
```

Capacity consumption plan needs to be at resource node, but is specified using a measure at a material node and hence throws the ERROR. If it is not mandatory, then it throws a warning and ignores the same. Note that if there is an ERROR in the log, the entire transaction is discarded and SCS run is ignored.

SCS Feature Functions

Planning Related

SCS is an order by order solver, which picks up one demand after another based on priority. It traverses the supply chain upstream consuming material and capacity in JIT (just-in-time) fashion. It adheres to the supply chain network rules modeled by the planner.

Generic Planning Behaviour

Business Need

Planners can model supply chain network as per their business and state the planning rules. A supply chain network shows the links between organizations and how information and materials flow between these links.

Functional Description

SCS follows an order by order heuristic, where it takes one demand (order or net forecast) at a time and checks for material and capacity availability upstream. If capacity/material is not available in JIT time, it will then check material and capacity in earlier time buckets, <one bucket at a time> until it reaches the build ahead limit. If the demand quantity is not fulfilled on primary path, it will explore other paths in the order of priority.

If SCS is unable to fulfill demands either in JIT time bucket or in earlier time buckets, it then looks for material /capacities to satisfy those demands late.

When checking for build late, SCS will check for all activities providing that material and for each such activity it will go bucket by bucket till build late limit is reached.

Simultaneously, capacity availability will be checked by SCS during this process. The least late supplies among all late supplies are selected. If supply is not available in these periods, then that portion of the demand will be shorted.

In a nutshell, SCS runs with the objective of satisfying the demands (generating a feasible plan) while simultaneously adhering to the material, capacity and network constraints in the supply chain.

Unconstrained Run

Business Need

Planners need a view of the total supply at each level of supply chain assuming there is no restriction on the availability of both capacity and material. This is usually used for material requirements planning (MRP) where one needs to provide the visibility to the supplier on the amount of material required from them.

Functional Description

In the unconstrained planning, SCS considers enough material or lot is available on the most upstream material node and there is no dearth of capacity, so that the demand is satisfied on the specified date. Lead Time, effectivity, holidays etc are honoured during the plan run.

The following combinations are possible.

- Unconstrained run - Both capacity and material are unconstrained [[Functional Test #1](#)]
- Material constrained, capacity unconstrained [[Functional Test #2](#)]
- Material unconstrained, capacity constrained [[Functional Test #3](#)]
- Constrained run - Both material and capacity are constrained. [[Functional Test #18](#)]

Configuration

Go to Designer→Edit model→ Rule → Plugins → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Material Constrained → check if “TRUE”, uncheck if FALSE

Material Constraint takes a boolean value of True/False. When set to FALSE indicates that material is unconstrained. This means infinite material supply is available at all the upstream most nodes which will enable the solver to plan demands.

Go to Designer→Edit model→ Rule → Plugins → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Parameters → Capacity Constrained → check if “TRUE”, uncheck if FALSE

If capacity constraint is true, Solver will solve for the demands by adhering to available capacity. If the capacity constraint is false, Solver will assume that there is infinite supply of capacity on all resources

An unconstrained run can be done for the following to solve demands in a specified scope.

- material, or

- | | |
|--|-------------------------------------|
| Material Constrained  | ☒ |
| Capacity Constrained  | ☐ |
| This run a material constrained capacity unconstrained plan | |
| • resources, or | |
| Material Constrained  | ☐ |
| Capacity Constrained  | ☒ |
| This runs a material unconstrained and capacity constrained plan | |
| • both material and resources | |
| Material Constrained  | ☐ |
| Capacity Constrained  | ☐ |
| This runs a material unconstrained and capacity unconstrained plan | |

Constrained Unconstrained

Business Need

Constraint plan provides the best use of resources to meet the demand. Planner need visibility on the demand satisfaction if constraints are relaxed. This will help them to evaluate outsourcing or building capacity in-house; getting more supplies from other suppliers etc. This would require them to utilize the resources of enterprise to the fullest before venturing out. Constrained unconstrained feature provides this capability as part of a single solver run.

Functional Description

In the constrained unconstrained planning, SCS first runs a fully constrained plan (material and capacity constrained) to satisfy the orders (on time and late). For only the short orders, it triggers an unconstrained run (material and capacity unconstrained) which will relax the material availability at the most upstream material node and relaxes the capacity to meet the order in the JIT bucket. This provides visibility to the planner as to how much more demand can be satisfied on time, the additional material required and the additional capacity required [[Functional Test #49](#)].

Configuration Steps

Goto Workspace→Designer→Edit

model→Rule→Plugin→Global→o9.GraphCubePlugins→SupplyChainSolver→SupplyChainSolver→Parameters → Constrained Unconstrained → check if “TRUE”, uncheck if FALSE

Constrained Unconstrained  ☒

Constrained Unconstrained is provided as a dynamic argument to the SCS Plugin. It can take Boolean values:

- TRUE: Existing settings of Material Constrained and Capacity Constrained gets overridden.
- FALSE (default): Existing settings of Material Constrained and Capacity Constrained gets honoured. NULL is equivalent to FALSE

Constrained Unconstrained: When this turns on (check box is ticked), it triggers a

constrained plan followed by an unconstrained plan. In this case, solver does a fully constrained plan (material and capacity constrained) to satisfy the orders. For the short orders, it triggers an unconstrained run (material and capacity unconstrained). This provides visibility to the planner as to how much more demand can be satisfied on time along with the additional material and capacity requirements. The default is false i.e. the checkbox is not ticked.

Parallel SCS Run

Business Need

Scenario planning is a structured way used by planners to evaluate different possibilities. This can help them trade off risks and opportunities. Scenarios can be with respect to upside/downside in demand; make vs buy for supply and so on. These are usually full dataset runs with different planning parameters. If indeed one of the scenario unfolds during execution phase, the planners can quickly tell the organization what can be done to contain the risks and what is the next best alternative at that point of time. Marketplace changes are happening at a rapid pace and this is a good way to evaluate different options. Running scenarios in parallel helps cut down the time so that the results for comparison are available sooner than later.

Functional Description

SCS plan for separate scenarios using separate SCS instances can be enabled through action button. An example for two scenarios running two instances is provided below. In this case, 2 separate transactions will be spawned and plan for each scenario will be run in parallel, thereby eliminating the need to trigger these separately one after the other. Plans will be created for the selected scenarios.

Configuration

1. Create a picklist which has all instances of Supply Chain Solver:
Goto Models (offline) → Manager Picklist → Add new record

The screenshot shows the 'Manage Pick Lists' interface. A table lists various picklists, including 'SCS Instance PL'. An 'Edit' modal is open for this picklist, showing fields for 'PickList Name' (SCS Instance PL), 'Description' (SCS Instance PL), 'IsMultiSelectAllowed' (checkbox), and 'Data Type' (string). The modal has 'Update' and 'Cancel' buttons.

PickList Name ↑	Description	IsMultiSelectAllowed	Data Type	
SCS Instance PL			string	Edit Delete
Child	Child		string	Edit Delete
EABOX	EABOX		string	Edit Delete
FGSub	FGSub		string	Edit Delete
FulfillmentPolicy	Fulfillment Policy		string	Edit Delete
SCS Instances	SCS Instances		string	Edit Delete
Unconstrained	Unconstrained		string	Edit Delete
Version Category List	Version Category List		string	Edit Delete

Now, add all instances of SCS for which you want to run Parallel SCS by clicking 'Add new record':

The screenshot shows the 'PICK LIST VALUES' table. It has columns for 'Display Name', 'Value', 'Display Position ↑', and actions. Two records are shown, both for 'Supply Chain Solver Constrained'.

Display Name	Value	Display Position ↑	
[Supply Chain Solver Constrained Day]	[Supply Chain Solver Constrained Day]	0	Edit Delete
[Supply Chain Solver Constrained]	[Supply Chain Solver Constrained]	1	Edit Delete

Note: All the picklist values should be enclosed in []

2. Configure an action button as follows:
Goto Designer → Edit Reports→ Action Buttons → Add New Action Button

Options	Config	RuleOptions	FieldBindings	DataSources
Type	Rule			
Name	ParallelSCS			
Tooltip	Parallel SCS run			
Global	☒			
Icon	align-justify			
Position	10			
Notify All <i>i</i>	☐			
Auto Execute <i>i</i>	☐			
Is Async	☒			
Has Members Updates <i>i</i>	☐			
Compact Display <i>i</i>	☐			
Two Column Display <i>i</i>	☐			
✓ Save ⌂ Cancel				

Options	Config	RuleOptions	FieldBindings	DataSources
☐ Validate				
SubmitButton				
Label	Submit			
ProgressMessage				
SuccessMessage				
ErrorMessage				
ConfirmMessage				
ResetButton				
Label	Reset			
CloseButton				
Label	Close			
ConfirmMessage	Unsaved changes will be lost. Continue?			
✓ Save ⌂ Cancel				

Options	Config	RuleOptions	FieldBindings	DataSources	Navigation
IBPLRules +					
exec parallel begin {{#if Instance1}} block begin exec plu... template ☐ input ☐ output <pre> exec parallel begin {{#if Instance1}} block begin exec plugin instance {{Instance1}} for </pre>					
JavascriptRules +					

The details of the query typed in the template section is as follows:

```
exec parallel begin
```

```
    {{#if Instance1}}
    block begin
    exec plugin instance {{Instance1}} for Measures {[Demand Quantity]}
    using scope (Version.[Version Name]{{#filter Version1}} * Order.[Order
    Name] );
    end
    {{/if}}
```

```
    {{#if Instance2}}
    block begin
    exec plugin instance {{Instance2}} for Measures {[Demand Quantity]}
    using scope (Version.[Version Name]{{#filter Version2}} * Order.[Order
    Name] );
    end
    {{/if}}
```

```
end;
```

Options Config RuleOptions **FieldBindings** DataSources

FieldBindings

+ picklist ▼

▶ Instance1	▲ ▼ ✕
▶ Version1	▲ ▼ ✕
▶ Instance2	▲ ▼ ✕
▶ Version2	▲ ▼ ✕

✓ Save ✕ Cancel

Select source as 'picklist' from drop-down picklist

Options Config RuleOptions **FieldBindings** DataSources

FieldBindings

+ picklist ▼

Now, values of Instance1 and Instance2 can be set as follows:

Instance1

fieldName: Instance1

title: Instance1

Description:

type: string

fieldGroup:

selectionType: single

select:

source: picklist

key: SCS Instance PL

visible: ☒

isEditable: ☒

validation

maxLength: 0

required: ☒

Here, set the 'key' to the picklist that we created in Step 1.

Now, select source as 'members' from drop-down picklist

Options Config RuleOptions **FieldBindings** DataSources Navigation

FieldBindings + members

Now, values of Version1 and Version2 can be set as follows:

Version1

fieldName: Version1

title: version for Instance1

Description:

type: picklist

fieldGroup:

selectionType: single

select:

source: members

DimensionName: Version

LevelAttributeName: Version Name

FilterExpression:

visible: ☒

isEditable: ☒

validation

maxLength: 0

required: ☒

filterBinding:

Note: The FieldBindings string is case-sensitive, only enter those strings which are a part of IBPL command in 'IBPLRules' section. For eg: if you define 'instance1' and 'version1' in FieldBindings, Action Button will error out, you should define 'Instance1' and 'Version1'

- Click on  to save the action button.

4. Refresh the page and click on 'ParallelSCS' action button to run the query. Select the relevant Instances and Versions and 'Submit' to run the action button.

ParallelSCS

Instance1 *
[Supply Chain Solver Constrained Day]

Instance2 *
[Supply Chain Solver Constrained]

Version for Instance1 *
CurrentWorkingView

Version for Instance2 *
ParallelSCS

Submit Reset Close

Note: The plans should be run on different scenarios. Running it on the same scenario causes conflict in LiveServer

Pegging

Business Need

Planners need the capability to identify the supply path for a particular order and how that got planned in the supply chain. This means one should be able to see what all material nodes and capacity nodes are linked to an order. Pegging provides an ability to link the demand to the supply, be it material or capacity.

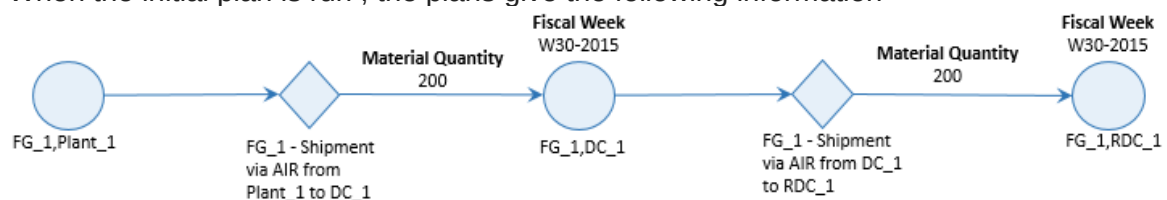
Functional Description

Pegging links demand to its incoming supply. SCS uses FIFO (first-in-first-out) pegging to match the producers to consumers on any node(material or capacity).

Consider 2 orders.

- Order_1 → FG_1,RDC_1 of 60 units on w30-2015
- Order_2 → FG_1,RDC_1 of 140 units on w30-2015

When the initial plan is run , the plans give the following information



However, planners not only need information on the aggregated flow of goods across various material nodes, but also need information on how many “units” are pegged to individual orders. This information can be viewed in pegging tables.

Pegging is a post process that is computed after the planning run and before the export.

[\[Functional Test #1\]](#)

Sometimes , the available inventory remains unutilized, due to various reasons such as excess stock or date effectivity. In that case, the unutilized inventory is pegged to Inventory Demand in the pegging tables. [\[Functional Test #3\]](#).

Planners can decide if they wish to see pegging reports or not. In such a case, they can flag the plan thereby restricting SCS to export pegging reports. [\[Functional test #270\]](#).

The effect of the flag is seen when plan is run on a large dataset. With export of pegging reports restricted, planning is done relatively faster which in turn leads to better performance.

Configuration

Pegging information can be maintained by configuring the following relationship types.

- Capacity_Pegging
- Consumption_Pegging
- Production_Pegging

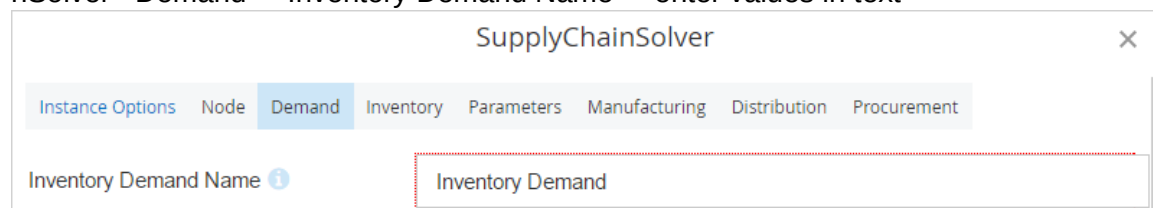
If Inventory is more than demands, then Planners may see that the inventory left after satisfying all demands is pegged to Inventory Demand.

Inventory Demand is configured as a member in Customer and Order dimension.

It can be configured in the SCS config as follows:

Goto Workspace→Designer→Edit

model→Rule→Plugin→Global→o9.GraphCubePlugins→SupplyChainSolver→SupplyChainSolver→Demand → Inventory Demand Name → enter values in text



The screenshot shows a window titled 'SupplyChainSolver' with a close button (X) in the top right corner. Below the title bar is a tabbed interface with tabs labeled 'Instance Options', 'Node', 'Demand', 'Inventory', 'Parameters', 'Manufacturing', 'Distribution', and 'Procurement'. The 'Demand' tab is currently selected. Within this tab, there is a label 'Inventory Demand Name' followed by a text input field. The input field contains the text 'Inventory Demand' and is highlighted with a red dashed border.

In the SCS Configuration wizards, planners have the flexibility to see various flavours of pegging reports through the use of flags.

No Pegging Flag:

Goto Workspace→Designer→Edit

model→Rule→Plugin→Global→o9.GraphCubePlugins→SupplyChainSolver→SupplyChainSolver→Parameters → No Peggings → Check if TRUE or Uncheck if FALSE [By default its Unchecked i.e FALSE]



The screenshot shows a section of the configuration window with the label 'No Peggings' and a checked checkbox. There is a small information icon (i) to the left of the label.

No Peggings is provided as a dynamic argument to the SCS Plugin. It can take Boolean values:

- TRUE:Pegging information is not exported after the plan is run. However, constraint report is exported for orders that are not built in JIT along primary path.
- FALSE (default):All the pegging information is exported after plan is run. NULL is equivalent to FALSE

Note : For more information on Pegging tables please refer to “Output of the Solver” under “SCS Configuration”

SCS provides flexibility to planners on whether or not they want to view [constraint report information](#) of all the partially/fully short/late orders.

Goto Workspace→Designer→Edit

model→Rule→Plugin→Global→o9.GraphCubePlugins→SupplyChainSolver→SupplyChainSolver→Parameters → No Constraints Report→ Check if TRUE or Uncheck if FALSE [By default its Unchecked i.e FALSE]



The screenshot shows a section of the configuration window with the label 'No Constraints Report' and an unchecked checkbox. There is a small information icon (i) to the left of the label.

No Constraints Report is provided as a dynamic argument to the SCS Plugin. It can take Boolean values:

- TRUE: [Constraint Report information](#) is not exported after the plan is run. However, pegging report is exported for orders that are partially built in JIT along primary path.
- FALSE (default): [Constraint Report information](#) is exported after plan is run. NULL is equivalent to FALSE

Incremental Planning (Demand Side)

Business Need

Supply chains need to be responsive to stay competitive in the marketplace. One of the common uncertainties is around demand. It could be changes to forecast based on intelligence input received after the batch run or a rush order arriving at the last moment. The planner needs to check what can be done about the same and act in an incremental manner on top of the existing plan.

Functional Description

When it comes to manipulate demand, one can expect the following

1. Order addition
2. Order quantity change
3. Order deletion
4. Order date change

Note :Order has Time dimension as one of its granularities. Modifying granularity data in LiveServer is not yet implemented. Therefore, Order Date change can be done by deleting an order of previous due date and creating the same order with the new due date.

The basic rule of incremental (net change) planning is to handle exceptions while minimizing the impact on existing commitments.

Configuration Steps

Configure second instance of SupplyChainSolver with the following changes:

Go to Designer→ edit model → Rule → -> Plugins → Global →
o9.GraphCubePlugins → SupplyChainSolver→ SupplyChainSolver1 →
Parameters → Plan Type → enter values in textbox

Plan Type 

INCREMENTAL_PLAN

Incremental Planning should be done on modified or new orders. SCS then keeps previous plan under consideration while planning the aforementioned orders.

Planners should mark the orders on which incremental planning is to be done. For this a new measure named Incremental_Allowed needs to be configured. This measure has the same granularity as Demand Quantity and hence can be defined in the same measure group. Incremental_Allowed measure takes boolean values.

- True : incremental planning is allowed on the corresponding demands.
- False: Demand does not participate in incremental planning

Note: False behaves same as NULL

This measure should also be specified in the SCS config as follows :

Go to Designer→ edit model → Rule → -> Plugins → Global →
o9.GraphCubePlugins → SupplyChainSolver→ SupplyChainSolver1 → Demand

→ Is Eligible for Incremental Plan Measure Name → choose measure from the drop down menu

Is Eligible For Incremental Plan Measure Name  Incremental_Allowed 

Limitation

- SCS Incremental Planning is on supply side changes are restricted to expected receipts and capacity. Also only basic supply chain structure is supported and not advanced capabilities.
- Incremental plan does not touch stocks created during full plan. Thus those stocks are not available for supplying demands during incremental plan.

Incremental Planning (Supply Side)

Business Need

Just like there are uncertainties around demand, one can expect inconsistencies/uncertainties in the supply {those which participate in demand fulfillment}. This may be due to factors both internal or external to the supply chain such as defective or damaged raw materials or improper coordination between planning and execution. Planners need to have the visibility as to what is the effect of increase/decrease of existing supply on existing demands thereby acting in an incremental manner on top of existing plan. Planners frequently use this to expedite the supply to meet existing demands which are short/late.

Functional Description

The basic rule of incremental (net change) planning is to handle small changes while minimizing the impact on existing plan.

When it comes to supply changes, one can expect the following:

1. Supply Addition
2. Supply Deletion
3. Supply Quantity Change(increase/decrease)
4. Supply Date Change

Supply can be any of the following:

1. Material Quantity
 - a. Expected Receipts
 - b. BOH
2. Capacity Availability

Note: Supply has Time dimension as one of its granularities. Modifying granularity data in LiveServer is not yet implemented. Therefore Supply Date change can be done by deleting an order of previous date and creating the same order with new date.

Planners can now also keep a track of all the demands that were replanned incrementally as a result of supply changes. This is only for incremental planning.

Marking Demands in the presence of Supply changes

There are scenarios when the capacity/material supply is not available to satisfy the demands. Demands are thus partially or fully shorted in such scenarios. In such a scenario planners increase supply (material / supply) and mark them. Additionally, they also need to mark demands that were partially or fully short initially. When incremental

plan is run, the marked demands are thus replanned and planners can check if the increase in supply was sufficient enough to satisfy the demand.

Configuration Steps

Configure second instance of SupplyChainSolver with the following changes:

Goto Workspace→Designer→Edit

model→Rule→Plugin→Global→o9.GraphCubePlugins→SupplyChainSolver→SupplyChainSolver1 → Parameters → Plan Type → enter values in textbox

Plan Type ⓘ

INCREMENTAL_PLAN

Incremental Planning should be done on modified or new supplies. SCS then keeps previous plan under consideration while planning the modified or new supplies.

Planners should mark the supplies participating in the incremental plan. The following measures need to be configured.

- For changes made to existing Expected Receipts and BOH
 - Measure
 - Supply Changed:
This measure takes boolean values:
 1. True : indicates that Supply (ER or BOH) on a material node for a specified time bucket has changed and hence should take part in incremental planning
 2. False (default): indicates that Supply(ER or BOH) on a material node for a specified time bucket should not take part in incremental planning.
Note: False behaves same as NULL
 - Granularity
 - Material Node
 1. Item.Item Name
 2. Location.Location Name
 - Time.Fiscal Week
 - Version.Version Name
- For Capacity Availability changes
 - Measure
 - Capacity Changed
This measure takes boolean values:
 1. True : indicates that Capacity Availability in the specified time bucket has changed and hence should take part in incremental planning
 2. False: indicates that Capacity Availability should not take part in incremental planning.
Note: False behaves same as NULL
 - Granularity
 - Capacity Node
 1. Resource.Resource Name
 - Time.Fiscal Week
 - Version.Version Name
- For tracking the demands that were replanned as a part of supply changes
 - Measure

- Participated in Incremental Plan: This measure is the output of the Incremental Plan. It takes boolean values.
 1. TRUE: If an order has been replanned after the supply change, then the value of this measure for the corresponding order is set as TRUE
 2. FALSE/NULL: If an order remains untouched from being incrementally planned, then the value of this measure for the corresponding order is set as FALSE.
- Granularity (Same as Delivery plan as it is an output measure)
 - Demand Node
 1. Order.Order Name
 2. Time.Fiscal Week
 3. Customer.Customer Name
 - Material Node
 1. Item.Item Name
 2. Location.Location Name
 - Version.Version Name

Note: If one runs the full plan by setting [Delete SCS Plan Data For Version] to true, then this gets reset to null. If only incremental plans are run in sequence, then it gets updated only for the demand that are replanned.

The measure Participated in Incremental Plan should be set only in Delivery Plan measure group and not in Demand Quantity measure group. Placing this measure in the Demand Signal measure group will cause the SCS plugin to error out.

These measures should also be specified in the SCS config as follows :

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver1 → Inventory → Supply Modified Measure Name → choose measure from the drop down menu

Supply Modified Measure Name  Supply Changed ▼

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver1 → Manufacturing → Capacity Modified Measure Name → choose measure from the drop down menu

Capacity Modified Measure Name  Capacity Changed ▼

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver1 → Edit → Demand → Re-Planned By Interactive Plan Measure Name → choose measure from the drop down menu

Re-Planned By Interactive Plan Measure Name  PIP ▼

The supply and capacity measures need to be used along with demand measure in the SCS plugin command for the incremental plan to consider those changes. For example,

*exec plugin instance SupplyChainSolver for Measures {[Demand Quantity], [Expected Receipts], [BOH], [Capacity Availability]} using scope ([Version].[Version Name].[CurrentWorkingView] **

*Time.[Fiscal Week] * Order.[Order Name]) arguments {[Material Constrained], "True"}, ([Capacity Constrained], "True"), ([Plan Type], "INCREMENTAL_PLAN")};*

Limitations

Supply changes with respect to the following are not yet supported in incremental planning

1. WIP
2. Lock Plans

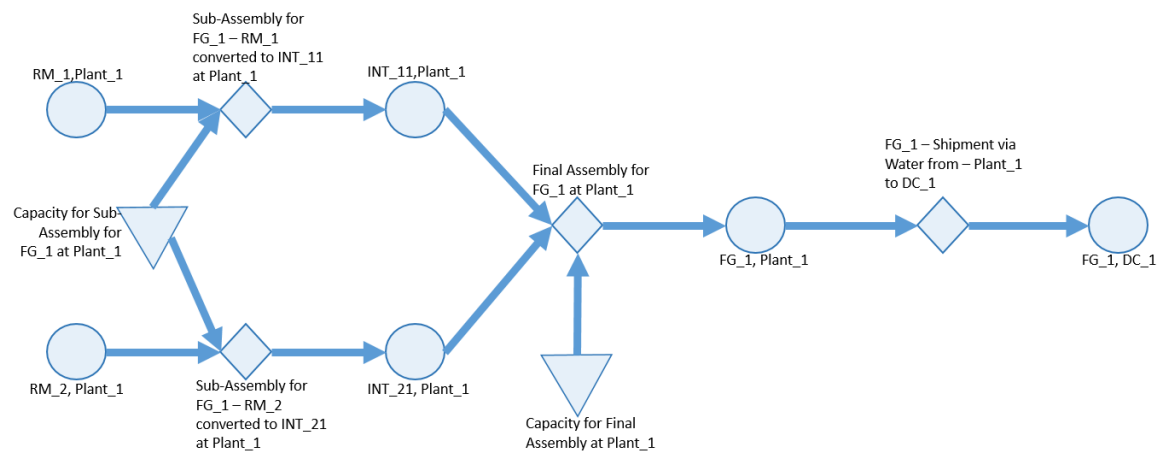
Incremental Plan Vs Full Plan

Demand Change

Demand Decrease (without Lotsize)

([FT#97](#), [FT#99](#))

Consider the following supply chain



Demands are as follows:

Order Name	Item	Location	Data	W13-2015
☒ Order_1	☒ FG_1	DC_1	Demand Quantity	10
			Demand Priority	1
☒ Order_2	☒ FG_1	DC_1	Demand Quantity	20
			Demand Priority	2
☒ Order_3	☒ FG_1	DC_1	Demand Quantity	30
			Demand Priority	3
☒ Order_4	☒ FG_1	DC_1	Demand Quantity	40
			Demand Priority	4

Supplies are as follows:

Item	Location	Lot ID	Data	W01-2015
RM_1	Plant_1	SCS BOH	BOH	80
RM_2	Plant_1	SCS BOH	BOH	80

Activity Lead times are as follows

Activity	Activity Lead Time
Sub-Assembly for FG_1 - RM_2 converted to INT_21 at Plant_1	1
Sub-Assembly for FG_1 - RM_1 converted to INT_11 at Plant_1	1
Final Assembly for FG_1 at Plant_1	1
FG_1 - Shipment via Water from - Plant_1 to DC_1	2

Planning behavior when plan type = FULL PLAN

When material constrained plan is run on the above data, demands are sorted according to their respective priorities.

Of the available inventory of 80 units

Order_1 is planned and 10 units are allocated to it.

Effective Inventory now becomes 70 units

Order_2 is planned and 20 units get allocated to it

Effective Inventory now becomes 50 units

Order_3 is planned and 30 units get allocated to it

Effective Inventory now becomes 20 units

Order_4 is planned and the remaining 20 units get allocated to it

Effective Inventory now becomes 0 units

Thus Order_4 is partially short.

Delivery plan for the above 4 orders looks as follows:

Order Name	Customer Name	Item	Location	Data	W13-2015
Order_1	Cust_1	FG_1	DC_1	Total MetOnTime Quantity	10
				Total Short Quantity	
Order_2	Cust_1	FG_1	DC_1	Total MetOnTime Quantity	20
				Total Short Quantity	
Order_3	Cust_1	FG_1	DC_1	Total MetOnTime Quantity	30
				Total Short Quantity	
Order_4	Cust_1	FG_1	DC_1	Total MetOnTime Quantity	20
				Total Short Quantity	20

Planning behavior when plan type = INCREMENTAL PLAN (FT #097)

Demand of Order_3 is changed from 30 units to 10 units and this demand is marked for incremental planning as shown below.

Order Name	Item	Location	Data	W13-2015
☒ Order_3	☒ FG_1	DC_1	Demand Quantity	10
			Incremental_Allowed	☒

Incremental Plan is run on the above model

SCS picks all the orders which are flagged for incremental planning.

In this case incremental planning demand set will have Order_3.

Order_3 is unplanned .

Consequently 30 units of inventory is released.

Effective inventory now becomes 30 units

Order_3 is replanned.

Of the available inventory 10 units are allocated to Order_3

Effective inventory now becomes 20 units

The remaining 20 units are allocated to stock.

Since Order_4 for not marked , it is not picked up for incremental planning thereby remaining partially short.

Delivery plan for Order_3 and Order_4

Order Name	Customer Name	Item	Location	Data	W13-2015
☒ Order_1	☒ Cust_1	☒ FG_1	DC_1	Total MetOnTime Quantity	10
				Total Short Quantity	
☒ Order_2	☒ Cust_1	☒ FG_1	DC_1	Total MetOnTime Quantity	20
				Total Short Quantity	
☒ Order_3	☒ Cust_1	☒ FG_1	DC_1	Total MetOnTime Quantity	10
				Total Short Quantity	
☒ Order_4	☒ Cust_1	☒ FG_1	DC_1	Total MetOnTime Quantity	20
				Total Short Quantity	20

Pegging for excess inventory

SKU	from. [Location]. [Location]	from. [Activity]. [Activity]	Supply Timebucket	from.[Lot]. [Lot ID]	to. [Location]. [Location]	to.[Item]. [Item]	to.[Order].[Order Name]	Order Timebucket	Material Production Pegged Quantity
RM_2	Plant_1	BOH Activity	W01-2015	SCS BOH	Plant_1	RM_2	Inventory De...	W52-2015	20.00
RM_1	Plant_1	BOH Activity	W01-2015	SCS BOH	Plant_1	RM_1	Inventory De...	W52-2015	20.00

If both Order_3 and Order_4 are marked for incremental planning. [\(FT# 099\)](#)

Order Name	Item	Location	Data	W13-2015
☐ Order_3	☐ FG_1	DC_1	Demand Quantity	10
			Incremental_Allowed	☒
☐ Order_4	☐ FG_1	DC_1	Demand Quantity	40
			Incremental_Allowed	☒

SCS picks all the orders that have been marked for incremental planning.

In this case incremental planning demand set will have Order_3 and Order_4

Orders are sorted on the basis their priorities

Unplanning of orders are done on the basis of their priorities starting from lower priority orders and then moving on it higher priority orders

Order_4 has a priority 4 (lower than that of Order_3) and thus is unplanned 1st.

Inventory allocated to it is released.

Effective inventory now becomes 20 units

After Order_4 is unplanned, Order_3 is unplanned.

Inventory allocated to it is released.

Effective inventory now becomes 20(from Order_4) + 30(from Order_3) = 50 units

Planning of orders are done on the basis of their priorities starting from higher priority orders and then moving on it lower priority orders

Order_3 has a priority 3 (higher than that of Order_4) and thus is planned 1st

Inventory of 10 units are allocated to it.

Effective inventory now becomes 40 units

After Order_3 is planned, Order_4 is planned.

Inventory of 40 units are allocated to it.

Effective inventory now becomes 0 units

Delivery plan looks as follows:

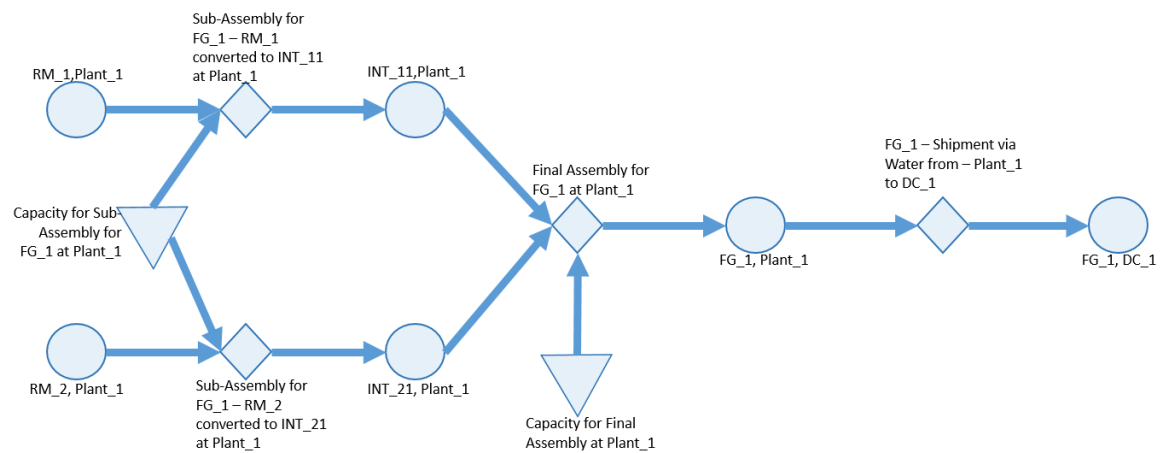
Order Name	Customer Name	Item	Location	Data	W13-2015
☐ Order_1	☐ Cust_1	☐ FG_1	DC_1	Total MetOnTime Quantity	10
				Total Short Quantity	
☐ Order_2	☐ Cust_1	☐ FG_1	DC_1	Total MetOnTime Quantity	20
				Total Short Quantity	
☐ Order_3	☐ Cust_1	☐ FG_1	DC_1	Total MetOnTime Quantity	10
				Total Short Quantity	
☐ Order_4	☐ Cust_1	☐ FG_1	DC_1	Total MetOnTime Quantity	40
				Total Short Quantity	

Order_4 is no longer short.

Demand Increase (without Lotsize)

(FT #093, FT#095)

Consider the following supply chain



Demands are as follows:

Order Name	Item	Location	Data	W13-2015
Order_1	FG_1	DC_1	Demand Quantity	10
			Demand Priority	1
Order_2	FG_1	DC_1	Demand Quantity	20
			Demand Priority	2
Order_3	FG_1	DC_1	Demand Quantity	30
			Demand Priority	3
Order_4	FG_1	DC_1	Demand Quantity	40
			Demand Priority	4

Supplies are as follows:

Item	Location	Lot ID	Data	W01-2015
RM_1	Plant_1	SCS BOH	BOH	80
RM_2	Plant_1	SCS BOH	BOH	80

Activity Lead times are as follows

Activity	Activity Lead Time
Sub-Assembly for FG_1 - RM_2 converted to INT_21 at Plant_1	1
Sub-Assembly for FG_1 - RM_1 converted to INT_11 at Plant_1	1
Final Assembly for FG_1 at Plant_1	1
FG_1 - Shipment via Water from - Plant_1 to DC_1	2

Planning behavior when plan type = FULL PLAN

When material constrained plan is run on the above data, demands are sorted according to their respective priorities.

Of the available inventory of 80 units

Order_1 is planned and 10 units are allocated to it.

Effective Inventory now becomes 70 units

Order_2 is planned and 20 units get allocated to it

Effective Inventory now becomes 50 units

Order_3 is planned and 30 units get allocated to it

Effective Inventory now becomes 20 units

Order_4 is planned and the remaining 20 units get allocated to it

Effective Inventory now becomes 0 units

Thus Order_4 is partially short.

Delivery plan for the above 4 orders looks as follows:

Order Name	Customer Name	Item	Location	Data	W13-2015
☐ Order_1	☐ Cust_1	☐ FG_1	☐ DC_1	Total MetOnTime Quantity	10
				Total Short Quantity	
☐ Order_2	☐ Cust_1	☐ FG_1	☐ DC_1	Total MetOnTime Quantity	20
				Total Short Quantity	
☐ Order_3	☐ Cust_1	☐ FG_1	☐ DC_1	Total MetOnTime Quantity	30
				Total Short Quantity	
☐ Order_4	☐ Cust_1	☐ FG_1	☐ DC_1	Total MetOnTime Quantity	20
				Total Short Quantity	20

Planning behavior when plan type = INCREMENTAL PLAN [\(FT#093\)](#)

Demand of Order_3 is changed from 30 units to 50 units and this demand is marked for incremental planning as shown below.

Order Name	Item	Location	Data	W13-2015
☐ Order_3	☐ FG_1	☐ DC_1	Demand Quantity	50
			Incremental_Allowed	☒

Incremental Plan is run on the above model

SCS picks all the orders which are flagged for incremental planning.

In this case incremental planning demand set will have Order_3.

Order_3 is unplanned .

Consequently 30 units of inventory is released.

Effective inventory now becomes 30 units

Order_3 is replanned.

Of the available inventory 30 units are allocated to Order_3

Effective inventory now becomes 0 units

20 units of Order_3 still remains satisfied.

Thus Order_3 is partially short.

Since Order_4 for not marked , it is not picked up for incremental planning thereby remaining partially short.

Delivery Plan looks as follows

Order Name	Customer Name	Item	Location	Data	W13-2015
Order_1	Cust_1	FG_1	DC_1	Total MetOnTime Quantity	10
				Total Short Quantity	
Order_2	Cust_1	FG_1	DC_1	Total MetOnTime Quantity	20
				Total Short Quantity	
Order_3	Cust_1	FG_1	DC_1	Total MetOnTime Quantity	30
				Total Short Quantity	20
Order_4	Cust_1	FG_1	DC_1	Total MetOnTime Quantity	20
				Total Short Quantity	20

If both Order_3 and Order_4 are marked for incremental planning. ([FT#095](#))

Order Name	Item	Location	Data	W13-2015
Order_3	FG_1	DC_1	Demand Quantity	50
			Incremental_Allowed	☒
Order_4	FG_1	DC_1	Demand Quantity	40
			Incremental_Allowed	☒

SCS picks all the orders that have been marked for incremental planning.

In this case incremental planning demand set will have Order_3 and Order_4

Orders are sorted on the basis their priorities

Unplanning of orders are done on the basis of their priorities starting from lower priority orders and then moving on it higher priority orders

Order_4 has a priority 4 (lower than that of Order_3) and thus is unplanned 1st.
Inventory allocated to it is released.

Effective inventory now becomes 20 units

After Order_4 is unplanned, Order_3 is unplanned.

Inventory allocated to it is released.

Effective inventory now becomes 20(from Order_4) + 30(from Order_3) =
50 units

Planning of orders are done on the basis of their priorities starting from higher priority orders and then moving on it lower priority orders

Order_3 has a priority 3 (higher than that of Order_4) and thus is planned 1st
Inventory of 50 units are allocated to it.

Effective inventory now becomes 0 units

After Order_3 is planned, Order_4 is planned.

There is no inventory left to be allocated to Order_4 and thus this demand is fully short.

Effective inventory remains as 0 units

Delivery plan looks as follows:

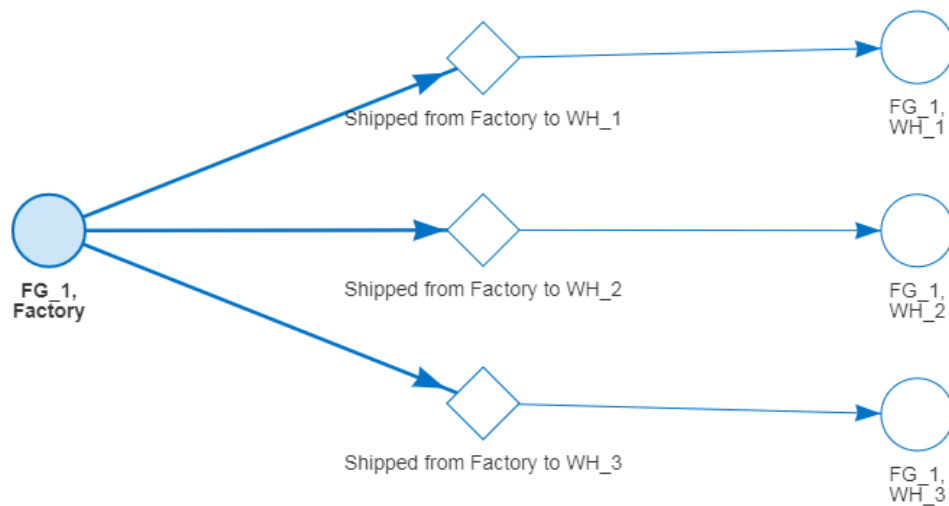
Order Name	Customer Name	Item	Location	Data	W13-2015
Order_1	Cust_1	FG_1	DC_1	Total MetOnTime Quantity	10
				Total Short Quantity	
Order_2	Cust_1	FG_1	DC_1	Total MetOnTime Quantity	20
				Total Short Quantity	
Order_3	Cust_1	FG_1	DC_1	Total MetOnTime Quantity	50
				Total Short Quantity	
Order_4	Cust_1	FG_1	DC_1	Total MetOnTime Quantity	
				Total Short Quantity	40

Supply Change

If only one BOH/ER is decreased

(FT#062)

Consider the following supply chain



The direction of arrows depict the flow of materials

Material node {FG_1,WH_1} will procure from {FG_1, Factory}. Similarly Material node {FG_1,WH_2} will procure from {FG_1, Factory} and Material node {FG_1,WH_3} will procure from {FG_1, Factory}.

Supply is as follows:

Item	Location	Lot ID	Data	W01-2015
FG_1	Factory	SCS BOH	BOH	150

Activity lead times are as follows:

Activity	Activity Lead Time
Shipped from Factory to WH_1	0
Shipped from Factory to WH_2	1
Shipped from Factory to WH_3	2

Demands are as follows:

Order Name	Item	Location	Data	W03-2015
Order_1	FG_1	WH_1	Demand Quantity	50
			Demand Priority	1
Order_2	FG_1	WH_2	Demand Quantity	50
			Demand Priority	2
Order_3	FG_1	WH_3	Demand Quantity	50
			Demand Priority	3

Planning behavior when plan type = FULL PLAN

When a Full Plan is run on the above scenario, Consumption pegging looks as follows:

from. [Item]. [Item]	from. [Location]. [Location]	from.[Lot]. [Lot ID]	from.[Time]. [Fiscal Week]	from.[Activity].[Activity]	to.[Order]. [Order Name]	to.[Item]. [Item]	to. [Location]. [Location]	to.[Time]. [Fiscal Week]	Material Consumption Pegged Quantity
FG_1	Factory	000SCS_LOT	W01-2015	Shipped from Factory to WH_3	Order_3	FG_1	WH_3	W03-2015	50.00
FG_1	Factory	000SCS_LOT	W02-2015	Shipped from Factory to WH_2	Order_2	FG_1	WH_2	W03-2015	50.00
FG_1	Factory	000SCS_LOT	W03-2015	Shipped from Factory to WH_1	Order_1	FG_1	WH_1	W03-2015	50.00

Planning behavior when plan type = INCREMENTAL PLAN

BOH supply are changed and the changed supply is marked as follows:

Item	Location	Lot ID	Data	W01-2015
FG_1	Factory	SCS BOH	BOH	100

Item	Location	Data	Value
FG_1	Factory	Supply Is Changed	☒

The demands remain unchanged .

Inventory is changed from 150 units to 100 units. The reduction in supply is 50 units.

This reduction in supply creates an NOH of 50 units in week 3, the supply to which higher priority demand Order_1 is pegged.

This BOH of 150 units was initially pegged to 3 orders.

These orders are picked up by SCS and added to incremental planning demand set.

The orders are sorted in the order of their priorities. SCS starts unplanning the orders in the reverse order of priorities, i.e starting by unplanning the lowest priority order 1st. Based on the priorities, Order_3 is unplanned and inventory allocated to it is released.. SCS keeps on unplanning the demands until reduction in supply is accounted for. In this case inventory released after unplanned Order_3 is equal to the reduction in supply.

Thus Order_3 is shorted in incremental plan.

Since Order_3 was unplanned, it would have been picked up for planning in the incremental planning mode, but it did not get picked up because there was no supply. Thus this order got shorted.

Order_1 and Order_2 due to higher priorities , remain fulfilled.

After incremental plan is run on the above scenario, the consumption pegging looks as follows:

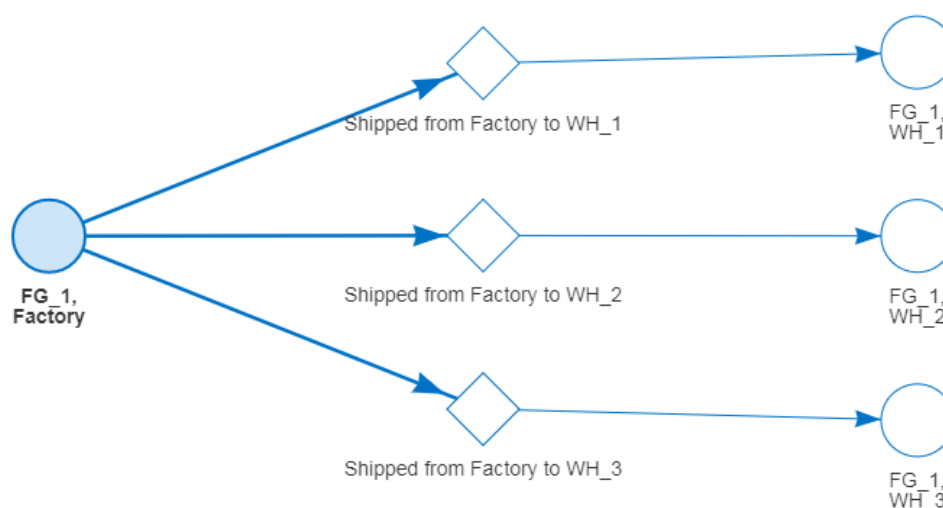
from. [Item]. [Item]	from. [Location]. [Location]	from.[Lot]. [Lot ID]	from.[Time]. [Fiscal Week]	from.[Activity].[Activity]	to.[Order]. [Order Name]	to.[Item]. [Item]	to. [Location]. [Location]	to.[Time]. [Fiscal Week]	Material Consumption Pegged Quantity
FG_1	Factory	000SCS_LOT	W03-2015	Shipped from Factory to WH_1	Order_1	FG_1	WH_1	W03-2015	50.00
FG_1	Factory	000SCS_LOT	W02-2015	Shipped from Factory to WH_2	Order_2	FG_1	WH_2	W03-2015	50.00

It is evident that while incremental planning, the order of the lowest priority was unplanned in order to account for the change in supply.

If only one BOH/ER is increased

([FT #063](#))

Consider the following supply chain



The direction of arrows depict the flow of materials

Material node {FG_1,WH_1} will procure from {FG_1, Factory}. Similarly Material node {FG_1,WH_2} will procure from {FG_1, Factory} and Material node {FG_1,WH_3} will procure from {FG_1, Factory}.

Supply is as follows:

Item	Location	Lot ID	Data	W01-2015
FG_1	Factory	SCS BOH	BOH	90

Activity lead times are as follows:

Activity	Activity Lead Time
Shipped from Factory to WH_1	0
Shipped from Factory to WH_2	1
Shipped from Factory to WH_3	2

Demands are as follows:

Order Name	Item	Location	Data	W03-2015
Order_1	FG_1	WH_1	Demand Quantity	50
			Demand Priority	1
Order_2	FG_1	WH_2	Demand Quantity	50
			Demand Priority	2
Order_3	FG_1	WH_3	Demand Quantity	50
			Demand Priority	3

Planning behavior when plan type = FULL PLAN

When a Full Plan is run on the above scenario, Consumption pegging looks as follows:

from. [Item]. [Item]	from. [Location]. [Location]	from.[Lot]. [Lot ID]	from.[Time]. [Fiscal Week]	from.[Activity].[Activity]	to.[Order]. [Order Name]	to.[Item]. [Item]	to. [Location]. [Location]	to.[Time]. [Fiscal Week]	Material Consumption Pegged Quantity
FG_1	Factory	000SCS_LOT	W03-2015	Shipped from Factory to WH_1	Order_1	FG_1	WH_1	W03-2015	50.00
FG_1	Factory	000SCS_LOT	W02-2015	Shipped from Factory to WH_2	Order_2	FG_1	WH_2	W03-2015	40.00

Demands are sorted on the basis of their given priorities .

Out of the 90 units of BOH available, 50 units are allocated to Order_1 and 40 units are allocated to Order_2.

Thus on the basis of priority

- Order_1 gets fully satisfied,
- Order_2 gets partially satisfied
- Order_3 gets fully short

Planning behavior when plan type = INCREMENTAL PLAN

BOH supply are changed and the changed supply is marked as follows:

Item	Location	Lot ID	Data	W01-2015
FG_1	Factory	SCS BOH	BOH	140

Item	Location	Data	Value
FG_1	Factory	Supply Is Changed	☒

The demands remain unchanged .

The demands that are fully and partially short are marked as shown below. Marking the demands ensures that they participate in incremental planning.

Order Name	Item	Location	Data	W03-2015
☒ Order_2	FG_1	WH_2	Incremental_Allowed	☒
☒ Order_3	FG_1	WH_3	Incremental_Allowed	☒

The demands that were marked for incremental planning were sorted in the order of their priorities.

Inventory is changed from 90 units to 140 units.

Therefore for incremental planning

Effective demand available = 60 units(10 units for Order_2 and 50 units for Order_3

Effective supply available = 50 units.

Based on the priorities, Order_2 is unplanned.

The supply available now is

50(increased by planner) + 40(available after unplanning demands) = 90.

Effective demand available for incremental planning now becomes 100 units.(Order_2 + Order_3)

Demand fulfillment is now done based on the priorities.

Order_2 is planned 1st and 50 units of supply is allocated to it.

Order_3 is planned 2nd and remaining 40 units of supply are allocated to it.

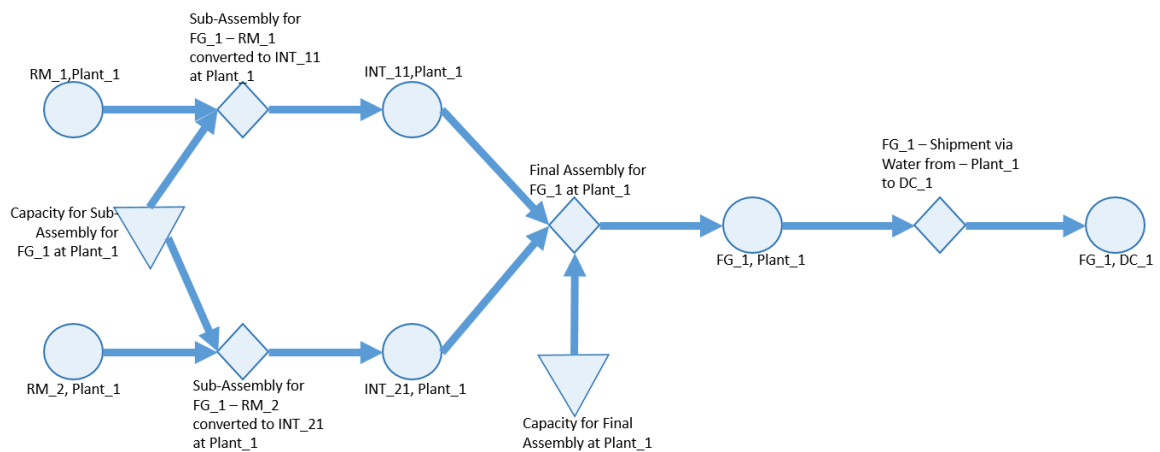
Thus after incremental plan is run, the consumption pegging looks as follows:

from. [Item]. [Item]	from. [Location]. [Location]	from.[Lot]. [Lot ID]	from.[Time]. [Fiscal Week]	from.[Activity]. [Activity]	to.[Order]. [Order Name]	to.[Item]. [Item]	to.[Location]. [Location]	to.[Time]. [Fiscal Week]	Material Consumption Pegged Quantity
FG_1	Factory	000SCS_LOT	W03-2015	Shipped from Factory to WH_1	Order_1	FG_1	WH_1	W03-2015	50.00
FG_1	Factory	000SCS_LOT	W02-2015	Shipped from Factory to WH_2	Order_2	FG_1	WH_2	W03-2015	50.00
FG_1	Factory	000SCS_LOT	W01-2015	Shipped from Factory to WH_3	Order_3	FG_1	WH_3	W03-2015	40.00

If supply is increased and marked but the demands that need to participate in incremental plan are not marked, then SCS pegged the increased supply to Stock.

If only one capacity is decreased ([FT#064](#))

Consider the following supply chain



Demands are as follows:

Order Name	Item	Location	Data	W13-2015
Order_1	FG_1	DC_1	Demand Quantity	10
			Demand Priority	1
Order_2	FG_1	DC_1	Demand Quantity	20
			Demand Priority	2
Order_3	FG_1	DC_1	Demand Quantity	30
			Demand Priority	3
Order_4	FG_1	DC_1	Demand Quantity	40
			Demand Priority	4

Supplies are as follows:

Location	Item	Lot ID	Data	W01-2015
Plant_1	RM_1	SCS BOH	BOH	100
	RM_2	SCS BOH	BOH	100

Capacities available are as follows:

Resource	Data	W09-2015	W10-2015
Capacity for Sub-Assembly for FG_1 at Plant_1	Capacity Availability	1000	
Capacity for Final Assembly at Plant_1	Capacity Availability		100

Activity Lead times are as follows

Activity	Activity Lead Time
Sub-Assembly for FG_1 - RM_2 converted to INT_21 at Plant_1	1
Sub-Assembly for FG_1 - RM_1 converted to INT_11 at Plant_1	1
Final Assembly for FG_1 at Plant_1	1
FG_1 - Shipment via Water from - Plant_1 to DC_1	2

Planning behavior when plan type = FULL PLAN

When a material and capacity constrained plan is run on the above model, capacity consumption pegging for the resource {Capacity for Final Assembly at Plant_1} looks as follows:

to.[Customer]. [Customer Name]	to. [Item]. [Item]	to. [Location]. [Location]	to.[Order]. [Order Name]	to.[Time]. [Fiscal Week]	from.[Lot].[Lot ID]	from.[Resource].[Resource]	Supply Bucket	Data	Value
☐ Cust_1	☐ FG_1	☐ DC_1	☐ Order_1	☐ W13-2015	☐ 000SCS_LOT	☐ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	10.00
			☐ Order_2	☐ W13-2015	☐ 000SCS_LOT	☐ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	20.00
			☐ Order_3	☐ W13-2015	☐ 000SCS_LOT	☐ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	30.00
			☐ Order_4	☐ W13-2015	☐ 000SCS_LOT	☐ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	40.00

When material constrained plan is run on the above data, demands are sorted according to their respective priorities.

Of the available capacity of 100 units

Order_1 is planned and 10 units are allocated to it.

Effective Capacity now becomes 90 units

Order_2 is planned and 20 units get allocated to it

Effective Capacity now becomes 70 units

Order_3 is planned and 30 units get allocated to it

Effective Capacity now becomes 40 units

Order_4 is planned and the remaining 40 units get allocated to it

Effective Capacity now becomes 0 units

Delivery Plan looks as follows:

Order Name	Customer Name	Item	Location	Data	W13-2015
☐ Order_1	☐ Cust_1	☐ FG_1	☐ DC_1	Total MetOnTime Quantity	10
☐ Order_2	☐ Cust_1	☐ FG_1	☐ DC_1	Total MetOnTime Quantity	20
☐ Order_3	☐ Cust_1	☐ FG_1	☐ DC_1	Total MetOnTime Quantity	30
☐ Order_4	☐ Cust_1	☐ FG_1	☐ DC_1	Total MetOnTime Quantity	40

Planning behavior when plan type = INCREMENTAL PLAN

Capacity Availability is changed and the changed supply is marked as follows:

Resource	Data	W10-2015
☐ Capacity for Final Assembly at Plant_1	Capacity Availability	50
	Capacity Changed	☒

The above capacity had been initially pegged to 4 orders.
SCS picks up these 4 orders and sorts them in the order of their priority.
The orders are unplanned in the order of their priority (lowest 1st) until the reduction in capacity is accounted for

1st Order_4 is unplanned and 40 units of resources allocated to it is released.

Effective capacity availability now becomes 40 units

Then Order_3 is unplanned and 30 units of resources allocated to it is released.

Effective capacity availability now becomes $40 + 30 = 70$ units

But the reduction in capacity has only been 50 units.

Thus the OFL is resolved.

SCS now starts planning the unplanned orders in the order of their priority(highest priority 1st)

In this case, 2 orders i.e Order_3 and Order_4 were unplanned, since Order_3 is of a higher priority, it is planned 1st.

Of the 30 units of demand for Order_3, 20 units are allocated, because that is the amount of excess capacity available.

Effective excess capacity now becomes 0 units

The remaining 10 units are short.

Order_4 is planned and no resources can be allocated to it. Thus Order_4 is shorted

Thus after incremental plan is run, the capacity pegging looks as follows:

to.[Customer]. [Customer Name]	to.[Item]. [Item]	to.[Location]. [Location]	to.[Order]. [Order Name]	to.[Time]. [Fiscal Week]	from.[Lot].[Lot ID]	from.[Resource].[Resource]	Supply Bucket	Data	Value
⊟ Cust_1	⊟ FG_1	⊟ DC_1	⊟ Order_1	⊟ W13-2015	⊟ 000SCS_LOT	⊟ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	10.00
			⊟ Order_2	⊟ W13-2015	⊟ 000SCS_LOT	⊟ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	20.00
			⊟ Order_3	⊟ W13-2015	⊟ 000SCS_LOT	⊟ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	20.00

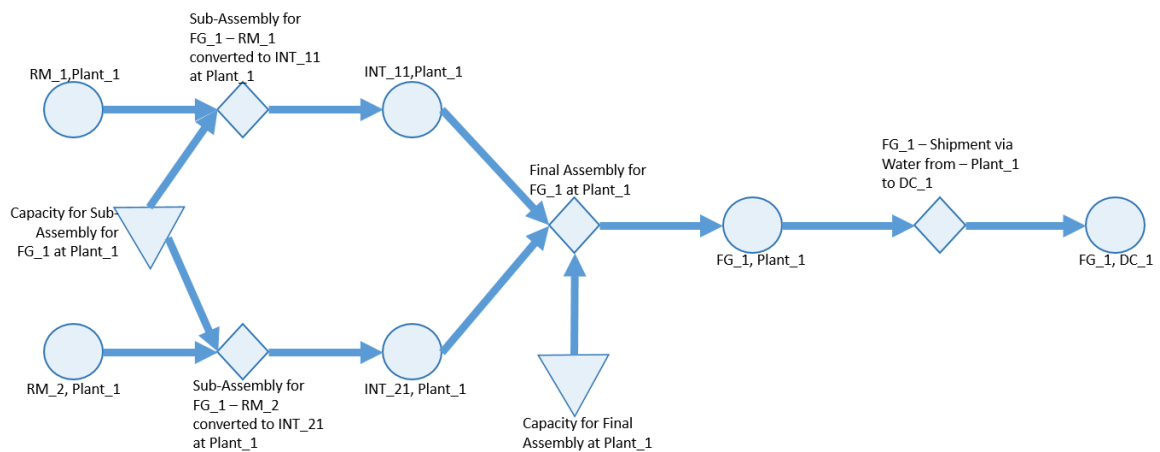
Constraint Report for short orders looks as follows:

to.[Customer]. [Customer Name]	to.[Item]. [Item]	to.[Location]. [Location]	to.[Order]. [Order Name]	to.[Time]. [Fiscal Week]	from.[Lot].[Lot ID]	from.[Resource].[Resource]	Supply Bucket	Data	Value
⊟ Cust_1	⊟ FG_1	⊟ DC_1	⊟ Order_3	⊟ W13-2015	⊟ 000SCS_LOT	⊟ Capacity for Final Assembly at Plant_1	W10-2015	CapacityConstraintQty	10.00
								CapacityConstraintType	CapacityShort
			⊟ Order_4	⊟ W13-2015	⊟ 000SCS_LOT	⊟ Capacity for Final Assembly at Plant_1	W10-2015	CapacityConstraintQty	40.00
								CapacityConstraintType	CapacityShort

If only one capacity is increased

([FT #065](#))

Consider the following supply chain



Demands are as follows:

Order Name	Item	Location	Data	W13-2015
Order_1	FG_1	DC_1	Demand Quantity	10
			Demand Priority	1
Order_2	FG_1	DC_1	Demand Quantity	20
			Demand Priority	2
Order_3	FG_1	DC_1	Demand Quantity	30
			Demand Priority	3
Order_4	FG_1	DC_1	Demand Quantity	40
			Demand Priority	4

Supplies are as follows:

Location	Item	Lot ID	Data	W01-2015
Plant_1	RM_1	SCS BOH	BOH	100
	RM_2	SCS BOH	BOH	100

Capacities available are as follows:

Resource	Data	W09-2015	W10-2015
Capacity for Sub-Assembly for FG_1 at Plant_1	Capacity Availability	1000	
Capacity for Final Assembly at Plant_1	Capacity Availability		50

Activity Lead times are as follows

Activity	Activity Lead Time
Sub-Assembly for FG_1 - RM_2 converted to INT_21 at Plant_1	1
Sub-Assembly for FG_1 - RM_1 converted to INT_11 at Plant_1	1
Final Assembly for FG_1 at Plant_1	1
FG_1 - Shipment via Water from - Plant_1 to DC_1	2

Planning behavior when plan type = FULL PLAN

When a material and capacity constrained plan is run on the above model, capacity consumption pegging for the resource {Capacity for Final Assembly at Plant_1} looks as follows:

to.[Customer]. [Customer Name]	to.[Item]. [Item]	to.[Location]. [Location]	to.[Order]. [Order Name]	to.[Time]. [Fiscal Week]	from.[Lot]. [Lot ID]	from.[Resource]. [Resource]	Supply Bucket	Data	Value
⊟ Cust_1	⊟ FG_1	⊟ DC_1	⊟ Order_1	⊟ W13-2015	⊟ 000SCS_LOT	⊟ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	10.00
			⊟ Order_2	⊟ W13-2015	⊟ 000SCS_LOT	⊟ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	20.00
			⊟ Order_3	⊟ W13-2015	⊟ 000SCS_LOT	⊟ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	20.00

And the constraint report looks as follows:

to.[Customer]. [Customer Name]	to.[Item]. [Item]	to.[Location]. [Location]	to.[Order]. [Order Name]	to.[Time]. [Fiscal Week]	from.[Lot]. [Lot ID]	from.[Resource]. [Resource]	Supply Bucket	Data	Value
⊟ Cust_1	⊟ FG_1	⊟ DC_1	⊟ Order_1	⊟ W13-2015	⊟ 000SCS_LOT	⊟ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	10.00
			⊟ Order_2	⊟ W13-2015	⊟ 000SCS_LOT	⊟ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	20.00
			⊟ Order_3	⊟ W13-2015	⊟ 000SCS_LOT	⊟ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	20.00

When material constrained plan is run on the above data, demands are sorted according to their respective priorities.

Of the available capacity of 50 units

Order_1 is planned and 10 units are allocated to it.

Effective Capacity now becomes 40 units

Order_2 is planned and 20 units get allocated to it

Effective Capacity now becomes 20 units

Order_3 is planned and 20 units get allocated to it

Effective Capacity now becomes 0 units

Thus Order_3 is partially short by 10 units

Order_4 is planned but no more capacity is left for this demand

Thus Order_4 is short by 40 units

Delivery Plan looks as follows:

Order Name	Customer Name	Item	Location	Data	W13-2015
Order_1	Cust_1	FG_1	DC_1	Total MetOnTime Quantity	10
				Total Short Quantity	
Order_2	Cust_1	FG_1	DC_1	Total MetOnTime Quantity	20
				Total Short Quantity	
Order_3	Cust_1	FG_1	DC_1	Total MetOnTime Quantity	20
				Total Short Quantity	10
Order_4	Cust_1	FG_1	DC_1	Total MetOnTime Quantity	
				Total Short Quantity	40

Planning behavior when plan type = INCREMENTAL PLAN

Capacity Availability is changed and the changed supply is marked as follows:

Resource	Data	W10-2015
Capacity for Final Assembly at Plant_1	Capacity Availability	90
	Capacity Changed	☒

The demands that are fully and partially short are marked as shown below. Marking the demands ensures that they participate in incremental planning.

Order Name	Customer Name	Item	Location	Data	W13-2015
Order_3	Cust_1	FG_1	DC_1	Incremental_Allowed	☒
Order_4	Cust_1	FG_1	DC_1	Incremental_Allowed	☒

Capacity is changed from 50 units to 90 units.

Therefore for incremental planning

Effective demand available = 50 units(10 units for Order_3 and 40 units for Order_4)

Effective Capacity available = 40 units.

Based on the priorities, Order_3 is unplanned.

The capacity available now is

40(increased by planner) + 20(available after unplanning demands) = 60.

Effective demand available for incremental planning now becomes 70 units.

Demand fulfillment is now done based on the priorities.

Order_3 is planned 1st and 30 units of capacity is allocated to it.

Order_4 is planned 2nd and remaining 30 units of capacity are allocated to it.

Thus after incremental plan is run, the capacity pegging looks as follows:

to.[Customer]. [Customer Name]	to.[Item]. [Item]	to. [Location]. [Location]	to.[Order]. [Order Name]	to.[Time]. [Fiscal Week]	from.[Lot].[Lot ID]	from.[Resource].[Resource]	Supply Bucket	Data	Value
⊟ Cust_1	⊟ FG_1	⊟ DC_1	⊟ Order_1	⊟ W13-2015	⊟ 000SCS_LOT	⊟ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	10.00
			⊟ Order_2	⊟ W13-2015	⊟ 000SCS_LOT	⊟ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	20.00
			⊟ Order_3	⊟ W13-2015	⊟ 000SCS_LOT	⊟ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	30.00
			⊟ Order_4	⊟ W13-2015	⊟ 000SCS_LOT	⊟ Capacity for Final Assembly at Plant_1	W10-2015	Capacity Pegged Quantity	30.00

Constraint Report for short order looks as follows:

to.[Customer]. [Customer Name]	to.[Item]. [Item]	to.[Location]. [Location]	to.[Order]. [Order Name]	to.[Time]. [Fiscal Week]	from.[Lot].[Lot ID]	from.[Resource].[Resource]	Supply Bucket	Data	Value
⊟ Cust_1	⊟ FG_1	⊟ DC_1	⊟ Order_4	⊟ W13-2015	⊟ 000SCS_LOT	⊟ Capacity for Final Assembly at Plant_1	W10-2015	CapacityConstraintQty	10.00
								CapacityConstraintType	CapacityShort

Constraint Report

Business Need

A constraint is any obstacle or planned control that limits throughput of the system. The constraints on the supply chain could be related to material, capacity or the network itself. The constraint report helps the planner understand why a demand is late or short. If materials are short, they can negotiate better with their suppliers. If capacity is short, they can see if some processes need to be outsourced etc. It provides visibility on the root causes as to why the demand are not planned on the preferred path.

Functional Description

Pegging determine how an order is linked to material and capacity nodes in the supply chain. The edge represents the pegged quantity. When a demand is not planned, it is now written into following two edge properties namely

- Constraint Quantity (Type: Numeric)
- Constraint Type (Type: string)

1. Constraint Quantity

- a. Material Constraint Quantity: Units required in this bucket for a demand to be met on time
- b. Capacity Constraint Quantity : Capacity required in this bucket for a demand to be met on time
- c. Storage Constraint Quantity : Storage space required in this bucket for a demand to be met on time.

2. Constraint Type

- a. Material Constraint Type

Can take any of the below values

1. **MaterialShort:** created for orders that are Fully / partially short or Fully/ Partially Build Late
[\[Functional Test #57\]](#)
2. **LeadtimePlanBeforeCurrent** : Demand is between Start and Current time horizon hence late.
[\[Functional Test #49\]](#)
3. **FrozenPeriod:** Activity is frozen during the period hence demand is short or late
[\[Function Test # 28\]](#)
4. **Effectiveness:** Due to Activity Effectivity when demand is short or late
[\[Functional Test #77\]](#)

5. **FrozenLocked**: Due to locked WIP when production cannot happen and demand is short or late
[\[Functional Test #34\]](#)
6. **NoBuild**: Demand is short or late due to No Build
[\[Functional Test #100\]](#)
7. **ActivityNotAvailable**: If demand are short or late when a network has revised priority of 0.
[\[Functional Test #144\]](#)
8. **Holiday**: Applicable only for Additive RCA when demand is short or late due to Activity Holiday
[\[Functional Test #51\]](#)
9. **MaxStock**: Demand is short or late due to Max Stock violation
[\[Functional Test #259\]](#)
10. **ActivityMarkerConstraint**: Demand is short or late due to activity marker
[\[Functional Test #12\]](#)
11. **ProportionalActivityNotAvailable**: Demand is short or late due to split ratio for any activity is set as 0 or null
[\[Functional Test #52.3\]](#)
12. **ProductionHoliday**: Demand short due to Production Holiday
13. **ConsumptionHoliday**: Demand is short due to consumption holiday
14. **DemandActivityEffectiveness**: Due to demand effectivity when the demand is short or late

b. Capacity Constraint Type

Can take the value

1. **CapacityShort**: Demand is short/late even after regular capacity or JIT and Late time bucket
[\[Functional Test #76\]](#)
2. **RegularCapacityShort**: When an order is built late &/ shorted even after using regular and overtime capacities at JIT &/ late time bucket, then capacity constraint is reported in terms of the units of Regular Capacity at JIT time bucket required to build the order on time.
[\[Functional Test #318\]](#),
[\[Functional Test #298\]](#)

c. Storage Constraint Type

Can take the value

1. **StorageConstraint**: Due to Storage Violation
[\[Functional Test #374\]](#)
2. **RegularStorageConstraint**: When an order is built late &/ shorted even after using regular and overtime storages at JIT &/ late time bucket, then storage constraint is reported in terms of the units of Regular Storage at JIT time bucket required to build the order on time.
[\[Function Test # 50\]](#)

Other tests:

Material Short & Effectiveness for 1 order[[Functional test 79](#)]

Material Short for 1 order and material Short & Effectiveness for 2nd order[[Functional Test 80](#)]

Configuration

Goto Workspace → Designer → Edit Model → Graph → Add the following relationship types

1. Capacity_Pegging
 - a. Edge properties
 - i. CapacityConstraintQty
 - ii. CapacityConstraintType
 - b. From Nodes
 - i. Resource Node
 - Resource.[Resource]
 - ii. Activity Node
 - Activity.[Activity Node]
 - iii. Time.[Fiscal Week]
 - c. To Nodes
 - i. Demand Node
 - Order.[Order Name]
 - Time.[Fiscal Week]
 - Customer.[Customer Name]
 - ii. Material Node
 - Item.[Item Node]
 - Location.[Location Name]
2. Production_Pegging
 - a. Edge Properties
 - i. MaterialConstraintQty
 - ii. MaterialConstraintType
3. From Nodes
 - a. Material Node
 - i. Item.[Item Name]
 - ii. Location.[Location Name]
 - b. Activity Node
 - i. Activity.[Activity Node]
 - c. Time.[Fiscal Week]
4. To Nodes
 - a. Demand Node
 - i. Order.[Order Name]
 - ii. Time.[Fiscal Week]
 - iii. Customer.[Customer Name]
 - b. Material Node
 - i. Item.[Item Name]
 - ii. Location.[Location Name]

All the above mentioned edge properties should be configured as follows:

MaterialConstraintQty

Name *

MaterialConstraintQty

Description *

MaterialConstraintQty

Data Type

Number

Aggregation

Sum

☐ Editable

Format

(0,000 a)

Following diagram demonstrates the same. The above mentioned edge properties should be configured in the SCS config as well. If not provided, the information will not be exported

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → <Select measures from the drop down menu as follows>

Material Production Constraint Type Property i	MaterialConstraintType
Material Production Constraint Qty Property i	MaterialConstraintQty
Capacity Consumption Constraint Type Property i	CapacityConstraintType
Capacity Consumption Constraint Qty Property i	CapacityConstraintQty
Storage Constraint Type Property i	StorageConstraintType
Storage Constraint Qty Property i	StorageConstraintQty

Note: Set Constraint Type aggregation as SUM (current limitation: It will not work if it is set to NONE or NULL)

Note: Constraint Report is populated even if No Pegging flag is set to TRUE.

Limitation

1. Constraints are considered only on the primary path.
2. Values are not seen for Capacity and Material shortness in the constraint report if planners are running an unconstrained plan.
3. Constraint Report is not populated if demand is met ON TIME using alternate activities.

Flow Plan Synchronizer

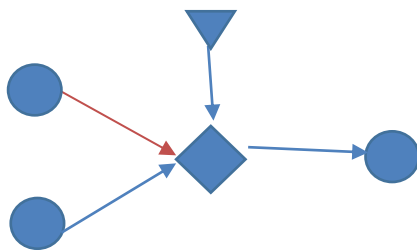
Business Need

Users need to manually override the plans and see local (one level or stage) synchronization of all plans immediately.

Functionality Description

Flow Plan Synchronizer (FPS) is an active plugin, which synchronizes all the flows connected with a particular activity in that bucket, when any one of them is edited (overridden). For example, in the schematic diagram shown below, when user overrides the consumption plan for one of the components (represented by the orange arrow), all the other flows connecting that activity (another component consumption, capacity consumption and production flow) are updated.

Image : FPS_001



The plug-in takes the configuration specified in the supply chain solver on-demand plug-in to update the measures.

Note: BOH and Expected receipts have different activities and won't change due to FPS. The total production quantity at an item-location node is the summation of all the production plan from activities producing into it.

Following are rules with respect to FPS in presence holidays defined on activities (stretch lead time)

- If consumed plan quantity is updated on a holiday bucket, SCS will not do any changes to the producing plans.
- If produced plan quantity is updated on a holiday bucket, SCS will not do anything as it is marked as a holiday. SCS will take care of it only in the full/incremental run and not in FPS.
- If x is consumption time bucket and $x+2$ is production time bucket, consumed quantity at x is updated and $x+2$ is marked as holiday, then, consumed quantity of x will be synced up with the produced quantity of $x+\text{leadtime}+\text{holiday}$ which is $x+2+1 = x+3$.
- If x is consumption time bucket and $x+2$ is production time bucket, and $x+1$ is updated as holiday and consumed quantity at x or produced quantity at $x+2$ is updated, then, consuming and producing buckets are synced up. That is, consumed quantity changed at x will update produced quantity at $x+3$. Produced quantity updated at $x+2$ will update consumed quantity at $x-1$.

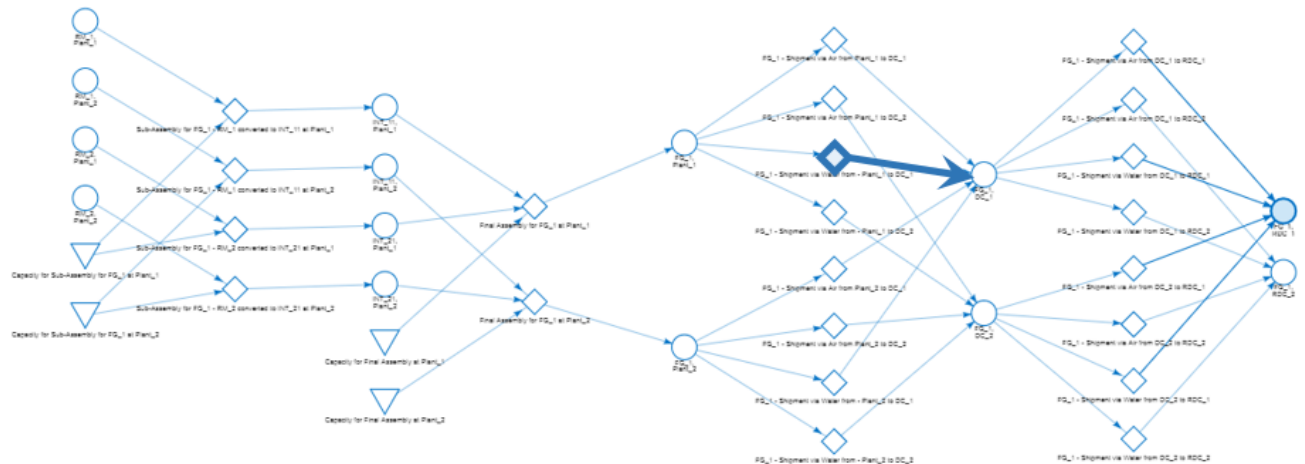
Configuration Steps

Consider the Activity in bold in image FPS_002 i.e **FG_1-Shipment Via Water from Plant_1 to DC_1**. This activity is maintained in both

- Material Production Plan

- o Material Quantity
- Material Consumption Plan
- o Material Consumption Quantity

Image : FPS_002



These 2 measures are the output of the Solver. However, the values of these measures for a particular time bucket can be edited by the respective planners. If the planner edits any one of the measures related to this activity, FPS is invoked (active plugin), which synchronizes all the immediate edge properties (measures in this case) based on the edited value.

Goto Workspace → Designer → Edit model → Rule → Active Rules→ FPS →Edit

After clicking the Edit button, the following pane is seen.

Active Rules

Comment : Migrated rule from scriptid: FPS

Comment: FPS

Plugin Name * : SupplyChainSolver Add Measures and Scope

Measures : Material Quantity x Material Consumption Quantity x Capacity Consumption Plan x

Scope : [Version].[Version Name].[CurrentWorkingView] * [Time].[Fiscal Week] * [Activity].[Activity] * [Item].[Item] * [Location].[Location] * [Resource].[Resource]

Arguments : Add Argument

Enter the following in that order

1. Comment : FPS
2. Plugin Name : Select SupplyChainSolver(or name of the active supplychainsolver plugin in the tenant) from the Drop Down Menu
3. Click on **Add Measures and Scope** to add measures and scope:
 - a. Upon clicking <Add Measures and Scope> the following form can be seen

Measures : Select Measures... **Box 1**

Scope : Enter Scope... **Box 2**

- b.
- c. Following measures can be selected in box 1 through drop down list:
 - i. Material Quantity

- ii. Material Consumption Quantity
 - iii. Capacity Consumption Plan
4. Box 2 is a text box space where rules are written, the following rule should be entered
- ```
[Version].[Version Name].[CurrentWorkingView] * [Time].[Fiscal Week] *
[Activity].[Activity] * [Item].[Item] * [Location].[Location] *
[Resource].[Resource]
```

Measure and dimension names should be in sync with those present in the tenant.  
User can also have member sets or namesets in the scope statement.

Click on Save to activate the plugin.

### **Limitations**

This is local in nature and only updates plans which are connected by that particular activity. It will not traverse the supply chain upstream or downstream. FPS has local view of the state. If any plan is changed at any point, then FPS will update plans connected to that particular activity.

## **Segmentation and Parallel Plan**

### **Business Need**

Today's corporations are large (especially retail volumes). They need faster results with optimized memory usage, thereby containing the hardware costs.

### **Functionality Description**

Supply chains are said to be independent if they do not share any capacity or material. These supply chains can be configured as segments. Segments are created by giving an end item level a unique segment ID. The solver solves these segments based on the CPUs availability one after the other. This memory is contained. If there are 10 segments and 5 threads; it will plan 5 segments in parallel and when each thread gets freed, it picks up the next segment.

### **Configuration Steps**

A measure 'Segment' needs to be configured. It can be included in the measure group "Demand Quantity". It is of type integer. Static supply chain's segments can be defined by specifying different numbers for different segments (at the end item level).

Goto Workspace → Designer → Edit Model → Plan → Supply Chain Plan and Graph → Click on the "Demand Quantity" measure group and add the measure "Segment".

*Image: SEG\_001*

DIMENSION

PLAN

GRAPH

RULE

FOLDER

Q Search

+

Demand Quantity

Demand BAL

Demand BLL

Demand Priority

Demand Quantity

Demand Trace

Incremental\_Allowed

Quantity Fulfilled

Quantity Incrementally fulfilled

Segment

Image:SEG\_002

Edit Reports

Edit Model

Segment

Editing Measure 'Segment'

Segment

Name \*

Segment

Description \*

Segments

ToolTip

Data Type

Number

Type

Regular

Editable

☒

Granularity

Order Name

Version Name

Customer Name

Item

Fiscal Week

Location

Measure Rules

- One or more Items can have the same Segment ID.
- One or more Locations can have the same segment ID.

Data can be uploaded as an excel file or directly entered using Excel add-in.

In SCS plugin configure this measure as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Demand

Demand Segment Measure Name ?

Segment

SCS plugin has an optional argument named 'Number of Planning Threads' (N). Its value determines the segments that can run in parallel. Default is one.

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Number of Planning Threads → Set Number of Planning Threads = 10

Number of Planning Threads ⓘ

A segment gets processed by a single thread. At most, N segments will be processed in parallel. The guideline is to have the number of threads to be less than the number of CPUs made available for this process on the server.

*Note: The configuration parameter “Number of Export Threads Per Segment” determines the threads used to write out SCS plans. It can be configured under the same tab “Parameters”.*

Number of Planning Threads ⓘ

Number of Export Threads Per Segment ⓘ

**Number of Export Threads Per Segment:** This feature exports SCS plans in parallel, thereby enabling multi threading to improve run-time and/or memory performance. The default value for this parameter is 1, which means that by default the export is single threaded.

### Limitations

If the entire supply chain is just one connected supply chain, this will not help. Also users are expected to provide the segments as inputs to the solver rather than solver automatically detecting it.

## Segment Identification

### Business Need

Supply chain solver has the capability to run independent supply chains in parallel to provide performance benefits and also scalability. SCS earlier needed to provide these segment information as input, where user marked the demand nodes falling into each segment with unique value. This is now enhanced so that users can run the plugin itself to identify such segments rather than providing it as an input.

### Functional Description

A new PlanType option IDENTIFY\_SEGMENTS has been introduced in SupplyChainSolver. It is meant for identifying existing segments in the network. When SCS plugin is invoked with this option, the static supply chain structure is loaded into memory. Different disconnected components in the network are identified and marked with an ID (1, 2, 3, etc). The material nodes' segment id is written out. Segment id of each demand record could be set with that of the material. Demand segment is to be set outside of the plugin[[Functional Test #360](#)].

Apart from identifying segments, nodes that could be split are also identified. The heuristic is - a node with the largest number of connections is split. Activity nodes are never split. Capacity, Storage and Material nodes are split in that order.

A couple of flags control the split. 'Segment Split Min Edge Count Limit' specifies the minimum edge count a node must have to be considered for splitting. 'Segment Split Max Node Count Limit' specifies the maximum number of nodes to split.

### Configuration

Following measure group needs to be maintained

1. Material Node parameters
  - a. Measure



- i. **Material Segment** : it takes numeric values. This measure is the output of the plugin. Material nodes belonging to the same supply chain have same segment ID. For eg, let material node M1 and M2 belong to supply chain S1 and material node M3 belong to supply chain S2 such that supply chain S1 and S2 are in no way connected to each other. Then after segment identification the measure group and the corresponding measure will have data as follows:

| Material node | Version            | Material Segment |
|---------------|--------------------|------------------|
| M1            | CurrentWorkingView | 1                |
| M2            | CurrentWorkingView | 1                |
| M3            | CurrentWorkingView | 2                |

b. **Granularity**

- i. **Material Node**
  1. Item.[Item]
  2. Location.[Location]
- ii. **Version.[Version Name]**

A copy of the instance SupplychainSolver can be created with the name Identify\_Segments(recommended).

After the above mentioned measures are configured, same should be specified in the SCS config as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →

o9.GraphCubePlugins → SupplyChainSolver → Identify\_Segments→ Inventory→ Material Segment Measure Name → choose measure from the drop down menu

Material Segment Measure Name ⓘ  ▼

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →

o9.GraphCubePlugins → SupplyChainSolver → Identify\_Segments → Parameters →Plan Type → enter values in text

Plan Type ⓘ

Similar things can be configured for activity, capacity and storage nodes

After the above configuration is done, the following code can be added to action button in UI/Excel or can be executed via meru:

```
exec plugin instance Identify_Segments for Measures {[Material Segment]} using
scope ([Version].[Version Name].[CurrentWorkingView]);
```

As soon as the above code is executed following information can be seen in the logs:

|                         |
|-------------------------|
| Segment 11 : 1 Nodes    |
| Segment 10 : 1 Nodes    |
| Segment 5 : 4 Nodes     |
| Segment 4 : 22 Nodes    |
| Segment 9 : 11 Nodes    |
| Segment 8 : 13 Nodes    |
| Segment 7 : 17 Nodes    |
| Segment 6 : 8 Nodes     |
| Segment 3 : 27 Nodes    |
| Segment 2 : 28 Nodes    |
| Segment 1 : 123 Nodes   |
| Number of segments : 11 |

After the query execution , one can query the material node properties to view segment IDs of unique material nodes.

A couple of flags control the split.

'Segment Split Min Edge Count Limit' specifies the minimum edge count a node must have to be considered for splitting.

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → Identify\_Segments → Parameters →Segment Split Min Edge Count Limit→Plan Type → enter values in text <500 by default>

Segment Split Min Edge Count Limit ⓘ

'Segment Split Max Node Count Limit' specifies the maximum number of nodes to split.

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → Identify\_Segments → Parameters →Segment Split Max Node Count Limit→Plan Type → enter values in text <5000 by default>

Segment Split Max Node Count Limit ⓘ

## Deletion of Previous Plans

### Business Need

Many industries still follow batch planning cycles. The data from previous planning cycle is completely deleted and new data is obtained on which plans are run. This feature aids the implementers need of clearing all the plan data in a particular version in an efficient way.

### Functional Description

SCS has the capability of

1. Planning for new demands after overriding or appending the existing flow plans

2. Planning for new demands after erasing previous flow plans.  
End user can opt for either of the 2 scenarios

**Delete SCS Plan Data For Version:** When set to true, all the output data for the specific version is truncated first (deleted if it is not a materialized version) and then the demand planning is done.

### Configuration Steps

SCS Plugin has a dynamic argument: Delete SCS Plan Data For Version, which takes boolean values.

When TRUE, every time an instance of the solver runs, all previous flow plans are deleted and new flow plans for demands within the scope are generated.[\[Functional Test #87\]](#).

When FALSE, every time an instance of the solver runs, previous flow plans belonging to that segment (connected tree) only are deleted and new flow plans for demands within that scope is created[\[Functional Test #88\]](#). Note that it will not delete plans and pegging information that are not in scope of this segment.[\[Functional Test #90\]](#)

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Delete SCS Plan Data For Version → Check if TRUE or Uncheck if FALSE [By default its Unchecked i.e FALSE]

*NOTE: If the graph association measure is zero, the corresponding plans on those activities are deleted. SCS re-run will not consider these activities as the association measure is zero. This behaviour is seen when Delete SCS Plan Data For Version is set to TRUE.*

When SCS Plan is run on any Scenario (any version with Key < 0) with Flag {Delete SCS Plan Data For Version} set as True, then we see the following logs

Truncate not allowed for version -1. Switching to delete.

The above log only means that for the scenario with key = -1, the previous plans of a previous SCS run on the same Scenario got deleted instead of getting truncated.

## Delete Plan

### Business Need

The data from previous planning cycle is completely deleted and without having the need to generate new plans. This feature aids the implementers need of clearing all the plan data in a particular version in an efficient way.

### Functional Description

This only completely deletes the solver output data from previous planning cycles, without having the need to run the solver i.e. does not produce new output.

### Configuration Steps

SCS Plugin has a dynamic argument: Plan Type, which takes values.

When **DELETE\_PLAN**, every time an instance of the solver runs, all previous flow plans are deleted

Not exposed in the UI, can be passed in the exec plugin command.

Sample: *exec plugin instance SCS for Measures {[Capacity Changed]} using scope ([Version].[Version Name].[CurrentWorkingView]) \* Resource.[Resource].filter(#.Name in {[R1],[R2]})) arguments {[Material Constrained], "true"), ([Capacity Constrained], "true"), ([Plan Type], "Delete\_Plan")};*

## Solver Command Line Signature

### Business Need

mPower users dynamically change some arguments and pass it to the solver when it runs. These act as an override to those specified in the config file.

### Functional Description

Planners have the liberty to run the Solver for a

1. Set of Orders or [\[Functional Test #38\]](#)
2. Set of Items or [\[Functional Test #36\]](#)
3. Set of Locations or [\[Functional Test #37\]](#)
4. Combination of Item, Locations or [\[Functional Test #34\]](#)
5. Combination of Orders, Items [\[Functional Test #39\]](#)

The following code snippet with respect to SCOPE should be a part of Action Button while invoking the Solver:

1. If the planner wants to run the Solver for a set of Orders
  - scope ([Version].[Version Name].[CurrentWorkingView] \* Order.[Order Name].filter(#.Name in {...})
2. If the planner wants to run the Solver for a set of Items
  - scope ([Version].[Version Name].[CurrentWorkingView] \* Item.[Item].filter(#.Name in {...})
3. If the planner wants to run the Solver for a set of Location
  - scope ([Version].[Version Name].[CurrentWorkingView] \* Location.[Location].filter(#.Name in {...})
4. If the planner wants to run the Solver for a combination of Item, Locations
  - scope ([Version].[Version Name].[CurrentWorkingView] \* Item.[Item].filter(#.Name in {...}) \* Location.[Location].filter(#.Name in {...})
5. If the planner wants to run the Solver for a combination of Order, Items
  - scope ([Version].[Version Name].[CurrentWorkingView] \* Order.[Order Name].filter(#.Name in {...}) \* Item.[Item].filter(#.Name in {...})

### Configuration Steps

*Note: in the above queries membersets can be replaced by namesets . eq. &CWV for [Version].[Version Name].[CurrentWorkingView] and so on. These name sets have to be created as IBPL active rules that should run at GraphCube Server Startup.*

## Direct Inserts (DI)

### Business Need

User wants superior performance, especially in writing data to measure group or graph tables in mPower.

## Functionality Description

Direct Inserts are an approach to adding data to a measure-group or graph-table where the entire contents of the table are being replaced by a fresh set of rows, i.e. the table is being truncated and repopulated. Plugins such as Supply-Chain-Solver perform such operations on their output tables, for example pegging export tables. Today such updates are made through a specialized version of the upsert operator that maintains the updated data as deltas to the base tables. Future queries on these tables require a merge between the base tables and the delta.

The Direct-Insert operator performs the truncate-and-insert operation differently. It writes all the new data for a table to disk, and switches LiveServer to use the newly created “base” tables instead of the original tables. The advantage is that the new data in tables is immediately persisted on disk and future queries do not need to do a merge between the base table data and delta to retrieve the correct values for individual rows.

Additionally, if there are no more changes to the table before the next save, then the on-disk tables can be directly used instead of doing a full save for the “direct-insert”ed tables, saving time.

The direct insert-operator also is more memory efficient, since the new rows of the table are directly persisted to disk and are not retained in memory (for the InMem btree mode). For very large tables, this can provide significant memory benefits.

Currently, the Direct-Insert capability has only been used in the pegging-table exports of the Supply-Chain Solver plugin. It is disabled by default.

Direct-Insert can be enabled through a dynamic parameter **DirectInsertEnabled** as shown below for a single SCS run:

```
exec plugin instance [SCS Weekly Plan] for Measures {[Demand Quantity]}
 using scope (Version.[Version Name].[CurrentWorkingView] * Demand.[Demand])
 arguments {
 ([Material Constrained], "True"),
 ([Capacity Constrained], "True"),
 ([Inventory Plan Type], "QUANTITY"),
 ([No Write Execution], "False"),
 ([Number of Planning Threads], 1),
 ([Number of Export Threads Per Segment], 5),
 ([Delete SCS Plan Data For Version], "True"),
 ([DirectInsertEnabled], "True")
 };
```

Note that both “Delete SCS Plan Data For Version” and “**DirectInsertEnabled**” need to be true for the Direct-Insert operator to be used for pegging table exports. This is because direct-insert is essentially a full replacement of the table data.

## Limitations

- Currently, on benchmark data-sets, Direct-Insert has been measured to be about 30% slower than the standard upserts for InMem btree mode, and approximately 50% faster than LSM btree mode.

## Planning Horizon

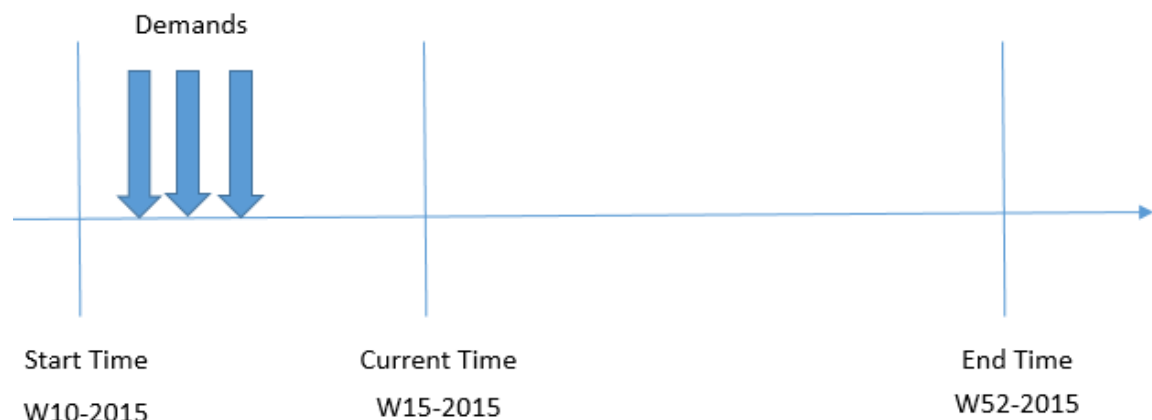
### Business Need

Planners require to honour near term commitments and create a plan that provide visibility into the future. The planning horizon denotes the total duration over which the plan is to be created. These vary across industry and business functions. Example, for electronic industry the average planning horizon is one year. If SCS is used for distribution planning, the planning horizon can be as short as 2 weeks as well to understand the deployments.

### Functional Description

Implementer's can run the solver for a specific time frame. It can be for a week, month, quarter or year. SCS currently supports fixed planning horizon, where in planners precisely define the start and end date.

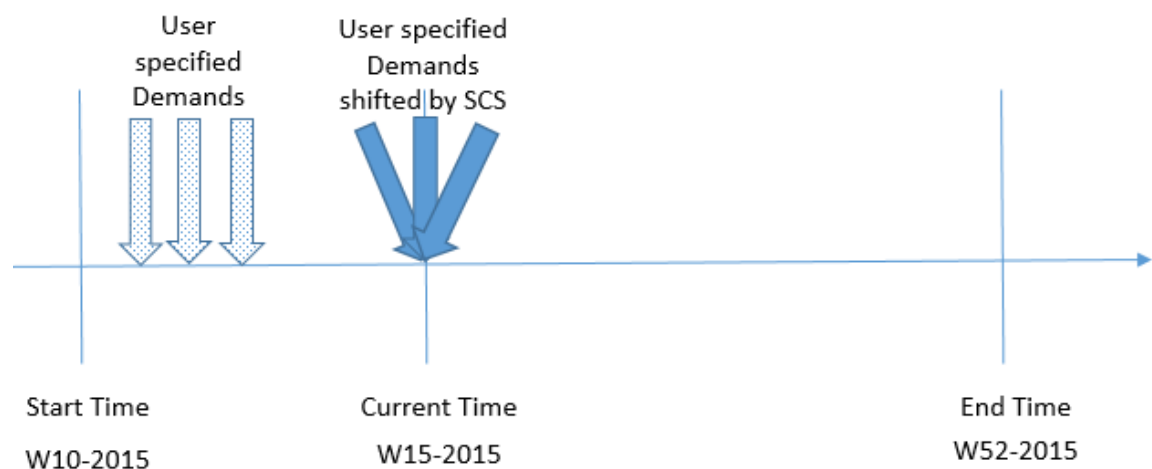
If planners provide demands that are between the Start Time bucket and current time bucket as follows:



SCS shifts all those demands to Current Time Bucket as follows and begins planning thereby honoring the build late limit (if specified)

Beginning on Hand defined earlier than current bucket are ignore.

Expected Receipts we assume that supply is coming in on that date and for planning the cumulative supplies are available on plan.current.



Eg:

Plan.start: W10

Plan.current: W15

Plan.End: W52

#### Case (1)

##### User Input

Demand D1 for 100 units in W11 with BLL=0  
BOH = 100 units available on W15.

##### SCS Output

Demand is met in full on W15

#### Case (2)

##### User Input

Demand D1 for 100 units in W11 with BLL=0  
BOH=0, ER=100 in W16

##### SCS Output

Demand is short (as it cannot late it further from plan.current)

#### Case (3)

##### User Input

Demand D1 for 100 units in W11 with BLL=1  
BOH=0, ER=100 in W16

##### SCS Output

Demand is met in full on W16 (BLL of 1 is computed from plan.current)

### Configuration Steps

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters →  
Start Time Bucket Name → enter values in text box

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters →  
Current Time Bucket Name → enter values in text box

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → End  
Time Bucket Name → enter values in text box

SupplyChainSolver

| Instance Options           | Node                                              | Demand | Inventory | Parameters | Manufacturing | Distribution | Procurement |
|----------------------------|---------------------------------------------------|--------|-----------|------------|---------------|--------------|-------------|
| Time Attribute Name ⓘ      | <input type="text" value="[Time].[Fiscal Week]"/> |        |           |            |               |              |             |
| Start Time Bucket Name ⓘ   | <input type="text" value="W01-2015"/>             |        |           |            |               |              |             |
| Current Time Bucket Name ⓘ | <input type="text" value="W10-2015"/>             |        |           |            |               |              |             |
| End Time Bucket Name ⓘ     | <input type="text" value="W52-2015"/>             |        |           |            |               |              |             |

Demands with due dates that fall outside the above mentioned Planning Horizon are strictly ignored by SCS. However Demands falling between Start Date and Current Date, are planned with due date as Current Date (via alternate activities with the least lead time thereby minimizing lateness).

The value in the text box should be in a format identical to the format as entered in the Time Dimension. The data mentioned in the Start Time bucket Name and End Time bucket Name should be synonymous to the Time Attribute Name mentioned in the config.

If Time Attribute Name = [time].[Fiscal Week] then Start Time Bucket Name = [any Week value mentioned in time dimension that planners want to start with].

If Time Attribute Name = [time].[Fiscal Month] then Start Time Bucket Name = any [Month value mentioned in time dimension that planners want to start with]

### **Limitations**

If Planning Horizon is not defined, the entire time horizon is passed to the solver to plan. If current time bucket is not specified it will take the current bucket specified in Graph Cube Service.

Max number of buckets supported in Planning Horizon is 1022 Buckets.

## **Demand Supply Synchronization (Violation\_Resolution)(Deprecated)**

### **Business Need**

Large businesses usually run unconstrained plan as it is scalable, parallelizable and can get results quickly. One of the requirements is to use the flow plans created from this unconstrained plan as starting point, highlight the nodes where there are violations (capacity violation on a resource node, material NOH, storage infeasibility etc) and create a demand supply balance that is feasible. For this, one needs to surgically remove the infeasibility to create a material-capacity constrained plan using unconstrained plan as a starting point. Violation resolution workflow helps achieve the same.

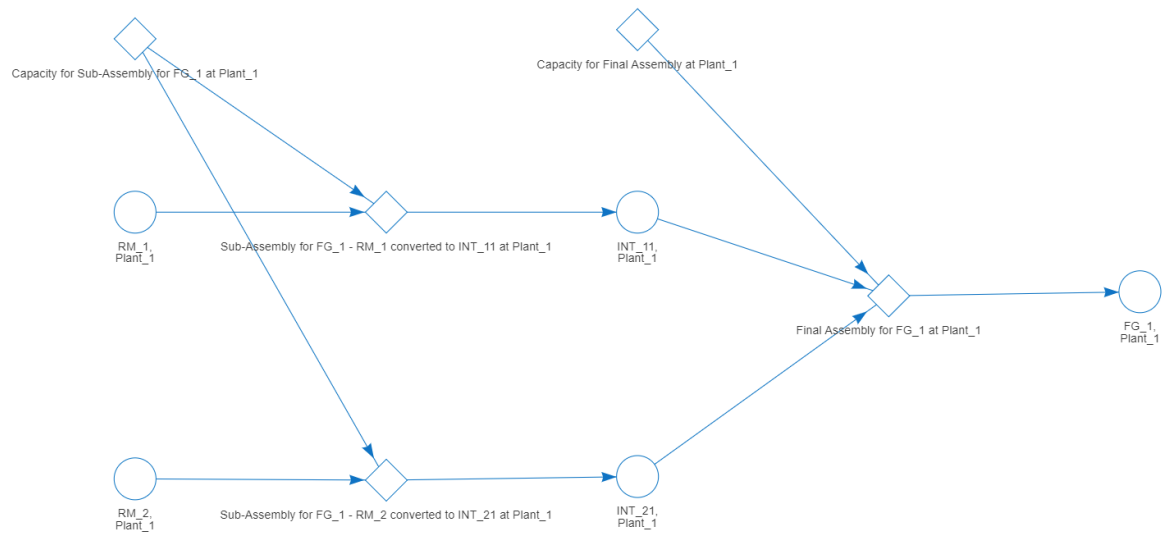
### **Functional Description**

The solver would overload the capacity and use extra material than what is present in the supply chain during unconstrained plan (both material & capacity unconstrained) to satisfy demands completely. This overloading of capacity or material over usage would be happened in JIT bucket.

The violation resolution workflow is a constrained planning workflow. Solver identifies the nodes where infeasibility exists and tries to resolve those surgically. Solver updates the measure to zero '0' for material consumption or capacity consumption for the buckets which were used earlier during unconstrained planning but were not used during violation resolution. This provides visibility to the user on the buckets which were overloaded or used earlier to satisfy all demands in unconstrained planning.



Example:



The final assembly capacity has 70 & 80 in JIT and JIT-1 buckets respectively.  
Demands for FG\_1 are 60 and 100 on W05 (JIT) bucket.  
The sub-assembly capacity has 400 in W01 bucket.  
In unconstrained plan both the demands get planned. The sub-assembly capacity and the assembly capacity will be overloaded in JIT for 160 on W03 and W04 buckets respectively (assume the activity time is 1).  
After violation resolution, we have 10 units of demand shortage.  
The sub-assembly capacity gets loaded 150 on W01 (build ahead bucket).  
The assembly capacity gets loaded 70 & 80 in JIT and JIT-1 buckets respectively.  
You will see the sub-assembly capacity will load 0 on W03 bucket.

## Configuration

Specify “Plan Type” parameter as “Violation\_Resolution” as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Plan Type.

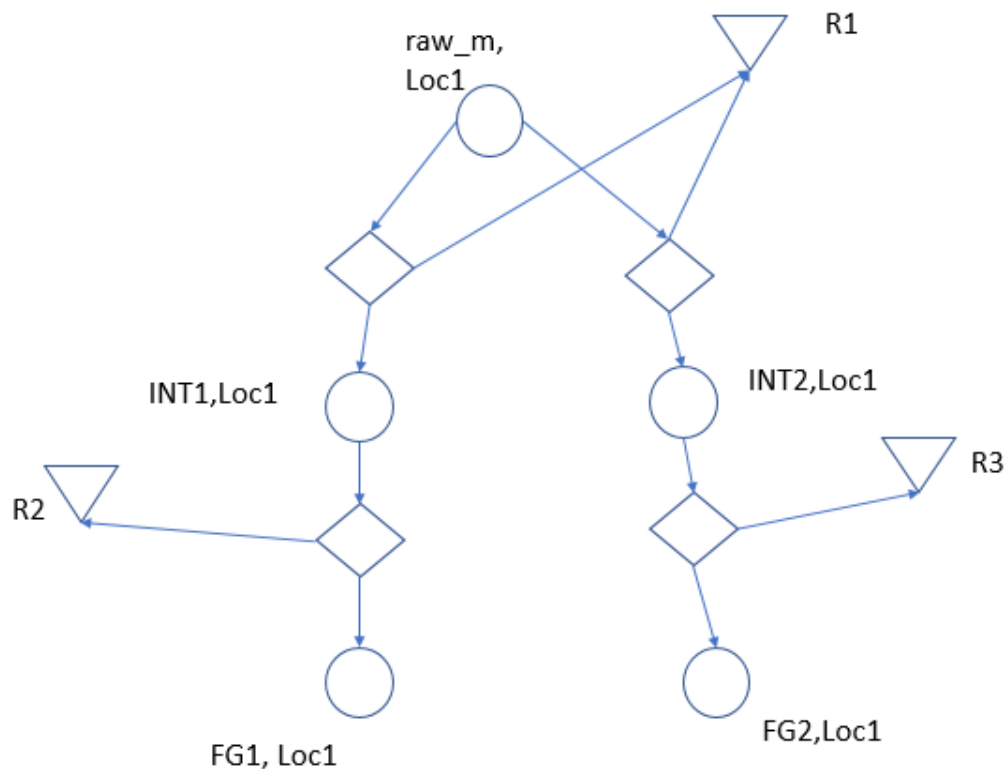
Run with the same scope (as unconstrained planning earlier) with capacity and material constraints as true and see that the flow plans are corrected and the peggings are written out for the demands those participated in this run.

During violation resolution workflow the parameter “Delete SCS Plan Data For Version” is ignored.

## Command line signature for running violation resolution

You can selectively choose the capacity or material to be considered during violation resolution instead of considering all capacity and material nodes. This can give solver run time benefit.

Let's take below example.



Let's say we run an unconstrained plan and then realized the resources R1 and/or R2 got overloaded. The violation resolution workflow will be used to resolve these infeasibilities.

Here first we need to mark those resources which have violation. Please note this is required to surgically do the violation resolution plan. One can mark all the resources as well, but that tells solver to consider all those and check if violation exists and then do the necessary corrections. User may be interested for a few critical or strategic capacities where violation exists and want to run violation resolution workflow for those. The surgically marking of those capacities help to achieve that use case.

In this example we can do this as below through IBPL commands

```

update Measure.[Capacity Changed]@(Resource.[Resource].[R1], [Version].[Version Name].[CurrentWorkingView])=1;
update Measure.[Capacity Changed]@(Resource.[Resource].[R2], [Version].[Version Name].[CurrentWorkingView])=1;
Or
scope:([Version].[Version Name].[CurrentWorkingView] *
Resource.[Resource].[R1]);Measure.[Capacity Changed]=1;end scope;
scope:([Version].[Version Name].[CurrentWorkingView] *
Resource.[Resource].[R2]);Measure.[Capacity Changed]=1;end scope;

```

Now run Violation Resolution Plan

```
exec plugin instance SCS for Measures {[Capacity Changed]} using scope
([Version].[Version Name].[CurrentWorkingView]) * Resource.[Resource].filter(#.Name in
{[R1],[R2]}) arguments {[Material Constrained], "true"}, ([Capacity Constrained], "true"),
([Plan Type], "Violation_Resolution");
```

The same can be done for material violation too.

Let's say raw\_m,Loc1 has material violation.

Indicate this by updating Measure [Supply Modified] like below.

```
update Measure.[Supply Modified]@(Item.[Item].[raw_m], Location.[Location].[Loc1],
[Version].[Version Name].[CurrentWorkingView])=1;
```

Or

```
scope:([Version].[Version Name].[CurrentWorkingView] * Item.[Item].[raw_m] *
Location.[Location].[Loc1]);Measure.[Supply Modified]=1;end scope;
```

Now run violation resolution plan for both material and capacity violations like this.

```
exec plugin instance SCS
for Measures {[Capacity Changed]} using scope ([Version].[Version
Name].[CurrentWorkingView] * Resource.[Resource].filter(#.Name in {[R1],[R2]}))
for Measures {[Supply Modified]} using scope ([Version].[Version
Name].[CurrentWorkingView] * Item.[Item].filter(#.Name in {[raw_m]}))
arguments {[Material Constrained], "true"}, ([Capacity Constrained], "true"), ([Plan Type],
"Violation_Resolution");
```

### Algorithm

- 1) From violating nodes do a downward pegging and find out impacted demands (Using [Dynamic memoryless supply-demand pegging algorithm](#))
- 2) Load flow plans for the impacted Supply Chain.
- 3) Release the supply for the identified demands (identified in step 1).
- 4) Re-plan all the impacted demands again.

#### Note:

a) *Measure.[Capacity Modified] is grained by Resource and Version.  
Measure.[Supply Modified] is grained by Item, Location and Version.*

*When Plan Type is INCREMENTAL\_PLAN  
these measures are grained by Time dimension too.*

b) *One can mark [Capacity Modified] and [Supply Modified] for all the capacity and material too  
before violation resolution plan.*

*In such case in the exec plugin command remove the exact filter for item and resource. Just use  
Item.[Item] and Resource.[Resource] in such case.*

## Concurrent Plan

### Business Need

SCS should be able to scale to large volumes (in terms of billions) and also be able to provide superior performance by streaming demands to the planning system and intelligently processing them in parallel.

Concurrent plan is only applicable for Hadoop/Hive implementations for distributed supply chain which does not have BOM.

### Functionality Description

- There is continuous flow of demands (from external system like Hadoop to mPower system) in predefined order say priority, time, item, location. This order is configurable.
- Define the number of parallel threads to be used for concurrent planning.
- Define the demand chunk size (demands to be planned in one thread). This optimizes the demand set (concurrent plan object) to be planned in a given thread. The idea is to have relatively large demand sets get planned concurrently through parallel threads.
- Users can have  $n$  number of concurrent plan objects where  $n$  is the number of threads configured. The plan starts as soon as one concurrent plan object is created. In parallel next concurrent plan object is being created and planned.
- In case demand priority or time changes, it will wait till the current ongoing concurrent objects finish the plan in order to honour strict priority and earlier time buckets getting higher priority than later time buckets.

### Configuration

Two parameters need to be set

- “Concurrent Planning” to “True”
- “Number of Planning Threads” to more than 1

User can define these in the SCS plugin instance configuration in the tab “Parameters”

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters  
Concurrent Planning → check if “TRUE”, uncheck if FALSE

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters  
Number of Planning Threads → Set Number of Planning Threads = 10



- These parameters can be used as command line arguments as shown below (example)

exec plugin instance [Supply Chain Solver] for Measures {[Demand Quantity]}  
using scope (Version.[Version Name].[CurrentWorkingView] \* Demand.[Demand])  
arguments {[[Concurrent Planning], "True"),

```
([Demand Chunk Size], 1000),
([Delete SCS Plan Data For Version], "True"),
([Number of Planning Threads], 10),
}
```

## Incremental plan in concurrent run

### Business Need

SCS should be able to scale to large volumes (in terms of billions) and also be able to provide superior performance by streaming demands to the planning system and intelligently processing them in parallel. At the same time business requires to see the impact of any incremental changes to supplies and/or demand efficiently.

### Functionality Description

Through Concurrent Planning, SCS addresses the initial problem of planning large volume. For addressing the incremental changes, the incremental planning of SCS has been enhanced to support item based filters. Starting from this release Item members can be passed into the SCS plugin scope filter. Note that the plan it at item level attribute level and not at the demand level.

### Configuration

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Plan Type → enter values in text

Plan Type ⓘ

INCREMENTAL\_PLAN

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters Concurrent Planning → check if "TRUE", uncheck if FALSE

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters Number of Planning Threads → Set Number of Planning Threads = 10

Concurrent Planning ⓘ



Number of Planning Threads ⓘ

1

Concurrent Planning needs to set it true to plan SCS concurrently  
Number of Planning Threads, it determines the number of CPUs made available for the SCS process to plan the segment (independent supply chains) in parallel to gain performance.

PlanType determines where it is a FULL\_PLAN or INCREMENTAL\_PLAN or VIOLATION\_RESOLUTION or IDENTIFY\_SEGMENTS, and for this functionality user needs to set it to INCREMENTAL\_PLAN.

Users can provide the command as an argument, For example

```
exec plugin instance SupplyChainSolver for Measures {[Demand Quantity]}
using scope (&CurrentWorkingView * &PlanningWeeks *&ItemSet)
arguments {[Plan Type], [INCREMENTAL_PLAN]}, ([Concurrent Planning], True)
([Number of Planning Threads], 4));
```

Here the incremental planning will be triggered for the Item members those are part of the Named Set "ItemSet".

Briefly in this type, incremental planning at item level is done as follows

- Load relevant minimal supply chain required for incremental planning
- Delete the relevant data in output tables.
- Correct values of remaining available capacity / storage.
- Concurrently plan the demands for the items as per the filter.
- Export data

## Incremental Planning with cumulative profile

### Business Need

Supply chains need to be responsive to stay competitive in the marketplace. One of the common uncertainties is around demand. It could be changes to forecast based on intelligence input received after the batch run or a rush order arriving at the last moment. The planner needs to check what can be done about the same and act in an incremental manner on top of the existing plan. Similarly, one can expect inconsistencies/uncertainties in the supply {those which participate in demand fulfillment}. This may be due to factors both internal or external to the supply chain such as defective or damaged raw materials or improper coordination between planning and execution. Planners need to have the visibility as to what is the effect of increase/decrease of existing supply on existing demands thereby acting in an incremental manner on top of existing plan. Planners frequently use this to expedite the supply to meet existing demands which are short/late.

### Functionality Description

The basic rule of incremental (net change) planning is to handle exceptions while minimizing the impact on existing commitments.

Planners can keep a track of all the demands that were replanned incrementally as a result of supply changes.

When it comes to manipulate demand, one can expect the following

1. Order addition
2. Order quantity change
3. Order deletion
4. Order date change

*Note :Order has Time dimension as one of its granularity. Modifying granularity data in LiveServer is not yet implemented. Therefore, Order Date change can be done by deleting an order of previous due date and creating the same order with the new due date.*

When it comes to supply changes, one can expect the following:

1. Supply/Lot Addition
2. Supply/Lot Deletion

3. Supply/Lot Quantity Change(increase/decrease)
4. Supply/Lot Date Change

Supply can be any of the following:

1. Material Quantity
  - a. Expected Receipts
  - b. BOH
2. Capacity Availability

## Configuration

Configure second instance of SupplyChainSolver with the following changes:

Goto Workspace→Designer→Edit

model→Rule→Plugin→Global→o9.GraphCubePlugins→SupplyChainSolver→SupplyChainSolver1 → Parameters → Plan Type → enter values in textbox

Plan Type ⓘ

INCREMENTAL\_PLAN2

Incremental Planning should be done on modified or new supplies and/or demands. SCS then keeps previous plan under consideration while planning the modified or new supplies and/or demands

Planners should mark the orders on which incremental planning is to be done. For this a new measure named Incremental\_Allowed needs to be configured. This measure has the same granularity as Demand Quantity and hence can be defined in the same measure group. Incremental\_Allowed measure takes boolean values.

- True : incremental planning is allowed on the corresponding demands.
- False: Demand does not participate in incremental planning

*Note: False behaves same as NULL*

This measure should also be specified in the SCS config as follows :

Go to Designer→ edit model → Rule → -> Plugins → Global → o9.GraphCubePlugins → SupplyChainSolver→ SupplyChainSolver1 → Demand → Is Eligible for Incremental Plan Measure Name → choose measure from the drop down menu

Is Eligible For Incremental Plan Measure Name ⓘ

Incremental\_Allowed

Planners should mark the supplies participating in the incremental plan. The following measures need to be configured.

- For changes made to existing Expected Receipts and BOH
  - Measure
    - Supply Changed:
 

This measure takes boolean values:

      1. True : indicates that Supply (ER or BOH) on a material node for a specified time bucket has changed and hence should take part in incremental planning

2. False (default): indicates that Supply(ER or BOH) on a material node for a specified time bucket should not take part in incremental planning.  
*Note: False behaves same as NULL*
- Granularity
    - Material Node
      1. Item.Item Name
      2. Location.Location Name
    - Version.Version Name
  - For Capacity Availability changes
    - Measure
      - Capacity Changed  
This measure takes boolean values:
        1. True : indicates that Capacity Availability in the specified time bucket has changed and hence should take part in incremental planning
        2. False: indicates that Capacity Availability should not take part in incremental planning.  
*Note: False behaves same as NULL*
    - Granularity
      - Capacity Node
        1. Resource.Resource Name
      - Time.Fiscal Week
      - Version.Version Name
  - For tracking the demands that were replanned as a part of supply changes
    - Measure
      - Participated in Incremental Plan: This measure is the output of the Incremental Plan. It takes boolean values.
        1. TRUE: If an order has been replanned after the supply change, then the value of this measure for the corresponding order is set as TRUE
        2. FALSE/NULL: If an order remains untouched from being incrementally planned, then the value of this measure for the corresponding order is set as FALSE.
    - Granularity (Same as Delivery plan as it is an output measure)
      - Demand Node
        1. Order.Order Name
        2. Time.Fiscal Week
        3. Customer.Customer Name
      - Material Node
        1. Item.Item Name
        2. Location.Location Name
      - Version.Version Name
- Note: If one runs the full plan by setting [Delete SCS Plan Data For Version] to true, then this gets reset to null. If only incremental plans are run in sequence, then it gets updated only for the demand that are replanned.*

The measure Participated in Incremental Plan should be set only in Delivery Plan measure group and not in Demand Quantity measure group. Placing this



measure in the Demand Signal measure group will cause the SCS plugin to error out.

These measures should also be specified in the SCS config as follows :

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver1 → Inventory →  
Supply Modified Measure Name → choose measure from the drop down menu

Supply Modified Measure Name 

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver1 → Manufacturing →  
Capacity Modified Measure Name → choose measure from the drop down menu

Capacity Modified Measure Name 

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver1 → Edit → Demand  
→ Re-Planned By Interactive Plan Measure Name → choose measure from the drop  
down menu

Re-Planned By Interactive Plan Measure Name 

The supply and capacity measures need to be used along with demand measure in the SCS plugin command for the incremental plan to consider those changes. For example,

*exec plugin instance SupplyChainSolver for Measures {[Demand Quantity], [Expected Receipts], [BOH], [Capacity Availability]} using scope ([Version].[Version Name].[CurrentWorkingView] \* Time.[Fiscal Week] \* Order.[Order Name]) arguments {[Material Constrained], "True"), ([Capacity Constrained], "True"), ([Plan Type], "INCREMENTAL\_PLAN")};.*

### Limitations

Supply changes with respect to the following are not yet supported in incremental planning

1. Lock Plans
2. Match code Planning

## WIP changes in incremental planning

### Business Need

WIP denotes work in process which includes procurement order, manufacturing order, distribution orders to name a few. Due to the operational nature, there is a higher likelihood that there can be changes - It could be changes to the date or the quantity. Users might call the vendor and ask to expedite some purchase orders, some manufacturing order might have to be pushed out due to unscheduled maintenance etc. Planners also want to see the impact of WIP changes to reduce the overall impact or do cost-benefit trade-offs. They would also want to evaluate the impact in an incremental manner by doing local search.

### Functionality Description

In the current release incremental planning of Supply Chain Solver (SCS) has been enhanced to support WIP changes in the model. User can increase or decrease the existing WIP and mark the changes for the solver to take it as an input. After the

changes, when user performs the incremental planning, the solver adjusts the existing plan based on the revised WIP quantity.

## Configuration

User needs to configure a Boolean measure, grain same as WIP to mark whether the WIP is changed or not post batch planning. User can configure an active rule for the same. Configure a measure called *Material Production WIP Changed* in model designer.

The screenshot shows the 'Material Production Graph Planned WIP' configuration window. On the left, there is a sidebar with the following fields: Name (Material Production WIP Changed), Description (Material Production WIP Changed), ToolTip (Material Production WIP Changed), Data Type (Boolean), and Type (Regular). The 'Editable' checkbox is checked. The main area is divided into two tabs: 'Measures' and 'Rules'. The 'Measures' tab is active, showing a list of measures: 'Material Production Planned WIP', 'Prod Consume Outbound Capacity For WIP', and 'Material Production WIP Changed'. The 'Material Production WIP Changed' measure is selected.

If the WIP quantity is changed for a certain WIP, above measure corresponding to the WIP should be set to True or 1.

User also needs to mark the demands to be considered in the subsequent incremental planning by setting following measure to True or 1.

In case of decrease in WIP quantity, demands that are pegged to this WIP are automatically selected for re-plan.

In the SCS config, a measure needs to be linked to its corresponding parameter in the Manufacturing tab as shown below.

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Prod Graph WIP Change Measure Name → Select corresponding measure from the drop down menu

The screenshot shows the 'Prod Graph WIP Change Measure Name' dropdown menu. The dropdown is open, showing the selected measure 'Material Production WIP Changed'.

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Is Eligible For Incremental Plan Measure Name → Select corresponding measure from the drop down menu

The screenshot shows the 'Is Eligible For Incremental Plan Measure Name' dropdown menu. The dropdown is open, showing the selected measure 'Incremental\_Allowed'.

Following SCS plugin command needs to be executed to consider the changed WIP during incremental planning. User can configure this command execution in an action button.

*exec plugin instance SupplyChainSolver for Measures {[Demand Quantity], [Material Production WIP Changed]} using scope ([Version].[Version Name].[CurrentWorkingView]) arguments {[Plan Type], [INCREMENTAL\_PLAN]};*

## Additive Demand RCA

### Business Need

Planners want to select a set of short and late demands and understand the root constraints at the aggregate level. Planners want to see Root Cause Analysis (RCA) of shorted and/or late demands for the constrained plan batch run as well. This RCA needs to be additive, i.e. when considering all late and short demands in the system, the total constraint quantity should be equal to the needed quantity of all late and short demands.

### Functionality Description

This is conceptually equivalent to running the unconstrained plan for the shorted demands post constrained run (regular demand planning) and doing the order-by-order (OBO) RCA. To prevent double counting of the left-over capacity/material by more than one demand (to preserve additivity), the leftover unused capacity is allocated to the shorted demands in prioritized order on the primary path only.

Users will get material, capacity and storage constraints related to short demand nodes in the RCA output. One can drop the demand node granularity from the RCA output to get the cumulative needed quantity for the material/capacity/storage nodes for a bucket.

This RCA is profile based which does inventory reallocation during RCA phase and reports the constraints appropriately unlike the conventional pegging-based RCA.

In the lotsize case, the sum of the needed quantity will not be same as sum of short demand quantity. This is because demand RCA considers the surplus material produced due to lotsize while doing RCA for higher priority demand and RCA for current short demand is reported only for the net short quantity.

In some cases, there will be no RCA records correspond to some short demands. This can happen when lotsize is defined in the model. The current short demand (low priority) can use the excess leftover during RCA plan for high priority demand and get planned. In such case, there will be no RCA for the current short demand.

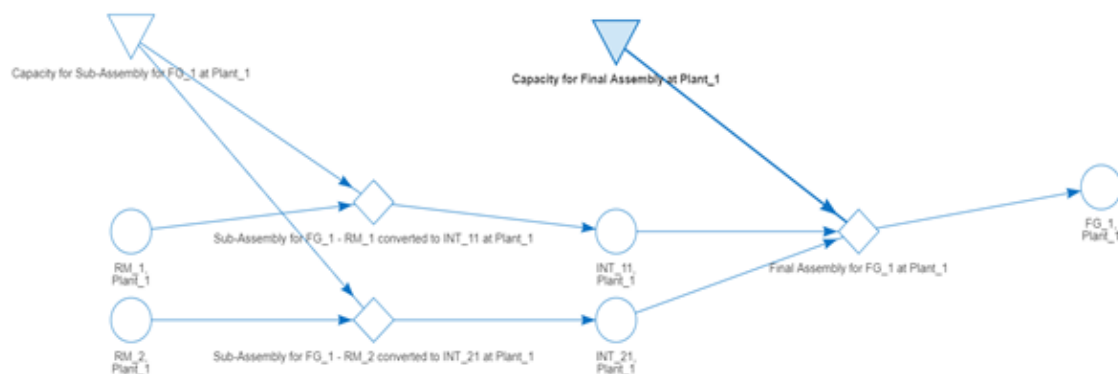
Also, sometimes the demand will be short during demand planning due to build ahead limit impact. However, during SS planning, it could produce at some earlier bucket (as activity is effective then). While doing RCA pass, it gets planned using SS planned flow plan quantity and hence does not report RCA. A hypothetical example is given below to illustrate this point.

- Demands have build ahead limit specified. Assume a (fake) demand D8 due in bucket 20 with BAL of 2.
- Demand D8 becomes short based on upstream activities available up until bucket 18
- Since demand D8 is deemed short during demand planning, it contributes to the TotalShortQty
- Next, during safety stock planning, assume there is a need to meet safety stock demand in bucket 12
- Now, the available activities for demands in bucket 12 are such that it is possible to meet the demand in bucket 12 !!
- Next, when RCA pass is in progress, the originally short demand D8 can be now met by consuming from the safety stock production.

In other words, combination of build ahead limit, available activities and safety stock planning result in this behavior.

In case of lead time/effectivity problem, it reports material problem at the immediate downstream material node in JIT bucket.

### Example



### Case1: Non-lotsize case

Consider the supply chain shown above. Here the final assembly capacity and the upstream raw materials are constrained. The sub assembly capacity is sufficient to plan the demands.

Here is the supply and demand picture.

The demands are placed for FG\_1 at Plant\_1 with build late limit 0.

The activity lead time is 1 Week.

### Demands

| Demands | Due bucket | Demand Quantity |
|---------|------------|-----------------|
| Order_1 | W3         | 30              |
| Order_2 | W4         | 40              |
| Order_3 | W5         | 30              |
| Order_4 | W7         | 20              |

#### Assembly capacity

| Capacity Bucket | Available Capacity |
|-----------------|--------------------|
| W0              | 50                 |
| W1              | 50                 |
| W2              | 50                 |
| W3              | 50                 |
| W4              | 15                 |
| W5              | 15                 |
| W6              | 50                 |
| W7              | 50                 |

#### Raw material supply (same for both RM\_1 and RM\_2)

| Expected Receipt Bucket | Supply Quantity |
|-------------------------|-----------------|
| W0                      | 50              |
| W3                      | 25              |
| W5                      | 50              |

#### Planning Results

- Order\_1 gets planned fully. RM supply of 30 units consumed on W1, 20 remains.
- Order\_2 gets planned partially 20 units due to RM constrained (20 units available on W2). 20 units Order\_2 shorted.
- Order\_3 gets planned 15 units due to capacity constrained (on W4). 15 units shorted. RM material left on W3 is 10 units.
- Order\_4 gets fully planned. From the pegging point of view it consumes the 10 units RM left over and remaining 10 units from next supply. So from pegging point of view there is no material left on W3 bucket.

#### RCA Results

- Requires 15 units assembly capacity on W4 for Order\_3 to satisfy fully
- Requires 20 units of RM on W2 for Order\_2 to satisfy fully

- Requires 5 units of RM on W3 for Order\_3 to satisfy fully. In case of conventional pegging based RCA, one would see 15 units need quantity on W3 although there is material available latter to satisfy the future demands.

#### Case2: Lotsize case

Let's consider a scenario where the order is short but RCA is not reported. Consider the same supply chain as above. Let's assume there is no material constraint and the only constraint is the assembly capacity. It has 100 units per week capacity available. The assembly activity has lotsize multiple of 25. Thus, any production to the FG\_1 at Plant\_1 will be in the multiple of 25.

Following is the demand picture

- Order\_1: 110 @Plant\_1 due on W5, Priority 1
- Order\_2: 115 @Plant\_1 due on W7, Priority 2

#### Planning Results

- Order1 gets planned first and 100 is planned. 10 shorted.
- Order2 gets planned next and 100 is planned. 15 shorted.

#### RCA Results

- In RCA phase it first picks Order\_1 (high priority) and plans the short quantity. For 10 short demand it needs to produce 25 to respect lotsize. So RCA need quantity of 25 is reported.
- Out of this 25 produced, 10 is used and 15 FG\_1 at Plant\_1 remains as residual.
- There will be no RCA for Order\_2. This is because the 15 excess generated by RCA previously will be used for Order\_2 shortage.

#### Case3: Demand late but no RCA

Following is the demand picture

- Order1=400, Due=W19
- Order2=100, Due=W20

The producing activity to demand node is linked to Resource R1

- R1=100 for W18, W19, W20, W21, W22, W23

#### Planning Results

- Order1 Plan:
  - JIT=200 (W18, W19 capacity consumed)
  - Delay=200 (W20, W21 capacity consumed)
- Order2 Plan:
  - JIT=0
  - Delay=100 on W22 (W22 capacity is consumed)

#### RCA Results

- Order1 processes first.

- It unplans the delay part - W20 and W21. The capacity is released.
- Flags 200 capacity deficit on W19 (JIT).
- Order2 processes next.
  - This time it find capacity on W20 and get planned. So no RCA is reported.
  - Thus, Order2 is late but there is no RCA reported.

## Configuration

Users need to configure 3 graph tables for SCS to write the RCA results correspond to Material, Capacity and Storage.

- Material RCA Graph
  - From Node → <Material Node Attrs>, <Time Attr (constraint bucket)>
  - To Node → <Demand Attrs>, <Time Attr (Due bucket)>
  - Edge Property → Required Material Qty

Example:

Material Demand RCA Graph Name   ✔ Loaded 0 Rules.

---

From Nodes Item Location Fiscal Week

---

To Nodes Fiscal Week Order Name Customer Name Item Location

---

Edges + Rules

Material Demand RCA Qty Property

- Capacity RCA Graph
  - From Node → <Resource Node Attrs>, <Time Attr (constraint bucket)>
  - To Node → <Demand Attrs>, <Time Attr (Due bucket)>
  - Edge Property → Required Capacity Qty

Example:

Capacity Demand RCA Graph Name   Loaded 0 Rules.

From Nodes Resource Fiscal Week

To Nodes Customer Name Order Name Item Location Fiscal Week

Edges + Rules

Capacity Demand RCA Qty Property

- Storage RCA Graph
  - From Node → <Storage Node Attrs>, <Time Attr (constraint bucket)>
  - To Node → <Demand Attrs>, <Time Attr (Due bucket)>
  - Edge Property → Required Storage Qty

### Example

Storage Demand RCA Graph Name   Loaded 0 Rules.

From Nodes Storage ID Fiscal Week

To Nodes Customer Name Order Name Item Location Fiscal Week

Edges + Rules

Storage Demand RCA Qty Property

After the graphs are modeled, these need to be configured in solver config (Manufacturing Tab) as shown in example below.

|                                                                                                                      |                                                               |
|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Material Demand RCA Graph Name    | <input type="text" value="Material Demand RCA Graph Name"/>   |
| Capacity Demand RCA Graph Name    | <input type="text" value="Capacity Demand RCA Graph Name"/>   |
| Storage Demand RCA Graph Name     | <input type="text" value="Storage Demand RCA Graph Name"/>    |
| Material Demand RCA Qty Property  | <input type="text" value="Material Demand RCA Qty Property"/> |
| Capacity Demand RCA Qty Property  | <input type="text" value="Capacity Demand RCA Qty Property"/> |
| Storage Demand RCA Qty Property   | <input type="text" value="Storage Demand RCA Qty Property"/>  |

Capacity Demand RCA Graph Name → Select the Capacity RCA Graph

Material Demand RCA Graph Name → Select the Material RCA Graph

Storage Demand RCA Graph Name → Select the Storage RCA Graph

Material Demand RCA Qty Property → Select the edge for needed material quantity

Capacity Demand RCA Qty Property → Select the edge for needed capacity quantity

Storage Demand RCA Qty Property → Select the edge for needed storage quantity



Finally, add the following parameter in the plugin execution command for demand RCA to run.

`([Demand RCA], "True")`

Example:

`exec plugin instance SupplyChainSolver for Measures {[Demand Quantity]} using scope ([Version].[Version Name].[CurrentWorkingView] * Time.[Fiscal Week] * Order.[Order Name]) arguments {[Material Constrained], "True"}, ([Capacity Constrained] , "True"), ([Delete SCS Plan Data For Version], "True"), ([Demand RCA], "True");`

You can refer some of the demand RCA functional test cases in [Jenkins](#) (Test case 24 through 35).

**Note:**

- For frozen period, locked wip, lead time, activity date effectivity case, solver does not traverse upstream and reports material constraint at the producing material node.
- The RCA is reported along the JIT effective activity. If the primary activity is not effective, solver will check next alternate activity that is effective during JIT and reports constraints along that instead of reporting effectivity constraint along the primary activity.

**Limitation**

Lateness RCA is not reported for hard pegged demands.

## RCA Constraint Type As Dimension Member

### Business Need

In some of the customer supply planning workflow, one of the postprocessing step after SCS run is to grain RCA outputs by root cause dimension in order to show root cause mix. This process (edge property to convert to dimension) takes quite some time. Instead if solver itself exports the RCA constraint type as dimension member the post processing step could be avoided.

### Functional Description

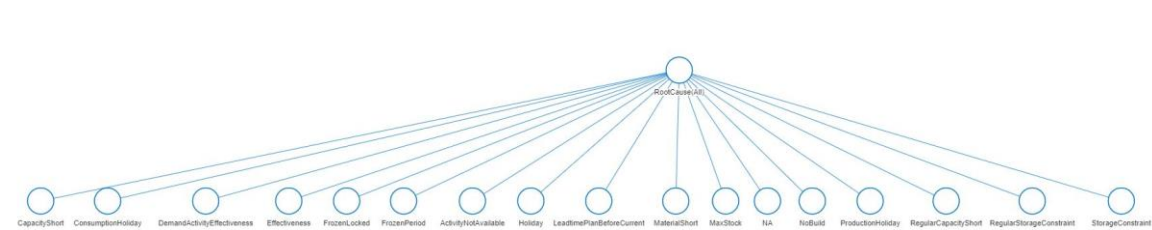
In this release SCS has been enhanced to export Demand RCA data with constraint type as a dimension member.

Users will find a system defined dimension **RootCause** with level attribute as **Problem Type**

◀  RootCause

 Problem Type

Following image shows all the supported members for **Problem Type LA**.



**i** Showing 17 rows of data.

| Problem Type                                                                      |
|-----------------------------------------------------------------------------------|
|  |
| CapacityShort                                                                     |
| ConsumptionHoliday                                                                |
| DemandActivityEffectiveness                                                       |
| Effectiveness                                                                     |
| FrozenLocked                                                                      |
| FrozenPeriod                                                                      |
| Holiday                                                                           |
| LeadtimePlanBeforeCurrent                                                         |
| MaterialShort                                                                     |
| MaxStock                                                                          |
| NA                                                                                |
| NoBuild                                                                           |
| ProductionHoliday                                                                 |
| RegularCapacityShort                                                              |
| RegularStorageConstraint                                                          |
| StorageConstraint                                                                 |

**Configuration**

1. Create following new graphs for Demand RCA. The only difference w.r.t. the existing demand RCA graph is the problem type defined as dimension instead of edge property.

- a. Material RCA Graph

From Node: <Material Node> , <Time>, <Rootcause>  
To Node: <Demand Node>, <Material Node>  
Edge Property: RCA Qty (Number)

- b. Capacity RCA Graph

From Node: <Capacity Node> , <Time> , <Rootcause>  
To Node: <Demand Node>, <Material Node>  
Edge Property: RCA Qty (Number)

- c. Storage RCA Graph

From Node: <Storage Node>, <Time>, <Rootcause>  
To Node: <Demand Node>, <Material Node>  
Edge Property: RCA Qty (Number)

### Example

Additive Material RCA

Loaded 0 Rules.

From Nodes

Item

Location

Fiscal Week

Problem Type

To Nodes

Customer Name

Order Name

Item

Location

Fiscal Week

Edges

+

Material RCA Qty

Rules

Additive Capacity RCA

Loaded 0 Rules.

From Nodes

Resource

Problem Type

Fiscal Week

To Nodes

Customer Name

Order Name

Item

Location

Fiscal Week

Edges

+

Capacity RCA Qty

Rules

Additive Storage RCA

Loaded 0 Rules.

From Nodes

Storage ID

Fiscal Week

Problem Type

To Nodes

Customer Name

Order Name

Item

Location

Fiscal Week

Edges

+

Storage RCA Qty

Rules

In the plugin instance config use these graphs for solver to consider during RCA exports.

| Instance Options                                                                                                        | Parameters              | Node | Demand | Inventory | Manufacturing | Distribution | Procurement |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------|------|--------|-----------|---------------|--------------|-------------|
| Material Demand RCA Graph Name         | Additive Material RCA ▼ |      |        |           |               |              |             |
| Capacity Demand RCA Graph Name         | Additive Capacity RCA ▼ |      |        |           |               |              |             |
| Storage Demand RCA Graph Name          | Additive Storage RCA ▼  |      |        |           |               |              |             |
| Material Demand RCA Qty Property       | Material RCA Qty ▼      |      |        |           |               |              |             |
| Material RCA Constraint Type Property  | ▼                       |      |        |           |               |              |             |
| Capacity Demand RCA Qty Property       | Capacity RCA Qty ▼      |      |        |           |               |              |             |
| Capacity RCA Constraint Type Property  | ▼                       |      |        |           |               |              |             |
| Storage Demand RCA Qty Property        | Storage RCA Qty ▼       |      |        |           |               |              |             |
| Storage RCA Constraint Type Property   | ▼                       |      |        |           |               |              |             |

In the Parameters tab, enable the **Demand RCA** check box.

Demand RCA 



## RCA with Activity as grain

### Business Need

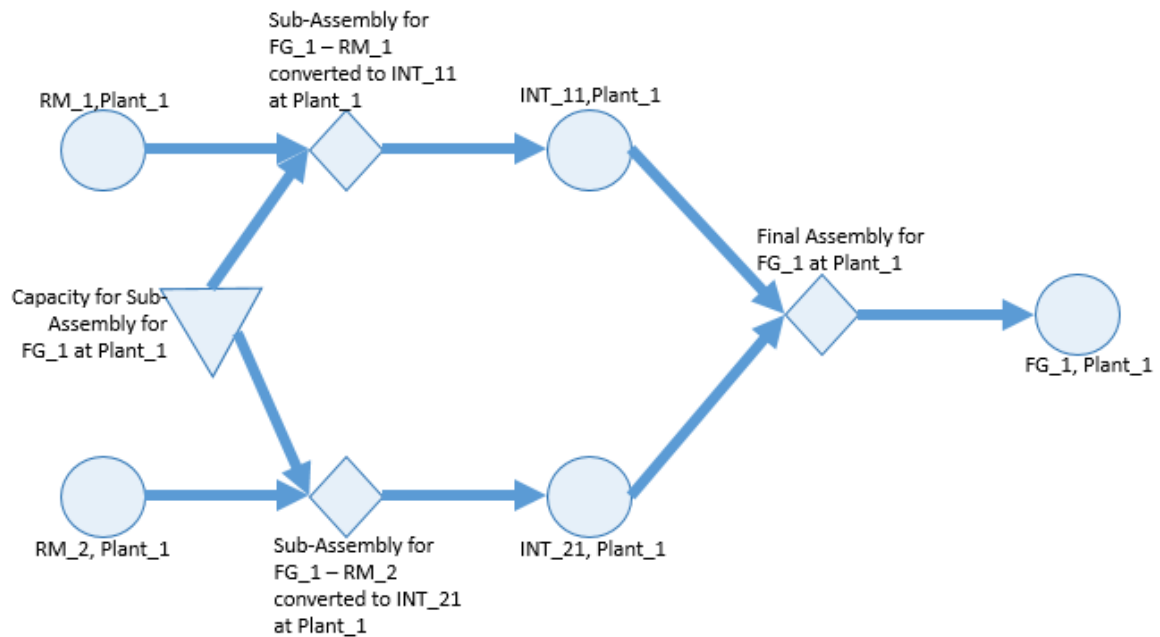
Planners want to see the activity ,while viewing the Root Cause Analysis (RCA) of shorted demands for Capacity/Material's constraints. This is beneficial in cases when capacity/material nodes are connected to multiple activities, planners will be able to exactly trace due to which activity there is shortage.

### Functional Description

Capacity RCA feature of SCS has been enhanced to show the root causes by Activities. Material RCA can also be enhanced to have activity in the from Node. It will show the producing activity incident on the material node where deficit is reported.

Example:

Consider the following supply chain where sub assembly capacity is shared between multiple activities.



Post SCS run if there is any demand shortage due to capacity/material constraints. User can see the Capacity/Material RCA which will show the activity due to which there is shortage.

### Configuration Steps

Change the Granularity of the below graphs to include “**Activity**” in the **From Node**

Capacity RCA Graph
⚙️
🔍
✔ Loaded 0 Rules.

From Nodes
Resource
Fiscal Week
Activity

To Nodes
Customer Name
Order Name
Item
Location
Fiscal Week

---

Material RCA Graph
⚙️
🔍
✔ Loaded 0 Rules.

From Nodes
Item
Location
Fiscal Week
Activity

To Nodes
Order Name
Customer Name
Item
Location
Fiscal Week

## OBO RCA

### Business Need

In case of constrained supply planning, users are interested to know the root cause for demands which are short or late. This will help them to visualize what are the real constraints exist in the current plan which caused the demand late/short. This is an advancement of pegging RCA to cover those cases where demands are short / late but not reported as constraints in pegging graphs.

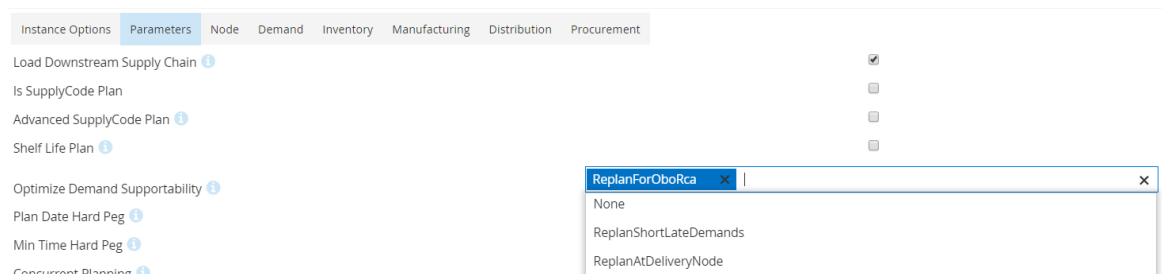
### Functionality

In Supply Chain Solver (SCS), there are different ways to generating and reporting demand late and short constraints. One of the approaches is Order By Order (OBO) RCA (root cause analysis). The existing OBO RCA uses pegging approach which gives some sub-optimal results due to the fact that planning and pegging output can be different. Due to this sometimes even if demand is late or short, users may not find any RCA for those. To avoid this SCS has been enhanced and a new workflow for OBO RCA introduced.

### Configuration

#### a) Defining [Optimize Demand Supportability]

Goto Workspace → Designer → Edit model → Rule → Plugin → Global o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameter → Optimize Demand Supportability → ReplanForOBORCA



If [Optimize Demand Supportability] is **ReplanForOboRca** then it will skip pegging OBO rca (where constraints reported in pegging graphs).

Note: [Optimize Demand Supportability] is a multi picklist now with values:

- 1) ReplanForOboRca
- 2) ReplanShortLateDemands
- 3) ReplanAtDeliveryNode
- 4) None (default)

If 4) is selected with any other option, 4) is ignored

If 2) and 3) are selected together, 3) is ignored

If 1) is selected with 2) or 3), first 2) or 3) is executed, then 1) is executed

One can use this parameter in the solver plugin command as arguments as:  
([Optimize Demand Supportability], [ReplanShortLateDemands, ReplanForOboRca])

b) Defining Graphs for writing out the RCA data as shown below.

**Material Production OBO RCA Graph Name**

From Nodes: <Activity Node>, Problem Type, <Material Node>, Time, Lot Attr

To Nodes: Demand, Item, Location, Time

Edge Property: Constraint Qty

**Capacity Consumption OBO RCA Graph Name**

From Nodes: <Resource Node>, <Activity>, Problem Type, Time, Lot Attr

To Nodes: Demand, Item, Location, Time

Edge Property: Constraint Qty

**Storage Consumption OBO RCA Graph Name**

From Nodes: <Storage Node>, Problem Type, <Material Node>, Time, Lot Attr

To Nodes: Demand, Item, Location, Time

Edge Property: Constraint Qty

*Note: If any of the graphs are not defined and the flag is set to ReplanForOboRca, it will just skip exporting to that table and will log the missing table.*

|                                                |            |      |        |           |               |                              |             |
|------------------------------------------------|------------|------|--------|-----------|---------------|------------------------------|-------------|
| Instance Options                               | Parameters | Node | Demand | Inventory | Manufacturing | Distribution                 | Procurement |
| Consumer to Consumer From Pegged Property Name |            |      |        |           |               |                              | ▼           |
| Consumer To Consumer To Pegged Property Name   |            |      |        |           |               |                              | ▼           |
| Producer To Producer From Pegged Property Name |            |      |        |           |               |                              | ▼           |
| Producer To Producer To Pegged Property Name   |            |      |        |           |               |                              | ▼           |
| Material Production OBO RCA Graph Name         |            |      |        |           |               | Material Production OBO RCA  | ▼           |
| Material Production OBO RCA Qty Property       |            |      |        |           |               | Material OBO RCA Qty         | ▼           |
| Capacity Consumption OBO RCA Graph Name        |            |      |        |           |               | Capacity Consumption OBO RCA | ▼           |
| Capacity Consumption OBO RCA Qty Property      |            |      |        |           |               | Capacity OBO RCA Qty         | ▼           |
| Storage Consumption OBO RCA Graph Name         |            |      |        |           |               |                              | ▼           |
| Storage Consumption OBO RCA Qty Property       |            |      |        |           |               |                              | ▼           |

If the model has distribution activity modeled separately, then following graphs need to be configured.

| Instance Options                                            | Parameters | Node | Demand | Inventory | Manufacturing | Distribution | Procurement                                 |
|-------------------------------------------------------------|------------|------|--------|-----------|---------------|--------------|---------------------------------------------|
|                                                             |            |      |        |           |               |              |                                             |
| Demand Distribution Activity End Date Measure Name ⓘ        |            |      |        |           |               |              | ▼                                           |
| Distribution Activity Proportion Measure Name ⓘ             |            |      |        |           |               |              | ▼                                           |
| Distribution Load Plan Graph Name ⓘ                         |            |      |        |           |               |              | ▼                                           |
| Distribution Load Plan Capacity Consumption Property Name ⓘ |            |      |        |           |               |              | ▼                                           |
| Distribution Load Plan Material Production Property Name ⓘ  |            |      |        |           |               |              | ▼                                           |
| Distribution Plan Graph Name ⓘ                              |            |      |        |           |               |              | ▼                                           |
| Distribution Plan Material Production Property Name ⓘ       |            |      |        |           |               |              | ▼                                           |
| Distribution Plan Material Consumption Property Name ⓘ      |            |      |        |           |               |              | ▼                                           |
| Distribution Material Production OBO RCA Graph Name ⓘ       |            |      |        |           |               |              | Material Production Distribution OBO RCA ▼  |
| Distribution Material Production OBO RCA Qty Property ⓘ     |            |      |        |           |               |              | Material Distribution OBO RCA Qty ▼         |
| Distribution Capacity Consumption OBO RCA Graph Name ⓘ      |            |      |        |           |               |              | Capacity Consumption Distribution OBO RCA ▼ |
| Distribution Capacity Consumption OBO RCA Qty Property ⓘ    |            |      |        |           |               |              | Capacity Distribution OBO RCA Qty ▼         |

If the model has supplier activity modeled separately, then following graphs need to be configured.

| Instance Options                                        | Parameters | Node | Demand | Inventory | Manufacturing | Distribution | Procurement                                |
|---------------------------------------------------------|------------|------|--------|-----------|---------------|--------------|--------------------------------------------|
|                                                         |            |      |        |           |               |              |                                            |
| Supplier Activity Segment Measure Name ⓘ                |            |      |        |           |               |              | ▼                                          |
| Demand Supplier Activity Priority Measure Name ⓘ        |            |      |        |           |               |              | ▼                                          |
| Demand Supplier Activity Start Date Measure Name ⓘ      |            |      |        |           |               |              | ▼                                          |
| Demand Supplier Activity End Date Measure Name ⓘ        |            |      |        |           |               |              | ▼                                          |
| Supplier Activity Proportion Measure Name ⓘ             |            |      |        |           |               |              | ▼                                          |
| Supplier Load Plan Graph Name ⓘ                         |            |      |        |           |               |              | ▼                                          |
| Supplier Load Plan Capacity Consumption Property Name ⓘ |            |      |        |           |               |              | ▼                                          |
| Supplier Load Plan Material Production Property Name ⓘ  |            |      |        |           |               |              | ▼                                          |
| Supplier Material Production OBO RCA Graph Name ⓘ       |            |      |        |           |               |              | Material Production Procurement OBO RCA ▼  |
| Supplier Material Production OBO RCA Qty Property ⓘ     |            |      |        |           |               |              | Material Procurement OBO RCA Qty ▼         |
| Supplier Capacity Consumption OBO RCA Graph Name ⓘ      |            |      |        |           |               |              | Capacity Consumption Procurement OBO RCA ▼ |
| Supplier Capacity Consumption OBO RCA Qty Property ⓘ    |            |      |        |           |               |              | Capacity Procurement OBO RCA Qty ▼         |

## Cascading RCA

### Business Need

Current RCA facilitates planners to list down the reasons why each unit quantity could not be met on time (due date) but with Cascading RCA, planners can also list down the reasons why each unit quantity could not be met in later buckets as well.

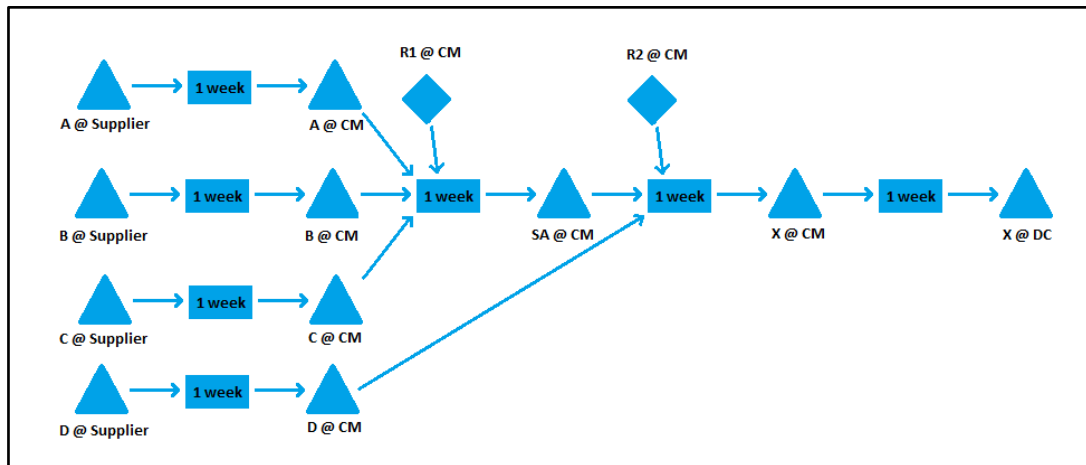
### Functionality Description

RCA would not only be listed wrt JIT bucket but also wrt JIT bucket + 1, JIT bucket + 2, JIT bucket + 3 and so on, till the user specified limit.

The root cause analysis (RCA) output of SCS answers the question "why was demand XYZ not met on-time?". It does not answer the question "why was demand XYZ planned 6 weeks late?". If a demand is planned 6 weeks late, then SCS shouldn't just calculate RCA on the due-date of the demand, but also on due-date plus 1 week, due-date plus 2 weeks, etc., upto due-date plus 5 weeks.

### Example





- Item X is sold out of location DC. X is sourced from CM with a one week lead-time.
- X is made at CM from sub-assembly SA and component D using resource R2 with a one week lead-time.
- SA is made at CM from components A, B, and C using resource R1 with a one week lead-time.
- Components A, B, C and D are sourced from location supplier, each with a one week lead-time.
- Components at supplier are infinite material nodes.
- Capacity for R1 and R2 is zero in the first two weeks, and 5 per week thereafter.
- All quantity-pers are 1.
- There is a single demand for X at DC for quantity 10 in the current week (W1).

If we run SCS for this demand, the RCA information will say that the root cause is LeadtimePlanBeforeCurrent for X at DC in week W1 for quantity 10. The demand will be planned for quantity 5 in week W6 and quantity 5 in week W7. The RCA explains why the demand couldn't be planned in week W1, but it doesn't really explain why it is planned 5-6 weeks late.

Now, let's pretend we changed the due-date to W2 and reran SCS. The RCA would then say that the root cause is LeadtimePlanBeforeCurrent for X at CM in week W1 for quantity 10.

If we continued moving the due-date out one week at a time and rerunning SCS, the result would be:

For due-date W3:

The root causes are LeadtimePlanBeforeCurrent for D at CM in week W1 for quantity 10 and LeadtimePlanBeforeCurrent for SA at CM in week W1 for quantity 10 and CapacityShort for R2 at CM in week W1 for quantity 10

For due-date W4:

The root causes are LeadtimePlanBeforeCurrent for A at CM in week W1 for quantity 10 and LeadtimePlanBeforeCurrent for B at CM in week W1 for quantity 10 and LeadtimePlanBeforeCurrent for C at CM in week W1 for quantity 10 and CapacityShort for R2 at CM in week W2 for quantity 10 and CapacityShort for R1 at CM in week W1 for quantity 10

For due-date W5:

The root causes are CapacityShort for R2 at CM in week W3 for quantity 5 and CapacityShort for R1 at CM in week W2 for quantity 10

For due-date W6:

The root cause is CapacityShort for R1 at CM in week W3 for quantity 5

If we put this information together in the following way, we would get a picture like this:

| Constraint Type           | Resource | Item | Location | W1       | W2       | W3       | W4       | W5       | W6      |
|---------------------------|----------|------|----------|----------|----------|----------|----------|----------|---------|
| LeadtimePlanBeforeCurrent |          | X    | DC       | 10 in W1 |          |          |          |          |         |
| LeadtimePlanBeforeCurrent |          | X    | CM       |          | 10 in W1 |          |          |          |         |
| LeadtimePlanBeforeCurrent |          | D    | CM       |          |          | 10 in W1 |          |          |         |
| LeadtimePlanBeforeCurrent |          | SA   | CM       |          |          | 10 in W1 |          |          |         |
| LeadtimePlanBeforeCurrent |          | A    | CM       |          |          |          | 10 in W1 |          |         |
| LeadtimePlanBeforeCurrent |          | B    | CM       |          |          |          | 10 in W1 |          |         |
| LeadtimePlanBeforeCurrent |          | C    | CM       |          |          |          | 10 in W1 |          |         |
| CapacityShort             | R2       |      | CM       |          |          | 10 in W1 | 10 in W2 | 5 in W3  |         |
| CapacityShort             | R1       |      | CM       |          |          |          | 10 in W1 | 10 in W2 | 5 in W3 |

## Construct

1. It is an extension of OBO RCA (not Additive RCA) as constraint quantities reported are not additive across demands.
2. OBO RCA is a universal flag for Cascading RCA, unless OBO RCA is enabled, Cascading RCA will not work.
3. When OBO RCA is enabled, OBO RCA will print JIT RCAs in OBO RCA Graphs and Cascading RCA will print RCAs for JIT, JIT+1, JIT+2 and so on in Cascading RCA Graphs. Therefore, JIT constraints will be reported in both Graphs.

Eg: if there are 2 demands A and B. For demand A, Cascading RCA Limit is 0 and for demand B, Cascading RCA Limit is 2:

Then, OBO RCA graphs will show outputs for

- JIT bucket of A
- JIT bucket of B

And, Cascading RCA graphs will show outputs for

- JIT bucket of B
- JIT+1 bucket of B
- JIT+2 bucket of B

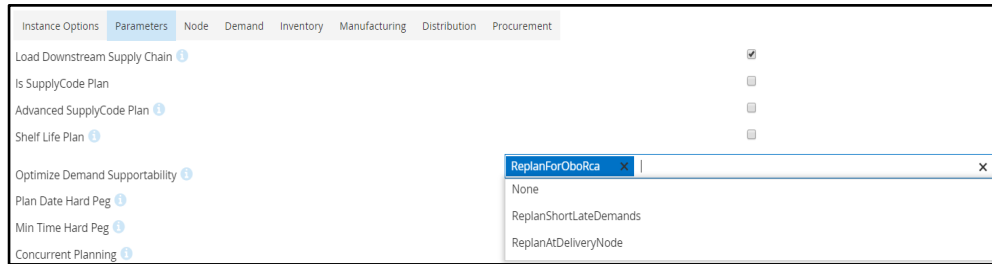
## Behavior Expected for Short & Late Demands

Let us assume we have demand D1 with BLL = 5 of 100 units demand

1. If D1 is short, then  
Cascading RCA would give results for JIT, JIT+1, JIT+2, JIT+3, JIT+4, JIT+5
2. If D1 is late planned at JIT+4, then  
Cascading RCA would give results for JIT, JIT+1, JIT+2, JIT+3
3. If D1 is 50 units short and 50 units late planned at JIT+4  
Cascading RCA would give results for 100 units at JIT, JIT+1, JIT+2, JIT+3, 50 units at JIT+4, JIT+5

## Configuration

1. Cascading RCA will only work when OBO RCA is enabled



Go to *Workspace* → *Designer* → *Edit model* → *Rule* → *Plugin* → *Global* → *o9.GraphCubePlugins* → *SupplyChainSolver* → *SupplyChainSolver* → *Parameter* → *Optimize Demand Supportability* → *ReplanForOBORCA*

It is a multi-select picklist, ReplanForOBORCA can also be selected along with ReplanShortLateDemands or ReplanAtDeliveryNode as well, or it can be enabled as SCS plugin query. For example:

```
exec plugin instance SupplyChainSolver for Measures {[Demand Quantity]} using
scope ([Version].[Version Name].[CurrentWorkingView] * Time.[Fiscal Week] *
Order.[Order Name]) arguments {[Material Constrained], "True"}, {[Capacity
Constrained] , "True"}, ([Optimize Demand Supportability], "ReplanForOboRca");
```

## 2. A measure 'Cascading RCA Limit' at a demand level

- Measure

- Cascading RCA Limit

It takes integer values. The value denotes the limit of Cascading RCA time horizon. If the value input here is x, then Cascading RCA will compute till JIT+x buckets or JIT+BLL buckets whichever is lesser. If value is 0, then regular OBO RCA will be computed and generated for only JIT bucket.

- Default value would be NULL or 0

- Granularity

- Demand Node

- Order.Order Name
    - Customer.Customer Name
    - Time. Fiscal Week

- Material Node

- Item.Item Name
    - Location.Location Name

- Version.Version Name

Same should be specified in the SCS config as follows:

Demand

Go to *Workspace* → *Designer* → *Edit model* → *Rule Plugin* → *Global* → *o9.GraphCubePlugins* → *SupplyChainSolver* → *SupplyChainSolver* → *Demand* → *Cascading RCA Limit Measure Name* → *select corresponding measure name from the list*



## 3. Creating new Timelike dimension

Edit Reports
Edit Model
Add new Dimension.

Add a Dimension

Name \*  
ReportedTime

Description \*  
Time bucket for which Cascading RCA is reported

☒ Time like

Aliases  
No content

- Go to Workspace → Designer → Edit model → Dimension → Create a new Dimension and select 'Time like'
- Now, create a Level Attribute inside this Dimension which is corresponding to the primary time dimension in the model
- Export your primary Time dimension LA data and import the same in this 2nd Time dimension LA.

Instance Options
Parameters
Node
Demand
Inventory
Manufacturing
Distribution
Procurement

Time Attribute Name ⓘ

Fiscal Week

DimensionName  
Attribute Name

Time  
Fiscal Week

Telescopic Attribute Name ⓘ

Cascade Time Attribute Name ⓘ

ReportingWeek

DimensionName  
Attribute Name

CascadingTime  
ReportingWeek

Your level attribute 'Fiscal Week' members should be identically same as 'ReportingWeek' members

- Same should be specified in the SCS config as follows:  
Parameters  
Go to *Workspace* → *Designer* → *Edit model* → *Rule* → *Global* → *o9.GraphCubePlugins* → *SupplyChainSolver* → *SupplyChainSolver* → *Parameters* → *Cascading Attribute Name* → *Select the respective DimensionName and AttributeName*

#### 4. Creating new Cascading RCA graphs

In total, Cascading rca is being exported to 7 graphs configured in the config.

- Material Production Cascading RCA Graph Name.
- Distribution Material Production Cascading RCA Graph Name.
- Supplier Material Production Cascading RCA Graph Name.
- Capacity Consumption Cascading RCA Graph Name.
- Distribution Capacity Consumption Cascading RCA Graph Name.
- Supplier Capacity Consumption Cascading RCA Graph Name.
- Storage Consumption Cascading RCA Graph Name.

Users need to create such configuration of graph tables for SCS to write the RCA results correspond to Material, Capacity and Storage:

- Material Cascading RCA Graphs
  - From Nodes: Activity, Problem Type, Item, Location, Time, Lot Attr
  - To Nodes: Demand, Item, Location, Time, ReportedTime
  - Edge Property: Constraint Qty
- Capacity Cascading RCA Graphs
  - From Nodes: Activity, Resource, Problem Type, Time, Lot Attr
  - To Nodes: Demand, Item, Location, Time, ReportedTime
  - Edge Property: Constraint Qty

|                                                   |                            |
|---------------------------------------------------|----------------------------|
| Storage Consumption OBO RCA Qty Property ⓘ        |                            |
| Storage Consumption Cascading RCA Graph Name ⓘ    |                            |
| Storage Consumption Cascading RCA Qty Property ⓘ  |                            |
| Material Production Cascading RCA Graph Name ⓘ    | Material Cascading RCA     |
| Material Production Cascading RCA Qty Property ⓘ  | Material Cascading RCA Qty |
| Capacity Consumption Cascading RCA Graph Name ⓘ   | Capacity Cascading RCA     |
| Capacity Consumption Cascading RCA Qty Property ⓘ | Capacity Cascading RCA Qty |

## Lot and Supply Code planning

| Parameter<br>Is SupplyCode<br>Plan | Parameter<br>Advanced<br>SupplyCode<br>Plan | Inventory<br>Attribute<br>Lot Attribute<br>Name | Inventory<br>Attribute<br>Lot Type<br>Attribute<br>Name | Comment                                                                                                                |
|------------------------------------|---------------------------------------------|-------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| False                              | False                                       | Empty                                           | Empty                                                   | Non Lot based<br>Planning-<br><b>Deprecated</b><br>This was the<br>old functionality<br>which is<br>deprecated         |
| True                               | False                                       | [Lot].[Lot ID]                                  | Empty                                                   | Previous Match<br>Code Planning-<br><b>Deprecated</b>                                                                  |
| False                              | False                                       | [Lot].[Lot ID]                                  | Empty                                                   | Lot Based<br>Planning for<br>those who do<br>not want to use<br>Supply Code<br>Planning.<br>Refer <a href="#">here</a> |
| True                               | True                                        | [Lot].[Lot ID]                                  | [Lot].[Lot Type]                                        | Supply Code<br>Planning-<br>Refer <a href="#">here</a>                                                                 |

## Lot Identifier

### Business Need

Incoming supply can be a part of various order types such as purchase order, distribution order, stock transfer order. These orders can be termed as Lot IDs or Document IDs. These identifiers will help identify flow of material at any material node. Users will also be able to differentiate between end demands of different Lot IDs.

### Functionality Description

Inventory maintained at any material node will now also have a Lot ID or Document ID as a mandatory attribute. For example, Beginning on Hand inventory can belong to SCS BOH lot. Similarly Expected Receipts can belong to SCS ER lot. The member data in the Lot ID are to be explicitly created by the planners.

| Item | Location | Lot ID  | Data              | W01-2015 | W10-2015 |
|------|----------|---------|-------------------|----------|----------|
| FG_1 | Factory  | SCS BOH | BOH               | 100      |          |
|      |          |         | Expected Receipts |          |          |
|      |          | SCS ER  | BOH               |          |          |
|      |          |         | Expected Receipts |          | 100      |

Let's consider the following supply chain



The lead time for the activities is 1 bucket.

Demand is specified on material node {FG\_1, VMI\_1} as follows:

| Order Name | Customer Name | Item | Location | Data            | W30-2015 |
|------------|---------------|------|----------|-----------------|----------|
| Order_1    | Cust_1        | FG_1 | VMI_1    | Demand Quantity | 200      |

When we run a material constrained plan on the above input data, one may see plans and peggings as follows:

### Material consumption plan

| Lot ID     | Item | Location | Activity                     | Data                          | W28-2015 | W29-2015 |
|------------|------|----------|------------------------------|-------------------------------|----------|----------|
| 000SCS_LOT | FG_1 | Factory  | Shipped from Factory to WH_1 | Material Consumption Quantity | 200      |          |
|            |      | WH_1     | Shipped form WH_1 to VMI_1   | Material Consumption Quantity |          | 200      |

### Material Production Plan

| Lot ID     | Item | Location | Activity                     | Data              | W29-2015 | W30-2015 |
|------------|------|----------|------------------------------|-------------------|----------|----------|
| 000SCS_LOT | FG_1 | WH_1     | Shipped from Factory to WH_1 | Material Quantity | 200      |          |
|            |      | VMI_1    | Shipped form WH_1 to VMI_1   | Material Quantity |          | 200      |

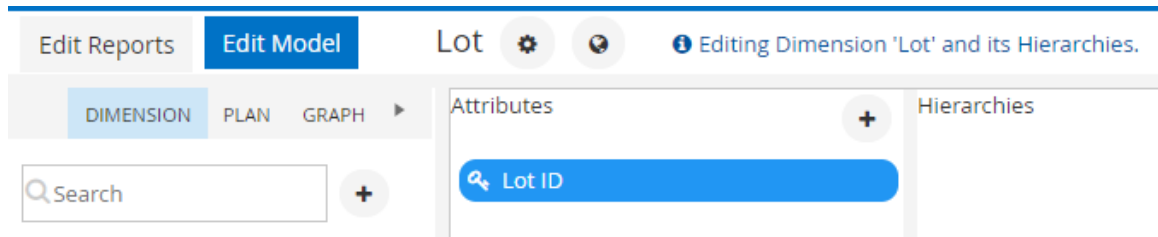
### Production Pegging

| to.<br>[Customer].<br>[Customer] | to.[Item].<br>[Item] | to.<br>[Location].<br>[Location] | to.[Order].<br>[Order Name] | Order<br>Timebucket | from.[Activity].[Activity]    | SKU  | from.<br>[Location].<br>[Location] | from.[Lot].[Lot<br>ID] | Supply<br>Timebucket | Material<br>Production<br>Pegged Quantity |
|----------------------------------|----------------------|----------------------------------|-----------------------------|---------------------|-------------------------------|------|------------------------------------|------------------------|----------------------|-------------------------------------------|
| Cust_1                           | FG_1                 | VMI_1                            | Order_1                     | W30-2015            | BOH Activity                  | FG_1 | Factory                            | SCS BOH                | W01-2015             | 100.00                                    |
|                                  |                      |                                  |                             |                     | Expected Reciept Activity(1)  | FG_1 | Factory                            | SCS ER                 | W10-2015             | 100.00                                    |
|                                  |                      |                                  |                             |                     | Shipped from WH_1 to VMI_1    | FG_1 | VMI_1                              | 000SCS_LOT             | W30-2015             | 200.00                                    |
|                                  |                      |                                  |                             |                     | Shipped from Factory to WH... | FG_1 | WH_1                               | 000SCS_LOT             | W29-2015             | 200.00                                    |

Thus we see that after demand fulfillment, flow plans and pegging tables are stamped with the corresponding Lot ID.

### Configuration

An additional dimension [Lot].[Lot ID] needs to be configured.



Maintaining hierarchies is optional. Hierarchies are advised only when a dimension has more than one level attributes.

The above configured [Lot].[Lot ID] should be configured in the following measure groups:

1. Inventory [ Input]
2. Material production graph planned WIP
3. Material Production Plan
4. Material consumption graph planned WIP
5. Material consumption plan
6. Capacity Graph Planned WIP <optional, only if supply chain has capacity modelled>
7. Capacity Consumption Plan <optional, only if supply chain has capacity modelled>

and in the following graphs:

1. Production\_Pegging (From Node)
2. Consumption\_Pegging (From Node)
3. Capacity\_Pegging (From Node) <optional, only if supply chain has capacity modelled>

When the above additions are made, the following changes should be made in the SCS config.

### Notes

- The SCS sorts the Lot ID in alphabetical order and selects the first Lot ID member that appears in the list (000SCS\_LOT in the above example. Flow plans and pegging tables are tagged with this Lot ID. This solver generated lot can be configured. User can have a member in LOTID dimension and use that in the SCS config as shown below.

SCS Lot Name SCS\_LOT

With this all the solver generated flow plans will be linked to SCS\_LOT lotid.

Define unique LOTID for the WIPs modeled along an activity.

This measure “Lot” can be specified in SCS config as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Demand → Acceptable Lot Sequence Measure Name → Select corresponding measure name from the drop down menu



Acceptable Lot Sequence Measure  
Name ⓘ

Lot

Planners have been provided with the flexibility to specify that an order can source from multiple lots {member data in dimension [Lot].[Lot ID]}. In such a case, they can specify those multiple lots names in measure “Lot” separated by a delimiter. This delimiter can be any special character which should be specified in SCS UI config as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →

o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Lot Attribute Name → Enter value in text {default value is tilde ~}

Lot Value Delimiter ⓘ

~

Maximum members supported in Lot Dimension is 4194303.

## Supply Code Planning

### Business Need

Certain businesses have requirements to match the demand with supply lots having certain attribute characteristics.

Incoming supply can be a part of purchase order, distribution order, stock transfer order and is represented as Lot IDs or Document IDs. These lots can now be classified as supply code and demand can have specific codes which has the requisite attributes.

The business use cases could be that certain demands have factory for supplier classification and can only take lots produced in those locations. Certain attributes can be imparted to the supply and demand need to specifically look at those attributes. Other use case can be PC assembly scenario where a higher priced PC can use only chips of a minimum speed. A lower priced PC would preferably use chips of lower speed; but if not available the business can opt to use next higher speed. All these use cases can be represented using the supply code planning.

### Functionality Description

Supply Code is a property of demand, which forms the basis of planning logic whereas Lot ID or Document IDs is the property of incoming supply.

A demand can optionally specify a sequence of SupplyCode values associated with it, and each inventory lot (on-hand, intransit, WIP) with a LotId also has an associated SupplyCode. The demand should be met by supplies of corresponding SupplyCode values, in the order of the demand SupplyCode sequence.

Let us see how to Set up Supply code along with few use cases

### Modelling Changes

[Lot Dimension Changes](#)

[Lot Dimension Data](#)

[Lot to Supply Code Mapping Graph](#)

[Lot To Supply Code Mapping Graph Data](#)

[Enable Supply Code Planning](#)

[Demand Measure for Supply Code Planning](#)

[Demand Measure for Supply Code Planning Data](#)

[Grain Changes in Measure groups and Graph](#)

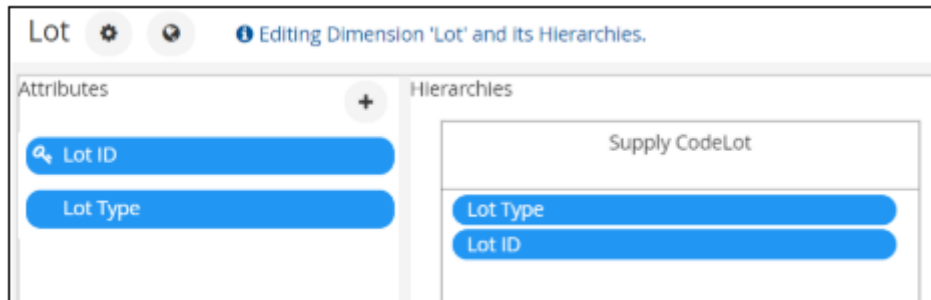
[Non Critical component Modeling](#)

### [Flow Plan and Pegging Stamping in Supply Code Simple Use Cases](#)

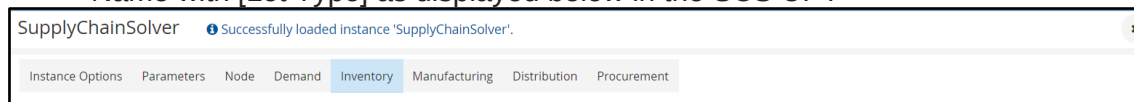
[Non BOM three Use cases](#)  
[BOM Use cases](#)  
[Limitations](#)  
[Link to Functional Use Cases in Jenkins](#)

## Lot Dimension Changes

1. Create a [Lot Type] attribute in Lot Dimension.
2. Create Hierarchy in Lot Dimension such that [Lot Type] is parent of [Lot ID]



3. Populate parameters Lot Attribute Name with [Lot ID] and Lot Type Attribute Name with [Lot Type] as displayed below in the SCS UI :



**Lot Attribute Name**

Lot ID

DimensionName

Lot

AttributeName

Lot ID

**Lot Type Attribute Name**

Lot Type

DimensionName

Lot

AttributeName

Lot Type

4. Create the special attribute member: [Lot Type].[SupplyCode]. Supply Codes are modeled as Lot ID attribute members with [SupplyCode] as the [Lot Type] parent.
5. SCS configurable parameter i.e. [SCS Lot Name] needs to map to a Supply Code member name (i.e. member is related to Lot Type [SupplyCode]).

A user can configure this value in the SCS Parameter UI by changing the SCS Lot Name value.

SupplyChainSolver

Instance Options Parameters Node Demand Inventory Manufacturing Distribution Procurement

SCS Lot Name

000SCS\_LOT

6. Create the special reserved Lot attribute members: [Lot ID].[INVALID] with Lot Type as [SupplyCode]. User cannot create [Lot ID].[ANY] which is a reserved word. The solver will ignore Lot ID or Supply Code ANY. This will be developed in future for supply code.

### Lot Dimension Data

Let us look at how the Lot Dimension Data will look like , how the solver interprets the same

| Lot Type          | Lot ID     | Comments                                                                                                                                                         |
|-------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SupplyCode</b> | SC_1       | SC_1 is a Supply Code since Lot Type uses Reserved word SupplyCode                                                                                               |
| <b>SupplyCode</b> | 000SCS_Lot | <a href="#">All the unmapped lots are mapped implicitly mapped to this supply code</a>                                                                           |
| <b>SupplyCode</b> | INVALID    | In case the demand is specified with invalid/incorrect/ineligible Supplycode, the RCA will be reported at that node i.e., the solver will not traverse upstream. |
| <b>SupplyCode</b> | ANY        | Currently not supported. Solver will ignore this                                                                                                                 |
| <b>NSC</b>        | Lot1       | Solver will treat Lot1 as Lot ID since the Lot Type does not match the reserved word SupplyCode                                                                  |
| <b>NMC</b>        | Lot2       | Solver will treat Lot2 as a Lot ID since the Lot Type does not match the reserved word SupplyCode                                                                |

|                                                                                     |                                                                                                                            |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
|  | We have a hard limit on the sum of distinct possible values of SupplyCode and LotId values in Lot Dimension : 4, 194, 302. |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|

### Lot to Supply Code Mapping Graph

We need to create a Graph as follows to map Supply Code with Lot ID.

Granularity: From: [Lot ID]; To: [Lot ID]

Edge : Create one Boolean Edge property. Example: LotSupplyCode Map

|                                                                                     |                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  | All From Nodes should be Lot members and All To Nodes should be Supply Code members. |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|

Map the Graph and Edge Property in SCS Plugin as follows:

## Lot To Supply Code Mapping Graph Data

Mapping between Lot ID and SupplyCode will be stored in this graph as follows:

| Lot ID [From] | Lot ID [To] | Lot SupplyCode Map | Comments                                                                                                                                                                                               |
|---------------|-------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lot1          | SC_1        | true               | Lot1 is mapped to Supply Code SC_1                                                                                                                                                                     |
| Lot2          | SC_1        | true               | Lot 2 is mapped to Supply code SC_1<br><br>One supply code can be mapped to multiple Lot ID i.e. SC_1 is mapped to both Lot1 and Lot2.                                                                 |
| Lot1          | SC_2        | true               | Ignored.<br>One Lot ID can map to only one supply code. In this case, Lot1 is already mapped to SC_1.<br><br>Any such subsequent maps for the same Lot ID to any other supply code are ignored by SCS. |
| Lot 3         | SC_3        | false              | Lot3 is not mapped to supply code SC_3, because the value is false                                                                                                                                     |
| Lot4          |             |                    | <b>Unmapped Lots are called Generic Lots</b><br><br><a href="#">Unmapped Lots are implicitly mapped to the configurable supply code [000SCS_Lot]</a>                                                   |



Input Supply needs to be specified using only mention Lot ID, otherwise the demand will be shorted.

## Enable Supply Code Planning

Enable these two flags in the SCS Parameter screen for Supply Code Planning. This will help Solver to find that supply code planning needs to be respected

SupplyChainSolver Successfully loaded Instance 'SupplyChainSolver'.

Instance Options Parameters Node Demand Inventory Manufacturing Distribution Procurement

Is SupplyCode Plan ☒

Advanced SupplyCode Plan i ☒



- The advanced supply code plan is a temporary flag to facilitate the feature migration.

## Demand Grain Measure for Supply Code

Create a string type measure in the demand grain which will be used to specify supply code in demand. Example: LotSupplyCode.

Edit Reports Edit Model Demand Quantity

LotSupplyCode ☒ ☐

Name \*  
LotSupplyCode

Description \*  
LotSupplyCode

ToolTip  
LotSupplyCode

Data Type  
String

Type  
Regular

Granularity Fiscal Week Order

Measures

- Demand BAL
- Demand BLL
- Demand Priority
- Demand Quantity
- Demand Target Date
- Demand Trace
- FulFilmentPolicy
- Incremental\_Allowed
- LotSupplyCode

Add the above measure in the SCS Plugin Schema for Acceptable LotSequence Measure Name.

Instance Options Parameters Node Demand Inventory Manufacturing Distribution Procurement

Acceptable Lot Sequence Measure Name i LotSupplyCode

Define LotValue Delimiter parameter if you want to use the multiple Supply code values in demand.

|                  |            |      |        |           |               |              |             |
|------------------|------------|------|--------|-----------|---------------|--------------|-------------|
| Instance Options | Parameters | Node | Demand | Inventory | Manufacturing | Distribution | Procurement |
|------------------|------------|------|--------|-----------|---------------|--------------|-------------|

|                                                                                                       |   |
|-------------------------------------------------------------------------------------------------------|---|
| Lot Value Delimiter  | ~ |
|-------------------------------------------------------------------------------------------------------|---|

|                                                                                   |                                                                                                                             |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
|  | <p>Only Supply Code's i.e. [Member of Lot Dimension with Lot Type as SupplyCode] can be used as input for this measure.</p> |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|

## Demand Measure for Supply Code Planning Data

Let us look at the this measures data and its interpretation during Demand Planning

| LotSupplyCode<br>[Supply Code Measure in Demand]                   | What Happens during Demand Planning                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SC_1                                                               | Demand will be fulfilled by only Lots mapped to SC_1.                                                                                                                                                                                                                                                                        |
| 000SCS_LOT<br>[configurable supply code]<br>Or<br>NULL<br>Or<br>"" | <p>Demand will be satisfied by Generic Lot's i.e. lots which are not explicitly mapped to any Supply Code</p> <p><b>When the demand is specified with the Supply Code as 000SCS_Lot and it does not find enough supply. It is not going to consume from other lots which are explicitly mapped to other supply codes</b></p> |
| SC_1 ~ SC_2                                                        | <p>Demand will first try to fulfill by lots mapped to SC_1.</p> <p>If not satisfied fully/partially will then try to satisfy using lots mapped to Supply code SC_2</p>                                                                                                                                                       |
| SC_1~<br><b>Or</b><br>SC_1~000SCS_LOT                              | <p>Demand will first try to fulfill by lots mapped to SC_1.</p> <p>If not satisfied fully/partially will then try to satisfy using generic lots i.e. lots not mapped to any Supply Code</p>                                                                                                                                  |
| SC_1~ ~SC_2                                                        | <p>Demand will first try to fulfill by lots mapped to SC_1.</p> <p>If not satisfied fully/partially will then try to satisfy using generic lots i.e. lots not mapped to any Supply Code</p> <p>If still not satisfied fully/partially will then try to satisfy using lots mapped to supply code SC_2</p>                     |
| SC_1~SC_2~SC_3                                                     | <p>Demand will first try to fulfill by lots mapped to SC_1.</p> <p>If not satisfied fully/partially will then try to satisfy using lots mapped to supply code SC_2</p> <p>If still not satisfied fully/partially will then try to satisfy using lots mapped to supply code SC_3</p>                                          |

## Grain Changes for Measure Group and Graphs

LotID should be included in the measure group

The above configured [Lot].[Lot ID] should be configured in the following measure groups:

- Inventory [ Input]
- Material production graph planned WIP
- Material Production Plan
- Material consumption graph planned WIP
- Material consumption plan
- Capacity Graph Planned WIP
- Capacity Consumption Plan
- Storage Plan
- Inventory Plan [Output]
- Inventory Inputs (like Constant Safety Stock Measure, Constant WOS Target, Max Stock Grain, Time Varying safety stock Measure's).

and in the following graphs:

- Production\_Pegging (From Node)
- Consumption\_Pegging (From Node)
- Capacity\_Pegging (From Node) <optional, only if supply chain has capacity modelled>
- Demand Fulfilment Report
- Capacity Pegging
- Storage Pegging

### Limitation:

There might be very few cases of sub-optimal planning as perceived from the business perspective. It is a trade-off between enforcing SupplyCode sequence priority vs consuming existing inventories while violating SupplyCode priority.

ANY option is not supported at the moment

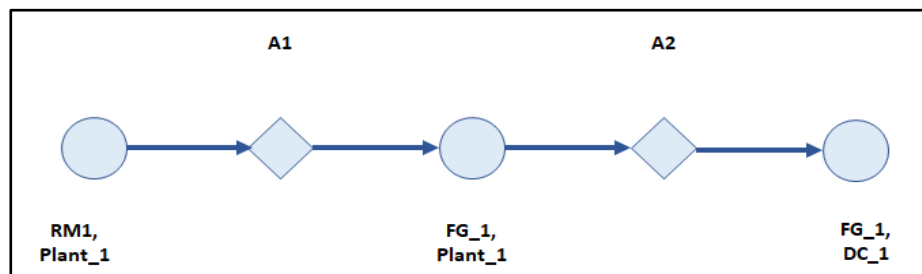
## Reports, Flow Plan and Pegging Stamping in Supply Code world

|                          | Comments                                                                                  |
|--------------------------|-------------------------------------------------------------------------------------------|
| Demand Fulfilment Report | Stamped with Supply Code                                                                  |
| Capacity Flow Plan       | <b>Capacity Flow plan</b> stamped with <b>configurable Supply Code</b> example 000SCS_LOT |
| Capacity Pegging         | <b>Capacity pegging</b> stamped with <b>configurable Supply Code</b> example              |

|                             |                                                                                                                                                                                                                                             |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | 000SCS_LOT                                                                                                                                                                                                                                  |
| Storage Flow Plan           | <b>Storage Flow plan</b> stamped with <b>configurable Supply Code</b> example 000SCS_LOT                                                                                                                                                    |
| Storage Pegging             | <b>Storage pegging</b> stamped with <b>configurable Supply Code</b> example 000SCS_LOT                                                                                                                                                      |
| Consumption Flow Plan       | Stamped with <b>Supply code</b><br>Except the Node where <b>WIP</b> is defined, that is stamped with <b>LotID</b>                                                                                                                           |
| Consumption Pegging         | Stamped with <b>Supply code</b><br>Except the Node where WIP is defined, that is stamped with <b>LotID</b>                                                                                                                                  |
| Constraint Report [OBO RCA] | Stamped with <b>Supply Code</b><br><br>Incase of Invalid or Ineligible supply code, stamped with <b>INVALID</b> .<br>At the Node where Demand is Incident i.e. Solver does not go upstream and errors out right there                       |
| Production Flow Plan        | Stamped with <b>Supply code</b><br>Except the Node where WIP is defined, that is stamped with <b>LotID</b>                                                                                                                                  |
| Production Pegging          | <b>Stamped with Lot ID:</b><br>Most Upstream activity i.e. BOH Activity and ER Activity<br>Node where WIP is defined<br><br><b>Stamped with Supply Code</b><br>All other activities<br>System generated expected receipts or magical supply |

### Non BOM Use Cases

For the use cases below, we will use the below supply chain



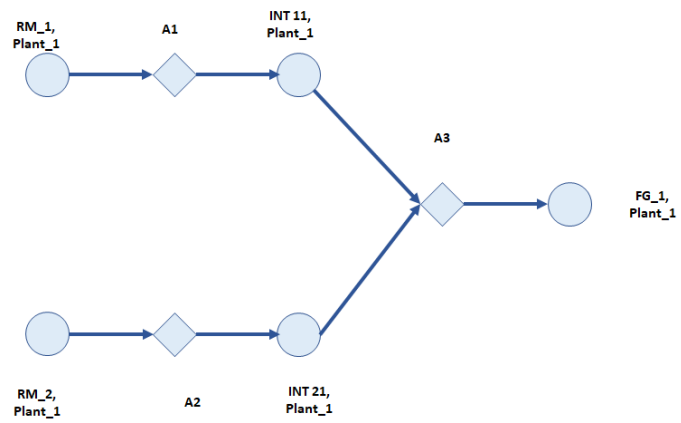
| Use Case | Input Condition | Demand Fulfilment |
|----------|-----------------|-------------------|
|----------|-----------------|-------------------|



|                                                                                                                          |                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case 1:</b><br><br><b>Strict Supply Code Match</b><br><br>i.e. demand can only be fulfilled using mapped Lot Id's | Lot 1 and Lot 2 Mapped to Supply code SC_1<br><br><b>Input Supply</b><br>Lot1 =50<br>Lot2=50<br><br><b>Demand</b><br>Supply Code SC_1<br>Qty=150                                                                                         | <b>Constrained Plan</b> : Fulfilment is Partial i.e. 100 Units using 50 from Lot1 and 50 from Lot2<br><br><b>UnConstrained Plan</b> : Demand is fully satisfied: Using 50 from Lot1, 50 from Lot2 and 50 using magical supply or system generated Expected Receipts |
| <b>Use Case 2:</b><br>Supply Code not mapped to any lots                                                                 | SC_2 supply code Not mapped to any lots<br><br><b>Input Supply</b><br>Lot1 =150<br><br><b>Demand</b><br>Supply Code SC_2<br>Qty=150                                                                                                      | <b>Constrained Plan</b> : Demand is short<br><br><b>UnConstrained Plan</b> : Fulfilment is fully satisfied using magical supply or system generated Expected Receipts                                                                                               |
| <b>Use Case 3:</b><br><br>Generic Lots                                                                                   | 000SCS_LOT is mapped to SCS Lot Name in Parameter<br><br>Lot3 not mapped to any supply code<br>Lot1 mapped to SC_1<br><br><b>Input Supply</b><br>Lot3 =100<br>Lot1 =10<br><br><b>Demand</b><br>Supply Code<br>000SCS_LOT~SC_1<br>Qty=150 | <b>Constrained Plan</b> : Demand is partially Fulfilled using Lot3 and Lot1<br><br><b>UnConstrained Plan</b> : Fulfilment is fully satisfied using 110 units of lot3 and Lot1 .Rest is with magical supply or system generated Expected Receipts                    |

### BOM Use Cases

For the use cases below, we will use the below supply chain



| Use Case | Input Condition | Demand Fulfilment |
|----------|-----------------|-------------------|
|----------|-----------------|-------------------|

|                                                                                                                                 |                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Use Case 1:</b></p> <p><b>Strict Supply Code Match</b></p> <p>i.e. demand can only be fulfilled using mapped Lot Id's</p> | <p>Lot 1 and Lot 2 Mapped to Supply code SC_1</p> <p><b>Input Supply</b><br/> Lot1 =50 on RM_1 and RM_2<br/> Lot2=50 on RM_1 and RM_2<br/> Lot3 =100 on RM_1 and RM_2</p> <p><b>Demand</b><br/> Supply Code SC_1<br/> Qty=150</p>                       | <p><b>Constrained Plan</b> : Fulfilment is Partial i.e. 100 Units using 50 from Lot1 and 50 from Lot2</p> <p><b>UnConstrained Plan</b> : Demand is fully satisfied: Using 50 from Lot1, 50 from Lot2 and 50 using magical supply or system generated Expected Receipts</p> |
| <p><b>Use Case 2:</b></p> <p>Strict Supply Code Match Changed</p>                                                               | <p>Lot 1 and Lot 2 Mapped to Supply code SC_1</p> <p><b>Input Supply</b><br/> Lot1 =100 on <b>RM_1</b><br/> Lot2= 150 on <b>RM_2</b></p> <p><b>Demand</b><br/> Supply Code SC_1<br/> Qty=150</p>                                                        | <p><b>Constrained Plan</b> : Fulfilment is Partial i.e. 100 Units using 100 from RM_1@Lot1 and 100 fromRM_2@ Lot2</p> <p><b>UnConstrained Plan</b> : Demand is fully satisfied:Using existing BOH and rest using magical supply or system generated Expected Receipts</p>  |
| <p><b>Use Case 3:</b></p> <p>Generic Lots</p> <p>000SCS_LOT is mapped to SCS Lot Name in Parameter</p>                          | <p>Lot3 not mapped to any supply code<br/> Lot1 mapped to SC_1</p> <p><b>Input Supply</b><br/> Lot3 =100 at RM_1 and 10 at RM_2<br/> Lot1 =100 at RM_2 and 7 at RM_1</p> <p><b>Demand</b><br/> Supply Code<br/> <b>000SCS_LOT~SC_1</b><br/> Qty=100</p> | <p><b>Constrained Plan</b> : Demand is partially Fulfilled using Lot3 &amp; Lot1 i.e. 17units</p> <p><b>UnConstrained Plan</b> : Demand is fully satisfied using 10 units of Lot3 and Rest is with magical supply or system generated Expected Receipts</p>                |

### Non Critical Component Modeling

Non Critical Material node is a practical requirement for using SupplyCode for any non-trivial model.

To enable this user need to **create a Measure with Boolean Data Type** in the **Material Node Measure group** i.e. Grain is [Item], [Location] and [VersionName].

Link this measure to **"Universal SupplyCode Measure Name"** which is present in Inventory Tab of SCS Config UI. Default measure name is: **"Universal SupplyCode"**

| Universal SupplyCode Values | What does it Mean                                                                                                                                                                                                                                                                                              |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                           | Default, the material node is critical for Supply Code. Such a node will respect Supply Code checks.                                                                                                                                                                                                           |
| 1                           | The material node is assumed to be non-critical for supply code purposes. Such a material node will ignore supply code checks for all planning/pegging decisions.<br><b>The assumption of all upstream nodes of a non-critical node also needing to be non-critical is a hard constraint for this to work.</b> |

## Non Critical Component Use Case

### Use Case 1:

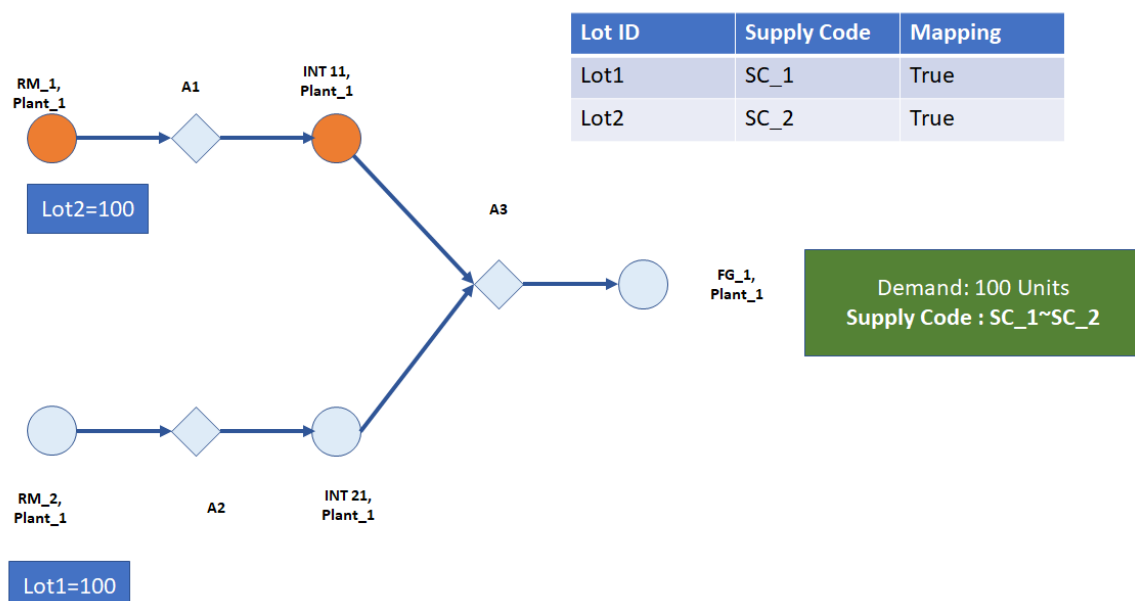
Consider the Supply Chain Below.

Demand has multiple Supply Code i.e. SC\_1 first preference followed by SC\_2.

Lot1 is mapped to SC\_1 and similarly Lot 2 is mapped to SC\_2.

RM\_1 has 100 units of Lot2

RM2 has 100 units of Lot1.



| Material Node RM_1 and INT_11 are Universal Supply Code enabled | What happens                                                                                                                                                                                        |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| False                                                           | Demand is short cannot be met                                                                                                                                                                       |
| True                                                            | Demand is met, because during the First pass i.e. SC_1<br>It see RM_2 has Lot1 which is mapped to SC_1<br>And RM_1 and INT_11 is universal supply code enabled hence ignores the supply code check. |

**Note :** If either RM\_1 or INT\_11 if any of them is not UniversalSupplyCode enabled then as well this will not work.

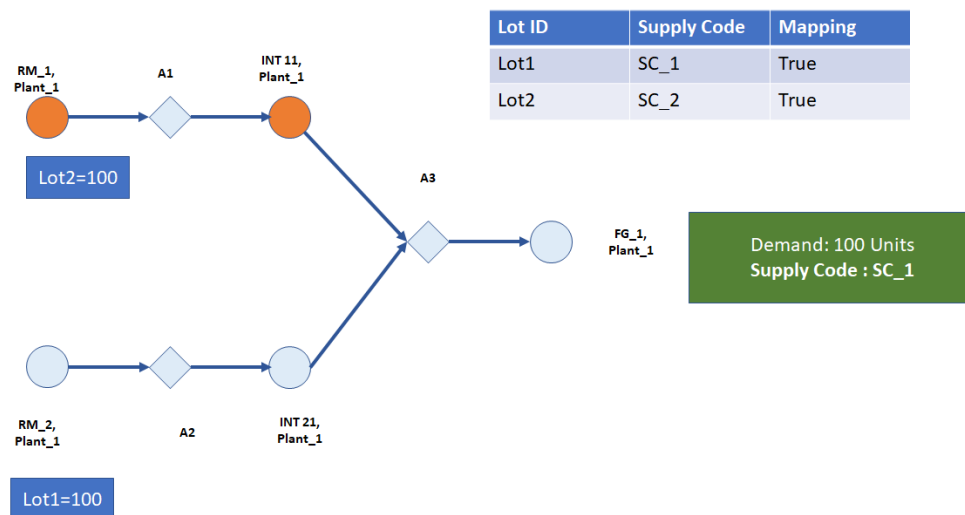
#### Use Case 2:

Consider the Supply Chain Below.

Demand has single Supply Code i.e. SC\_1

Lot1 is mapped to SC\_1 and similarly Lot 2 is mapped to SC\_2.

RM\_1 has 100 units of Lot2 and RM2 has 100 units of Lot1.



| Material Node RM_1 and INT_11 are Universal Supply Code enabled | What happens                  |
|-----------------------------------------------------------------|-------------------------------|
| False                                                           | Demand is short cannot be met |
| True                                                            | Demand is short.              |

|  |                                                                                                                                             |
|--|---------------------------------------------------------------------------------------------------------------------------------------------|
|  | Because Universal supply code enabled will only help to fulfil demand if it has the right supply-coded supply available to match the demand |
|--|---------------------------------------------------------------------------------------------------------------------------------------------|

## Lot Size

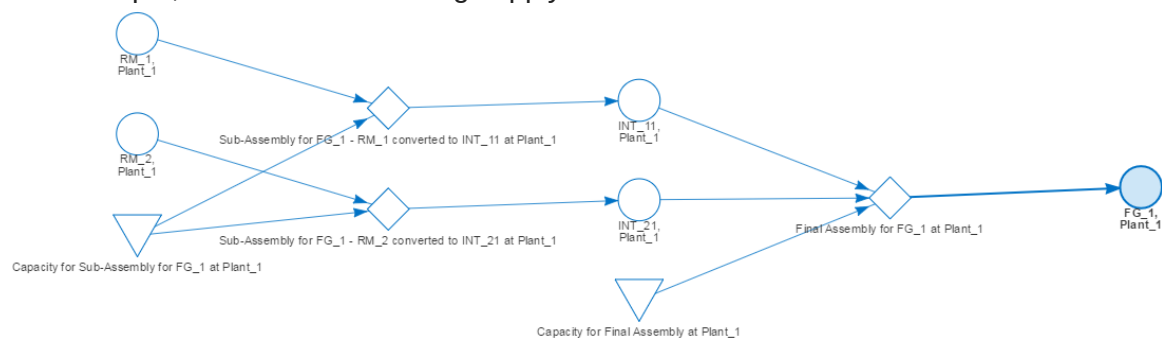
### Business Need

Certain supply chain processes have restrictions on the quantity to be processed, which can be in the form of minimum and multiple. These restrictions on procuring, producing and distributing specified quantities are termed as lot sizing. For example, biscuits can be packed in a carton of dozen.

### Functionality Description

SCS can model minimum and multiple quantity desired on an activity. An activity can either transform or transport material between two material nodes. In either case, the activity is consuming from a material node and producing into another material node. Lot Size restriction is specified on activity. It can be specified if an activity has to produce a minimum quantity on a material node irrespective of the order quantity. Similarly, whether an activity has to produce in multiples of any number can also be specified.

LotSize takes precedence over the yield in the production flow. [\[Functional Test #108\]](#)  
For example, consider the following supply chain



There is a demand of 124 units at FG\_1, Plant\_1

Activity {Final Assembly for FG\_1 at Plant\_1} has the following lot-size

- Minimum Qty = 100
- Multiple Qty = 10

Production and consumption yield for the same activity i.e {Final Assembly for FG\_1 at Plant\_1} is defined as follows:

- Consumption Qty of 4 units from INT\_11, Plant\_1
- Consumption Qty of 1 units from INT\_21, Plant\_1
- Production Qty per of 0.8 units at FG\_1, Plant\_1

This means for Activity {Final Assembly for FG\_1 at Plant\_1} to produce 1 units of FG\_1 at Plant\_1 it requires 5 units of INT\_11 and 1.25 units of INT\_21 at Plant\_1.

**When SCS plan is run on the above model:**

First and foremost, SCS tries to adhere to the Lotsize as follows:

User Specified Demand = 124

Minimum Qty = 100

Multiple Qty = 10

SCS has to produce such that it honors both lotsize and demand.

Therefore New demand =  $13 * \text{Multiple}(10) = 130$ .

The rest 6 units are pegged to Stock.

**Thus SCS converts the original demand of 124 to lot-sized demand of 130.**

After Lotsizing the demand, SCS incorporates yield in its calculations.

The production yield is defined as 0.8, therefore to successfully produce 130, SCS has to actually create a demand for 162.5 ( $130/0.8 = 162.5$ ) and propagate the same upstream.

**Thus the initial demand of 124 which was converted to lotsized demand of 130 and finally converted a demand of 162.5 thereby honoring production yield.**

In order to make 162.5 units, SCS now has to honor the Consumption Quantity Per activity {Final Assembly for FG\_1 at Plant\_1}. Thus this activity consumes

- 650 units from INT\_11, Plant\_1 { $162.5 * 4 = 650$ }
- 162.5 units from INT\_21, Plant\_1 { $162.5 * 1 = 162.5$ }

Thus,

the production flow looks as follows :

| Activity                           | Item | Location | Lot ID     | Data              | W30-2015 |
|------------------------------------|------|----------|------------|-------------------|----------|
| Final Assembly for FG_1 at Plant_1 | FG_1 | Plant_1  | 000SCS_LOT | Material Quantity | 130      |

The consumption flow looks as follows:

| Activity                                                    | Item | Location | Lot ID     | Data                          | W28-2015 |
|-------------------------------------------------------------|------|----------|------------|-------------------------------|----------|
| Sub-Assembly for FG_1 - RM_1 converted to INT_11 at Plant_1 | RM_1 | Plant_1  | 000SCS_LOT | Material Consumption Quantity | 650.0    |
| Sub-Assembly for FG_1 - RM_2 converted to INT_21 at Plant_1 | RM_2 | Plant_1  | 000SCS_LOT | Material Consumption Quantity | 162.5    |

However, Pegging tables show the actual quantities from the production/consumption flows pegged to demand.

**Production Pegging**

| Supply Timebucket | from.[Activity].[Activity]         | SKU  | from.[Location].[Location] | from.[Lot].[Lot ID] | to.[Order].[Order Name] | to.[Item].[Item] | to.[Location].[Location] | Order Timebucket | Material Production Pegged Quantity |
|-------------------|------------------------------------|------|----------------------------|---------------------|-------------------------|------------------|--------------------------|------------------|-------------------------------------|
| W30-2015          | Final Assembly for FG_1 at Plant_1 | FG_1 | Plant_1                    | 000SCS_LOT          | Order_1                 | FG_1             | Plant_1                  | W30-2015         | 124.00                              |
| W30-2015          | Final Assembly for FG_1 at Plant_1 | FG_1 | Plant_1                    | 000SCS_LOT          | Inventory Demand        | FG_1             | Plant_1                  | W52-2015         | 6.00                                |

**Consumption Pegging**

| from.[Time].[Fiscal Week] | from.[Activity].[Activity]         | from.[Item].[Item] | from.[Location].[Location] | from.[Lot].[Lot ID] | to.[Order].[Order Name] | to.[Item].[Item] | to.[Location].[Location] | to.[Time].[Fiscal Week] | Material Consumption Pegged Quantity |
|---------------------------|------------------------------------|--------------------|----------------------------|---------------------|-------------------------|------------------|--------------------------|-------------------------|--------------------------------------|
| W29-2015                  | Final Assembly for FG_1 at Plant_1 | INT_11             | Plant_1                    | 000SCS_LOT          | Inventory Demand        | FG_1             | Plant_1                  | W52-2015                | 30.00                                |
| W29-2015                  | Final Assembly for FG_1 at Plant_1 | INT_21             | Plant_1                    | 000SCS_LOT          | Inventory Demand        | FG_1             | Plant_1                  | W52-2015                | 7.50                                 |
| W29-2015                  | Final Assembly for FG_1 at Plant_1 | INT_11             | Plant_1                    | 000SCS_LOT          | Order_1                 | FG_1             | Plant_1                  | W30-2015                | 620.00                               |
| W29-2015                  | Final Assembly for FG_1 at Plant_1 | INT_21             | Plant_1                    | 000SCS_LOT          | Order_1                 | FG_1             | Plant_1                  | W30-2015                | 155.00                               |

To summarize, when Lotsize and Yields both are specified for an activity, the production quantity is always Lotsized. This quantity when propagated upstream accounts for yield in its calculation.

**Configuration Steps**

Lot Size is specified in Material Production Graph measure group.

It has the following measures

1. Material Production Min Qty: It takes numeric values. The number specifies the minimum quantity an activity should produce into a material node in each time bucket irrespective of the requested quantity. [\[Functional Test #07\]](#)
2. Material Production Multiple Qty: It takes numeric values. The number specifies that the activity will always produce in multiples of that number across each time bucket irrespective of the requested quantity. [\[Functional Test #08\]](#)

If both the above mentioned measures are specified [\[Functional Test #09\]](#) SCS flow quantity is always a multiple of multiple lot size and greater than min lot size

When Min Qty is less than Multiple Qty solver ignores the Min Qty and produces Multiple Qty.

Please check the table below for Solver Behaviour with varying and Min and Multiple lot condition

| Min Lot, Multiple Lot | Demand | Output Production/Flow Quantity |
|-----------------------|--------|---------------------------------|
| 10,0                  | 9      | 10                              |
| 10,0                  | 11     | 11                              |
| 0,10                  | 9      | 10                              |
| 0,10                  | 11     | 20                              |
| 24,7                  | 23     | 28                              |
| 24,7                  | 31     | 35                              |
| 6,20                  | 5      | 20                              |
| 6,20                  | 21     | 40                              |

Material Production Graph has the following Granularity

1. Activity Node
  - a. Activity.[Activity]
2. Material Node
  - a. Item.[Item]
  - b. Location.[Location]
3. Version.[Version Name]



Material Production Min Qty and Material Production Multiple Qty are default names for configuring Lotsize. However, planners can specify other names, which in turn need to be configured in SCS as follows

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →

o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Select corresponding measure name from drop-down menu.

|                                   |                                  |   |
|-----------------------------------|----------------------------------|---|
| Prod Graph Min Qty Measure Name   | Material Production Min Qty      | ▼ |
| Prod Graph Mul Qty Measure Name ⓘ | Material Production Multiple Qty | ▼ |

## Lot based planning- **Deprecated**

Note : Till Supply Code stabilizes please refer to this [document](#) if you want to enable the below functionality

### Business Need

Certain businesses have requirements to match demand with inventory having certain attribute characteristics. For example, in apparel, inventory lots with supplier characteristics needs to be pegged to demand which have the same lot characteristics. Some demands might specify preferred lots, which means they would search for supplies which satisfy the criteria of these lots. For example, the user might say that he would like a blue shirt and if that is not available a green shirt and nothing else. Some demand might just say give me shirt of any colour. Also, to account for the demand supply variability one might specify desired safety stock levels on the material nodes for these lots. This feature can be used as attribute based planning, where the attribute imparted on inventory at upstream is propagated through the supply chain network and is pegged to the demand.

**Note:** This is not yet supported for Document ID based planning.

### Functional Description

An extra dimension i.e. Lot should be specified as one of the components of the Inventory. Demands are incident on material node having dimensions Item and Location. An item can be distinguished based on specific attribute, eg. Style, Color or Size. The “Lot” component of demand signal corresponds to these attributes. This “Lot” component can be specified by the planners as a measure which in turn is grained by demand signal.

Consider the following supply chain:



Planners can have a demand as follows:

| Order Name | Customer Name | Item | Location | Data            | W30-2015        |
|------------|---------------|------|----------|-----------------|-----------------|
| Order_1    | Cust_1        | FG_1 | VMI_1    | Demand Quantity | 100             |
|            |               |      |          | Lot             | Lot 1           |
| Order_2    | Cust_1        | FG_1 | VMI_1    | Demand Quantity | 100             |
|            |               |      |          | Lot             | Lot 1~Lot 2     |
| Order_3    | Cust_1        | FG_1 | VMI_1    | Demand Quantity | 100             |
|            |               |      |          | Lot             | Lot 3~Lot 2     |
| Order_4    | Cust_1        | FG_1 | VMI_1    | Demand Quantity | 100             |
|            |               |      |          | Lot             | Lot 4~Lot 2~ANY |

Here Order\_1 has specifically asked for Lot 1 and nothing else. Order 2 request is for Lot 1 and if that is not available, is to provide Lot 2 but nothing else. However Order 4 request is for Lot 4 and if that is not available to provide supply having Lot 2 and even if that is not available, can provide “ANY” lot.

Note that “ANY” is a key word which means it can search for all the available lots in the system.

And inventory in the form of BOH is maintained at the most upstream node as follows:

| Item | Location | Lot ID | Data | W01-2015 |
|------|----------|--------|------|----------|
| FG_1 | Factory  | Lot 1  | BOH  | 10       |
|      |          | Lot 2  | BOH  | 10       |
|      |          | Lot 3  | BOH  | 10       |
|      |          | Lot 4  | BOH  | 10       |
|      |          | Lot 5  | BOH  | 10       |

The above data can interpret as follows:

For a demand that is incident on material node {FG\_1, VMI\_1}, the corresponding most upstream material node {FG\_1, Factory}.

BOH for 5 different lots is maintained at the most upstream node.

SCS plans the orders based on the available lots.

During Constrained planning:

- For Order\_1: SCS looks for available inventory for the specific Lot 1 only
- For Order\_2: SCS looks for available inventory for Lot 1 and tries to fulfil demands using Lot 1. If available inventory of Lot 1 is unable to fulfill the demand, then SCS looks for available inventory in Lot 2 and tries to satisfy the remaining quantity of Order\_2 using Lot 2.
- For Order\_3: SCS looks for available inventory for Lot 3 and tries to fulfil demands using Lot 3. If available inventory of Lot 3 is unable to fulfill the demand, then SCS looks for available inventory in Lot 2 and tries to satisfy the remaining quantity of Order\_3 using Lot 2.

- For Order\_4: SCS looks for available inventory for Lot 4 and tries to fulfil demands using Lot 4. If available inventory of Lot 3 is unable to fulfill the demand, then SCS looks for available inventory in Lot 2 and tries to satisfy the remaining quantity of Order\_4 using Lot 2. If Lot 2 is unable to fulfill the demand then SCS looks for ANY other lot {Lot 1, Lot 3 and Lot 5} whose corresponding inventory can satisfy the remaining part of the demand.

When no lot is specified as a measure for a demand, i.e. mean measure “Lot” for a demand is set to NULL, SCS tries to fulfill that demand using all the available

When “Lot” for a demand is set to ANY and its BOM components {say A and B for example} are available such that component A is only available in the form of Lot 1 and component B is only available in the form of Lot 2, SCS shorts the demand. This is because SCS searches for same lots for the components A and B to meet the demand.[\[FT#2\]](#)

#### Constraint Report Creation:

Based on the input data given by the planners, SCS becomes aware of the preferred lots for each of the above orders which are as follows:

| Demand  | Preferred Lot |
|---------|---------------|
| Order_1 | Lot 1         |
| Order_2 | Lot 1         |
| Order_3 | Lot 3         |
| Order_4 | Lot 4         |

Thus if orders are partially or fully short, then Constraint Report for the corresponding short quantity is created on the preferred lot of that Order. For example, based on the above data, BOH pegging and constraint report for Order\_4 is as follows. One can see that demand is 80 units short and this constraint is shown on Lot 4[\[FT#7\]](#)

| from.[Item],<br>[Item] | from.[Location],<br>[Location] | from.[Activity],[Activity]   | from.[Time],<br>[Fiscal Week] | to.[Item],<br>[Item] | to.[Location],<br>[Location] | to.[Time],<br>[Fiscal Week] | to.[Order],<br>[Order Name] | from.[Lot],<br>[Lot ID] | Data                                | Value         |
|------------------------|--------------------------------|------------------------------|-------------------------------|----------------------|------------------------------|-----------------------------|-----------------------------|-------------------------|-------------------------------------|---------------|
| FG_1                   | Factory                        | BOH Activity                 | W01-2015                      | FG_1                 | VMI_1                        | W30-2015                    | Order_4                     | Lot 4                   | Material Production Pegged Quantity | 10.00         |
|                        |                                |                              |                               |                      |                              |                             |                             |                         | MaterialConstraintQty               |               |
|                        |                                |                              |                               |                      |                              |                             |                             |                         | MaterialConstraintType              |               |
|                        |                                |                              |                               |                      |                              |                             |                             | Lot 5                   | Material Production Pegged Quantity | 10.00         |
|                        |                                |                              |                               |                      |                              |                             |                             |                         | MaterialConstraintQty               |               |
|                        |                                |                              |                               |                      |                              |                             |                             |                         | MaterialConstraintType              |               |
|                        |                                | Expected Receipt Activity... | W28-2015                      | FG_1                 | VMI_1                        | W30-2015                    | Order_4                     | Lot 4                   | Material Production Pegged Quantity |               |
|                        |                                |                              |                               |                      |                              |                             |                             |                         | MaterialConstraintQty               | 80.00         |
|                        |                                |                              |                               |                      |                              |                             |                             |                         | MaterialConstraintType              | MaterialShort |

When no lot is specified as a measure for a demand, i.e. the measure “Lot” for a demand is set to NULL[\[FT#3\]](#), SCS tries to fulfill that demand using all the available lots, Even then if the demand is partially or fully short, then corresponding constraint report for this demand is created on the first lot attribute that happens to be in the list maintained by SCS {Lot 1 in this case}. This list is based on the ASCII names. The lot attributes are sorted by name[\[FT#3\]](#).

#### Lot ID in Infinite Material Node:

When specifying a material node as infinite material node, planners can put a dummy Lot ID in the scope as follows:

| Item | Location | Lot ID | Infinite Inventory                  |
|------|----------|--------|-------------------------------------|
| FG_1 | Factory  | Lot 1  | <input checked="" type="checkbox"/> |

SCS assumes that the material node {FG\_1, Factory} can supply an infinite amount of material for all possible lots that planners have configured in the system i.e Lot 1 through Lot 5 in this case. Note that Lot 1 is just a placeholder and does not have any significance. Thus if a material constrained plan is run on the same supply chain model for the following demands

| Order Name | Customer Name | Item | Location | Fiscal Week | Demand Quantity | Lot   |
|------------|---------------|------|----------|-------------|-----------------|-------|
| Order_1    | Cust_1        | FG_1 | VMI_1    | W03-2015    | 100             | Lot 1 |
| Order_2    | Cust_1        | FG_1 | VMI_1    | W03-2015    | 100             | Lot 2 |

One may find that both the above demands get fulfilled and pegging report is seen as follows:

| from.[Item] | from.[Location] | from.[Activity].[Activity]   | from.[Time].[Fiscal Week] | to.[Item] | to.[Location] | to.[Time].[Fiscal Week] | to.[Order] | to.[Customer] | from.[Lot] | Data                                | Value  |
|-------------|-----------------|------------------------------|---------------------------|-----------|---------------|-------------------------|------------|---------------|------------|-------------------------------------|--------|
| FG_1        | Factory         | Expected Receipt Activity(1) | W01-2015                  | FG_1      | VMI_1         | W03-2015                | Order_1    | Cust_1        | Lot 1      | Material Production Pegged Quantity | 100.00 |
|             |                 |                              |                           |           |               |                         | Order_2    | Cust_1        | Lot 2      | Material Production Pegged Quantity | 100.00 |

Thus specifying Lot ID in Infinite Material Node is a misnomer [\[FT#11\]](#)

### Configuration Steps

A dimension named Lot is to be configured. It has Lot ID as the level attribute.

Lot

Editing Attribute 'Lot ID'.

Lot ID

Name \*

Lot ID

Description \*

for preferred lots or Generic lot

Order By

Name

☒ Key

Key Column Data Type

Integer

Key Generation

Auto

Attributes

Lot ID

Hierarchies

The level attribute [Lot].[Lot ID] has to be added as one of the members in the **granularity** of the following measure groups:

1. Inventory
  - a. Granularity
    - i. Material Node
      - Item.[Item Name]
      - Location.[Location Name]
    - ii. Lot.[Lot Id]
    - iii. Time.[Fiscal Week]
    - iv. Version.[Version Name]
2. Material Production Plan
  - a. Granularity
    - i. Material Node
      - Item.[Item Name]
      - Location.[Location Name]
    - ii. Activity Node
      - Activity.[Activity]
    - iii. Lot.[Lot Id]
    - iv. Time.[Fiscal Week]
    - v. Version.[Version Name]
3. Material Consumption Plan
  - a. Granularity
    - i. Material Node
      - Item.[Item Name]
      - Location.[Location Name]
    - ii. Activity Node
      - Activity.[Activity]
    - iii. Lot.[Lot Id]
    - iv. Time.[Fiscal Week]
    - v. Version.[Version Name]
4. Capacity Consumption Plan
  - a. Granularity
    - i. Resource Node
      - Resource.[Resource]
    - ii. Activity Node
      - Activity.[Activity]
    - iii. Lot.[Lot Id]
    - iv. Time.[Fiscal Week]
    - v. Version.[Version Name]

[Lot].[Lot ID] has to be added as to the **from\_members** component of the following relationship types:

1. Demand\_Fulfilment\_Report
  - a. From Node
    - i. Time.[Fiscal Week]
    - ii. Lot.[Lot Id]
  - b. To Node
    - i. Material Node
      - Item.[Item Name]
      - Location.[Location Name]
    - ii. Demand Node
      - Order.[Order Name]
      - Time.[Fiscal Week]
      - Customer.[Customer Name]
2. Capacity\_Pegging
  - a. From Node

- i. Resource Node
      - Resource.[Resource]
    - ii. Activity Node
      - Activity.[Activity Name]
    - iii. Time.[Fiscal Week]
    - iv. Lot.[Lot Id]
  - b. To Node
    - i. Material Node
      - Item.[Item Name]
      - Location.[Location Name]
    - ii. Demand Node
      - Order.[Order Name]
      - Time.[Fiscal Week]
      - Customer.[Customer Name]
- 3. Consumption\_Pegging
  - a. From Node
    - i. Material Node
      - Item.[Item]
      - Location.[Location]
    - ii. Activity Node
      - Activity.[Activity Name]
    - iii. Time.[Fiscal Week]
    - iv. Lot.[Lot Id]
  - b. To Node
    - i. Material Node
      - Item.[Item Name]
      - Location.[Location Name]
    - ii. Demand Node
      - Order.[Order Name]
      - Time.[Fiscal Week]
      - Customer.[Customer Name]
- 4. Production\_Pegging
  - a. From Node
    - i. Material Node
      - Item.[Item]
      - Location.[Location]
    - ii. Activity Node
      - Activity.[Activity Name]
    - iii. Time.[Fiscal Week]
    - iv. Lot.[Lot Id]
  - b. To Node
    - i. Material Node
      - Item.[Item Name]
      - Location.[Location Name]
    - ii. Demand Node
      - Order.[Order Name]
      - Time.[Fiscal Week]
      - Customer.[Customer Name]

Note that Lot is not added to the **to\_members** of the above graphs

[Lot].[Lot ID] can be specified in SCS config as follows:  
 Goto Plans → Global Plugins → SupplyChainSolver → Edit → Inventory → Lot Attribute  
 Name → Enter value in text

Lot Attribute Name ⓘ

[Lot].[Lot ID]

Following measure group is required.

### Measure Groups

1. Demand Quantity
  - a. Measure
    - i. Lot: The values that planners specify “Lot” measure group should be in sync with the member list provided in dimension [Lot].[Lot ID]. By any chance if they don't match, running a plan for such a scenario would give the following error:  
*at o9.GraphCube.Plugins.SupplyChainSolver.plan.Plan.GetLotId(String name)*
  - b. Granularity
    - i. Material Node
      1. Item.[Item]
      2. Location.[Location]
    - ii. Demand Node
      1. Order.[Order Name]
      2. Time.[Fiscal Week]
      3. Customer.[Customer Name]
    - iii. Version Name

## Activity Markers with Lot based Planning

### Business Need

The business rule can be to use all the inventory in the system upto a certain level/stage before inducing any fresh start from upstream to meet the demand. The business rule can also be to match demand with inventory having certain attribute characteristics. Depending on the business need one can use

- (i) all lots up to a stage or
  - (ii) for a given lot use inventory and then induce fresh start and then go to the next lot.
- There is a need to provide flexibility based on the business objective.

### Functional Description

SCS is capable of demand planning based on Lots and capable of demand planning based on activity marker exclusion list. However, for a tenant in which planners have modelled both activity marker and lots, the 2 corresponding flavours of demand planning can be merged. In such a case, planners have the flexibility to decide whether or not demand planning based on Lots takes precedence over demand planning based on activity marker exclusion list and vice versa.

Consider the following supply chain



Material Node {FG\_1, VMI\_1} can get its supply from material node {FG\_1, WH\_1} via activity {Shipped form WH\_1 to VMI\_1}.

Material node {FG\_1, WH\_1} can gets its supply from material node {FG\_1, Factory} via activity {Shipped from Factory to WH\_1}

Let the above activity be marked as Fresh as follows:

| Activity                     | Activity Marker |
|------------------------------|-----------------|
| Shipped from Factory to WH_1 | Fresh           |

Let the demand of 100 units of FG\_1 be placed at VMI\_1. This order can be fulfilled by Lot 1 as well as Lot 2, with Lot 1 being the preferred lot for demand fulfillment.

| Order Name | Customer Name | Item | Location | Data            | W03-2015    |
|------------|---------------|------|----------|-----------------|-------------|
| Order_1    | Cust_1        | FG_1 | VMI_1    | Demand Quantity | 100         |
|            |               |      |          | Lot             | Lot 1~Lot 2 |

Infinite Inventory is set to "True" for material node {FG\_1, Factory} for all lots

| Item | Location | Lot ID | Infinite Inventory                  |
|------|----------|--------|-------------------------------------|
| FG_1 | Factory  | Lot 1  | <input checked="" type="checkbox"/> |
|      |          | Lot 2  | <input checked="" type="checkbox"/> |
|      |          | Lot 3  | <input checked="" type="checkbox"/> |
|      |          | Lot 4  | <input checked="" type="checkbox"/> |
|      |          | Lot 5  | <input checked="" type="checkbox"/> |

Inventory units are maintained as follows:

| Item | Location | Lot ID | Data              | W01-2015 |
|------|----------|--------|-------------------|----------|
| FG_1 | WH_1     | Lot 2  | BOH               | 10       |
|      |          |        | Expected Receipts |          |

### **Demand planning when objective rank list is set as [Lot, ActivityMarker] [FT#6]**

Objective Rank List 

Lot, ActivityMarker

#### **1st pass**

1. SCS traverses upstream via activity {Shipped from WH\_1 to VMI\_1} to material node {FG\_1, WH\_1}
2. SCS finds that Lot 1 for FG\_1 is not available at this material node.
3. SCS then traverses further upstream via activity {Shipped from Factory to WH\_1} to material node {FG\_1, Factory}
4. SCS finds that activity {Shipped from Factory to WH\_1} is marked as "Fresh". Thus this activity is excluded in the 1st pass of demand planning.

#### **2nd pass**

1. SCS relaxes the constraint of activity marker on activity {Shipped from Factory to WH\_1}
2. It traverses upstream via activity {Shipped from WH\_1 to VMI\_1} to material node {FG\_1, WH\_1}
3. SCS finds that Lot 1 for FG\_1 is not available at this material node
4. SCS then traverses upstream via activity {Shipped from Factory to WH\_1} to material node {FG\_1, Factory}
5. SCS finds that material node {FG\_1, Factory} can supply infinite inventory of FG\_1 of Lot 1.



6. SCS pegs 100 units of this Lot 1 to demand
7. The remaining 10 units of FG\_1 of Lot 2 at WH\_1 is pegged to Stock.

| from.[Item].<br>[Item] | from.[Location].<br>[Location] | from.[Activity].[Activity]   | from.[Time].<br>[Fiscal Week] | to.[Item].<br>[Item] | to.[Location].<br>[Location] | to.[Time].<br>[Fiscal Week] | to.[Order].[Order<br>Name] | from.[Lot].<br>[Lot ID] | Material Production<br>Pegged Quantity |
|------------------------|--------------------------------|------------------------------|-------------------------------|----------------------|------------------------------|-----------------------------|----------------------------|-------------------------|----------------------------------------|
| FG_1                   | Factory                        | Expected Receipt Activity... | W01-2015                      | FG_1                 | VMI_1                        | W03-2015                    | Order_1                    | Lot 1                   | 100.00                                 |
|                        | WH_1                           | BOH Activity                 | W01-2015                      | FG_1                 | WH_1                         | W01-2015                    | Inventory Demand           | Lot 2                   | 10.00                                  |

### **Demand planning when objective rank list is set as [ActivityMarker, Lot] [FT#5]**

Objective Rank List 

ActivityMarker, Lot

#### **1st pass**

1. SCS traverses upstream to via activity {Shipped from WH\_1 to VMI\_1} to material node {FG\_1, WH\_1}
2. SCS finds that Lot 1 of FG\_1 is not available at WH\_1
3. SCS traverses upstream to via activity {Shipped from Factory to WH\_1} to material node {FG\_1, Factory}
4. SCS finds activity {Shipped from Factory to WH\_1} is marked as "Fresh" and hence is not able to access the inventory of material node {FG\_1, Factory}.
5. SCS recurses back
6. SCS traverses upstream now using 10 units of Lot 2 from material node {FG\_1, WH\_1}..
7. SCS pegs 10 units of this Lot 2 to demand.

#### **2nd pass**

1. SCS relaxes the constraint of activity marker on activity {Shipped from Factory to WH\_1}
2. It traverses upstream via activity {Shipped from WH\_1 to VMI\_1} to material node {FG\_1, WH\_1}
3. SCS finds that Lot 1 of FG\_1 is not available at WH\_1
4. SCS traverses upstream via activity {Shipped from Factory to WH\_1} to material node {FG\_1, Factory}
5. SCS finds that material node {FG\_1, Factory} can supply infinite inventory of Lot 1.
6. SCS pegs 90 units of Lot 1 to demand.

| from.[Item].<br>[Item] | from.[Location].<br>[Location] | from.[Activity].[Activity]   | from.[Time].<br>[Fiscal Week] | to.[Item].<br>[Item] | to.[Location].<br>[Location] | to.[Time].<br>[Fiscal Week] | to.[Order].<br>[Order Name] | from.[Lot].<br>[Lot ID] | Material Production<br>Pegged Quantity |
|------------------------|--------------------------------|------------------------------|-------------------------------|----------------------|------------------------------|-----------------------------|-----------------------------|-------------------------|----------------------------------------|
| FG_1                   | Factory                        | Expected Receipt Activity... | W01-2015                      | FG_1                 | VMI_1                        | W03-2015                    | Order_1                     | Lot 1                   | 90.00                                  |
|                        | WH_1                           | BOH Activity                 | W01-2015                      | FG_1                 | VMI_1                        | W03-2015                    | Order_1                     | Lot 2                   | 10.00                                  |

### **Configuration**

There are 2 measure groups that need to be maintained

1. Material Node parameters
  - a. Measure
    - i. Infinite Inventory
  - b. Granularity
    - i. Material Node
      - Item.Item Name
      - Location.Location Name
    - ii. Version.Version Name
2. Activity Node Parameters.
  - a. Measure Name
    - i. Activity Marker
  - b. Granularity
    - i. Activity Node
      - [Activity].[Activity Name]

- [Version].[version Name]

Same should be specified in SCS config as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters →  
Activity Marker Exclusion List → enter comma separated values in textbox

Activity Marker Exclusion List ⓘ

**Activity marker exclusion list:** It is a comma separated list of activity markers, which tells the solver to exclude these mentioned activities in the order of priority given. It provides control on material search. It can be specified using wild card characters.

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters →  
Objective Rank List → enter comma separated values in textbox {either (ActivityMarker, Lot) or (Lot, ActivityMarker)}

Objective Rank List ⓘ

OR

Objective Rank List ⓘ

ActivityMarker and Lot are case-insensitive keywords.

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Inventory → Has  
Infinite Inventory Measure Name → Select corresponding measure from the drop down  
menu

Has Infinite Inventory Measure Name ⓘ

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing →  
Activity Marker Measure Name → Select corresponding measure from the drop down  
menu

Activity Marker Measure Name ⓘ

Another parameter **Late** has been added to control delay in demand due to Activity Marker constraint. Let's see how the solver treats Late

| Objective Rank List     | Output                                                                                               |
|-------------------------|------------------------------------------------------------------------------------------------------|
| ActivityMarker,Lot,Late | Constraint on Lateness is relaxed first<br>Followed by Lot before finally relaxing<br>ActivityMarker |
| ActivityMarker,Late,Lot | Constraint on Lot is relaxed first<br>Followed by Lateness before finally<br>relaxing ActivityMarker |
| Late,ActivityMarker,Lot | Constraint on Lot is relaxed first<br>Followed by ActivityMarker before finally<br>relaxing Lateness |
| Late,Lot,ActivityMarker | Constraint on ActivityMarker is relaxed                                                              |

|                         |                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------|
|                         | first<br>Followed by Lot before finally relaxing Lateness                                         |
| Lot,ActivityMarker,Late | Constraint on Lateness is relaxed first<br>Followed by ActivityMarker before finally relaxing Lot |
| Lot,Late,ActivityMarker | Constraint on ActivityMarker is relaxed first<br>Followed by Lateness before finally relaxing Lot |

### Limitation

Infinite inventory material node provides the lot that is asked for by the demand and the same would be shown in the pegging. Currently it cannot be differentiated whether or not it is a fresh start (shows as expected receipt activity in pegging). This needs to be inferred by the planner since it is from an infinite material node.

## Pegging Optimization when there are attribute based demands

### Business Need

In a master planning workflow there can be mix set of demands where some of them are associated with specific attributes like demand based alternate selection, plan date, min time etc and some of demands do not need any attributes. In such case, it was required to do inventory re-allocation between the demands which are not associated with any attributes. There should not be any inventory reallocation for demands that have attributes specified. This helps meet the planner specification and also increase supportability.

### Functionality Description

SCS used to do plan and peg (i.e. hard pegging) even if there is one demand with attribute. This usually resulted in less supportability. SCS has been enhanced to support both hard pegging and FIFO in the same plugin invocation.

By default, solver will do the pegging immediately after the demand planning for demands with attributes. The FIFO pegging will be performed at last for the remaining demands (the ones that don't have attributes) in the model.

### Configuration

One can change the default behaviour through following plugin parameters for two cases – plan date planning and min time planning.

| Instance Options | Parameters                                                                                           | Node | Demand | Inventory | Manufacturing | Distribution | Procurement                                                                         |
|------------------|------------------------------------------------------------------------------------------------------|------|--------|-----------|---------------|--------------|-------------------------------------------------------------------------------------|
|                  | Plan Date Hard Peg  |      |        |           |               |              |  |
|                  | Min Time Hard Peg   |      |        |           |               |              |  |

Plan Date Hard Peg – Set it to False (uncheck) if you do not want hard pegging for demands marked with plan date offset

Min Time Hard Peg – Set this to False (uncheck) if you do not want hard pegging for demands marked for min time planning

## Target Date/Backlog to Target

### Business Need

Planners need to check if an order can be promised prior to its due date.

### Functionality Description

SCS solves all such demands based on their respective due date. This is JIT planning. However, planners have now the feasibility to provide an alternate date, over the already provided due date. The due date and alternate date are usually called as commit date and target date respectively. The target date must be present between plan current date and due date (or commit date) . In such a case, the SCS first solves all the demands based on the due date. After this JIT planning, the orders for which an alternate date is provided is unplanned and planned again but this time based on this alternate date. This is done one demand at a time without violating the earlier promised dates.

[\[Test Case #14\]](#)

[\[Test Case #15\]](#)

### Configuration

Following measure needs to be maintained

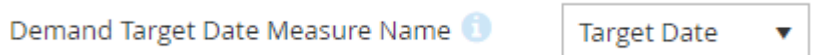
1. Measure
  - a. Demand Target Measure Name:
 

This measure takes datetime values. The datetime value is to specify the demand target date. Demand target is usually set such as it lies before the demand due date
2. Granularity
  - a. Demand Node
    - i. [Order].[Order Name]
    - ii. [Time].[Fiscal Week]
    - iii. [Customer].[Customer Name]
  - b. Material Node
    - i. [Item].[Item]

- ii. [Location].[Location]
- c. [Version].[Version Name]

Same should be specified in the SCS config as follows:

Goto Workspace→Designer→Edit  
model→Rule→Plugin→Global→o9.GraphCubePlugins→SupplyChainSolver→SupplyChainSolver→Demand → Demand Target Date Measure Name → Select corresponding measure from the drop down menu



Following measure needs to be configured in Delivery Plan measure group (solver output). This configuration is optional.

1. Measure
  - a. Commit Date Measure Name:
 

This measure takes datetime values. It is used for exporting target date when the demand is planned. In case of multiple delivery plans for an order (while doing backlog to target), this measure shows the latest date (last part of the delivery) on which supply is arriving for the demand.
2. Granularity
  - b. Demand Node
    - i. [Order].[Order Name]
    - ii. [Time].[Fiscal Week]
    - iii. [Customer].[Customer Name]
  - c. Material Node
    - iv. [Item].[Item]
    - v. [Location].[Location]
  - d. [Version].[Version Name]

For example,

Demand D1, Due Date=W20, Target Date=W10, Qty=100

After planning, we have delivery plans like this

W10=50, W15=30, W18=20

In this case the delivery plan commit date for the order will show W18.

The measure can be configured in SCS config as following

Goto Workspace→Designer→Edit  
model→Rule→Plugin→Global→o9.GraphCubePlugins→SupplyChainSolver→SupplyChainSolver→Demand → Commit Date Measure Name → Select corresponding measure from the drop down menu

## Limitations

1. This functionality is not supported with hard pegged demands (demand attribute based planning)

## Intermediate Peggings

### Business Need

The business requirement is regarding the pegging producing flow plans at one level of the supply chain for consuming flow plans at another level of the supply chain. Only certain combinations are needed for such exports and this allows control over which combinations are calculated and exported. This eventually helps in the performance impact of generating and writing this output.

### Functional Description

In this functionality, a user needs to specify a pair of material nodes where intermediate peggings are required in terms of “from node” and “to node” of a graph table. Producing or consuming flow plans at one node will be pegged for producing or consuming flow plans at other node based on the intermediate pegging type.

To enable this feature the following configurations are required.

### Configuration

1. Solver Input

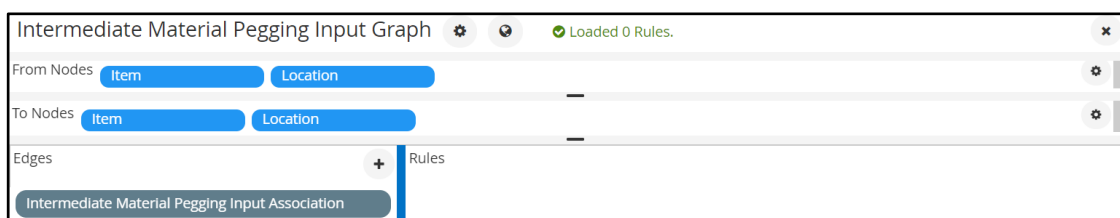
**"Intermediate Material Pegging Input Graph"** - input graph to SCS to define the material node combinations for which intermediate peggings should be calculated and exported; From Node as Material Node dimensions; To Node as Material Node dimensions; number property to represent association.

### Nodes

- From Node: <Material Node>
- To Node: <Material Node>

### Edge properties

- **"Intermediate Material Pegging Input Association"** - number (0/1) property of the graph
  - If 1, the record/association is active and intermediate pegging is required for that combination



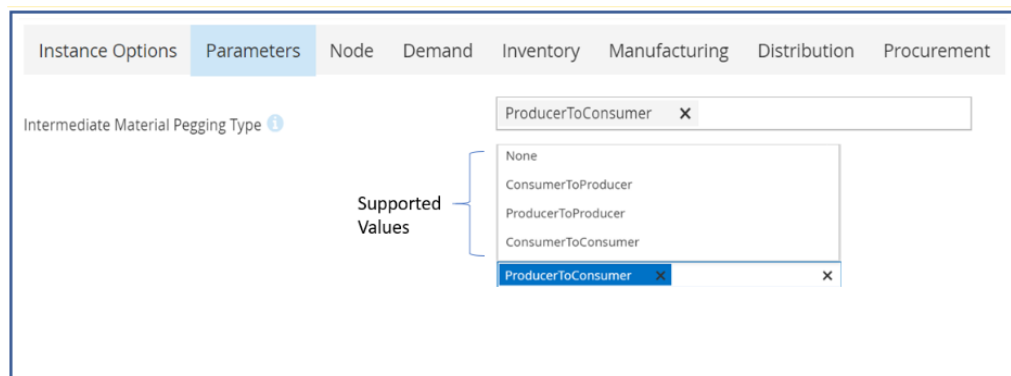
| Rules                                           |
|-------------------------------------------------|
| Intermediate Material Pegging Input Association |

## 2. Parameter: **Intermediate Material Pegging Type**

Type: Multi-select picklist

Supported Values:

- **None** - This is the default value. When this is used, solver will not do intermediate peggings. For intermediate peggings, users need to remove the “None” from the selection and pick any one or more from below mentioned values.
- **ProducerToConsumer** - Intermediate pegging is done between *producing* flow plan at “from node” to *consuming* flow plans at “to node” as mentioned in the input graph displayed above.
- **ConsumerToProducer** - Intermediate pegging is done between consuming flow plan at “from node” to producing flow plans at “to node” as mentioned in the input graph displayed above.
- **ProducerToProducer** - Intermediate pegging is done between producing flow plan at “from node” to producing flow plans at “to node” as mentioned in the input graph displayed above.
- **ConsumerToConsumer** - Intermediate pegging is done between consuming flow plan at “from node” to consuming flow plans at “to node” as mentioned in the input graph displayed above.



- If “from node” and “to node” material nodes are the same in the input graph, then the user needs to see the peggings for production and consumption at that material node. In that case, pegging happens only for pegging type ProducerToConsumer. For the remaining 3 types ConsumerToProducer, ProducerToProducer and ConsumerToConsumer, it does not make sense. In that case, there will be no exports done for the given input pairs.
- It is recommended to configure only the critical things for making solver performant and preferably only one setting (instead of multi select) to make solver performance better.

## 3. Solver Output

**"Intermediate Material Pegging Output Graph Name"** - output graph from SCS for the output of the intermediate pegging calculations (flow plan to flow plan pegging).

**Nodes:**

- From Node: <Activity> <Material Node> <LotID> <Time>
- To Node: <Activity> <Material Node> <LotID> <Time>

**Edge properties:**

- There is a need to have eight edge properties for showing pegging quantity and two for each different type of intermediate pegging types as

mentioned above. A user can define only relevant edge properties based on the pegging type used.

In the solver config, one needs to configure this as mentioned below:

| Instance Options                                  | Parameters | Node | Demand | Inventory | Manufacturing                                | Distribution | Procurement |
|---------------------------------------------------|------------|------|--------|-----------|----------------------------------------------|--------------|-------------|
| Intermediate Material Pegging Output Graph Name ⓘ |            |      |        |           | Intermediate Material Pegging Output Graph ▼ |              |             |
| Producer To Consumer From Pegged Property Name ⓘ  |            |      |        |           | ProducerToConsumerFromPegQty ▼               |              |             |
| Producer To Consumer To Pegged Property Name ⓘ    |            |      |        |           | ProducerToConsumerToPegQty ▼                 |              |             |
| Consumer To Producer From Pegged Property Name ⓘ  |            |      |        |           | ConsumerToProducerFromPegQty ▼               |              |             |
| Consumer To Producer To Pegged Property Name ⓘ    |            |      |        |           | ConsumerToProducerToPegQty ▼                 |              |             |
| Consumer To Consumer From Pegged Property Name ⓘ  |            |      |        |           | ConsumerToConsumerFromPegQty ▼               |              |             |
| Consumer To Consumer To Pegged Property Name ⓘ    |            |      |        |           | ConsumerToConsumerToPegQty ▼                 |              |             |
| Producer To Producer From Pegged Property Name ⓘ  |            |      |        |           | ProducerToProducerFromPegQty ▼               |              |             |
| Producer To Producer To Pegged Property Name ⓘ    |            |      |        |           | ProducerToProducerToPegQty ▼                 |              |             |

Intermediate Material Pegging Output Graph
Loaded 0 Rules.

From Nodes
Activity
Item
Location
Lot ID
Fiscal Week

To Nodes
Activity
Item
Location
Lot ID
Fiscal Week

Edges
+
Rules

ProducerToConsumerFromPegQty
ProducerToConsumerToPegQty
ConsumerToProducerFromPegQty
ConsumerToProducerToPegQty
ConsumerToConsumerToPegQty
ConsumerToConsumerFromPegQty
ProducerToProducerFromPegQty
ProducerToProducerToPegQty

If a user has configured the input and output graph and pegging type set properly, but the relevant edge properties are not configured, then the solver will throw error and exit. Users can see the following types of error messages:

```
2019-07-25 21:40:52.437 TenantId:11282 UserId:233893 (10.0.1.242) ERROR
LiveServer (61) Required config value [Producer To Consumer To Pegged
Property Name] not specified.
2019-07-25 21:40:52.437 TenantId:11282 UserId:233893 (10.0.1.242) ERROR
LiveServer (61) Required config value [Producer To Consumer From Pegged
Property Name] not specified.
```

#### 4. Deliver Consumer

To show delivery consumptions in the intermediate peggings, users need to create a member for delivery activity in Activity dimension. This will be used in pegging output.

In the solver config, mention this activity name as displayed below:

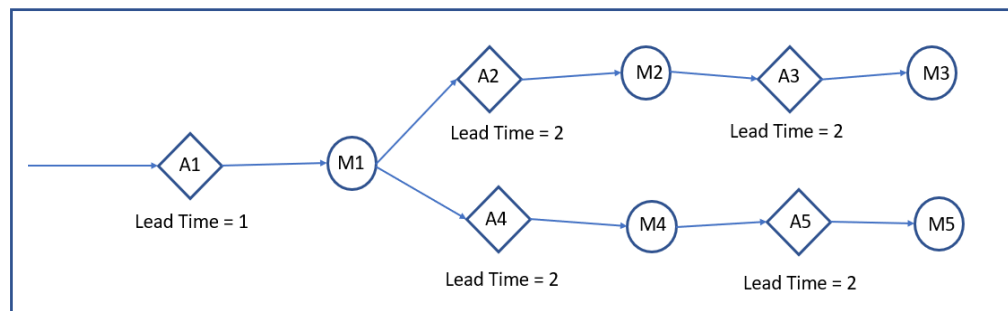


Instance Options
Parameters
Node
Demand
Inventory
Manufacturing
Distribution
Procurement

Delivery Activity ⓘ

### Example

Let us assume that upstream has sufficient material and it is capacity unconstrained plan. The delay demand is not allowed i.e., demand build late limit is 0.



Current bucket: W1

### Case1:

- Activity frozen window = 5 for A1
- Input Supply

| Activity | Material Node | Time | LotID | Producing WIP Qty |
|----------|---------------|------|-------|-------------------|
| A2       | M2            | W8   | Lot1  | 500               |

- Input Pair for Intermediate Pegging

| From Node | To Node | Association Property |
|-----------|---------|----------------------|
| M1        | M2      | 1                    |
| M1        | M3      | 1                    |

- Intermediate Material Pegging Type is set to ProducerToConsumer
- Demands

| DemandID | Material Node | Time | Demand Priority | Demand Qty |
|----------|---------------|------|-----------------|------------|
| Order1   | M2            | W9   | 2               | 800        |
| Order2   | M3            | W10  | 1               | 100        |

Order2 gets planned first and fully planned. Order1 could not do fresh production due to the frozen window and gets partially satisfied 400 units from WIP supply. Due to no delay demand it does not get delayed.

- Output of Intermediate Pegging

| From. Activity | From. Material Node | Form. Time | From.LotID | To.Activity | To.Material Node | To.Time | To.LotID | Producing PegQtyFromNode | Consuming PegQtyToNode |
|----------------|---------------------|------------|------------|-------------|------------------|---------|----------|--------------------------|------------------------|
| A1             | M1                  | W6         | SCS LOT    | A3          | M2               | W8      | SCS LOT  | 100                      | 100                    |
| A1             | M1                  | W6         | SCS LOT    | Deliver     | M2               | W9      | SCS LOT  | 400                      | 400                    |
| A1             | M1                  | W6         | SCS LOT    | Deliver     | M3               | W10     | SCS LOT  | 100                      | 100                    |

### Case2:

- Lotsize increment of 500 defined for A1
- Input pair for intermediate pegging

| From Node | To Node | Association Property |
|-----------|---------|----------------------|
| M1        | M1      | 1                    |
| M1        | M3      | 1                    |
| M1        | M5      | 1                    |

- Intermediate Material Pegging Type is set to ProducerToConsumer
- Demands

| DemandID | Material Node | Time | Demand Priority | Demand Qty |
|----------|---------------|------|-----------------|------------|
| Order1   | M3            | W10  | 2               | 500        |
| Order2   | M3            | W10  | 1               | 300        |

Order2 gets planned first and fully planned. This makes 500 production at M1 due to lotsize. Order1 gets planned next and triggers another 500 production at M1. Finally, both demands gets planned fully and 200 ending on hand remains at M1 till the end of the horizon.

- Output of Intermediate Pegging

| From. Activity | From. Material Node | Form. Time | From.LotID | To.Activity | To.Material Node | To.Time | To.LotID | Producing PegQtyFromNode | Consuming PegQtyToNode |
|----------------|---------------------|------------|------------|-------------|------------------|---------|----------|--------------------------|------------------------|
|                |                     |            |            |             |                  |         |          |                          |                        |

|    |    |    |            |         |    |     |            |     |     |
|----|----|----|------------|---------|----|-----|------------|-----|-----|
| A1 | M1 | W6 | SCS<br>LOT | A2      | M1 | W6  | SCS<br>LOT | 500 | 500 |
| A1 | M1 | W6 | SCS<br>LOT | A4      | M1 | W6  | SCS<br>LOT | 300 | 300 |
| A1 | M1 | W6 | SCS<br>LOT | Deliver | M3 | W10 | SCS<br>LOT | 500 | 500 |
| A1 | M1 | W6 | SCS<br>LOT | Deliver | M5 | W10 | SCS<br>LOT | 300 | 300 |

## Limitations

- Hard pegging is not supported yet in intermediate peggings. The intermediate pegging behavior is undefined with hard peggings. This support will be added in a future release.
- Resource node to material node intermediate peggings is not supported in this release. This will be added in a future release.
- This intermediate material pegging is not supported with the incremental plan. This will be added in future releases.
- This intermediate material pegging does not work with the Concurrent planning. If used when “Concurrent Planning” set to “True”, the behavior is undefined. The intermediate peggings may not be exported then.

## RePlan At Delivery Node

### Business Need

Sometimes due to the nature of order by order (OBO) heuristic planning, some demands are planned short or late. In such cases, if those demands are replanned at the end, there is a chance that some of those can be met and thus overall demand satisfaction will increase. In this phase these demands could get planned potentially due to change in supply picture over the course of planning demands one by one and then followed by other workflows like inventory planning, excess cleanups etc. However, this replanning typically a performance intensive workflow and users needed alternative quick approach which can address this problem without causing any performance bottleneck.

### Functional Description

The replanning of late and short demands at the end has an impact on overall run time for solver plugin. This is because it unplans and replans the late and short part of the demand like regular demand planning by traversing the whole supply chain. Instead of doing that solver is enhanced to do replanning at the delivery material node only. This may not give as optimal results one can get by doing complete replanning of late and short demands but it will be fast in terms of runtime.

This also solves some issues related to RCA. Sometimes users will see short and late demands with no RCA. Those cases might not be directly related to RCA rather the nature of order by order planning and subsequent workflow. By the time RCA workflow runs, it finds material/capacity and hence demand gets planned during RCA phase. Thus, users will not find RCA. This can happen due to lotsize production during demand planning or inventory planning that follows demand planning. If users do replan, those late and short

demands will get planned. Hence there will be no short and late demands without RCA. Thus, this replanning indirectly solves some of RCA issues.

With replan at delivery node, solver unplans the delivery flow plans at the material node and replans demands in the order of demand priority. When it unplans the delivery flows at a material node, the supplies will be available at that node for consumption by direct demands at that material node during replan. This helps planning some of earlier unplanned or short or late demands thus improving demand satisfaction. Since it is unplanning and planning at the delivery node only, it doesn't affect solver's performance in terms of run time. Thus it is not the deep scan (that full replan or incremental plan will do) rather local to the delivery node.

## Note

- The replan at delivery node may lead to violation of build ahead limit set for a demand.  
For example,
  1. A demand was planned short due to build ahead limit restriction.
  2. However at a later stage during inventory planning, solver has created some inventory in earlier bucket to meet safety stock target in that bucket.
  3. Now, all the delivery flow plans are removed and demands at the node will be re-planned in order of their priorities.
  4. In that case, above demand may get planned using the supply available at the node which was created during inventory planning in much earlier bucket.
  5. Finally, user will see the build ahead limit is violated for this demand (looking at the pegging output).

So use this feature with caution.

- The delivery flows for hard pegging demands remain untouched. They don't get unplanned.
- In case of delivery node is an intermediate node (i.e it has direct and indirect demands), only the delivery flows related to direct demands at the node are unplanned.

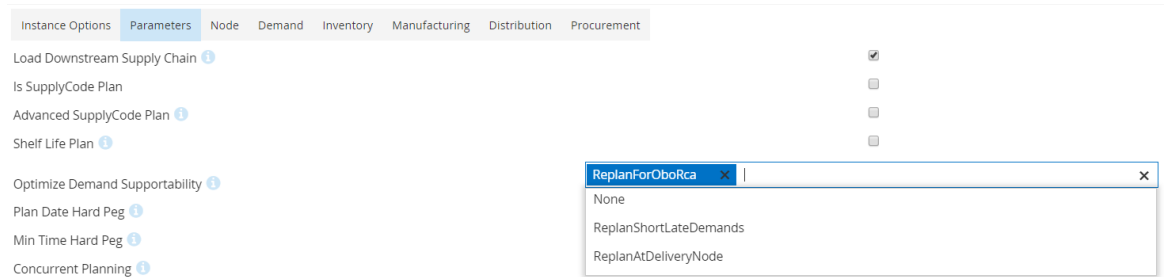
## Configuration

This functionality is controlled through a flag. This is an existing flag. Prior to this, it was a boolean flag. Now this has been made a multi picklist with following supported values.

Parameter: **Optimize Demand Supportability**

Type: Multi select picklist

Supported values:



- **None**

This is default selection. In the existing config, this correspond to False. One needs to change it to None and save the config. In future we will remove False internally. This has been kept for backward compatibility.

With this setting, there won't be any re-planning for short and late demands at the end by the solver.

- **ReplanAtDeliveryNode**

With this setting, solver unplans all the delivery flow plans and replans demands in the order of priority.

- **ReplanShortLateDemands**

With this setting, solver re-plans the late and short part of late and short demands. This will be complete unplanning of the late part and replanning. Thus this involves the complete traversal of supply chain as solver does during regular demand planning.

In the solver config UI, one can set this flag in the Parameters tab as shown below

- **ReplanForOboRca**

With this setting, solver unplans and replans short / late demands and generates [OBO RCA](#)

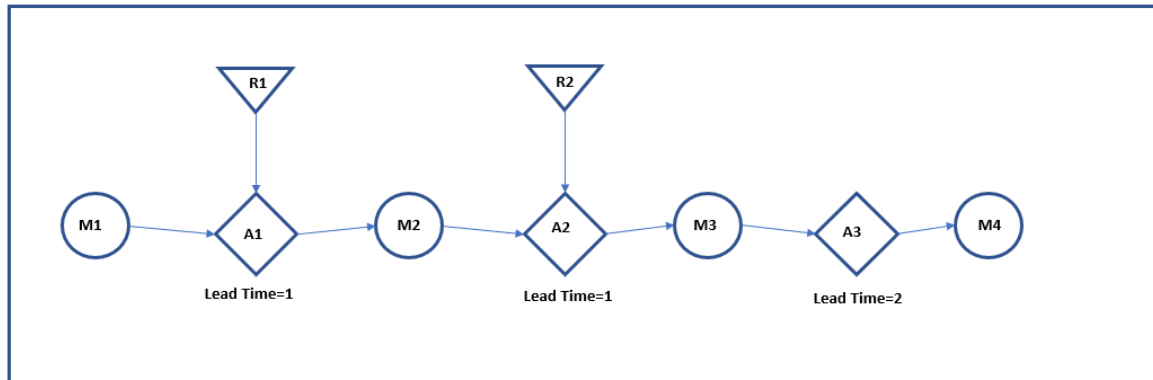
1. When **None** is selected with any other option, **None** is ignored.
2. When **ReplanShortLateDemands** and **ReplanAtDeliveryNode** are selected, **ReplanAtDeliveryNode** is ignored.
3. When **ReplanForOboRca** is selected with **ReplanShortLateDemands** or **ReplanAtDeliveryNode**, firstly **ReplanShortLateDemands** or **ReplanAtDeliveryNode** is executed, then **ReplanForOboRca** is executed.

One can also use this parameter in the solver plugin command as arguments as shown below example:

```
([Optimize Demand Supportability], [ReplanShortLateDemands,
ReplanAtDeliveryNode, ReplanForOboRca])
```

## Example

Let's consider following supply chain.



Current Bucket: W1

**Case1:** Lotsize, NoCarry delivery node, No safety stock planning

- M3 node is a no carry node
- A2 has lotsize min = 1000
- BOH at M1=2000
- Capacity for R1=2000 on W11
- Capacity for R2=1000 on W12
- Demands
  - D1 for M3, Quantity=500, Due=W15, Priority 1
  - D2 for M4, Quantity=300, Due=W17, Priority 2
- Planning
  - D1 is planned short. M3 is a no carry node and the producing activity A2 is having lotsize min of 1000. Hence, solver did not produce as it would lead to no carry violation. This kind of situation happens when delivery node is a no carry node.
  - D2 gets planned next and it produces 1000 at M3 node. Out of this it consumes 300 for demand. Thus one can see 700 as excess inventory sitting at M3 node violating no carry. Since there are no future demands to be planned (along the path), this material could not be flushed to downstream to reduce no carry violation.
- RCA
  - When demand RCA runs, it sees material present at M3 node and able to plan D1 thus no RCA reported. This is a case of short demand with no RCA.
- With **Optimize Demand Supportability** set to **[ReplanAtDeliveryNode]** D1 will be replanned.

**Case2:** With safety stock planning

- Freeze time bucket for A1 & A2 = 3
- Safety stock target=200 at M3
- BOH at M1=2000

- Capacity for R1
  - 1000 on W4
  - 1000 on W10
- Capacity for R2
  - 500 on W5
  - 500 on W11
- Demands
  - D1 for M3, Quantity=700, Due=W15, build ahead limit=3
- Planning
  - D1 is planned short by 200 units by utilizing R2 capacity on W11 due to build ahead.
- SS plan
  - It can make SS 200 in W6 bucket at M3 at the earliest due to activity freeze time bucket.
- RCA
  - When demand RCA runs, it sees material present at M3 node due to SS plan and able to plan D1 thus no RCA reported. This is another case of short demand with no RCA.
- With **Optimize Demand Supportability** set to **[ReplanAtDeliveryNode]** D1 will be replanned. However, as you see it violates the demand build ahead limit.

## Campaign Planning (Batch Consolidation)

### Business Need

Typically, the capacity intensive manufacturing industries deploy campaign planning to maintain the efficiency of the manufacturing process. In some cases one large manufacturing order feeding multiple demand orders is preferred to many small manufacturing orders feeding those demand orders. Most of the time, this kind of consolidated manufacturing orders are preferred for setup or process reasons, or to minimize the shop floor paperwork and reporting by avoiding many small manufacturing orders for the same item (or same product family). Thus, consolidation combines some of the manufacturing orders together into larger manufacturing orders over a period.

For example, there are three manufacturing orders for 100, 50 and 150 units of a specific item due on Monday, Wednesday and Friday, respectively. The consolidation interval for the item is one week. A single manufacturing order of 300 units can be created for this week (on Monday) to satisfy these three separate demand orders.

### Functionality Description

Following are the terms introduced in solver for campaign planning.

- **BCP** – Batch Consolidation Period, the time window where the consolidation occurs
- **Max Batch Size** – Maximum consolidated flow plan that solver can make in a given bucket
- **Activity Group** – Like product family. Flow plans for activities belong to same activity group will be consolidated to the same bucket in BCP range.
- **Production Slack** – BCP has to be more than zero to invoke BCP planning. In case solver is not able to consolidate flow plan in the BCP period as solver doesn't find any flow plan for same activity or activity group, instead of creating flow in JIT bucket it will offset by this value and create the flow plan in that bucket (earlier than JIT bucket).
- **Batch Consolidation Fence** - The consolidation of flow plans occur in the period current bucket plus this offset.

In Supply Chain Solver, key points of Campaign Planning are mentioned below.

- WIP flows are not moved-in.
- In the presence of WIP flow in the BCP range during demand planning, solver consolidates the current required flow (after consuming available material from WIP production) in the WIP produced bucket.
- During demand planning, solver checks BCP period for existence of any flow plans when encounters an activity with BCP specified.
  - If flow plan for same activity/group exists then it creates current flow plan in that bucket.
  - If there exists no flow plan in that bucket (JIT - BCP), solver tries next bucket (for existence of flow plan) and so on till JIT (finally creates flow there).



- If it is not able to plan completely in JIT, it will do build ahead. In the build ahead buckets, solver will again look within the BCP period for consolidation.
- In case of delay demand BCP planning occurs as well.
- When a flow plan is consolidated (moved-in or created) to an earlier bucket, solver checks future period in BCP range w.r.t. current bucket where consolidation just happened if any flow plan exists for the same activity/group. Then try to move-in that flow to the current bucket.
- In a given bucket the total flow plan is limited by max batch size constraint.
- While planning, if no flow plan found in BCP range and production slack is defined for the JIT bucket, then production happens in [JIT - slack bucket]. If not able to plan in that bucket or planned partially, then solver moves to JIT bucket and try to plan there.
- BCP planning respects storage constraints. It should not cause storage violation in any of the buckets.
- During SS planning, batch consolidation occurs as well
  - Move-In Phase

| There exists no flow plan in SS target bucket                                                                                                          | There exists some flow plan in SS target bucket                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| It will try to move-in flow plan from future bucket (where consolidated flow plan exists) to target bucket in whole (Solver won't split the flow plan) | Regular move-in (for the required quantity by splitting the flow plan between the source and target bucket)<br>In this case there is no special treatment for campaign planning. |

- Make-New Phase
  - If move-in fails or still SS not met for a bucket, make-new is needed during SS planning. It will try new production in target bucket considering BCP (as done during demand planning) i.e. it will see if there exists any flow plans in the bucket from [SS target bucket - BCP] to [SS target bucket] range and plan there.
  - At present, it does not consolidate the future flow plan (as there present might be consolidated flow plan) to this bucket during make-new for the cases where the flow plan movement already discarded due to Move-In failure (during Move-in phase) for some reason.

## Configuration

Following measures and parameters are required to be configured for enabling this feature in Supply Chain Solver.

- Measure: **BCP** - Batch Consolidation Period  
Granularity: <Activity Node>

**Batch Consolidation Period**     

Granularity **Activity** **Version Name**

In the solver config this needs to be configured as below

Instance Options Parameters Node Demand Inventory **Manufacturing** Distribution Procurement

Batch Consolidation Period Measure Name  Batch Consolidation Period ▼

- Measure: **Max Batch Size**  
Granularity: <Activity Node>

**Max Batch Size**      

Granularity **Activity** **Version Name**

In the solver config this needs to be configured as below

Instance Options Parameters Node Demand Inventory **Manufacturing** Distribution Procurement

Max Batch Size Measure Name  Max Batch Size ▼

- Measure: **Activity Group Association**  
Granularity: <Activity>, <GroupID>

**Activity Group Association**       Loaded C

Granularity **Version Name** **Activity** **Group ID**

GroupID is the dimension used to group the simultaneous resources. So same dimension can be reused here.

In the solver config this needs to be configured as below

Instance Options **Parameters** Node Demand Inventory Manufacturing Distribution Procurement

Activity Group Association Measure Name  Activity Group Association ▼

- Measure: **Production Slack**  
Granularity: <Activity>, <Time>

In the solver config this needs to be configured as below

- Parameter: **Batch Consolidation Fence**

Solver does consolidation of flow plans from current bucket till this fence. The default is 0 meaning no consolidation of flow plans.

In the solver config this needs to be configured as below

As shown above the consolidate will be done from W1 till  $[W1+30]$  where W1 is current bucket.

Functional test cases can be found [here](#).

## Alternate Resource Consolidation

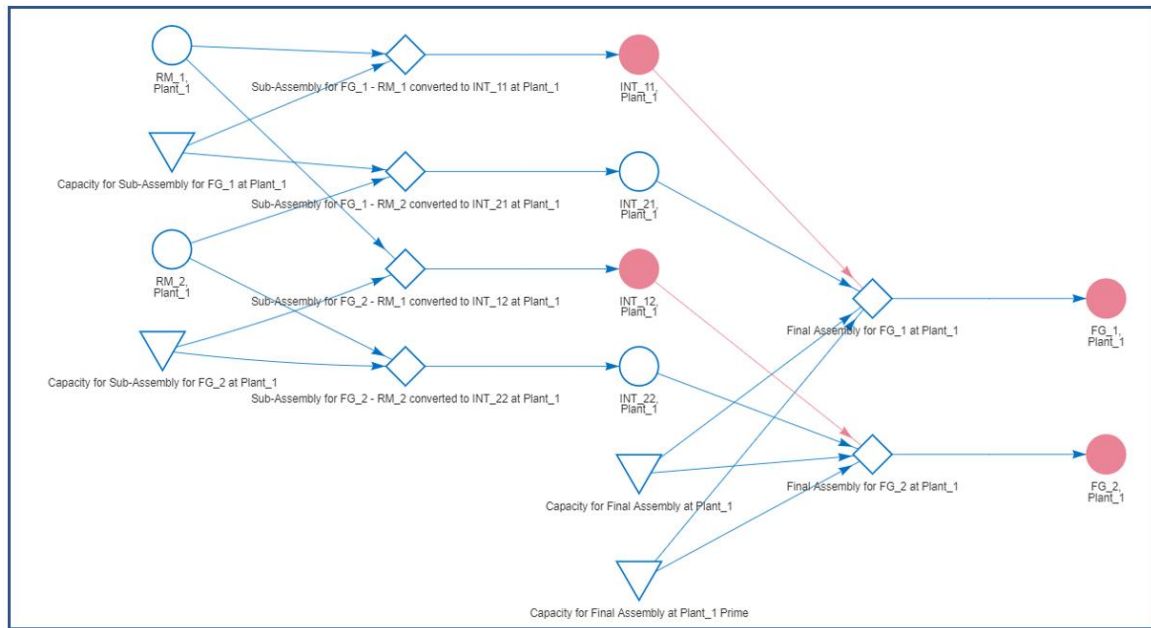
### Business Need

In capacity intensive manufacturing industry (especially in process industries) same resource is used to produce different products. Those products belong to the same family (the difference can be different packaging). There can be alternate resources used at the manufacturing process for such cases. The business requirement is to use one of the alternative resources in the same bucket while planning for products those belong to the same product family. This is a hard constraint i.e. the demands could potentially be shorted or planned late, when they could have been planned on time using the secondary resource, by splitting the plans for the items across resources.

### Functional Description

In Supply Chain Solver this is achieved by resource blocking strategy. The alternate resources are formed a group. This means for a same product family one of the resources of the group can be consumed in a given bucket (other resources are blocked for consumption in that bucket for the same product family).

Let's consider following supply chain.



- Product Family
  - FG\_1, Plant\_1
  - FG\_2, Plant\_1
- Alternate Resource Group
  - Capacity for Final Assembly at Plant\_1 (Primary resource), say R1
  - Capacity for Final Assembly at Plant\_1 Prime (Alternate resource), say R2
- Batch consolidation Period = 1 for Final assembly activities for FG\_1 and FG\_2
- Lead time = 0 for assembly activities
- Assume upstream sub assembly and raw material are sufficient
- Demands

| OrderID | Product | Bucket | Quantity | Priority |
|---------|---------|--------|----------|----------|
| Order1  | FG_1    | W12    | 200      | 1        |
| Order2  | FG_1    | W13    | 200      | 3        |
| Order3  | FG_1    | W14    | 200      | 5        |
| Order4  | FG_1    | W15    | 200      | 6        |
| Order5  | FG_1    | W16    | 200      | 8        |
| Order6  | FG_1    | W17    | 200      | 10       |
| Order7  | FG_2    | W12    | 200      | 2        |
| Order8  | FG_2    | W13    | 200      | 4        |
| Order9  | FG_2    | W15    | 200      | 7        |
| Order10 | FG_2    | W16    | 200      | 9        |

- Expected output for assembly resource consumption

|             | W12 | W13 | W14 | W15 | W16 | W17 |
|-------------|-----|-----|-----|-----|-----|-----|
| FG_1 demand | 200 | 200 | 200 | 200 | 200 | 200 |
| FG_2 demand | 200 | 200 |     | 200 | 200 |     |
| R1 (input)  | 200 |     |     | 600 | 600 |     |
| R2(input)   | 200 | 600 |     | 400 | 400 |     |
| R1(plan)    | 200 |     |     | 600 | 600 |     |
| R2(plan)    |     | 600 |     |     |     |     |

Order3 and Order7 will be delayed. Without this alternate resource consolidation, these two would have planned on due bucket.

## Configuration

- **Activity Group** – The produced items by the activities of the same group belong to one product family.

Measure: *Activity Group Association*

Granularity: <Activity Node>, <GroupID>

GroupID is the dimension used to group the simultaneous resources. So same dimension can be reused here.

In the solver config this needs to be configured as below

- **Resource Group** – The resources those are grouped treated as alternate resource group. One of the resources of the same group can be used in a given bucket for a product family.

Measure: *Resource Group Association*  
Granularity: <Resource Node>, <GroupID>

GroupID is the dimension used to group the simultaneous resources. So same dimension can be reused here.

In the solver config this needs to be configured as below

- **Batch Consolidation Fence** - This is the period from current bucket till when this restriction (of using one resource from alternate resource group in a bucket for a product family) holds true. Default is 0. Outside this period, planning behavior will be normal as usual.

In the solver config this needs to be configured as below

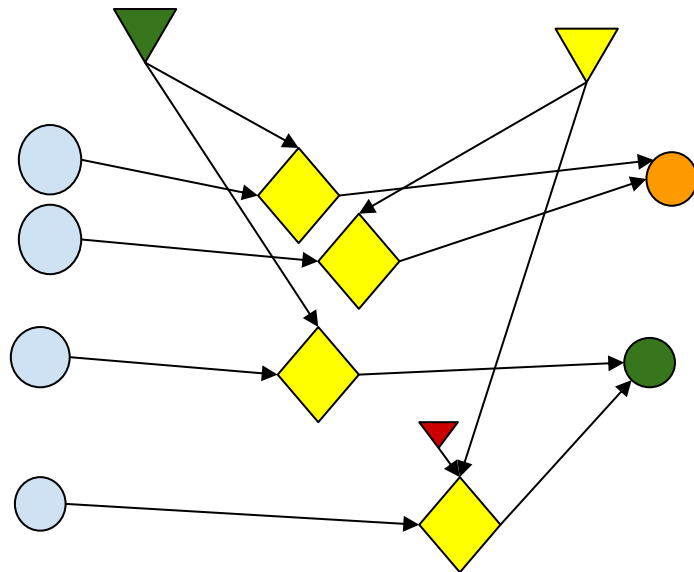
## Brief Algorithm for ARC

In ARC planning, change is only on the capacity consumption side, planning side there are no changes.

While planning when hit an activity which is grouped (same product family) and grouped capacity is attached to the same activity

- Check if there is any capacity consumption from grouped resources in that bucket
- If Yes: Available capacity is from that grouped resource (where there is already a consumption) on that particular bucket (Other grouped resources can't be loaded in the same bucket) along with non-grouped resources
- If No: pick the grouped resource with highest priority along with the ungrouped resources.

(When there are simultaneous capacity, capacity will be consumed from all the simultaneous capacity and treated as a single grouped resource)



In the above supply chain the four activities and the two capacities colored green and yellow are grouped. The capacity colored red is ungrouped. All the capacities are alternate to each other when attached to an activity. Upstream material node has enough material to satisfy the demand.

Capacity picture at green

| bucket   | 4   | 5   | 6   | 7   | 8   | 9   | 10  |  |
|----------|-----|-----|-----|-----|-----|-----|-----|--|
| capacity | 100 | 100 | 100 | 100 | 100 | 100 | 100 |  |

Capacity picture at yellow

| bucket   | 4  | 5  | 6  | 7  | 8  | 9  | 10 |  |
|----------|----|----|----|----|----|----|----|--|
| capacity | 50 | 50 | 50 | 50 | 50 | 50 | 50 |  |

Capacity picture on red capacity

| bucket   | 4  | 5  | 6 | 7 | 8 | 9 | 10 |  |
|----------|----|----|---|---|---|---|----|--|
| capacity | 50 | 50 | 0 | 0 | 0 | 0 | 0  |  |

Let there are two demands one on orange material node, D1(Due on 6th bucket of qty 100, BAL/BLL = infinite ) and D2 on green material node (Due on 6th bucket of qty 100, BAL/BLL = infinite)

During the planning of D1, solver asks activity "1" (1 has higher priority than 2), Activity "1" consumes from green capacity on 6th bucket of qty 100 and satisfy the demand.

During planning of D2, solver asks activity "4" (4 has higher priority than 3), but as the green and yellow capacities are grouped and there is already a consumption from green

capacity on 6th bucket (during planning of D1), on the same bucket the yellow one cant be loaded. IN this bucket the red one also has zero capacity, so on 6th bucket, demand satisfaction will be zero. Solver will build ahead to plan on 5th bucket, during the planning on 5th bucket, 50 capacity is available on yellow (grouped resource) and 50 on red resource(ungrouped resource). Solver will consume both capacity and satisfy the demand.

NOTE: During the planning of D2, though to the activity “4”, green capacity is not attached, solver check all the grouped (same group) capacities for any consumption, if there is a consumption on a given bucket, another grouped capacity can’t be loaded on same bucket.

## Network

### Time-varying Production & Consumption

#### Business Need

Manufacturing processes stabilize over a period of time and yields improve. Similarly yields can decrease due to infrastructure ageing over time. Hence, production and consumption rates in any supply chain can vary with time. Secondly, based on production efficiency improvements, capacity consumption can change over time. Therefore, planners should be able to set the values appropriately for any period of the manufacturing process.

#### Functional Description

Time Varying Production/Consumption can be specified for

- Material Production
- Material Consumption
- Capacity Consumption

#### Time Varying Yield for Material Production



An Activity produces at a Material Node. However, this produced quantity may vary over time. The above diagram denotes that for unit consumption at material node M1, activity A1 produces 0.9 units and 0.8 units on M2 on Week 30 and Week 31 respectively.

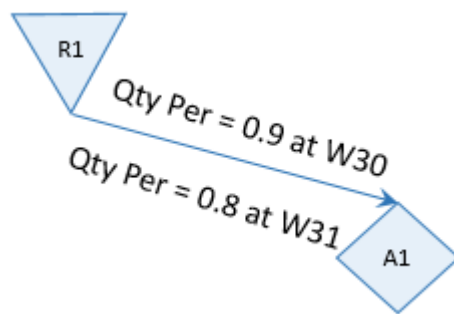
#### Time Varying Qty Per for Material Consumption



An Activity consumes from a Material Node. However, this consumed quantity may vary over time. The above diagram denotes that for unit production at a material node M2, activity A1 consumes 0.9 units and 0.8 units from M1 on Week 30 and Week 31 respectively.



### Time Varying Qty Per for Capacity Consumption



An Activity consumes from a Capacity Node. However, this consumed quantity may vary over time. The above diagram denotes that for unit production & consumption at a material node M2, activity A1 consumes 0.9 units and 0.8 units from R1 on Week 30 and Week 31 respectively.

### **Configuration Steps**

The following measure groups need to be maintained

1. Time Varying Material Production
  - a. Measure : Time Varying Material Production
  - b. Granularity
    - i. Activity Node
      1. Activity.[Activity]
    - ii. Material Node
      1. Item.[Item]
      2. Location.[Location]
    - iii. Time.[Fiscal Week]
    - iv. Version.[Version Name]
2. Time Varying Material Consumption
  - a. Measure : Time Varying Material Consumption
  - b. Granularity
    - i. Activity Node
      1. Activity.[Activity]
    - ii. Material Node
      1. Item.[Item]
      2. Location.[Location]
    - iii. Time.[Fiscal Week]
    - iv. Version.[Version Name]
3. Time Varying Capacity Consumption
  - a. Measure : Time Varying Capacity Consumption
  - b. Granularity
    - i. Activity Node
      1. Activity.[Activity Node]
    - ii. Capacity Node
      1. Resource.[Resource]
    - iii. Time.[Fiscal Week]
    - iv. Version.[Version Name]

Same should be specified in the SCS config as follows:

Goto Workspace→Designer→Edit  
model→Rule→Plugin→Global→o9.GraphCubePlugins→SupplyChainSolver→SupplyChainSolver→ Manufacturing → Prod Graph Time Yield Measure Name → Select the corresponding measure name from the drop-down menu.

Prod Graph Time Yield Measure Name  ▼

Goto Workspace→Designer→Edit  
model→Rule→Plugin→Global→o9.GraphCubePlugins→SupplyChainSolver→SupplyChainSolver→ Manufacturing → Consumption Graph Time Qty Per Measure Name → Select the corresponding measure name from the drop-down menu.

Consumption Graph Time Qty Per Measure Name  ▼

Goto Workspace→Designer→Edit  
model→Rule→Plugin→Global→o9.GraphCubePlugins→SupplyChainSolver→SupplyChainSolver→ Manufacturing → Capacity Graph Time Qty Per Measure Name → Select the corresponding measure name from the drop-down menu.

Capacity Graph Time Qty Per Measure Name  ▼

Time Varying Qty Per / Yields and Consumption / Production graphs bear the same granularity except that Time varying yields have an extra dimension Time added to its granularity. Values for a specific material node-activity combination in Time varying yield, if present, take precedence over values present for the same in material production/consumption graph. For example, consider activity-material node A1-M1 has yield = 1 in material production graph and yield = 2 at Week 30 in time varying material production. When SCS solves the demands, it encounters A1-M1 at Week 30 and takes into account the yield having a value of 2. For all other weeks it takes yield = 1(unless specified otherwise).[\[Functional Test #102\]](#)

## Proportional Sourcing

### Business Need

Manufacturers usually have mid-long term contracts with strategic suppliers, where they agree upon providing business some percentage of the overall raw material quantity procured. Usually these are negotiated at quarterly level and can change over time. For example, from the two suppliers, planners want to get supplies from supplier 1 and supplier 2 in 60:40 proportion in Q1 and can be 50:50 in Q2 and so on. Also in manufacturing they might want to produce in some ratio across different plants or different lanes within a plant.

### Functional Description

By Default,If two or more activities are producing to a material node, they are ordered by SCS for upstream requests on the basis of priorities set in Material Production Graph. However, if the activities are of the same priorities, then SCS picks up the activity by its key value. In short, SCS picks just one activity at a time to fulfill an order.

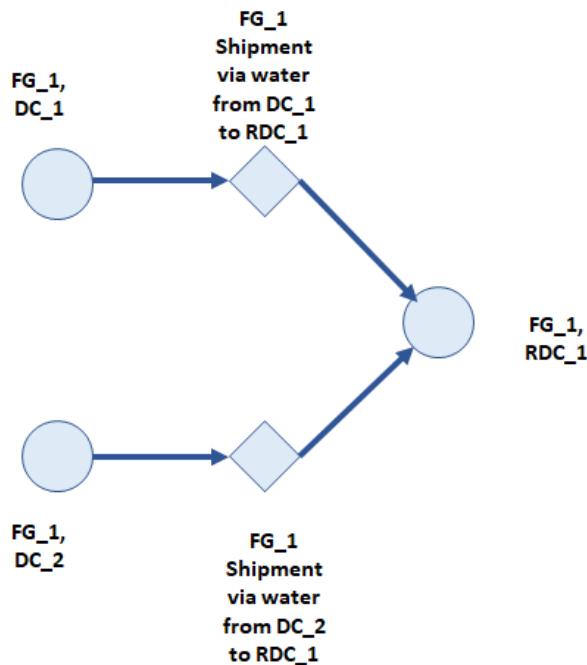
**Planners have now been provided with the flexibility of using all the activities with the same priority, producing at a material node in the desired proportion.**

They can specify ratio in which these activities can produce into the material node during demand fulfilment. SCS still picks just one activity at a time to fulfill the demand request,

by keeping track of the ratio example A1:A2 in the specified time frame and chooses the activity which moves it closer to the specified ratio.

Note : Proportional Spilt is a soft constraint, hence this can be violated to satisfy the demand.

Consider the below Supply Chain



Let Priority be same both of these activities:

| Activity                                     | Item | Location | Data                         | Value |
|----------------------------------------------|------|----------|------------------------------|-------|
| FG_1 - Shipment via Water from DC_1 to RDC_1 | FG_1 | RDC_1    | Material Production Priority | 1     |
| FG_1 - Shipment via Water from DC_2 to RDC_1 | FG_1 | RDC_1    | Material Production Priority | 1     |

Also Split Ratio is defined as follows:

| Activity                                     | Item | Location | Data        | W01-2015 | W11-2015 | W12-2015 |
|----------------------------------------------|------|----------|-------------|----------|----------|----------|
| FG_1 - Shipment via Water from DC_1 to RDC_1 | FG_1 | RDC_1    | Split Ratio | 0.60     | 0.70     | 0.50     |
| FG_1 - Shipment via Water from DC_2 to RDC_1 | FG_1 | RDC_1    | Split Ratio | 0.40     | 0.30     | 0.50     |

Let A1 be activity {FG\_1 - Shipment via Water from DC\_1 to RDC\_1} and A2 be activity {FG\_1 - Shipment via Water from DC\_2 to RDC\_1}.

Let M1 be material node FG\_1 @ RDC\_1.

SCS would interpret the above Split Ratio as follows.

- From **Week 1 to Week 10** material node M1 can source from both the activities A1 and A2 in the ratio of 60:40
- Week 11** material node M1 can source from both the activities A1 and A2 in the ratio of 70:30
- Form **Week 12 till end of planning horizon**, material node M1 can source from both the activities A1 and A2 in the ratio of 50:50.

Let there demand as follows for **Week 1 to Week 10**

| Order Name | Item | Location | Customer Name | Fiscal Week | Demand Quantity | Demand Priority |
|------------|------|----------|---------------|-------------|-----------------|-----------------|
| Order_1    | FG_1 | RDC_1    | Cust_1        | W01-2015    | 10              | 1               |
| Order_2    | FG_1 | RDC_1    | Cust_1        | W02-2015    | 10              | 1               |
| Order_3    | FG_1 | RDC_1    | Cust_1        | W03-2015    | 10              | 1               |
| Order_4    | FG_1 | RDC_1    | Cust_1        | W04-2015    | 10              | 1               |
| Order_5    | FG_1 | RDC_1    | Cust_1        | W05-2015    | 10              | 1               |
| Order_6    | FG_1 | RDC_1    | Cust_1        | W06-2015    | 10              | 1               |
| Order_7    | FG_1 | RDC_1    | Cust_1        | W07-2015    | 10              | 1               |
| Order_8    | FG_1 | RDC_1    | Cust_1        | W08-2015    | 10              | 1               |
| Order_9    | FG_1 | RDC_1    | Cust_1        | W09-2015    | 10              | 1               |
| Order_10   | FG_1 | RDC_1    | Cust_1        | W10-2015    | 10              | 1               |

When the Solver is run, then both the activities participate i.e. A1:A2 in the ratio of 60:40 in demand planning for Week 1 to Week 10. Output is as follows

| Activity                                     | Item | Location | Fiscal Week | Material Quantity |
|----------------------------------------------|------|----------|-------------|-------------------|
| FG_1 - Shipment via Water from DC_1 to RDC_1 | FG_1 | RDC_1    | W01-2015    | 10                |
|                                              |      |          | W03-2015    | 10                |
|                                              |      |          | W05-2015    | 10                |
|                                              |      |          | W06-2015    | 10                |
|                                              |      |          | W08-2015    | 10                |
|                                              |      |          | W10-2015    | 10                |
| FG_1 - Shipment via Water from DC_2 to RDC_1 | FG_1 | RDC_1    | W02-2015    | 10                |
|                                              |      |          | W04-2015    | 10                |
|                                              |      |          | W07-2015    | 10                |
|                                              |      |          | W09-2015    | 10                |

Let there be 10 demand in **Week 11**

| Order Name | Customer Name | Item | Location | Fiscal Week | Demand Quantity | Demand Priority |
|------------|---------------|------|----------|-------------|-----------------|-----------------|
| Order_1    | Cust_1        | FG_1 | RDC_1    | W11-2015    | 10              | 1               |
| Order_10   | Cust_1        | FG_1 | RDC_1    | W11-2015    | 10              | 1               |
| Order_2    | Cust_1        | FG_1 | RDC_1    | W11-2015    | 10              | 1               |
| Order_3    | Cust_1        | FG_1 | RDC_1    | W11-2015    | 10              | 1               |
| Order_4    | Cust_1        | FG_1 | RDC_1    | W11-2015    | 10              | 1               |
| Order_5    | Cust_1        | FG_1 | RDC_1    | W11-2015    | 10              | 1               |
| Order_6    | Cust_1        | FG_1 | RDC_1    | W11-2015    | 10              | 1               |
| Order_7    | Cust_1        | FG_1 | RDC_1    | W11-2015    | 10              | 1               |
| Order_8    | Cust_1        | FG_1 | RDC_1    | W11-2015    | 10              | 1               |
| Order_9    | Cust_1        | FG_1 | RDC_1    | W11-2015    | 10              | 1               |

When the Solver is run, then both the activities participate i.e. A1:A2 in the ratio of 70:30 in demand planning for Week 11. Output is as follows

| Activity                                     | Location | Item | Fiscal Week | Material Quantity |
|----------------------------------------------|----------|------|-------------|-------------------|
| FG_1 - Shipment via Water from DC_2 to RDC_1 | RDC_1    | FG_1 | W11-2015    | 30                |
| FG_1 - Shipment via Water from DC_1 to RDC_1 | RDC_1    | FG_1 | W11-2015    | 70                |

Let there demand as follows for **Week 12 till End of Horizon**

| Order Name | Customer Name | Item | Location | Fiscal Week | Demand Quantity | Demand Priority |
|------------|---------------|------|----------|-------------|-----------------|-----------------|
| Order_1    | Cust_1        | FG_1 | RDC_1    | W12-2015    | 10              | 1               |
| Order_10   | Cust_1        | FG_1 | RDC_1    | W21-2015    | 10              | 1               |
| Order_2    | Cust_1        | FG_1 | RDC_1    | W13-2015    | 10              | 1               |
| Order_3    | Cust_1        | FG_1 | RDC_1    | W14-2015    | 10              | 1               |
| Order_4    | Cust_1        | FG_1 | RDC_1    | W15-2015    | 10              | 1               |
| Order_5    | Cust_1        | FG_1 | RDC_1    | W16-2015    | 10              | 1               |
| Order_6    | Cust_1        | FG_1 | RDC_1    | W17-2015    | 10              | 1               |
| Order_7    | Cust_1        | FG_1 | RDC_1    | W18-2015    | 10              | 1               |
| Order_8    | Cust_1        | FG_1 | RDC_1    | W19-2015    | 10              | 1               |
| Order_9    | Cust_1        | FG_1 | RDC_1    | W20-2015    | 10              | 1               |

When the Solver is run, then both the activities participate i.e. A1:A2 in the ratio of 50:50 in demand planning for Week 12 till End of Horizon. Output is as follows

| Activity                                     | Location | Item | Fiscal Week | Material Quantity |
|----------------------------------------------|----------|------|-------------|-------------------|
| FG_1 - Shipment via Water from DC_2 to RDC_1 | RDC_1    | FG_1 | W21-2015    | 10                |
|                                              |          |      | W19-2015    | 10                |
|                                              |          |      | W17-2015    | 10                |
|                                              |          |      | W15-2015    | 10                |
|                                              |          |      | W13-2015    | 10                |
| FG_1 - Shipment via Water from DC_1 to RDC_1 | RDC_1    | FG_1 | W20-2015    | 10                |
|                                              |          |      | W18-2015    | 10                |
|                                              |          |      | W16-2015    | 10                |
|                                              |          |      | W14-2015    | 10                |
|                                              |          |      | W12-2015    | 10                |

Note :

- SCS proportionally splits demands, adhering to the split ratio as well as the concept of using just one activity to satisfy a demand. If there are constraints on one path, then we off-load the unfulfilled portion of the demand on the other path i.e. will no longer honour the proportionality.
- Production Split Ratio should add up to 1, in case it is not Solver will normalize
- While building ahead due to constraints SCS will try to build ahead on the chosen activity till the specified boundary and then go to alternate activity. At the boundary, it will re-evaluate the ratio to choose the activity. For e.g., if demand is on W40 and the chosen activity is A1, it will build ahead till W27 and from then on go to A2 trying to build ahead from W40 to W27. If demand is still not satisfied, SCS will again evaluate the current ratio of fulfilled demands and choose between A1 and A2. If it chooses A2 it will try to build ahead till W11 and then go to A1 and do a build ahead searching for supply from W26 to W11. Once SCS finishes build ahead or reaches the build ahead limit, it will then try to meet the demand by building late. During build late phase, it will try to minimize the lateness in a bucket when choosing the activity.
- Treatment of NULL and Zero please see the table below

| Proportional Sourcing on Entire Horizon | Case1 | Case2 | Case 3 | Case 4 |
|-----------------------------------------|-------|-------|--------|--------|
|                                         |       |       |        |        |

|                  |                                                                                                                          |                                                                                                                                                                                                                                         |                                                                          |                                                                                                                                                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A1               | NULL                                                                                                                     | NULL                                                                                                                                                                                                                                    | 1                                                                        | 0.8                                                                                                                                                                                                                                                                        |
| A2               | NULL                                                                                                                     | 0                                                                                                                                                                                                                                       | NULL                                                                     | 0.2                                                                                                                                                                                                                                                                        |
| A3               | NULL                                                                                                                     | 0                                                                                                                                                                                                                                       | 0                                                                        | NULL                                                                                                                                                                                                                                                                       |
| Solver Treatment | Solver treats as the case of No Proportional Sourcing defined .The solver takes the priority measure route for sourcing. | Solver consider proportional sourcing is defined as zero for all . Neither of the activities are available, solver will go to next priority activity<br><br>In this case Solver will write RCA stating ProportionalActivityNotAvailable | Solver will source only from A1<br>A2 & A3 will be unavailable to source | Solver will source the from only from A1& A2 in the ratio of 80%:20%<br>Say there are 10 demands 8 demand will be processed from A1 and 2 demands from A2. Each demand solver will try to ask any of the activity full quantity for planning<br><br>A3 will be unavailable |

## Configuration

A measure group Proportional Split can be maintained

- Measure
  - Split Ratio  
It takes decimal/numeric values. The values denote the yield ratio of the activities on a material node from that time bucket till time bucket where ratio is specified again
- Granularity
  - Material Node
    - Item.Item Name
    - Location.Location Name
  - Activity Node
    - Activity.Activity Name
  - Time.Fiscal Week
  - Version.Version Name

Same should be specified in the SCS config as follows

### Manufacturing

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing→ Activity Proportion Measure Name → select corresponding measure name from drop-down menu

Activity Proportion Measure Name ⓘ Split Ratio ▼

### Distribution

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Distribution→ Distribution Activity Proportion Measure Name → select corresponding measure name from drop-down menu

Distribution Activity Proportion Measure Name ⓘ ▼

## Procurement

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Procurement → Supplier Activity Proportion Measure Name → select corresponding measure name from drop-down menu

Supplier Activity Proportion Measure  
Name ⓘ

## Intra-Demand Proportional Sourcing

### Business Need

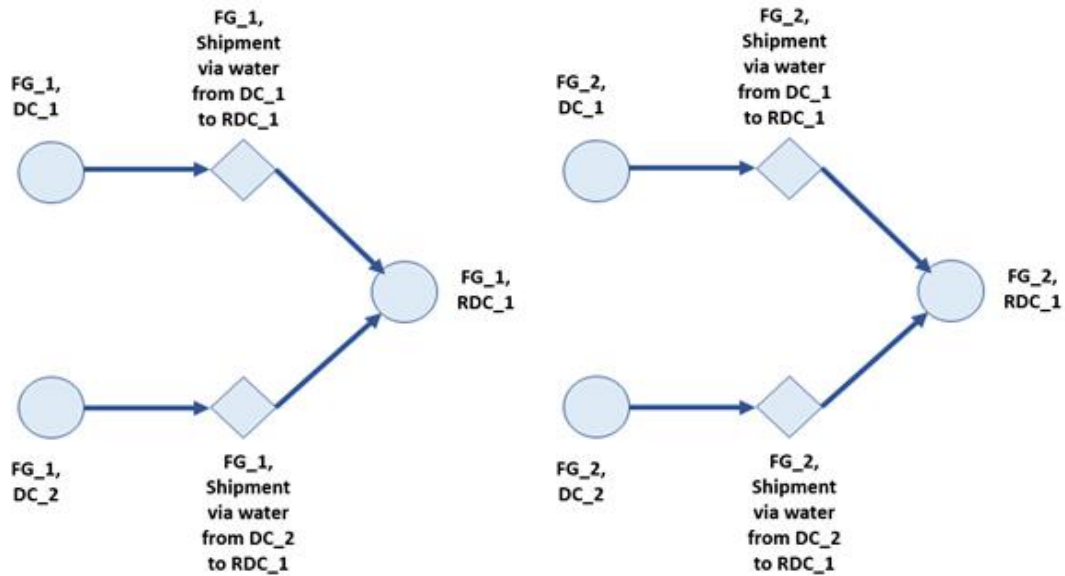
Supply chain (Procurement, Manufacturing and Distribution) sometimes, see a need for splitting the orders in a certain ratio to mitigate risk and to restrict business to a supplier or a warehouse or a plant. Sometimes, the demand quantity of each order is enormous, and the business needs to split the orders from different suppliers. In such cases, there is a need to split one order itself from multiple sources / lines in a certain ratio. Usually these are negotiated at quarterly level and can change over time. For example, for 1 huge order from the two suppliers, planners want to get supplies from supplier 1 and supplier 2 in 60:40 proportion in Q1 and can be 50:50 in Q2 and so on.

### Functional Description

It is similar to Proportional Sourcing as explained above, but in contrast, Planners now have been provided with a flexibility of splitting the demand quantity along the available paths based on the proportion ratio defined. Earlier, the same demand would have planned along one of the paths completely. In case of constraints along that it would switch to alternate. However with this functionality, while planning a particular demand itself the split occurs at the alternate structure and demand is planned along the paths based on the proportions. Once all the available paths are tried w.r.t. the proportion, solver plans the remaining quantity (if any) along the alternates one by one. At this time, there will be no demand quantity split at the alternate structure.

User can mark a demand if intra-demand split is required during proportional sourcing. There is a second level flag at the material node with producing alternate structure having proportional split defined that controls solver behavior whether to do intra-demand split or not.

Consider the below Supply Chain for 2 materials FG\_1 and FG\_2:



Let Priority be same both of these activities and Split Ratio is defined as follows:

| Item | Activity                                     | Location | Material Production Priority | Split Ratio |
|------|----------------------------------------------|----------|------------------------------|-------------|
| FG_1 | FG_1 - Shipment via Water from DC_1 to RDC_1 | RDC_1    | 1                            | 0.2         |
|      | FG_1 - Shipment via Water from DC_2 to RDC_1 | RDC_1    | 1                            | 0.8         |
| FG_2 | FG_2 - Shipment via Water from DC_1 to RDC_1 | RDC_1    | 1                            | 0.2         |
|      | FG_2 - Shipment via Water from DC_2 to RDC_1 | RDC_1    | 1                            | 0.8         |

Let A1 be activity {FG\_1 - Shipment via Water from DC\_1 to RDC\_1} and A2 be activity {FG\_1 - Shipment via Water from DC\_2 to RDC\_1}, A3 be activity {FG\_2 - Shipment via Water from DC\_1 to RDC\_1} and A4 be activity {FG\_2 - Shipment via Water from DC\_2 to RDC\_1}

Let M1 be material node FG\_1 @ RDC\_1, M2 be FG\_2 @ RDC\_1

Let demand of M1 and M2 at RDC\_1 is 10 each

Now, there are 2 boolean flags 'Demand Split Measure Name' and 'IL Demand Split Measure Name' which determine whether Intra-Demand Proportional split should happen or regular Proportional split should happen. One is at Demand level and at Material level. Only when both flags are TRUE, Intra-Demand Proportional split will occur.

If we have provided Proportional Sourcing Split ratios, then these are the various scenarios and their expected results:

| Demand Flag<br>'Demand Split Measure' | Material Flag<br>'IL Demand Split Measure' | Intra-Proportional Sourcing | Inter-Proportional Sourcing (or Proportional Sourcing) |
|---------------------------------------|--------------------------------------------|-----------------------------|--------------------------------------------------------|
|                                       |                                            |                             |                                                        |



|       |       |     |     |
|-------|-------|-----|-----|
| True  | True  | Yes | Yes |
| True  | False | No  | Yes |
| False | True  | No  | Yes |
| False | False | No  | Yes |

So, we have provided these flags for Order\_1, FG\_1 and FG\_2:

| Item | Location | Order Name | Demand Split                        | IL Demand Split                     |
|------|----------|------------|-------------------------------------|-------------------------------------|
| FG_1 | RDC_1    | Order_1    | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| FG_2 | RDC_1    | Order_1    | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |

SCS would interpret the above Split Ratio as follows.

- For FG\_1, material node M1 can source from only either of the activities because it will only honor Inter-Proportional Sourcing (or regular Proportional Sourcing). It picks A2 first due to higher proportional split defined (0.8). If demand is not able to meet completely along A2, it will source along A1.
- For FG\_2, demand at RDC\_1 both the intra-demand proportional split related measures set to true. So, while planning the requested quantity will be splitted at the material node M2 along A3 and A4 as per the ratio 20:80

Output for Material Consumption is as follows:

| Location | Item | Activity                                     | Data                          | W10-2015 |
|----------|------|----------------------------------------------|-------------------------------|----------|
| DC_2     | FG_1 | FG_1 - Shipment via Water from DC_2 to RDC_1 | Material Consumption Quantity | 10       |
| DC_1     | FG_2 | FG_2 - Shipment via Water from DC_1 to RDC_1 | Material Consumption Quantity | 2        |
| DC_2     | FG_2 | FG_2 - Shipment via Water from DC_2 to RDC_1 | Material Consumption Quantity | 8        |

**Note :**

- SCS Intra-demand proportionally splits demands, adhering to the split ratio as well as to the concept of using multiple activities to satisfy a demand. If there are constraints (like minimum lot-size quantity, multiple lot-size quantity, etc.) on the paths, it then comes close to the ratio as much as possible.
- 2 measures 'Demand Split' and 'IL Demand Split' can be maintained which guide the Intra-Proportional Sourcing functionality. Only if both flags are TRUE, Intra-demand proportional sourcing will occur, else Inter-demand proportional sourcing will occur.
- Split Ratio should add up to 1, in case it is not then Solver will normalize
- Let us assume that we have 3 activities A1, A2, A3 as alternate activities producing a material node. For treatment of NULL and Zero please see the table below:

| Proportional Sourcing on Entire Horizon | Case1                                                                                                                    | Case2                                                                                                                                                                                                                                     | Case 3                                                                 | Case 4                                                                                      |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| A1                                      | NULL                                                                                                                     | NULL                                                                                                                                                                                                                                      | 1                                                                      | 0.8                                                                                         |
| A2                                      | NULL                                                                                                                     | 0                                                                                                                                                                                                                                         | NULL                                                                   | 0.2                                                                                         |
| A3                                      | NULL                                                                                                                     | 0                                                                                                                                                                                                                                         | 0                                                                      | NULL                                                                                        |
| Solver Treatment                        | Solver treats as the case of No Proportional Sourcing defined, The solver takes the priority measure route for sourcing. | Solver consider proportional sourcing is defined as zero for all. If neither of the activities are available, solver will go to next priority activity<br><br>In this case Solver will write RCA stating ProportionalActivityNotAvailable | Solver will source only from A1. A2 & A3 will be unavailable to source | Solver will source the from only from A1& A2 in the ratio of 80%:20% A3 will be unavailable |

## Configuration

2 measures 'Demand Split' and 'IL Demand Split' can be maintained,

### 'Demand Split'

- Measure
  - Demand Split
 

It takes boolean values. The values denotes a flag for Intra-demand sourcing split. When it is TRUE, Intra-demand proportional split is enabled for the demand. Default is FALSE.
- Granularity
  - Demand Node
    - Order.Order Name
    - Customer.Customer Name
    - Time. Fiscal Week
  - Material Node
    - Item.Item Name
    - Location.Location Name
  - Version.Version Name

### 'IL Demand Split'

- Measure
  - IL Demand Split
 

It takes boolean values. The values denotes a flag for Intra-demand split.

When it is TRUE, Intra-Demand proportional sourcing split is enabled for the material node. Default is FALSE.

- Granularity
  - Material Node
    - Item.Item Name
    - Location.Location Name
  - Version.Version Name

Same should be specified in the SCS config as follows:

#### **Demand**

Goto Workspace → Designer → Edit model → Rule Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Demand →  
Demand Split Measure Name → select corresponding measure name from drop-  
down menu

Demand Split Measure Name ⓘ

#### **Inventory**

Goto Workspace → Designer → Edit model → Rule Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Inventory →  
IL Demand Split Measure Name → select corresponding measure name from  
drop-down menu

IL Demand Split Measure Name ⓘ

## **Stretched Lead Time**

### **Business Need**

Businesses have holidays and planned maintenance schedules. These vary across locations. This means the activities needs to be stretched over this holiday period.

### **Functional Description**

An activity cannot start in a bucket which is a holiday.[\[Functional Test #112\]](#).

An activity cannot end in a bucket which is a holiday.[\[Functional Test #113\]](#)

A holiday bucket between start and end cannot be counted towards lead time of the activity. Build ahead and build late ranges are agnostic to holidays that may fall in between.[\[Functional Test #343\]](#) WIP should not be specified on holidays. If specified, it will be ignored by the solver.[\[346, 347, 348, 349\]](#)

### **Configuration Steps**

A measure group by name 'Activity Node Time Parameters' needs to be created. This measure group is dimensioned by Activity, Time & Version. It has a measure by name 'Is Holiday'. This measure is of type Boolean. A value of 'false' (default) means that the activity is working (not holiday) in the specified time bucket. A value of 'true' means that the activity is on holiday in the specified time bucket. A value of null has the same meaning as false.

## Shipping and Receiving Holidays

### Business Need

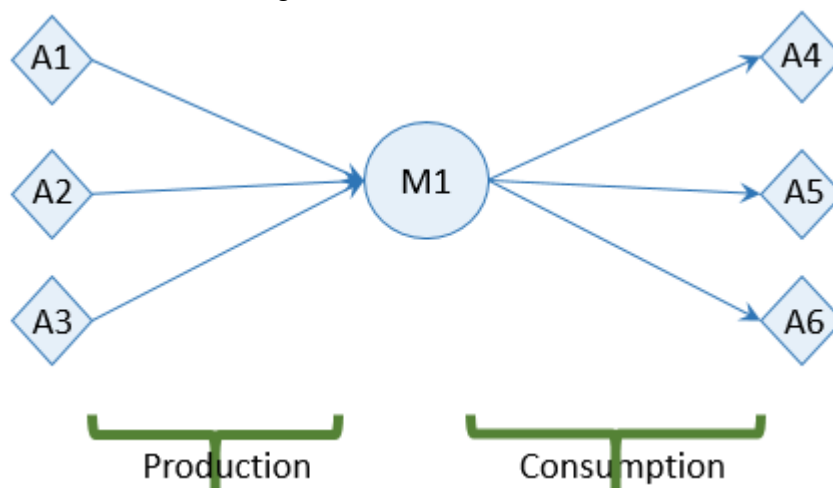
Certain businesses have restrictions when a particular material can arrive at or leave the facility. For eg., vendor A can provide his material only on a monday and Vendor B can provide it on tuesday, wednesday and so on. Also there might be cases when the facility is closed and no shipping or receiving is possible in certain time buckets. There are also business cases when a manufacturing location is on a holiday but open to shipping goods, as it is handled by a third party service provider.

### Functional Description

Holidays are defined on production and consumption edges. If an Activity and material node combination  $\{A_x, M_x\}$  on consuming graph (or associated measure group) is defined as a holiday at time bucket  $T_x$  then activity  $A_x$  cannot consume from this material node in that time bucket. Similarly  $\{A_y, M_y\}$  on producing graph is defined as a holiday at time bucket  $T_x$  then activity  $A_y$  cannot produce into this material node in that time bucket. For example, if lead is 3 days and the request is placed on friday, which is a holiday it would ask one day earlier. Shipping and receiving holidays won't stretch the lead time. It acts as Closed or Open for shipping and receiving purposes at material node.

Holidays can also be defined on activities {stretched lead time}, in which case it would stretch the lead time. For example, if lead time is 3 days the consumption which starts on a monday would end on a wednesday. If tuesday is a holiday for the activity, it would stretch it by a day and now start on a sunday. { [Functional Test #147](#) }

Consider the following Material node



If  $\{A1, M1\}$  is marked as on holiday on  $T1$ , then SCS assumes that Activity A1 cannot produce on material node M1 on time bucket  $T1$ . The production then takes place at time bucket  $\{T1-1\}$ . However, production and consumption schedules of other activities interacting with M1 remain unchanged. Similarly if  $\{A4, M1\}$  is marked on a holiday at  $T1$ , then SCS assumes that activity A4 cannot consume from material node M1 at time bucket  $T1$ . The consumption then takes place in earlier buckets after adhering to the build ahead limits and other supply chain constraints.

### Configuration Steps

Following measure groups have to be maintained to incorporate holiday calendars

- Material Node Consumption Holiday
  - Measure : Consumption Holiday

This measure takes boolean values :

1. TRUE: setting the measure as true for a specific granularity combination means the activity cannot consume from that material node in that time bucket
2. FALSE/NULL: setting the measure as false/null for a specific granularity combination means the activity is free to consume from that material node in that time bucket

- Material Node Production Holiday
  - Measure : Production Holiday

This measure takes boolean values :

1. TRUE: setting the measure as true for a specific granularity combination means the activity cannot produce at that material node in that time bucket
2. FALSE/NULL: setting the measure as false/null for a specific granularity combination means the activity is free to produce at that material node in that time bucket

Both the measure groups have the following granularity

1. Material Node
  - a. Item.[Item Name]
  - b. Location.[Location Name]
2. Activity Node
  - a. Activity.[Activity Name]
3. Time.[Fiscal Week]
4. Version.[Version Name]

Same should be specified in the SCS config as follows:

### Manufacturing

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Production Graph Holiday Measure Name → select corresponding measure name from drop-down menu

Prod Graph Holiday Measure Name ⓘ Production Holiday ▼

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Consumption Graph Holiday Measure Name → select corresponding measure name from drop-down menu

Consumption Graph Holiday Measure Name ⓘ Consumption Holiday ▼

### Distribution

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Distribution → Distribution Production Graph Holiday Measure Name → select corresponding measure name from drop-down menu

Distribution Prod Graph Holiday Measure Name ⓘ ▼

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Distribution → Distribution Consumption Graph Holiday Measure Name → select corresponding measure name from drop-down menu

Distribution Consumption Graph Holiday  
Measure Name ⓘ

## Procurement

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Procurement → Supplier Production Graph Holiday Measure Name → select corresponding measure name from drop-down menu

Supplier Prod Graph Holiday Measure  
Name ⓘ

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Procurement → Supplier Production Graph Holiday Measure Name → select corresponding measure name from drop-down menu

Supplier Consumption Graph Holiday  
Measure Name ⓘ

## Activity Exclusion/Inclusion

### Business Need

Certain sections of the supply chain network may not be active at a given time. For example, during new product introduction, they might model things but do not want to make it part of the system. There could be cases where supply from certain suppliers/factories are permitted only on an exception basis. The base plan should not plan along these paths.

### Functional Description

Activity Inclusion and Exclusion is implemented in SCS through Measure Groups. These measure groups include boolean measures. If the measure is set as true, then SCS assumes that the corresponding activity is active. Similarly, if the measure is set as false, then SCS assumes that corresponding activity is inactive.

### Configuration

Three measure groups are to be maintained

1. Material Production Graph
  - a. Measure Name
    - i. Material Production Graph Association
  - b. Granularity
    - i. Material Node
      1. Item.Item name
      2. Location.Location name
    - ii. Activity Node
      1. Activity.Activity name
    - iii. Version.Version Name

2. Material Consumption Graph
  - a. Measure Name
    - i. Material Consumption Graph Association
  - b. Granularity
    - i. Material Node
      1. Item.Item name
      2. Location.Location name
    - ii. Activity Node
      1. Activity.Activity name
    - iii. Version.Version Name
3. Capacity Consumption Graph
  - a. Measure name
    - i. Capacity Consumption Graph Association
  - b. Granularity
    - i. Capacity Node
      1. Resource.Resource Name
    - ii. Activity Node
      1. Activity.Activity Name
    - iii. Version.Version Name

*Note : Please go to the section “**Designing SCS-o9 Supply Chain Network through Graphs**” for further information on the graph configurations.*

## Date Effectivity

### Business Need

Supply Chain Networks evolve over time. This means a parts/network which are active in some time buckets may not be active in other time buckets. Supply chain network design should be flexible to handle these cases. A component in a BOM can be replaced by a new component, a new transportation route can get added, some parts can get discontinued etc.

### Functional Description

Certain routes/activities are valid only for a certain period of time. These depend on how businesses decide to use them. They can be modeled with start and end dates, so that it is active only within these period. For new routes they can specify a start period, indicating from when this route is operational. For cancelled routes, they can specify an end date, which indicates when till when this route is valid.

### Configuration

A measure group Activity Parameters is maintained with the following granularity

- Activity Node
  - [Activity].[Activity Name]
- [Version].[Version Name]

And has start date and end date defined as following measures.

- Solver Activity Start Date
- Solver Activity End Date

An activity is considered inactive before its corresponding Solver Activity Start Date and after its corresponding Solver Activity End Date. By default , an activity is considered to

be effective throughout the planning horizon. Flows are planned on an activity, when both start and end date are active. For example, if lead time is 5 buckets and start date is week 2 and end date is W5, then nothing is planned on this activity.

## Frozen Horizon

### Business Need

Plans in the near term are fixed as it is passed to execution system. Also one does not want to make any changes in these and termed as frozen. This is modeled as frozen horizon where the flows provide visibility to the planners on the actions that they have already taken and asked to execute. The solver should not trigger fresh starts in this period as it is very close to the current time and shop floor cannot adapt to changes in this period.

### Functional Descriptions

Frozen horizon can be set as input to the solver in terms of number of time buckets from the current time bucket. This can be specified at the activity level. No plans will be triggered in the frozen period. Build ahead will not build new supplies in this period. SCS is able to import the production and consumption flow plans (from WIP & Lock tables) and adhere to it, even if it is within the frozen period. Infeasibilities are removed by creating new supplies upstream. If it cannot get material or capacity this portion of infeasibility is treated as a planner override and is hidden from subsequent JIT/SS runs.

On completion of supply plan, it is reviewed and any necessary adjustments are done. Also certain flow plans are frozen (Eg: Plans released to shop floor, Plans scheduled but not yet released), which means that these flows are firmed up from this run and should be adhered in the next run as well. After roll forward, the solver needs to honour these plans within frozen horizon.

### Configuration Steps

A measure group Activity Parameters is maintained.

It has measure

- Activity Freeze Time Bucket Offset

And granularity

- Activity Node
  - [Activity].[Activity Name]
- [Version].[Version Name]

Activity Freeze Time Bucket offset measure is also given in the SCS config as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →

o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver →

Manufacturing→Activity Freeze Time Bucket Offset Measure Name → Select the corresponding measure name from the drop-down menu.

Activity Freeze Time Bucket Offset  
Measure Name 

Activity Freeze Time Bucket Offset





Measure Activity Freeze Time Bucket Offset takes numeric values. It specifies the number of time buckets from that current time bucket that the corresponding activity is frozen.

## Composite Activity Node

### Business Need

The supply chain process extends across procurement, manufacturing and distribution. The underlying constructs vary across these. There is a need to align better with the data that comes from the ERP or other external systems.

### Functional Description

mPower supported (and still supports) connecting material nodes via Activities, which is the most atomic level for transformations. Users needed to model activities appropriately. It has now been enhanced to be represented as a composite node. Activity nodes can be represented differently for different supply chain processes. Examples are provided below

1. Manufacturing: BOM ID, Routing (Composite node)
2. Procurement: Supplier ID, Transport Mode (Composite node)
3. Distribution: BOD ID, Transport Mode (Composite node)

Only 3 types of Activity Nodes are supported. The supply chain network should be represented using these building blocks. The composite node can consist of 1 or more dimensions. The users can model only one type and have just activity, which would be the basic form of representation

### Configuration Steps

Assuming dimensions of above are created, one can represent it in the SCS Config as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Node → Activity Node Components → enter values in text

- Existing Configuration

Activity Node Components ⓘ

[Activity].[Activity]

- For Manufacturing

Activity Node Components ⓘ

[BOM].[BOM ID], [Routing].[Routing ID]

- For Procurement

Supplier Activity Node Components ⓘ

[Supplier].[Supplier ID], [Transport].[Transport Mode]

- For Distribution

Distribution Activity Node Components ⓘ

[BOD].[BOD ID], [Transport].[Transport Mode]

*Note: Manufacturing uses existing configuration of Activity Node Components itself*

## Co-Products

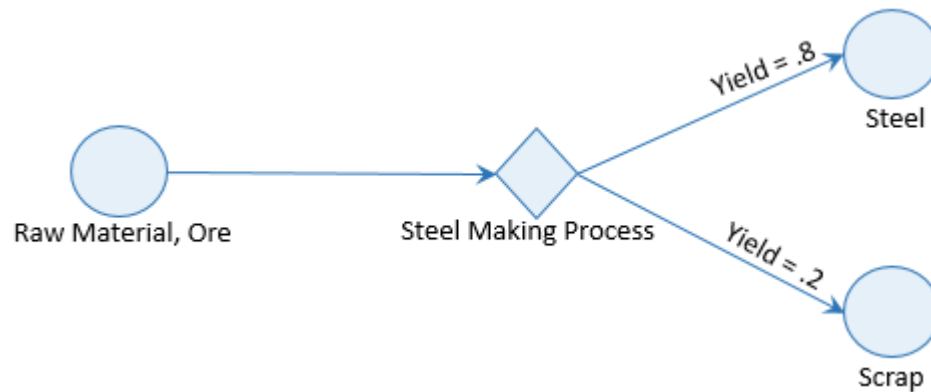
### Business Need

The production of a certain item can result in the production of another item (co-product or by-product). Production of a higher grade item may also lead to simultaneous production of a lower grade item. The lower grade material can be sold or processed further. Thus, the lower grade material can also have an independent demand. There

can also be restrictions on whether these demands can be propagated upstream when coming through the co-product item.

### Functional Description

Consider the following network



This means for every 100 units of {Raw Material , Ore} consumed by the {Steel Making Process}, 80 units of {Steel} are produced and 20 units of {Scrap} are produced. If there are demands for 80 units of {Steel} and 20 units of {Scrap}, then production pegging will show 80 units pegged to Steel demand and 20 units to that of Scrap. Consumption pegging will show two records (i) 100 units of {Raw Material , Ore} pegged to Steel demand of 80 units and (ii) 100 units of {Raw Material , Ore} pegged to Scrap demand of 20 units. Note that the same 100 units is pegged to both Steel and Scrap in the pegging. If the demand for Scrap is less, the surplus after demand fulfilment will be pegged to Stock (Inventory demand).

Activity {Steel Making Process} can be called both via orders incident on

- {Steel} Material Node and/or  
If demand = 80 units of steel  
then 80 units of steel are produced and 20 units of scrap are produced  
[\[Functional Test # 83\]](#)
- {Scrap} Material Node  
If demand = 80 units of Scrap  
Then 80 units of scrap are produced and 320 units of steel are produced  
[\[Functional Test #93\]](#)

Consequently, there may be a scenario where there is demand only for {Scrap}. Producing {Scrap} also produces {Steel} in large quantity, which is not prudent. In that case, when demand is incident on {Scrap} material node, planners can restrict the production of {Scrap} through the activity {Steel Making Process}. Demands on {Scrap} material node are thus fulfilled by the available material that was produced as a result of demands being incident on {Steel} Material Node only.

### Configuration

The co-product can be modeled in the production graph which connects the activity node with the material node. This is available only for manufacturing graphs. A new measure “No Build” can be specified in the Material Production Graph. This measure indicates whether upstream propagation for corresponding material node is desired or not. This measure takes boolean values

- TRUE:

When No Build = true for an activity (A1) - material node (M1) combination, then only for those orders that use A1-M1 in this primary path, the activity A1 will be restricted to produce at material node M1 where M1 is a co-product. However if alternate activities are configured, the same order will be solved using that path unless sufficient stock is already present from previous co-production. [\[Functional Test #97\]](#)

- FALSE[default]

When No Build = false for an activity (A1) - material node (M1) combination, then A1 is allowed to produce at M1 irrespective of orders being placed at any material node. A1 produces at M1 in a ratio specified in the Yield measure Group. [\[Functional Test #99\]](#)

*Note : A null value has the same behavior as false.*

Material Production Graph has the following granularity

1. Material Node
  - a. Item.[Item]
  - b. Location.[Location]
2. Activity Node
  - a. Activity.[Activity]
3. Version.[Version Name]

Subsequently No Build Measure name needs to be specified in the Solver config as follows

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Select corresponding measure name from drop-down menu.

Prod Graph No Build Measure Name ⓘ  ▼

In a scenario where demands are not getting fulfilled when NoBuild is set to true, the corresponding constraint is reflected in the Constraint Report as NoBuild.

## Limitations

Currently, time varying yields for co-products are not supported. A workaround is to model date effective alternate activities.

## Demand based Network Selection

### Business Need

Certain demand requests have very specific source from where the material can be procured or manufactured. For example, all requests have China as the primary source, but certain specific requests might not want to get the supplies from China but would prefer getting the supplies from Europe. In some businesses, planners need plan stability, where certain orders planned from certain location are always maintained. This requires the planners to override the static network priorities at the demand level. Demand based network selection enables configuring the priorities of the path at the demand level which overrides the static priorities.

### Limiting Safety Stock Planning

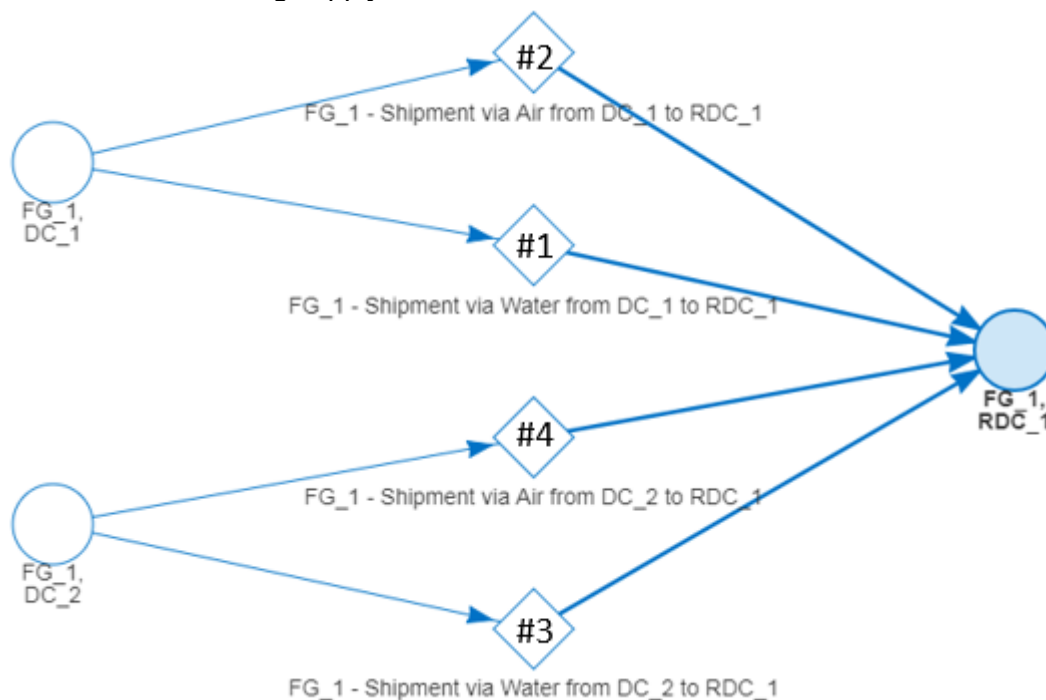
This concept can also be applied when safety stock planning should not be done along certain paths. For example, Target Inventory should never be procured via Air and

should only be done through Water/Land. Planners may have activities that cater to all the three routes . However, they can block the activities with respect to Air Procurement for safety stock planning.

### Functional Description

Consider a single level supply chain where a producing operation has 4 alternates (A1, A2, A3, A4) namely primary, secondary, tertiary and quaternary with priority 1, 2, 3, 4 respectively. Depending on the demand signal (usually represented by Order, customer, item, location, time) one needs to specify the priority in the network and also block certain routes which are not desirable. This could be at stage (level) 1 of supply chain. As it propagates upstream, there could be similar cases at different levels of the supply chain.

Consider the following supply chain:



A Demand incident at FG\_1, RDC\_1 can be fulfilled by the following activities (in that order of priority)

1. FG\_1 - Shipment via Water from DC\_1 to RDC\_1
2. FG\_1 - Shipment via Water from DC\_2 to RDC\_1
3. FG\_1 - Shipment via Air from DC\_1 to RDC\_1
4. FG\_1 - Shipment via Air from DC\_2 to RDC\_1

All orders incident on FG\_1, RDC\_1 across the planning horizon will use the above mentioned activities in the order of their priority unless specified otherwise.

However, planners might have some orders that should not use air transport and some orders that should procure from DC\_2 and not DC\_1. In that case , following are the 3 networks that planners can run in a single SCS run.

1. Generic network for all order across the planning horizon
2. Customized network for orders that don't use air transport
3. Customized network for orders that procure form DC\_2 and not DC\_1

In the presence of demand-attribute-based-path-selection, the pegging mechanism differs from that of the usual planning process.

|                                                                                                                                                                                  |                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Consists data with respect to demand and/or safety stock. This data would give visibility if the demand is to be planned via another path and not using the originally set path. | SCS plans and pegs this demand before moving on to the next demand                                                                        |
| Has no data{Empty table}                                                                                                                                                         | All the demands in the scope are planned first. After planning of demands is done, Pegging operation is performed on the planned demands. |

## Configuration Steps

A measure group Network Customization needs to be configured

### 1. Measure

#### a. RevisedPriority

This measure takes numeric values:

- 0 : the corresponding activity is blocked for the specified order [\[Functional Test #144\]](#)
- Null: takes the static priority described for material node - activity combination in material production graph
- n>1: priority of the corresponding activity is Material Production Graph is replaced by n, for this specific order [\[Functional Test #143\]](#)

### 2. Granularity

#### a. Demand Node

- Order.Order Name
- Time.Fiscal Week
- Customer.Customer Name

#### b. Material Node

- Item.Item Name<item that has been ordered>
- Location.Location Name

#### c. Activity Node

- Activity. Activity Name

#### d. Version.Version Name

Same should be specified in the SCS config as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Demand Activity Priority Measure Name → Select corresponding measure name from the drop down menu

Demand Activity Priority Measure Name

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Distribution → Demand Distribution Activity Priority Measure Name → Select corresponding measure name from the drop down menu

Demand Distribution Activity Priority Measure Name

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Procurement → Demand Supplier Activity Priority Measure Name → Select corresponding measure name from the drop down menu

Demand Supplier Activity Priority Measure Name 

For the above mentioned use case, data in the Network Customization measure group looks as follows:

- For Order\_5(does not use air transport)

| Order Name | Customer Name | Item | Location | Activity                                     | Data            | W30-2015 |
|------------|---------------|------|----------|----------------------------------------------|-----------------|----------|
| Order_5    | Cust_1        | FG_1 | RDC_1    | FG_1 - Shipment via Air from DC_1 to RDC_1   | RevisedPriority | 0        |
|            |               |      |          | FG_1 - Shipment via Air from DC_2 to RDC_1   | RevisedPriority | 0        |
|            |               |      |          | FG_1 - Shipment via Water from DC_1 to RDC_1 | RevisedPriority | 1        |
|            |               |      |          | FG_1 - Shipment via Water from DC_2 to RDC_1 | RevisedPriority | 2        |

- For Order\_2(procures form DC\_2 and not DC\_1)

| Order Name | Customer Name | Item | Location | Activity                                     | Data            | W40-2015 |
|------------|---------------|------|----------|----------------------------------------------|-----------------|----------|
| Order_2    | Cust_1        | FG_1 | RDC_1    | FG_1 - Shipment via Air from DC_1 to RDC_1   | RevisedPriority | 0        |
|            |               |      |          | FG_1 - Shipment via Air from DC_2 to RDC_1   | RevisedPriority | 2        |
|            |               |      |          | FG_1 - Shipment via Water from DC_1 to RDC_1 | RevisedPriority | 0        |
|            |               |      |          | FG_1 - Shipment via Water from DC_2 to RDC_1 | RevisedPriority | 1        |

When considering attribute based planning for safety stock , planners can use this existing Measure Group such that they set Order Name and Customer Name both as Inventory Demand. In such a case, when SCS will adhere to the revised priorities for planning safety stock at a material node.

For example, Safety Stock of 50 units is to be planned on Material node {FG\_1, RDC\_1} on time bucket {W32-2015}. A Safety Stock incident at FG\_1, RDC\_1 can be fulfilled by the following activities (in that order of priority)

1. FG\_1 - Shipment via Water from DC\_1 to RDC\_1
2. FG\_1 - Shipment via Water from DC\_2 to RDC\_1
3. FG\_1 - Shipment via Air from DC\_1 to RDC\_1
4. FG\_1 - Shipment via Air from DC\_2 to RDC\_1

However, planners may decide that safety stock procurement should only be done via location {DC\_2} and never from {DC\_1}. In that case, they enter the data in the Network Customization measure group as follows.

| Order Name       | Customer Name    | Item | Location | Activity                                     | Data            | W32-2015 |
|------------------|------------------|------|----------|----------------------------------------------|-----------------|----------|
| Inventory Demand | Inventory Demand | FG_1 | RDC_1    | FG_1 - Shipment via Air from DC_1 to RDC_1   | RevisedPriority | 0        |
|                  |                  |      |          | FG_1 - Shipment via Air from DC_2 to RDC_1   | RevisedPriority | 2        |
|                  |                  |      |          | FG_1 - Shipment via Water from DC_1 to RDC_1 | RevisedPriority | 0        |
|                  |                  |      |          | FG_1 - Shipment via Water from DC_2 to RDC_1 | RevisedPriority | 1        |

In the presence of demand attribute based path selection, the pegging mechanism differs from that of the full plan.

| Network Customization Measure Group                                                                                                                                    | Planning vs Pegging                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Consists data with respect to demand and/or safety stock. This data would give visibility if the demand is to be planned via another path and not using the originally | SCS plans and pegs a demand before moving on to the next demand |

|                     |                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------|
| set path.           |                                                                                                           |
| Has no data(0 rows) | All the demands are planned first. After planning, Pegging operation is performed on the planned demands. |

## Demand Attribute based date effectiveness

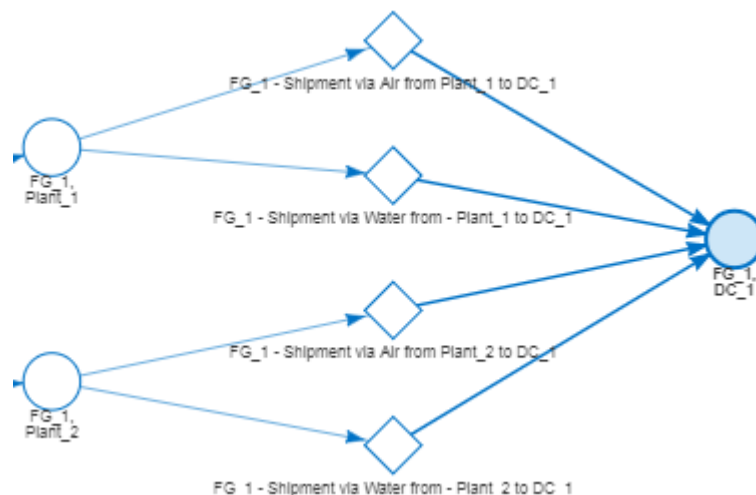
### Business Need

Consider a qualification use case where an item can be sourced from different factories. Depending on the customer, the qualification schedule on a factory can be different. For example, item1 can be produced from factory A and factory B. The factory B is a new one. Customer C1 has already qualified factory B. Customer C2 is going to qualify the factory B by the 4th week from the current time. Customer C3 is going to qualify the factory B on the 6th week from the plan current. These use cases are supported using demand attribute based date effectiveness on BOD, BOM and/or procurement.

### Functional Description

SCS enables the planners to assign date effectiveness on activities. Planners can assign start date and end date on an activity A1 thereby specifying that this Activity A1 is active only between its specified start time and end time. However, this date effectiveness setting on an activity remains static for all demands using this activity. Demand attribute based date effectiveness relaxes that assumption and now the user can specify the effectiveness of start and end dates of an activity at the demand level. In such a case, the initial static setting of date effectiveness on the activity gets overridden by the latter setting on the same activity but only for that demand.

For example, consider the following supply chain



A Demand incident at {FG\_1, DC\_1} can be fulfilled from supplies available at either Plant\_1 or Plant\_2 via the following 4 activities

1. FG\_1 - Shipment via Water from Plant\_1 to DC\_1
2. FG\_1 - Shipment via Water from Plant\_2 to DC\_1
3. FG\_1 - Shipment via Air from Plant\_1 to DC\_1
4. FG\_1 - Shipment via Air from Plant\_2 to DC\_1

Let the planning horizon be from W01-2015 to W52-2015



Planners can have a static setting such that Plant\_1 is active for the 2nd month in the planning horizon and Plant\_2 is not active within the planning horizon. In such a case static date effectivity looks as follows:

| Activity                                         | Solver Activity Start Date | Solver Activity End Date |
|--------------------------------------------------|----------------------------|--------------------------|
| FG_1 - Shipment via Air from Plant_1 to DC_1     | 01/02/2015                 | 01/03/2015               |
| FG_1 - Shipment via Air from Plant_2 to DC_1     | 01/01/2014                 | 01/01/2014               |
| FG_1 - Shipment via Water from - Plant_1 to DC_1 | 01/02/2015                 | 01/03/2015               |
| FG_1 - Shipment via Water from - Plant_2 to DC_1 | 01/01/2014                 | 01/01/2014               |

This static setting of date effectivity applies to all orders that are to be planned within the planning horizon.

However, **planners can override this static setting** such that different demands can source from the same material node at different time buckets.

| Order Name | Customer Name | Item | Location | Fiscal Week | Activity                                         | Demand Based Activity Start Date | Demand Based Activity End Date |
|------------|---------------|------|----------|-------------|--------------------------------------------------|----------------------------------|--------------------------------|
| Order_1    | Cust_1        | FG_1 | RDC_1    | W29-2015    | FG_1 - Shipment via Air from Plant_1 to DC_1     | 01/01/2014                       | 01/01/2014                     |
|            |               |      |          |             | FG_1 - Shipment via Air from Plant_2 to DC_1     | 01/02/2015                       | 01/03/2015                     |
|            |               |      |          |             | FG_1 - Shipment via Water from - Plant_1 to DC_1 | 01/01/2014                       | 01/01/2014                     |
|            |               |      |          |             | FG_1 - Shipment via Water from - Plant_2 to DC_1 | 01/02/2015                       | 01/03/2015                     |

It can be inferred from the above image that contrary to the static setting, for Order\_1, Plant\_2 is active for the 2nd month in the planning horizon and Plant\_1 is not active within the planning horizon.

Thus if there are 2 orders, both incident on material node as given below

| Order Name | Customer Name | Item | Location | Fiscal Week | Demand Quantity |
|------------|---------------|------|----------|-------------|-----------------|
| Order_1    | Cust_1        | FG_1 | RDC_1    | W29-2015    | 100             |
| Order_2    | Cust_2        | FG_1 | RDC_1    | W29-2015    | 100             |

And when unconstrained plan in run on the above orders,

Order\_1 sources from Plant\_2

| from.[Time]<br>[Fiscal Week] | from.[Item]<br>[Item] | from.[Location]<br>[Location] | from.[Activity].[Activity]                                  | to.[Item]<br>[Item] | to.[Location]<br>[Location] | to.[Time]<br>[Fiscal Week] | to.[Order]<br>[Order Name] | Data                                | Value  |
|------------------------------|-----------------------|-------------------------------|-------------------------------------------------------------|---------------------|-----------------------------|----------------------------|----------------------------|-------------------------------------|--------|
| W04-2015                     | RM_1                  | Plant_2                       | Expected Receipt Activity(1)                                | FG_1                | RDC_1                       | W29-2015                   | Order_1                    | Material Production Pegged Quantity | 100.00 |
|                              | RM_2                  | Plant_2                       | Expected Receipt Activity(1)                                | FG_1                | RDC_1                       | W29-2015                   | Order_1                    | Material Production Pegged Quantity | 100.00 |
| W05-2015                     | INT_11                | Plant_2                       | Sub-Assembly for FG_1 - RM_1 converted to INT_11 at Plant_2 | FG_1                | RDC_1                       | W29-2015                   | Order_1                    | Material Production Pegged Quantity | 100.00 |
|                              | INT_21                | Plant_2                       | Sub-Assembly for FG_1 - RM_2 converted to INT_21 at Plant_2 | FG_1                | RDC_1                       | W29-2015                   | Order_1                    | Material Production Pegged Quantity | 100.00 |
| W06-2015                     | FG_1                  | Plant_2                       | Final Assembly for FG_1 at Plant_2                          | FG_1                | RDC_1                       | W29-2015                   | Order_1                    | Material Production Pegged Quantity | 100.00 |
| W08-2015                     | FG_1                  | DC_1                          | FG_1 - Shipment via Water from - Plant_2 to DC_1            | FG_1                | RDC_1                       | W29-2015                   | Order_1                    | Material Production Pegged Quantity | 100.00 |
| W29-2015                     | FG_1                  | RDC_1                         | FG_1 - Shipment via Water from DC_1 to RDC_1                | FG_1                | RDC_1                       | W29-2015                   | Order_1                    | Material Production Pegged Quantity | 100.00 |

But Order\_2 sources from Plant\_1

| from.[Time]<br>[Fiscal Week] | from.[Item]<br>[Item] | from.[Location]<br>[Location] | from.[Activity].[Activity]                                  | to.[Item]<br>[Item] | to.[Location]<br>[Location] | to.[Time]<br>[Fiscal Week] | to.[Order]<br>[Order Name] | Data                                | Value  |
|------------------------------|-----------------------|-------------------------------|-------------------------------------------------------------|---------------------|-----------------------------|----------------------------|----------------------------|-------------------------------------|--------|
| W23-2015                     | RM_1                  | Plant_1                       | Expected Receipt Activity(1)                                | FG_1                | RDC_1                       | W29-2015                   | Order_2                    | Material Production Pegged Quantity | 100.00 |
|                              | RM_2                  | Plant_1                       | Expected Receipt Activity(1)                                | FG_1                | RDC_1                       | W29-2015                   | Order_2                    | Material Production Pegged Quantity | 100.00 |
| W24-2015                     | INT_11                | Plant_1                       | Sub-Assembly for FG_1 - RM_1 converted to INT_11 at Plant_1 | FG_1                | RDC_1                       | W29-2015                   | Order_2                    | Material Production Pegged Quantity | 100.00 |
|                              | INT_21                | Plant_1                       | Sub-Assembly for FG_1 - RM_2 converted to INT_21 at Plant_1 | FG_1                | RDC_1                       | W29-2015                   | Order_2                    | Material Production Pegged Quantity | 100.00 |
| W25-2015                     | FG_1                  | Plant_1                       | Final Assembly for FG_1 at Plant_1                          | FG_1                | RDC_1                       | W29-2015                   | Order_2                    | Material Production Pegged Quantity | 100.00 |
| W27-2015                     | FG_1                  | DC_1                          | FG_1 - Shipment via Water from - Plant_1 to DC_1            | FG_1                | RDC_1                       | W29-2015                   | Order_2                    | Material Production Pegged Quantity | 100.00 |
| W29-2015                     | FG_1                  | RDC_1                         | FG_1 - Shipment via Water from DC_1 to RDC_1                | FG_1                | RDC_1                       | W29-2015                   | Order_2                    | Material Production Pegged Quantity | 100.00 |

Order\_2 follows the static supply chain priorities whereas Order\_1 uses the demand attribute based date effectiveness.[FT#394]

## Configuration Steps

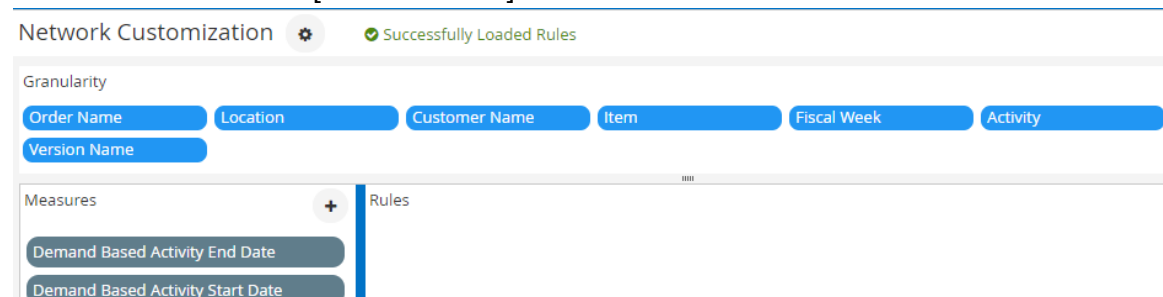


Demand Attribute Based Date Effectivity can be provided in the following measure group:

1. Measure
  - a. Demand based Activity Start Date
    - i. This measure takes datetime values. The value of this measure corresponding to a demand override the value of the measure given in Solver Activity Start date
  - b. Demand Based Activity End Date
    - i. This measure takes datetime values. The value of this measure corresponding to a demand override the value of the measure given in Solver Activity End date.

Values given for Demand Based Activity Start Date and Demand Based Activity End date for a demand take precedence over values for Solver Activity Start date and Solver Activity end Date respectively.

2. Granularity
  - a. Demand Node
    - i. Order.[Order Name]
    - ii. Customer.[Customer Name]
    - iii. Time.[Fiscal Week]
  - b. Activity Node
    - i. Activity.[Activity]
  - c. Material Node
    - i. Item.[Item]
    - ii. Location.[Location]
  - d. Version.[Version Name]



Same should be specified in the SCS config UI as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Demand Activity Start Date Measure Name → choose measure from the drop down menu

Demand Activity Start Date Measure Name Demand Based Activity Start Date

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Demand Activity End Date Measure Name → choose measure from the drop down menu

Demand Activity End Date Measure Name Demand Based Activity End Date

### Limitations

For a given demand, an activity can have only one value for measures Demand Activity Start Date and Demand Activity End Date respectively. For example for a given demand

Order\_1, activity “FG\_1 - Shipment via Water from - Plant\_2 to DC\_1” can only have 1 value for its corresponding measures Demand Based Activity Start Date and Demand Based Activity End Date. Even if planners provide multiple values for these measures using scope statements, GraphcubeServer assigns the most recently assigned values to these measures.

| Order Name | Customer Name | Item | Location | Fiscal Week | Activity                                     | Demand Based Activity Start Date | Demand Based Activity End Date |
|------------|---------------|------|----------|-------------|----------------------------------------------|----------------------------------|--------------------------------|
| Order_1    | Cust_1        | FG_1 | RDC_1    | W29-2015    | FG_1 - Shipment via Air from Plant_2 to DC_1 | 01/02/2015                       | 01/03/2015                     |

The workaround for this limitation would be to configure duplicate activities and then assign values to its corresponding Demand Based Activity Start Date and Demand Based Activity End Date.

## Attribute Based Mandatory Prebuild Business Need

In bucketized planning, there are implicit assumptions that all demands and supplies are feasible as long as they arrive or leave a material node at any point in the bucket. However, in reality, certain activities have specified scheduling constraints such that the supplies can consume or produce only in the beginning of bucket or end of bucket. For example, certain distribution BOD Lanes may have shipping schedules that only depart at the end of the bucket or arrive at the end of a bucket. This means that the bucketized plan when released for execution may be infeasible, as in reality the supply will be arriving too late for the demand to be met on time. For this case, the supply plans for the specified BOD Lanes need to be offset earlier by typically 1 bucket in order to have a feasible plan released to the execution system.

### Functionality Description

Supply Chain Solver considers the offset bucket as hard constraint. There will be certain activities (mostly distribution or procurement) which will carry the offset value specified through a measure. While planning, solver respects this offset and makes sure the production to the consuming material node happens before this.

For example,

<MN1> ----- <ACT1> ----- <MN2> -----<ACT2> -----<MN3>

Here <ACT2> has offset set to 1 bucket.

Leadtime for <ACT2> is 2 buckets.

Demand D1 at <MN3> is on W10 for 100 units.

Producing Flow Plans at <MN2>

| <Activity> | <Material Node> | <Time Bucket> | Produced Quantity |
|------------|-----------------|---------------|-------------------|
| ACT1       | MN2             | W7            | 100               |

Consuming Flow Plans at <MN2>

| <Activity> | <Material Node> | <Time Bucket> | Consumed Quantity |
|------------|-----------------|---------------|-------------------|
| ACT2       | MN2             | W8            | 100               |

### Production Pegging

| < Activity> | < Material Node> | <Time Bucket> | <Demand > | <Demand Due> | Production Pegged Quantity |
|-------------|------------------|---------------|-----------|--------------|----------------------------|
| ACT1        | MN2              | W7            | D1        | W10          | 100                        |

### Consumption Pegging

| < Activity> | < Material Node> | <Time Bucket> | <Demand > | <Demand Due> | Consumption Pegged Quantity |
|-------------|------------------|---------------|-----------|--------------|-----------------------------|
| ACT2        | MN2              | W8            | D1        | W10          | 100                         |

## Configuration

1. Define a measure with granularity same as activity node to specify the pre build offset value

2. Add the measure to the SCS config

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing →

Activity Mandatory Prebuild Offset Measure Name → Select corresponding measure from the drop down menu

|                  |            |      |        |           |               |              |             |
|------------------|------------|------|--------|-----------|---------------|--------------|-------------|
| Instance Options | Parameters | Node | Demand | Inventory | Manufacturing | Distribution | Procurement |
|------------------|------------|------|--------|-----------|---------------|--------------|-------------|

Activity Mandatory Prebuild Offset Measure Name ⓘ
 

Mandatory PreBuild Bucket Offset ▼

For distribution and procurement activities (if defined with different nodes), the measure needs to be linked to the corresponding config parameters as shown below.

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Distribution → Activity Mandatory Prebuild Offset Measure Name → Select corresponding measure from the drop down menu

Distribution Activity Mandatory Prebuild Offset Measure Name ⓘ

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Procurement → Activity Mandatory Prebuild Offset Measure Name → Select corresponding measure from the drop down menu

Supplier Activity Mandatory Prebuild Offset Measure Name ⓘ

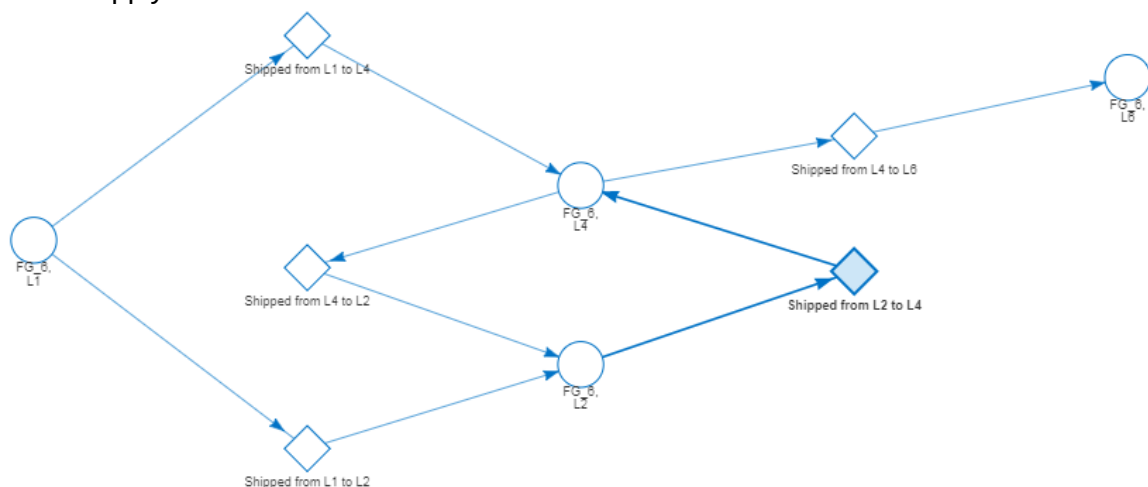
## Cyclicity

### Business Need

Planners can design supply chain network where goods can flow amongst locations at the same and/or different levels thereby giving rise to closed cyclical loops. Business may decide to move inventory from Warehouse 1 to Warehouse 2 or vice versa depending on where there is more stock and hence need to model it as part of the static supply chain structure.

### Functional Description

Consider the following supply chain. The red box in the diagram below encloses a cycle in the supply chain.



From this graph, SCS infers that if some stock is needed at material node {FG\_6, L4}, it can be satisfied from either of the material nodes {FG\_6, L1} or {FG\_6, L2}. Similarly if some stock is needed at {FG\_6, L2}, it can be satisfied from either of the material nodes {FG\_6, L1} and {FG\_6, L4}. When two or more material nodes are individually capable

enough to satisfy the stock requirement at {FG\_6, L4}, then SCS adheres to the priorities of the corresponding activities, picking activities of higher priorities first. When material is available such that two or more material nodes are collectively capable of satisfying the request coming from the material node, then SCS takes into account both these material nodes. For eg. if stock at {FG\_6,L1} and {FG\_6, L4} are both sufficient (but individually insufficient) to satisfy a stock requirement at {FG\_6, L2}, then SCS takes both. In short when SCS encounters a cycle in the supply chain, it breaks the cycle by backtracking the path until it reaches the same node and they go upstream on the next available path.

## **Configuration**

None

## **Demand**

One of the primary inputs into the solver is the demand signals. One can represent demand signal as a composite node which is incident on the material node. Usually we have seen that it is represented using Demand, Time, Customer, Item, Location as the keys.

## **Demand Priority**

### **Business Need**

Planners need the ability to say the relative priority of demands. This can be done based on channel, product, time etc. Demand prioritization is a separate process which can be fine tuned to obtain the priority of demands, which is passed as an input to the solver.

### **Functional Description**

SCS is an order by order solver, which picks up one demand at a time based on demand priority and traverses the supply chain based on the specified rules and consumes material and capacity along its path.

## **Configuration**

Demand Priority is defined as a measure. This Measure is maintained in the measure group Demand Quantity. The measure group in turn has the granularity as follows :

1. Demand Node
  - a. Demand ID/Order ID
  - b. Time
  - c. Customer
2. Material Node
  - a. Item
  - b. Location
3. Version

Demand Priority takes numeric values. Lower number denotes higher priority. For same priority, demand with earlier due dates are planned first. When both due date and priority are same, SCS sorts the orders by demand quantity (ascending).

Demands are sorted in the following order of the demand attributes:

1. Demand Priority
2. Demand Due Date: if demand priority measure group has no value(NULL) for demands or if 2 or more demands have the same priority then demands are sorted by their respective due date in ascending order

3. Demand Quantity: if 2 or more demands have the same priorities and the same due dates then demands are sorted in increasing order of the demand quantity.

If there are 2 demands incident on the

- Same material node (hence same supply chain)
- Same time bucket
- Same priority

But

- Unique demand quantity

Then

Orders are picked up on the basis of the demand quantity {in ascending order}[[Functional Test 341](#)]

If there are 2 demands incident on the

- Different material nodes (but same supply chain)
- Same time bucket
- Same priority

But

- Unique demand quantity

Then

Orders are picked up on the basis of the demand quantity {in ascending order}[[Functional Test 342](#)]

## Build Ahead Limit

### Business Need

There are occasions when demands cannot be met by planning JIT. In that case, usually the planners are fine with consuming material and capacity in earlier time buckets. There are also class of demands (perishable items) where one does not want to build it very early and would specify the limits at a global level or at demand specific level.

### Functional Description

SCS is capable of searching for material and capacity in earlier buckets. The specified time can be provided at a demand level in terms of buckets. Planners can define a build ahead limit that is universal for all demands or can specify individual build ahead limits for each demand. Build Ahead Limit is agnostic of Expected receipts present at upstream material nodes at earlier time buckets. This is because Expected Receipts, if not used, is always carried forward to the next time bucket for the same material node.[ft 361]

### Configuration

Global Build Ahead Limit

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Build Ahead Limit → Set value

Build Ahead Limit 

Build Ahead Limit (BAL) takes numeric values. The number denotes the maximum number of earlier time buckets an order can search for supplies before the JIT bucket after accounting for lead times. It can also be passed as a dynamic argument to the plan run of SCS. By default, BAL is infinite, which means it will build ahead until the current bucket is reached. Global Build ahead Limit applies to all demands given in the command line signature.

## Local Build Ahead Limit

Local build ahead limit are set for individual demands. This is done through measures.

- Measure
  - Demand Build Ahead Limit
- Granularity
  - Demand Node
    - Order
    - Time
    - Customer
  - Material Node
    - Item
    - Location
  - Version

Local Build Ahead Limit is a measure that has the same granularity as demand quantity. Therefore, Local Build Ahead Limit can be maintained in the same measure group as demand quantity. This measure takes numeric values. The number denotes the maximum number of earlier time buckets an order can search for supplies before the JIT bucket after accounting for lead times. Consider the following local build ahead limits:

| Order Name | Customer Name | Item | Location | Data            | W06-2015 | W10-2015 | W14-2015 |
|------------|---------------|------|----------|-----------------|----------|----------|----------|
| Order_1    | Cust_1        | FG_1 | RDC_1    | Demand Quantity | 100      |          |          |
|            |               |      |          | Demand BAL      | 0        |          |          |
| Order_2    | Cust_1        | FG_1 | RDC_1    | Demand Quantity |          | 100      |          |
|            |               |      |          | Demand BAL      |          | 5        |          |
| Order_3    | Cust_1        | FG_1 | RDC_1    | Demand Quantity |          |          | 100      |
|            |               |      |          | Demand BAL      |          |          |          |

Order\_1 has a Local BAL set to 0. This means SCS will only allow this demand to be built in JIT or late but never will this demand be build ahead of its due bucket.

Order\_2 has a Local BAL set to 0. SCS will try to solve this demand JIT. If not, then this demand will be build 1, 2, 3, 4 or at max 5 time buckets ahead of its due bucket {whichever bucket is closest to JIT bucket}. If demand is not fulfilled even 5 buckets ahead of JIT bucket, SCS won't proceed to solving this demand 6 time buckets ahead of its due bucket. SCS in turn would start building the order late until build late limit is reached.

Order\_3 has a Local BAL set to Null. This means if SCS can build this order in JIT bucket, then SCS is allowed to build the order any buckets ahead of its JIt bucket until start of planning horizon is reached.

If both Local and Global BAL are provided by the planners i.e if both parameters have non-Null values, then Local BAL takes precedence over Global BAL.

## Build Late Limit

### Business Need

There are occasions when demands cannot be met on time, due to constrained supply of material or capacity or both. Planners can specify the backlog limit on certain class of

demands. It denotes the number of buckets upto which the demands can be made late from the specified due date.

### Functional Description

SCS is capable of building an order later than the due date. The build late limit (BLL) is specified in terms of buckets. Planners can specify Build Late Limit specific to each demand or specify a universal Build Late Limit that applied to all demands. The former takes precedence over the latter.

### Configuration

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Build Late Limit → Set value

Build Late Limit ⓘ

Build Late limit takes numeric values. The number denotes the maximum time buckets an order can be built late from its due date. It can also be passed as a dynamic argument to the plan run of SCS.

## Delay Demand Planning Performance Optimizations

### Business Need

There are occasions when demands cannot be met on time, due to constrained supply of material or capacity or both. In that case these can be built late if delay is allowed. When delay demand planning happens, the enhancement provides opportunity for solver performance optimization .

### Functional Description

SCS is capable of building an order later than the due date. As of previous release it used to follow a late plan strategy called LINEAR. In the current release, two more new flavours are added as mentioned below. The optimal strategy can vary due to supply chain characteristics, ie cost of a successful probe vs cost of quick failure (to find supply), so some experimentation is necessary to find optimal strategy for a given supply chain.

#### LINEAR

This is the default behaviour. After planning on the JIT, if the demand is short or partially met, planner will plan on JIT+1, JIT+2... till the Build Late Limit or end of planning horizon is reached or demand quantity is fully met.

#### BINARY\_NEIGHBORHOOD

In this strategy, if the demand is

- Totally short on JIT:  
SCS will do a binary search from JIT+1 to JIT+Build\_Late\_Limit and find out the earliest bucket on the late planning horizon where some demand planning is possible and plan the demand on that bucket followed by a sequential planning and then switch to BINARY.
- Partially met on JIT:  
SCS will do a NEIGHBORHOOD planning from JIT+1 to



JIT+Build\_Late\_Limit until the demand is satisfied or on a bucket where there is zero demand satisfaction. At this point SCS will do a BINARY probe from the next bucket to JIT+Build\_Late\_Limit bucket.

#### **UNIFORM\_INTERVAL**

In this strategy an interval is chosen as the square root of the Build Late Limit value and probe is done on those intervals starting from JIT, if there is any satisfaction on the probe, planning will be done in a sequential manner from the start of that interval to the end of the interval otherwise the interval will be discarded.

Here probe means SCS will do a planning followed by unplanning on that bucket to find out if there is any demand satisfaction possible.

#### **Example**

Let there be a demand D1 of quantity 10 due on W5, Build Ahead Limit is 4 and Build Late Limit is 10. Let due to capacity constraint, following quantity satisfaction is possible across buckets W2: 2, W3: 2, W10:2, W11:2, W14:2. So JIT planning horizon is from W1 to W5 and Late planning horizon is from W6 to W15. Behaviour of JIT planning will be same for all the three above. Behaviour of late planning will be as follows:

#### **LINEAR**

After JIT planning, planner will plan on all the buckets from W6 to W14. Planner will not search on further buckets as the demand is fully met on W14.

#### **BINARY\_NEIGHBORHOOD**

As there is fulfillment of quantity 4 on JIT (Note though there is zero satisfaction on JIT bucket) planning, SCS will first try to plan on bucket W6 and will get zero fulfillment. Now it will do a binary search probe from bucket W7 to W15 (end of late planning horizon for the demand). This probe will return the bucket number 10 where some demand satisfaction is possible. Then SCS will plan on W10, at this point as planner gets some satisfaction (of qty 2), it will NOT do a binary search from W11 to W15 rather go for NEIGHBORHOOD and plan on W11. On W11 it will get satisfaction of quantity 2 and will again plan on W12 where it gets a satisfaction of zero. Now it will switch to binary probe mode. Binary probe will return the bucket W14, planning on this bucket, the demand will fully met.

#### **UNIFORM\_INTERVAL**

In this method an interval is computed as the square root of value of BLL. In this example the BLL value is 10, square root of 10 is 3 (rounded). Hence in this case interval value is 3. During late planning phase, SCS will probe on (JIT + 3) W8 with added Build Ahead Limit. It will get zero satisfaction on W8. So it will move to the end bucket of next interval i.e. W11. In this probe on W11th bucket, SCS will get

some satisfaction. Hence SCS will start planning on all the buckets of this interval i.e. W9 followed by W10 and W11. On bucket W9 there will be zero satisfaction, and 2 each on W10 and W11. As the demand is not fully satisfied till now, the probe will happen on the end bucket of next interval i.e. W14. As there is a satisfaction on 14th, planner will start planning on W12, W13 (zero satisfaction) and then 2 on W14. As the demand is satisfied fully on W14, SCS will exit now from the planning cycle for this demand.

## Configuration

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Late Plan Strategy → Select Value

Late Plan Strategy ⓘ

LINEAR ▼

Takes one of the three values i.e. LINEAR, UNIFORM\_INTERVAL or BINARY\_NEIGHBORHOOD

Define following plugin parameters in the SCS plugin invocation command as arguments  
 exec plugin instance [Supply Chain Solver] for Measures {[Demand Quantity Day]} using scope (Version.[Version Name].[CurrentWorkingView]) using arguments {[**Late Plan Strategy**],[**BINARY\_NEIGHBORHOOD**]};

## Demand Slice Factor

### Business Need

Business need to fairshare limited supply for demands of equal priority. When two or more competing demands cannot be differentiated by absolute priority, then the limited supplies should be fair shared to the extent possible.

### Functional Description

In this heuristic based Order-By-Order Solver, the demands are planned order by order. Thus demands with same priority are sliced based on a user specified factor called demand Slice factor (should be greater than 1), and one slice from each demand is processed first, before proceeding to the next slice.

Example: Say we have 2 orders

| Order_Name | Quantity |
|------------|----------|
| O1         | 100      |
| O2         | 100      |

By default, if the Demand slice factor is not mentioned by the user, the solver first solves for order O1 and then solves for Order O2, even if it is of the same priority. Now, say we have limited material for 100 units, then the Order O1 will be pegged to the material. Order O2 will be shorted as there isn't enough supply.

There are occasions when one needs to at least provide some supply to against requests from the limited supply. This is achieved by using the Demand Slice factor. If Demand Slice factor is 2, each order gets broken down into 2 sub-orders and solver solves for the sub orders in an order-by-order fashion.

| Order_Name | Quantity | Demand Slice factor = 2 | Order in which solver plans demand                   |
|------------|----------|-------------------------|------------------------------------------------------|
| O1         | 100      | 50                      | Solver solves 1st                                    |
|            |          | 50                      | Solver solves 3rd, but there isn't any material left |
| O2         | 100      | 50                      | Solver solves 2nd                                    |
|            |          | 50                      | Solver solves 4th, but there isn't any material left |

#### [\[Functional Test #43\]](#)

Say Demand Slice factor = 3. Each order gets broken down into 3 sub-orders and solver solves for broken Demands in an order-by-order fashion.

| Order_Name | Quantity | Demand Slice factor = 3 | Order in which solver plans demand. |
|------------|----------|-------------------------|-------------------------------------|
| O1         | 100      | 33.33                   | Solver solves 1st                   |
|            |          | 33.33                   | Solver solves 3rd                   |
|            |          | 33.33                   | 5th                                 |
| O2         | 100      | 33.33                   | Solver solves 2nd                   |
|            |          | 33.33                   | 4th                                 |
|            |          | 33.33                   | 6th                                 |

#### [\[Functional Test #44\]](#)

#### Configuration

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Demand Slice Factor → Set value

Demand Slice Factor    

Demand Slice Factor takes numeric values. The number denotes the total number of sub-orders a stand-alone order will be divided in order to achieve fairshare.

## Demand Slicing with Threshold

### Business Need

Planners need to fair share limited supply for demands of equal priority. However, small quantities of demands should not be included in fair share as those are minimal quantities ordered by some customers. This calls for slicing a demand for fair share only when the demand quantity is greater than a user specified threshold value.

### Functionality Description

Demand slicing should not happen for demands where quantity is less than or equal to the given threshold value. Users can provide a threshold value for a demand which can guide solver to slice the demand or not during fair share planning.

Example: Say we have 3 orders with same priority (sorted using key of the order dimension, in this case the order name)

| Order Name | Quantity |
|------------|----------|
| O1         | 100      |
| O2         | 10       |
| O3         | 100      |

By default, when slice factor is specified (for fair share planning), all the demands of same priority will be sliced and each slice gets planned one after the other (i.e. slice by slice planning).

Let's say slice factor=2 and slice threshold=10. Upstream supply is 100.

Here O2 will not be sliced and gets planned as a whole.  
O1 and O3 will be sliced to 50 each.

| Order Name | Quantity | Slice factor = 2 | Order in which solver plans demand                                        |
|------------|----------|------------------|---------------------------------------------------------------------------|
| O1         | 100      | 50               | Solver solves this order first                                            |
|            |          | 50               | Solver solves 4th, but there isn't any material left                      |
| O2         | 10       | 10               | Solver solves this 2nd. It does not get sliced due to slice threshold 10. |
| O3         | 100      | 50               | Solver solves 3rd. It plans 40 as there isn't any material left           |

|  |  |    |                                                      |
|--|--|----|------------------------------------------------------|
|  |  | 50 | Solver solves 5th, but there isn't any material left |
|--|--|----|------------------------------------------------------|

## Configuration

There is a Supply Chain Solver plugin parameter [Demand Slice Threshold] needs to be set to the required threshold value. User can specify this through command line argument or config UI as shown below.

**Demand Slice Threshold:** Demands with quantity less than or equal to the threshold value will not be sliced. Default is 0.

### Specifying the parameter as argument:

```
exec plugin instance SupplyChainSolver for Measures {[Demand Quantity]} using scope
([Version].[Version Name].[CurrentWorkingView] * Time.[Fiscal Week] * Order.[Order
Name]) arguments {[Material Constrained], "True"}, ([Capacity Constrained] , "False"),
([Delete SCS Plan Data For Version], "True"), ([Demand Slice Factor], 2), ([Intra Bucket
Slicing Only], "False"), ([Demand Slice Threshold], 10);
```

### Specifying the parameter in UI:

Provide the slice threshold value in the parameter on UI by accessing the plugin instance in the "Parameters" tab as shown below.

The screenshot shows a horizontal tab bar with the following tabs: Instance Options, Parameters (selected), Node, Demand, Inventory, Manufacturing, Distribution, and Procurement. Below the tabs, the 'Demand Slice Threshold' parameter is displayed with a value of 10. A small blue information icon is next to the parameter name. To the right of the value is a vertical slider control with up and down arrows.

## Intra bucket slicing

### Business Need

There are occasions when planners prefer to satisfy earlier demands first over demands with later due dates, even though the priority is the same.

### Functional Description

Intra bucket slicing is an extension of demand slice factor, to achieve the above business case. When Intra Bucket Slicing Only = FALSE (default value), all demands of same priority get broken down in the ratio of specified Demand Slice factor irrespective of the time buckets in which demand is due.

But when Intra Bucket Slicing Only = TRUE, Demands having same priority get broken down in the ratio of Demand Slice factor, but this time the slicing takes place Time-Bucket-wise. Therefore, earlier due date demands will implicitly get a higher priority.

**Intra Bucket Slicing Only:** It accepts Boolean values TRUE/FALSE (Default: False). When Intra Bucket Slicing Only = FALSE, Demands (having same priority) get broken

down in the ratio of the previously mentioned Demand Slice factor across all time buckets.

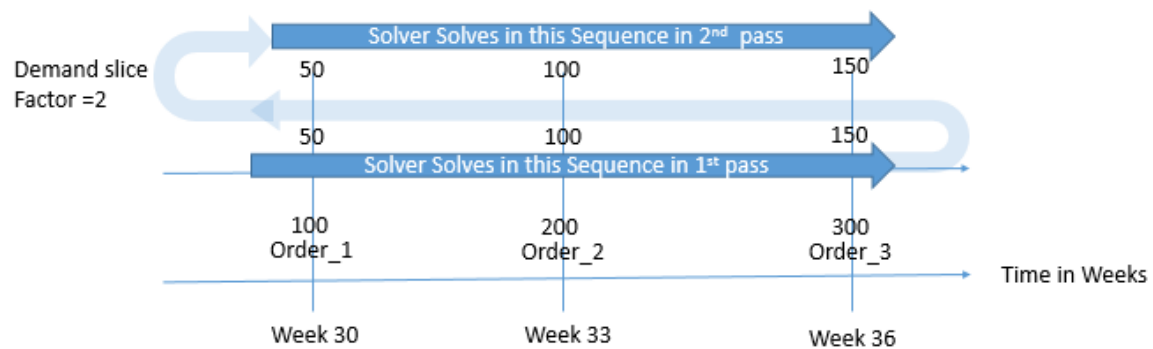
But when Intra Bucket Slicing Only = TRUE, Demands having the same priority get broken down in the ratio of Demand Slice factor, but this time the slicing takes place Time-Bucket-wise. Therefore, whichever demand is relatively due before will implicitly get a higher priority.

Consider there are 3 demands of same priorities in different time buckets.

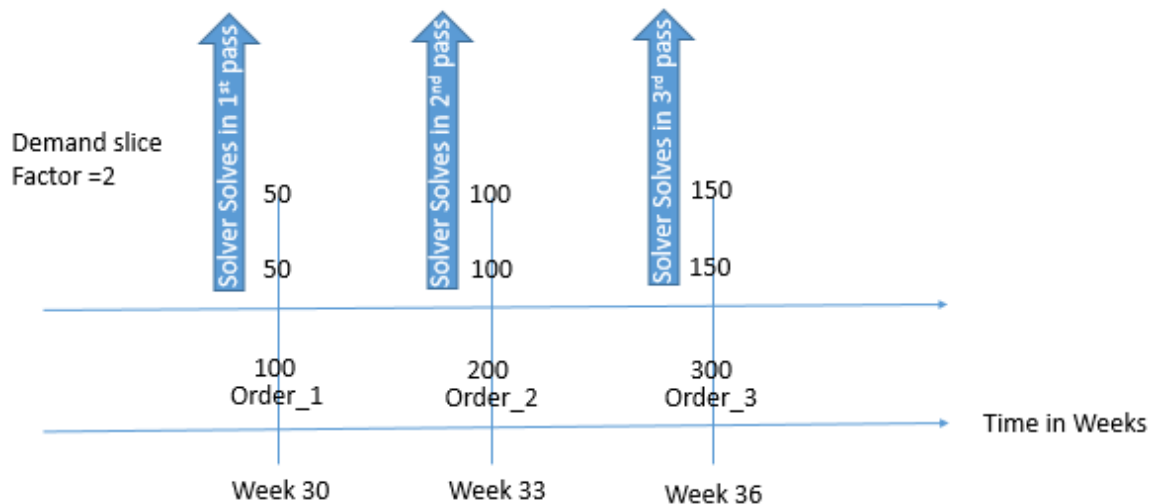
| Time Buckets   | Week 30 | Week 33 | Week 36 |
|----------------|---------|---------|---------|
| Order ID       | Order_1 | Order_2 | Order_3 |
| Order Quantity | 100     | 200     | 300     |

If Demand Slice Factor = 2 then depending on the value of Intra Bucket Slicing Only, SCS solves the orders in the following sequence

When Intra Bucket Slicing Only = FALSE



When Intra Bucket Slicing Only = TRUE



## Configuration

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Intra Bucket Slicing Only → Check if TRUE or Uncheck if False

Intra bucket slicing takes boolean values, TRUE or FALSE.

## Demand Trace

### Business Need

Implementers need to understand the behaviour of how the solver has traversed different paths and met/shorted the demand. This is a debug handle available to understand the solver behaviour as it plans for demands in an order by order fashion.

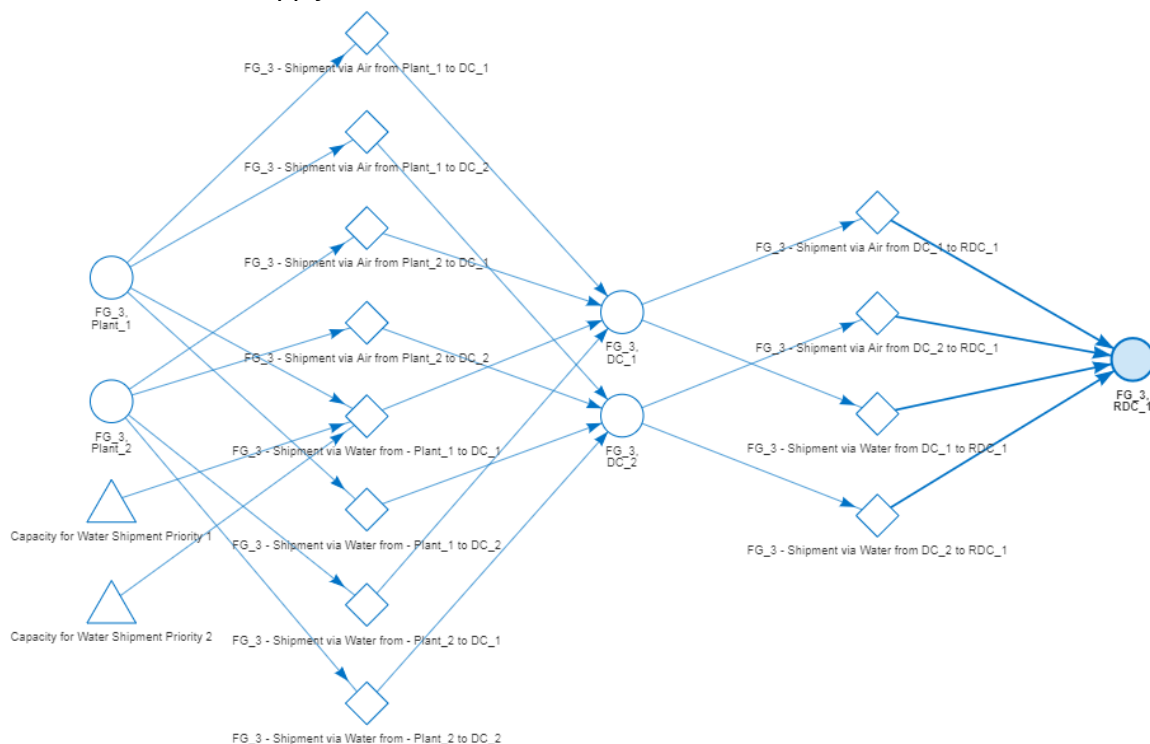
### Functional Description

SCS has a format for creating logs while solving demands. Nevertheless sometimes a more detailed log is needed for specific demands. This is modelled in the SCS in the form of Demand Trace, where just the specific demand or a group of demands can be traced for the path it traverses in the supply chain seeking material and capacity while honouring the network constructs.

### Configuration

Demand Trace is a measure that has the same granularity as demand quantity. Therefore, Demand Trace can be maintained in the same measure group as demand quantity. Demand Trace takes numeric values. If the Demand Trace for a Demand [Order\_1] is set to a value greater than 0, SCS will log the Supply Chain Traversal of that Order\_1 in detail unlike those demands whose Demand Trace is set to 0.

Consider a 3 level supply chain



And Demand Trace measure is defined as follows :

| Order Name | Customer Name | Item | Location | Data            | W30-2015 |
|------------|---------------|------|----------|-----------------|----------|
| Order...   | Cust_1        | FG_1 | VMI_1    | Demand Quantity | 100      |
|            |               |      |          | Demand Trace    | 1        |

When the Solver is run for the above Demand Quantity the log generated for the given demand is as follows:

|                         |                                                                 |                                                                                                          |
|-------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 2018-08-02 15:18:32.202 | TenantId:11250 UserId:192184 (10.0.1.242) INFO LiveServer (111) | [Segment : 0]. Finished plan generation                                                                  |
| 2018-08-02 15:18:32.202 | TenantId:11250 UserId:192184 (10.0.1.242) INFO LiveServer (111) | Completed Planning Demand]                                                                               |
| 2018-08-02 15:18:32.201 | TenantId:11250 UserId:192184 (10.0.1.242) INFO LiveServer (111) | Produced (100) units of (FG_3, RDC_1) @ Bucket (W30-2015)]                                               |
| 2018-08-02 15:18:32.201 | TenantId:11250 UserId:192184 (10.0.1.242) INFO LiveServer (111) | Produced (100) units at (FG_3 - Shipment via Water from DC_1 to RDC_1) @ Bucket (W30-2015)]              |
| 2018-08-02 15:18:32.201 | TenantId:11250 UserId:192184 (10.0.1.242) INFO LiveServer (111) | Produced (100) units of (FG_3, DC_1) @ Bucket (W25-2015)]                                                |
| 2018-08-02 15:18:32.201 | TenantId:11250 UserId:192184 (10.0.1.242) INFO LiveServer (111) | Produced (100) units at (FG_3 - Shipment via Water from - Plant_1 to DC_1) @ Bucket (W25-2015)]          |
| 2018-08-02 15:18:32.201 | TenantId:11250 UserId:192184 (10.0.1.242) INFO LiveServer (111) | Produced (100) units of (FG_3, Plant_1) @ Bucket (W20-2015)]                                             |
| 2018-08-02 15:18:32.200 | TenantId:11250 UserId:192184 (10.0.1.242) INFO LiveServer (111) | [Requested (100) units of (FG_3, Plant_1) @ Bucket (W20-2015)                                            |
| 2018-08-02 15:18:32.200 | TenantId:11250 UserId:192184 (10.0.1.242) INFO LiveServer (111) | [Requested (100) units at (FG_3 - Shipment via Water from - Plant_1 to DC_1) @ Bucket (W25-2015)         |
| 2018-08-02 15:18:32.200 | TenantId:11250 UserId:192184 (10.0.1.242) INFO LiveServer (111) | [Requested (100) units of (FG_3, DC_1) @ Bucket (W25-2015)                                               |
| 2018-08-02 15:18:32.200 | TenantId:11250 UserId:192184 (10.0.1.242) INFO LiveServer (111) | [Requested (100) units at (FG_3 - Shipment via Water from DC_1 to RDC_1) @ Bucket (W30-2015)             |
| 2018-08-02 15:18:32.200 | TenantId:11250 UserId:192184 (10.0.1.242) INFO LiveServer (111) | [Requested (100) units of (FG_3, RDC_1) @ Bucket (W30-2015)                                              |
| 2018-08-02 15:18:32.200 | TenantId:11250 UserId:192184 (10.0.1.242) INFO LiveServer (111) | [Planning Demand : (Order_1 for Material FG_3, RDC_1 and Quantity 100 with Priority 1 @ Bucket W30-2015) |
| 2018-08-02 15:18:32.199 | TenantId:11250 UserId:192184 (10.0.1.242) INFO LiveServer (111) | [Segment : 0]. Started plan generation                                                                   |

Order of Log vs Time

The above highlighted text when viewed in the logs extracted from env machine look indented and are as follows:

```

2017-03-10 09:55:59,758 (INFO) [4] (8891, 200698) - [Segment : 0]. Started plan generation
2017-03-10 09:55:59,758 (INFO) [4] (8891, 200698) - [Planning Demand : (Order_1 for IL FG_3 @ RDC_1 and Quantity 200 with Priority 1 @ Bucket W30-2015)
2017-03-10 09:55:59,758 (INFO) [4] (8891, 200698) - [Requested (200) units of (FG_3, RDC_1) @ Bucket (W30-2015)
2017-03-10 09:55:59,759 (INFO) [4] (8891, 200698) - [Requested (200) units at (FG_3 - Shipment via Water from DC_1 to RDC_1) @ Bucket (W30-2015)
2017-03-10 09:55:59,759 (INFO) [4] (8891, 200698) - [Requested (200) units of (FG_3, DC_1) @ Bucket (W25-2015)
2017-03-10 09:55:59,759 (INFO) [4] (8891, 200698) - [Requested (200) units at (FG_3 - Shipment via Water from - Plant_1 to DC_1) @ Bucket (W25-2015)
2017-03-10 09:55:59,759 (INFO) [4] (8891, 200698) - [Requested (200) units of (FG_3, Plant_1) @ Bucket (W20-2015)
2017-03-10 09:55:59,759 (INFO) [4] (8891, 200698) - [Requested (200) units of (FG_3, Plant_1) @ Bucket (W20-2015)]
2017-03-10 09:55:59,760 (INFO) [4] (8891, 200698) - Produced (200) units at (FG_3 - Shipment via Water from - Plant_1 to DC_1) @ Bucket (W25-2015)]
2017-03-10 09:55:59,760 (INFO) [4] (8891, 200698) - Produced (200) units of (FG_3, DC_1) @ Bucket (W25-2015)]
2017-03-10 09:55:59,760 (INFO) [4] (8891, 200698) - Produced (200) units at (FG_3 - Shipment via Water from DC_1 to RDC_1) @ Bucket (W30-2015)]
2017-03-10 09:55:59,760 (INFO) [4] (8891, 200698) - Produced (200) units of (FG_3, RDC_1) @ Bucket (W30-2015)]
2017-03-10 09:55:59,760 (INFO) [4] (8891, 200698) - Completed Planning Demand]
2017-03-10 09:55:59,761 (INFO) [4] (8891, 200698) - [Segment : 0]. Finished plan generation

```

The highlighted text shows the extra information corresponding to the order whose Demand Trace was set to 1 (in this case Order\_1)

## Limitations

This is performance intensive and should never be used in production.

## Delivery Plan

### Business Need

Demand satisfaction is one of the key metrics used for plan analysis. Delivery plan provides visibility into the various facets. It provides the quantity that is met on time in full, the quantity that is short and the quantity that is fulfilled late. This allows the planners to start their analysis at a higher level in the demand hierarchy and only then get into the details at lower level (by slicing and dicing of the data).

### Functional Description

SCS plan exports different plan measures. When a constrained plan is run, chances are that some demands may be short, or built late. SCS writes the corresponding data in these measures, thereby giving planners the visibility into the metrics of shortness and lateness for each demand. [\[Functional Test #56\]](#)

## Configuration



A measure group named Delivery Plan is created. It has the following measures

1. Total Met On Time Quantity  
Units of demands satisfied in JIT Bucket.
2. Total Late Quantity  
Units of demand that is built late (if the demand is not satisfied in JIT)
3. Maximum Lateness Buckets  
When a demand is split and met late, the maximum lateness from the bucket in which the demand is incident is reported. Ex: If demand D1 is incident on time bucket T1 and is met partially in T2, T3, T4. Then  $T4 - T1 = 3$  buckets is what gets reported in this measure.
4. Weighted Late Quantity  
This measure stores the  $\text{Sum}(\text{late qty} * \text{time bucket from current})$  for that particular demand. Ex: If 10 units is late by 1 bucket and 1 unit late by 2 buckets, Weighted late quantity =  $10 * 1 + 1 * 2 = 12$
5. Total Short Quantity  
If Demands are not satisfied, then this measure stores the quantity short for that demand.

Delivery Plan measure group has the following Granularity:

1. Demand Node
  - a. [Order].[Order Name]
  - b. [Time].[Fiscal Week]
  - c. [Customer].[Customer Name]
2. Material Node
  - a. [Location].[Location]
  - b. [Item].[Item]
3. [Version].[Version Name]

## Plan Date (Early Build)

### Business Need

Sometimes demands are expected to build as early as possible in order to utilize the available inventory, capacity usually at the Plant. One may also infer that planners are fine with holding inventory at a material node. This behavior is opposite to JIT planning done by SCS. Also, even if inventory is held at a material node, planners don't want order to be fulfilled earlier than requested. This is more of a contingent plan, where the higher priority order even though later in the horizon are planned earlier and held in stock so that the commitments are honoured.

### Functionality Description

The solver provides feasibility to stock material on earlier time buckets (rather than JIT) but only on reserved material nodes. Planners can specify the material nodes on which early build is to be allowed. This does not mean that orders will be available on the demand node before JIT. Order fulfillment on the downstream nodes will still adhere to JIT planning. Thus in the light of this feature, it is intended that SCS will produce at an earlier time bucket, then JIT and then build late instead of its usual behavior of planning JIT first, then build ahead and then build late.

### Configuration

Following measures are to be maintained

1. Measure

- a. Plan Date Offset: This measure takes numeric values. The measure specifies the number of buckets the plan date will be offset on the flagged material node.
- b. Granularity
  - i. Demand Node
    - 1. Demand ID/Order ID
    - 2. Time
    - 3. Customer
  - ii. Material Node
    - 1. Item
    - 2. Location
  - iii. Version

2. Measure

- a. Plan Date Material Node Marker: This measure takes boolean values.
  - i. TRUE : this means that the corresponding material node is a decoupling point where the ask date will be moved to earlier bucket (rather than JIT bucket) based on plan date offset value.
  - ii. FALSE : this means that the corresponding material will adhere to the default behavior in planning . This means production will 1st happen in the JIT bucket, and then Build Ahead and lastly in the late bucket
  - iii. NULL : null is synonymous with false
- b. Granularity
  - i. Material Node
    - 1. Item.Item
    - 2. Location.Location
  - ii. Version.Version Name

Same should be specified in the SCS config as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Demand → Plan Date Offset Measure Name → Select corresponding measure from the drop down menu

Plan Date Offset Measure Name ⓘ Plan Date ▼

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Inventory → Plan Date Material Node Marker Measure Name → Select corresponding measure from the drop down menu

Plan Date Material Node Marker Measure Name ⓘ Plan Date MN Marker ▼

**Notes:**

- Solver respects the build ahead limit while doing plan date offset based planning. If build ahead happens in the precedent paths, the target bucket will be adjusted based on (plan date offset - built ahead buckets used). Therefore,

one needs to set the build ahead limit properly to get the desired behavior (as how much early the productions should move-in).

- When there exists multiple producing activities to the material node with plan date marker is set, if the solver cannot produce the material in the desired bucket, it tries out the next bucket rather than trying out the desired bucket on the alternate activity.
- By default, hard pegging (planning followed by pegging for the demand) is used for plan date offset functionality. However, it is possible to turn off this option.

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Uncheck the checkbox to turn off the hard pegging for plan date offset demands.

Plan Date Hard Peg 



- The multi-stage setting of plan date offsets is supported. However, it does not do offset at each stage with plan date marker set. The offset happens at the first material node with plan date marker set when it encountered while traversing from downstream to upstream in the supply chain.
- The search sequence (for material production) at the plan date marker material node:
  - First try on plan date bucket as per below formula  

$$\text{plan date (revised ask date)} = \{\text{Current Ask date} - \max(0, [\text{plan date offset} - \text{cumulative built ahead buckets so far}])\}$$

Cumulative built ahead is the sum of built ahead buckets until this point of time. If the cumulative built ahead buckets used more than the plan date offset at the material node marked as plan date marker, then the solver will do JIT planning. There will be no further offset at the material node.

- Then try sequentially between plan date and JIT date bucket
- Next, if still demand not satisfied and if build ahead left, it can further ask earlier to plan date (till build ahead limit allows it).
- Then it can enter into delay demand phase. Here, the plan date is respected for the plan date offset demands as well (to the extent possible).
- Example:  
 Demand due date = W11, build ahead limit=6, plan date offset=4, lead time=0

While planning when it reaches at the material node marked for plan date planning, the build ahead used so far is 1. So the current JIT bucket at this material node is W10. So the revised ask date would be (considering the plan date offset)=  $W10 - \max(0, [4-1]) = W7$ . As you can see still 2 buckets of build ahead left w.r.t. the plan date (i.e. W7). In this case the search order should be

- ☐ Plan at plan date (W7)
- ☐ Then try till JIT bucket (W10) → W8, W9, W10
- ☐ Build ahead at plan date (W7) → W6, W5

☐ Delay demand

[\[Test Case: Plan date search pattern\]](#)

[\[Test Case: Plan date with finite build ahead limit\]](#)

## Plan Date Search Pattern Change in Case of Alternates

### Business Need

In Plan Date or Early Build planning there was a requirement to have a provision to switch the search pattern in case of alternate paths at the plan date marker material node.

### Functionality Description

In the default Plan Date planning, the solver does the early build at the plan date offset bucket along the primary path (in case of alternate producing activities at the plan date marker node). Then, it sequentially tries next buckets up to JIT bucket. If still the demand is not met, it tries build ahead buckets (if build ahead remains). Otherwise, it does similar asks along the alternate path/s sequentially until the demand gets planned fully.

With the current enhancement, user has the flexibility to guide the solver if it does above default planning or switch to available alternate path in the same bucket while doing the planning in the plan date offset range.

### Configuration

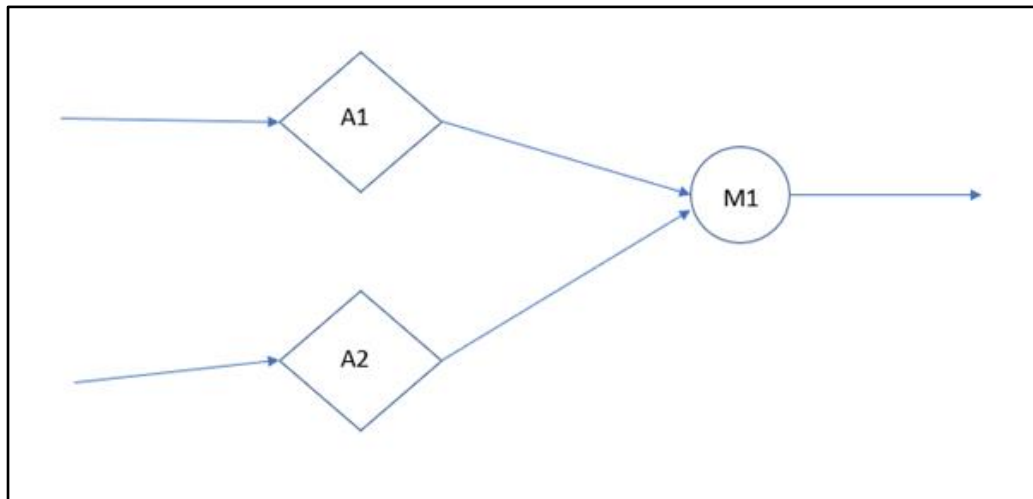
There is a boolean measure which a user needs to set to true to change the default behaviour. The granularity of the measure is material node. Basically, this measure needs to be set to true for the plan date marker material node if a user wants to change the default search pattern in the plan date offset period.

In the solver config UI, go to:

Workspace → Designer → Edit model → Rule → Plugin → Global o9.GraphCubePlugins → SupplyChainSolver → <Select solver Instance> → Inventory → Plan Date Alternate Search Node Marker Measure Name → Select the corresponding measure from the list

|                                                       |            |      |        |                  |                              |              |             |
|-------------------------------------------------------|------------|------|--------|------------------|------------------------------|--------------|-------------|
| Instance Options                                      | Parameters | Node | Demand | <b>Inventory</b> | Manufacturing                | Distribution | Procurement |
| Plan Date Alternate Search Node Marker Measure Name ⓘ |            |      |        |                  | Plan Date Search Alternate ▼ |              |             |

## Example



- M1 is a plan date marker node
- The JIT bucket (material required) at M1 = W10
- The plan date offset set for the current demand being planned = 3
- The demand build ahead limit = 5
- A1 is the primary path and A2 is the alternate path

In the default case, the solver will ask material at M1 in the following search pattern:

- Plan at W7 along A1 → W8 along A1 → W9 along A1 → W10 along A1 → W6 along A1 → W5 along A1 then
- Plan at W7 along A2 → W8 along A2 → W9 along A2 → W10 along A2 → W6 along A2 → W5 along A2

To change the above search pattern, a user needs to configure a measure and set it to true for M1. In that case, the solver will ask material at M1 in the following search pattern:

Plan at W7 along A1 → W7 along A2 → W8 along A1 → W8 along A2 → W9 along A1 → W9 along A2 → W10 along A1 → W10 along A2 → W6 along A1 → W6 along A2 → W5 along A1 → W5 along A2

[\[Reference Test Case\]](#)

## SS Planning during JIT (Min time along with JIT planning)

### Business Need

For certain demands, after planning a particular demand, need to plan SS (cover based, so that it can be pulled in) for that order immediately and hard peg it (should not give this supply or reallocate to lower priority demands). This is order by order, plan, pull-in (SS) and peg.

## Functionality Description

While planning a demand, solver checks if demand is marked to plan min time then move in the ask to earlier in time to respect min time target at the concerned material node. Basically the flow plans are created ahead in time to respect min time while planning a demand itself.

The pull in of flow plans happens due to min time does not consume demand build ahead limit. This means the build ahead is added to the min time and its search space increases. Here it asks on min time bucket and then before min time bucket till build ahead and after that it does ask after the min time bucket till JIT bucket.

[\[Test case for constant min time\]](#)

[\[Test case for time varying min time\]](#)

## Configuration

Following measures are to be maintained

- Plan Min Time During Demand Run Measure Name: This measure takes boolean values.

The default is False. When it is set to True, min time will be respected during demand run

- Granularity

1. Demand Node
  - a. Demand ID/Order ID
  - b. Time
  - c. Customer
2. Material Node
  - a. Item
  - b. Location
3. Version

- Min Time Target Measure Name: Time varying min time target defined at a material node. This takes numeric value. While demand planning if the demand is marked for “Plan Min Time During Demand Run Measure Name”, the ask date is pulled in to earlier bucket based on this “Min Time Target Measure Name” offset value.

- Granularity

1. Material Node
  - a. Item
  - b. Location
2. Time.[Fiscal Week]
3. Version

- Constant Min Time Target Measure Name: This takes numeric value. This is used to define constant min time target at material node.

- Granularity

1. Material Node

- a. Item
  - b. Location
2. Version

The above measures can be specified in SCS config as below

| Instance Options                                            | Parameters | Node | Demand | Inventory         | Manufacturing | Distribution | Procurement |
|-------------------------------------------------------------|------------|------|--------|-------------------|---------------|--------------|-------------|
| Plan Min Time During Demand Run Measure Name <span>i</span> |            |      |        | min time marker ▼ |               |              |             |

| Instance Options                                     | Parameters | Node | Demand | Inventory               | Manufacturing | Distribution | Procurement |
|------------------------------------------------------|------------|------|--------|-------------------------|---------------|--------------|-------------|
| Min Time Target Measure Name <span>i</span>          |            |      |        | Time varying min time ▼ |               |              |             |
| Constant Min Time Target Measure Name <span>i</span> |            |      |        | constant min time ▼     |               |              |             |

## Demand Fulfilment Policy

### Business Need

In case of delay in demand, sometimes organization do not want to receive multiple deliveries. They just need a consolidated delivery at a later stage in time post demand due bucket. There are also cases when businesses do not accept partial demand satisfaction and shorten the demand, completely.

### Functionality Description

Demand fulfillment policy is now included in the Supply Chain Solver through which a user can direct the solver to generate either multiple delivery or single delivery in case a demand is planned late.

In case the demand is not fully met (even post delay demand) user can ask the solver to shorten the demand.

Fulfilment Policy needs to be specified at the demand level.

### Configuration

There is a system defined picklist FulfilmentPolicy which is added to the Graph Cube Service for this functionality. The picklist have these values - MULTIPLE, INFULL, SINGLE which are available for all the tenants. Null is treated as MULTIPLE, since default value is MULTIPLE

| ▲ FulfilmentPolicy | Fulfilment Policy | true               | string |
|--------------------|-------------------|--------------------|--------|
| PICK LIST VALUES   |                   |                    |        |
| + Add new record   |                   |                    |        |
| Display Name       | Value             | Display Position ↑ |        |
| MULTIPLE           | MULTIPLE          | 0                  |        |
| SINGLE             | SINGLE            | 1                  |        |
| INFULL             | INFULL            | 2                  |        |

User needs to create a measure in demand grain as follows:

Note : Type is picklist and FulfilmentPolicy Picklist is selected

Edit Reports
Edit Model
Demand Quantity
Editing Measure 'FulFulmentPolicy'.

FulFulmentPolicy
Name \*
FulFulmentPolicy
Description \*
FulFulmentPolicy
ToolTip
FulFulmentPolicy
Data Type
Picklist
PickList Name
FulFulmentPolicy
Type
Regular
Reporting Measure
Editable

Granularity
Fiscal Week
Order Name
Version Name
Location
Customer Name
Item

Measures
Rules

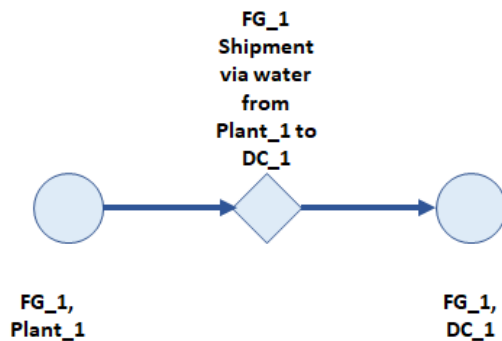
After the measure is created, a user must configure it in the solver config UI.

Instance Options
Parameters
Node
Demand
Inventory
Manufacturing
Distribution
Procurement

Fulfilment Policy Measure Name
DemandFulfilmentPolicy

## Example

Consider the below supply chain



Lead time be 2.

Consider BOH and ExpectedReceipts as follows

| Fiscal Week | Item | Location | BOH | Expected Receipts |
|-------------|------|----------|-----|-------------------|
| W01-2015    | FG_1 | Plant_1  | 50  |                   |



|          |      |         |  |    |
|----------|------|---------|--|----|
| W30-2015 | FG_1 | Plant_1 |  | 5  |
| W31-2015 | FG_1 | Plant_1 |  | 20 |

Consider there is demand on FG1@DC1 on W28-2015 of 100 units.

Let us how the demand fulfilment report varies based on the FulfilmentPolicy

**MULTIPLE/NULL:** This is a default setting (whether user defines it exclusively, or has left it undefined). In the example, we have the following demand satisfaction:

W28=50

W32=5

W33=20

The demand is short by 25 units.

The user gets 3 records in Demand Fulfilment Report.

**SINGLE:** In the same example if the SINGLE fulfilment policy is set for the demand, a user gets one record in Demand Fulfilment Report.

W33=75

The demand is short by 25 units.

**INFULL:** In the same example if this fulfilment policy is set for the demand, the demand is fully shortened (as it is not able to meet completely). Hence there will not be any record in demand fulfilment report.

Note : Spreading is supported.

**Limitations :** Storage with Single Fulfilment policy is not supported

## Demand Delivery Lot Size

### Business Need

There is a business requirement that when the sales order commit is generated in a partial delivery manner since the availability of the supply is not sufficient at the due date, the delivery is to be made as a multiple of lot size that is defined as sales order delivery parameter.

### Functionality Description

The solver plans the demand as-is and at the end of the demand planning, the delivery flow plans is adjusted to respect the delivery lot size. Any excess inventory left out at the demand item material node due to this adjustments is cleaned up during the regular excess cleanup phase which occurs at the end.

### Important Assumptions for this functionality to work

- The demand quantity has to be multiple of delivery lot size
- In case of lot size mismatch fulfillment, quantity will be of nearest multiple of lot size. However, the fulfillment plan is not more than the demand quantity. That is Solver will never produce more than demand (even in the case of unconstrained plan)

Examples:

Demand quantity is exact multiple of delivery lot size

Demand= 120

Delivery Lot size= 40

Due bucket= W10

| Demand Fulfilment Report  |     |     |     |     |
|---------------------------|-----|-----|-----|-----|
|                           | W10 | W11 | W12 | W13 |
| Without Delivery Lot size | 50  |     | 35  | 35  |
| With Delivery Lot Size    | 40  |     | 40  | 40  |

Demand quantity is not a multiple of delivery lot size

Demand= 100

Delivery Lot size= 30

Due bucket= W10

| Demand Fulfillment Report |     |     |     |     |
|---------------------------|-----|-----|-----|-----|
|                           | W10 | W11 | W12 | W13 |
| Without Delivery Lot size | 50  |     | 30  | 20  |
| With Delivery Lot Size    | 30  |     | 30  | 30  |

As you see 90 units fulfillment is completed. The demand is short by 10 units.

### Limitation:

- Future release the solver will be enhanced to report this short as a special constraint type in demand RCA.
- Storage will be enhanced to support this Demand Lot

## Configuration

A user needs to specify the delivery lot size multiple values for a given demand line. For this, one measure needs to be configured in the same grain as demand quantity.

**Measure:** Demand Delivery Lot Size (Type Integer)

**Granularity:** <Material Node>, <Demand Node>, <Version>

The screenshot shows the 'Edit Model' tab in the mPower configuration interface. The 'DemandLot' measure is selected, and its configuration is shown in the 'Rules' section. The 'Regular Scope' is defined as [Time].[Fiscal Week] \* [Order].[Order Name] \* [Item].[Item] \* [Location].[Location] \* [Customer].[Customer Name].[Version Name].[CurrentWorkingView]. The 'Formula' is Measure.[Quantity Fulfilled] = Edge.[Demand\_Fulfillment\_Report].[Demand\_Fulfillment\_Quantity]@([From].[Time].[Fiscal Week], [To].[Time].[Fiscal Week], [Order].[Order Name], [Item].[Item], [Location].[Location], [Customer].[Customer Name], [Version Name]). The 'DemandLot' measure is also listed in the 'Measures' section on the left.

Link the measure in the SCS config. Go to:

*Designer* → *Edit Model* → *Rule* → *Plugins* → *Global* → *o9.GraphCubePlugins* → *SupplyChainSolver* → *SupplyChainSolver* → *Demand Lot Size Multiple Measure Name*  
Select the corresponding measure name from the list.

SupplyChainSolver ✔ Successfully updated instance 'SupplyChainSolver'.

The screenshot shows the 'Demand Lot Size Multiple Measure Name' dropdown menu. The dropdown is open, showing the 'DemandLot' measure selected.

## Manufacturing

Manufacturing operations are where parts and components are transformed to create intermediates or finished goods. It refers to the MAKE operation in the SCOR process. The basic building blocks are BOM (bill of material) and Routing. One can have many combinations of this in the form of alternate BOM's, alternate components, date effective components, inter site BOM, time varying yields etc. All these can be modeled appropriately to build the manufacturing process of the supply chain.

### Bill of Material

#### Business Need

Bill of Material (BOM) specifies the produced part. BOM Components represents the consuming parts that are used in producing the produced part. For example, 4 legs and one top are the components that come together to form a table.

#### Functional Description

BOM is a dimension in mPower and the same can be specified as one of the composite activity node components.

#### Configuration Steps

Dimension BOM is configured as follows

**Designer**

Edit Reports BOM Editing Dimension 'BOM'.

BOM

Name \*  
BOM

Description \*  
Bill Of Material

Attributes

BOM ID

Hierarchies

**Designer**

Edit Reports BOM Editing Attribute 'BOM ID'.

BOM ID

Name \*  
BOM ID

Description \*  
Bill of material ID

Order By  
Name ▼

☒ Key

Key Column Data Type  
Integer ▼

Key Generation  
Auto ▼

Attributes

BOM ID

Hierarchies

Maintaining hierarchies is optional. Hierarchies are advised only when a dimension has more than one level attributes.

Following are the mandatory members under BOM ID

1. BOH Activity
2. Expected Receipt Activity

## Routing

### Business Need

On a shop floor, manufacturing of a part can be done using different process lanes which are called routes. Usually the characteristics of these processes and the resources attached are different.

### Functional Description

Routing is a dimension in mPower and the same can be specified as one of the composite node activity components.

### Configuration Steps

Dimension Routing is configured as follows

The image displays two screenshots of the mPower configuration interface. The top screenshot shows the 'Routing' dimension configuration. The 'Name' field is set to 'Routing' and the 'Description' field is also set to 'Routing'. The 'Attributes' section shows 'Routing ID' as the selected attribute. The bottom screenshot shows the 'Routing ID' attribute configuration. The 'Name' field is set to 'Routing ID', the 'Description' field is set to 'Routing ID', the 'Order By' dropdown is set to 'Name', the 'Key' checkbox is checked, the 'Key Column Data Type' dropdown is set to 'Integer', and the 'Key Generation' dropdown is set to 'Auto'.

**Routing Dimension Configuration**

| Field         | Value   |
|---------------|---------|
| Name *        | Routing |
| Description * | Routing |

**Routing ID Attribute Configuration**

| Field                | Value                               |
|----------------------|-------------------------------------|
| Name *               | Routing ID                          |
| Description *        | Routing ID                          |
| Order By             | Name                                |
| Key                  | <input checked="" type="checkbox"/> |
| Key Column Data Type | Integer                             |
| Key Generation       | Auto                                |

Maintaining hierarchies is optional. Hierarchies are advised only when a dimension has more than one level attributes.

Following are the mandatory members under Routing ID

1. BOH Activity
2. Expected Receipt Activity

## Activity Node Components

Activity node components in SCS config by default represents manufacturing activity. Specifying it as a composite node of BOM, Routing is optional and one can as well represent it as an activity dimension throughout the supply chain (including distribution and procurement).

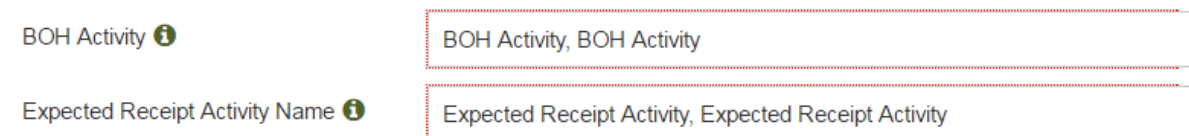
However, if one wishes to opt for BOM, Routing configuration in SCS, it can be done as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Node → Activity Node Components → [BOM dimension name].[Dimension Attribute], [Routing Dimension Name].[Dimension Attribute]



The screenshot shows the 'SupplyChainSolver1' configuration window with the 'Node' tab selected. The 'Activity Node Components' field is highlighted with a red dashed border and contains the text '[BOM].[BOM ID], [Routing].[Routing ID]'. The 'Material Node Components' field is also highlighted with a red dashed border and is currently empty.

Subsequently, BOH and Expected Receipts Activity name corresponding to BOM and Routing can be set in the SCS config as follows.



The screenshot shows two configuration fields. The 'BOH Activity' field is highlighted with a red dashed border and contains the text 'BOH Activity, BOH Activity'. The 'Expected Receipt Activity Name' field is also highlighted with a red dashed border and contains the text 'Expected Receipt Activity, Expected Receipt Activity'.

The comma separated values correspond to the mandatory data that is maintained in BOM and Routing dimensions.

## Distribution

Distribution refers to moving the item from one location to another. It refers to the DELIVER operation of the SCOR process. It includes the transportation and logistic activities for moving the supplies from one region to another. There could be restrictions around start and end date, different modes of transport used, effectivity and exclusion/inclusion of the routes. Primarily it can be presented by BOD (bill of distribution) and Transport mode. In short it is the connection between the source and the destination on a network. Specifying distribution components is optional in SCS.

### Bill of Distribution

#### Business Need

Bill of distribution (BOD) represents movement of item from source to destination. It can also be used to represent properties like lot size, effectivity etc.

#### Functional Description

BOD is a dimension in mPower and the same can be specified as one of the components of distribution activity node.

### Configuration Steps

Dimension BOD is configured as follows.

The screenshot shows the 'Editing Dimension 'BOD'' window. On the left, the 'BOD' dimension is listed with a checkmark and an 'x' icon. Below it, the 'Name' field contains 'BOD' and the 'Description' field contains 'Bill of Distribution'. On the right, the 'Attributes' section shows a search bar with 'BOD ID' entered. The 'Hierarchies' section is empty.

Maintaining hierarchies is optional. Hierarchies are advised only when a dimension has more than one level attributes.

Following are the mandatory members under BOD ID

1. BOH Activity
2. Expected Receipt Activity

## Transport Mode

### Business Need

The source and destination can be connected by multiple modes of transport namely truck, rail, ship and air. The lead times and cost of using different modes vary and some of these modes can be used only on an exception basis. The producing graph is used to represent the primary, secondary, tertiary modes of transport for receiving items at the material node.

### Functional Description

Transport Mode (Transmode) is a dimension in mPower and can be used as a composite activity node component.

### Configuration Steps

Dimension TransportMode is configured as follows.

The screenshot shows the 'Editing Dimension 'TransportMode'' window. On the left, the 'TransportMode' dimension is listed with a checkmark and an 'x' icon. Below it, the 'Name' field contains 'TransportMode' and the 'Description' field contains 'Mode of Transport'. On the right, the 'Attributes' section shows a search bar with 'Trans ID' entered. The 'Hierarchies' section is empty.

Maintaining hierarchies is optional. Hierarchies are advised only when a dimension has more than one level attributes.

Following are the mandatory members under Trans ID

1. BOH Activity
2. Expected Receipt Activity

## Distribution Activity Node Components

Distribution Activity node components in SCS config are optional. If required, it can be represented as a composite node. In the example here it represented as composite node of two dimensions namely BOD and Transport Mode. Configured as follows in SCS config.

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Node → Distribution Activity Node Components→

Distribution Activity Node Components

[BOD].[BOD ID], [TransportMode].[Trans ID]

Following are the required measure groups to be configured to incorporate BOD, Transport mode:

### 1. Distribution Parameters

- a. Measures:
  - i. Distribution LeadTime
- b. Granularity:
  - i. Activity Node
    - [BOD].[BOD ID]
    - [TransportMode].[Trans ID]
  - ii. Version.[Version Name]

### 2. Distribution Production Graph

- a. Measures:
  - i. Dist Prod Graph Association
  - ii. Distribution Priority
- b. Granularity:
  - i. Material Node
    - Item.[Item]
    - Location.[Location]
  - ii. Activity Node
    - [BOD].[BOD ID]
    - [TransportMode].[Trans ID]
  - iii. Version.[version Name]

### 3. Distribution Consumption Graph

- a. Measures:
  - i. Dist Cons Graph Association
- b. Granularity:
  - i. Material Node
    - Item.[Item]
    - Location.[Location]
  - ii. Activity Node
    - [BOD].[BOD ID]
    - [TransportMode].[Trans ID]
  - iii. Version.[Version Name]

### 4. Distribution Capacity Consumption Graph:

- a. Measures
  - i. Distribution Capacity Consumption Graph Association
  - ii. Distribution Capacity Consumption Qty Per
- b. Granularity
  - i. Material Node
    - Item.[Item]
    - Location.[Location]
  - ii. Activity Node
    - [BOD].[BOD ID]



- [TransportMode].[Trans ID]
  - iii. Version.[Version Name]
- 5. Distribution Consumption Plan**
  - a. Measures:
    - i. Distribution Consumption Quantity
  - b. Granularity:
    - i. Material Node
      - Item.[Item]
      - Location.[Location]
    - ii. Activity Node
      - [BOD].[BOD ID]
      - [TransportMode].[Trans ID]
    - iii. Time.[Fiscal Week]
    - iv. Version.[Version Name]

**6. Distribution Production Plan**

- a. Measures:
  - i. Distribution Production Quantity
- b. Granularity:
  - i. Material Node
    - Item.[Item]
    - Location.[Location]
  - ii. Activity Node
    - [BOD].[BOD ID]
    - [TransportMode].[Trans ID]
  - iii. Time.[Fiscal Week]
  - iv. Version.[Version Name]

**7. Distribution Capacity Consumption Plan:**

- a. Measures
  - i. Distribution Capacity Consumption Quantity
- b. Granularity
  - i. Resource Node
    - Resource.[Resource]
  - ii. Activity Node
    - [BOD].[BOD ID]
    - [TransportMode].[Trans ID]
  - iii. Time.[Fiscal Week]
  - iv. Version.[Version Name]

Following are the pegging tables:

**1. Dist\_Consumption\_Pegging**

- a. Property : Distribution Consumption Pegged Quantity
- b. From\_Node
  - i. Material Node
    - Item.[Item]
    - Location.[Location]
  - ii. Activity Node
    - BOD.[BOD ID]
    - TransportMode.[Trans ID]
  - iii. Time.[Fiscal Week]
- c. To\_Node
  - i. Demand Node
    - Order.[Order name]
    - Time.[Fiscal Week]
    - Customer.[Customer Name]
  - ii. Material Node
    - Item.[Item]

- Location.[Location]
- 2. Dist\_Production\_Pegging**
- a. Property : Distribution Production Pegged Quantity
  - b. From\_Node
    - i. Material Node
      - Item.[Item]
      - Location.[Location]
    - ii. Activity Node
      - BOD.[BOD ID]
      - TransportMode.[Trans ID]
    - iii. Time.[Fiscal Week]
  - c. To\_Node
    - i. Demand Node
      - Order.[Order name]
      - Time.[Fiscal Week]
      - Customer.[Customer Name]
    - ii. Material Node
      - Item.[Item]
      - Location.[Location]
- 3. Dist\_Capacity\_Pegging**
- a. Property : Distribution Capacity Pegged Quantity
  - b. From\_Node
    - i. Capacity Node
      - Resource.[Resource]
    - ii. Activity Node
      - BOD.[BOD ID]
      - TransportMode.[Trans ID]
    - iii. Time.[Fiscal Week]
  - c. To\_Node
    - i. Demand Node
      - Order.[Order name]
      - Time.[Fiscal Week]
      - Customer.[Customer Name]
    - ii. Material Node
      - Item.[Item]
      - Location.[Location]

The above mentioned measures, pegging table names and corresponding properties can be specified in the SCS config as follows

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Distribution→

|                                                     |                                   |
|-----------------------------------------------------|-----------------------------------|
| Distribution Lead Time Measure Name                 | Distribution LeadTime             |
| Distribution Production Graph Measure Name          | Dist Prod Graph Association       |
| Distribution Prod Graph Priority Measure Name       | Distribution Priority             |
| Distribution Consumption Graph Measure Name         | Dist Cons Graph Association       |
| Distribution Material Production Plan Measure Name  | Distribution Production Quantity  |
| Distribution Material Consumption Plan Measure Name | Distribution Consumption Quantity |

|                                                      |                                          |
|------------------------------------------------------|------------------------------------------|
| Distribution Material Production Pegging Graph Name  | Dist_Production_Pegging                  |
| Distribution Material Production Pegging Property    | Distribution Production Pegged Quantity  |
| Distribution Material Consumption Pegging Graph Name | Dist_Consumption_Pegging                 |
| Distribution Material Consumption Pegging Property   | Distribution Consumption Pegged Quantity |
| Distribution Capacity Graph Measure Name             | DistCapCons Association                  |
| Distribution Capacity Graph Qty Per Measure Name     | DistCapCons Qty Per                      |
| Distribution Capacity Consumption Plan Measure Name  | Dist Cap Cons Qty                        |
| Distribution Capacity Consumption Pegging Graph Name | Dist_Capacity_Pegging                    |
| Distribution Capacity Consumption Pegging Property   | DCC Pegged Qty                           |

*Note: Names of measure groups and measures can be specified as desired by the user and provided in the SCS configuration. The ones provided here are representative and corresponds to the tenant '[testing\\_SCS](#)' available on genie. One needs to define distribution activity node components and have the same components in flow plans and pegging tables.*

## Procurement

Procurement refers to the purchase of raw materials or components from vendors or suppliers. It refers to the SOURCE operation in the SCOR process. Specifying procurement components are optional in SCS.

## Supplier

### Business Need

Procurement is an important business function and refers to the amount of material purchased from suppliers. Planners run an MRP run to understand the quantities that they need to request from different suppliers. The commitment provided by suppliers are input as confirmed purchase orders.

### Functional Description

Supplier can be modeled as a dimension in mPower. A composite activity node of Supplier and transport mode can present the procurement process in entirety.

### Configuration Steps

Dimension Supply ID is configured as follows.

Maintaining hierarchies is optional. Hierarchies are advised only when a dimension has more than one level attributes.

Following are the mandatory members under Supply ID

1. BOH Activity
2. Expected Receipt Activity

## Supplier Activity Node Components

Supplier Activity node components in SCS config are optional. If required, it can be represented as a composite node of Supply, Transport Mode and configured as follows in SCS config.

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Node → Supplier Activity Node Components → enter value

Supplier Activity Node Components ⓘ

[Supply].[Supply ID], [TransportMode].[Trans ID]

Following are the required measure groups to be configured to incorporate Supply, Transport mode:

### 1. Supplier Parameters

- a. Measures:
  - i. Supplier LeadTime
- b. Granularity:
  - i. Supplier Activity Node
    - Supply.[Supply ID]
    - TransportMode.[Trans ID]
  - ii. Version.[Version Name]

### 2. Supplier Production Graph

- a. Measures:
  - i. Supp Prod Association
  - ii. Supp Prod Priority
- b. Granularity:
  - i. Supplier Activity Node
    - Supply.[Supply ID]
    - TransportMode.[Trans ID]
  - ii. Material Node
    - Item.[Item]
    - Location.[Location]
  - iii. Version.[Version Name]

### 3. Supplier Consumption Graph

- a. Measures:

- i. Supp Cons Association
  - b. Granularity:
    - i. Supplier Activity Node
      - Supply.[Supply ID]
      - TransportMode.[Trans ID]
    - ii. Material Node
      - Item.[Item]
      - Location.[Location]
    - iii. Version.[Version Name]
- 4. Supplier Capacity Consumption Graph**
  - a. Measures:
    - i. Supp Cap Cons Association
  - b. Granularity:
    - i. Supplier Activity Node
      - Supply.[Supply ID]
      - TransportMode.[Trans ID]
    - ii. Capacity Node
      - Resource.[Resource]
    - iii. Version.[Version Name]
- 5. Supplier Production Plan**
  - a. Measures:
    - i. Supply Produced
  - b. Granularity:
    - i. Supplier Activity Node
      - Supply.[Supply ID]
      - TransportMode.[Trans ID]
    - ii. Material Node
      - Item.[Item]
      - Location.[Location]
    - iii. Time.[Fiscal Week]
    - iv. Version.[Version Name]
- 6. Supplier Consumption Plan**
  - a. Measures:
    - i. Supply Consumed
  - b. Granularity:
    - i. Supplier Activity Node
      - Supply.[Supply ID]
      - TransportMode.[Trans ID]
    - ii. Material Node
      - Item.[Item]
      - Location.[Location]
    - iii. Time.[Fiscal Week]
    - iv. Version.[Version Name]
- 7. Supplier Capacity Consumption Plan**
  - a. Measures:
    - i. Supplier Capacity Consumed
  - b. Granularity:
    - i. Supplier Activity Node
      - Supply.[Supply ID]
      - TransportMode.[Trans ID]
    - ii. Capacity Node
      - Resource.[Resource]
    - iii. Time.[Fiscal Week]
    - iv. Version.[Version Name]

Following are the pegging tables:

### 1. Supplier\_Consumption\_Pegging

- a. Property : Supplier Consumption PQ
- b. From\_Node
  - i. Material Node
    - Item.[Item]
    - Location.[Location]
  - ii. Supplier Activity Node
    - Supply.[Supply ID]
    - TransportMode.[Trans ID]
  - iii. Time.[Fiscal Week]
- c. To\_Node
  - i. Demand Node
    - Order.[Order name]
    - Time.[Fiscal Week]
    - Customer.[Customer Name]
  - ii. Material Node
    - Item.[Item]
    - Location.[Location]

### 2. Supplier\_Production\_Pegging

- a. Property : Supplier Production Pegged Quantity
- b. From\_Node
  - i. Material Node
    - Item.[Item]
    - Location.[Location]
  - ii. Supplier Activity Node
    - Supply.[Supply ID]
    - TransportMode.[Trans ID]
  - iii. Time.[Fiscal Week]
- c. To\_Node
  - i. Demand Node
    - Order.[Order name]
    - Time.[Fiscal Week]
    - Customer.[Customer Name]
  - ii. Material Node
    - Item.[Item]
    - Location.[Location]

### 3. Supplier\_Capacity\_Consumption\_Pegging

- a. Property : Supplier Capacity Consumption Pegged Quantity
- b. From\_Node
  - i. Capacity Node
    - Resource.[Resource]
  - ii. Supplier Activity Node
    - Supply.[Supply ID]
    - TransportMode.[Trans ID]
  - iii. Time.[Fiscal Week]
- c. To\_Node
  - i. Demand Node
    - Order.[Order name]
    - Time.[Fiscal Week]
    - Customer.[Customer Name]
  - ii. Material Node
    - Item.[Item]
    - Location.[Location]

The above mentioned measures, pegging table names and corresponding properties can be specified in the SCS config as follows

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Procurement→

|                                                  |                                               |
|--------------------------------------------------|-----------------------------------------------|
| Supplier Lead Time Measure Name                  | Supplier LeadTime                             |
| Supplier Production Graph Measure Name           | Supp Prod Association                         |
| Supplier Prod Graph Priority Measure Name        | Supp Prod Priority                            |
| Supplier Consumption Graph Measure Name          | Supp Cons Association                         |
| Supplier Material Production Plan Measure Name   | Supply Produced                               |
| Supplier Material Consumption Plan Measure Name  | Supply Consumed                               |
| Supplier Material Production Pegging Graph Name  | Supplier_Production_Pegging                   |
| Supplier Material Production Pegging Property    | Supplier Production Pegged Quantity           |
| Supplier Material Consumption Pegging Graph Name | Supplier_Consumption_Pegging                  |
| Supplier Material Consumption Pegging Property   | Supplier Consumption Pegged Quantity          |
| Supplier Capacity Graph Measure Name             | Supp Cap Cons Association                     |
| Supplier Capacity Consumption Plan Measure Name  | Supplier Capacity Consumed                    |
| Supplier Capacity Consumption Pegging Graph Name | Supplier_Capacity_Consumption_Pegging         |
| Supplier Capacity Consumption Pegging Property   | Supplier Capacity Consumption Pegged Quantity |

*Note: Names of measure groups and measures can be specified as desired by the user and provided in the SCS configuration. The ones provided here are representative and corresponds to the tenant 'testing\_SCS' available on genie. One needs to define supplier activity node components and have the same components in flow plans and peggings.*

## Inventory

Inventory determines the stock in the supply chain. It could be in the form of production orders in manufacturing, distribution orders in distribution, purchase orders in procurement. It could also be the onhand inventory at any material node at the start of the planning bucket. It is also used to specify the in-transit lots which will arrive at the material node in certain time bucket.

## Beginning on Hand

### Business Need

The inventory starting condition is represented by beginning on hand (BOH). It specifies the amount of stock available at plan current on the material node. The BOH in other

time buckets are computations based on the supply that comes in and the demand that goes out of the material node.

### Functionality Description

BOH is maintained as a measure. SCS reads this BOH field from this Inventory measure group only in the plan current time bucket. BOH stock if not used in the plan current gets carried forward to the next time bucket until it gets exhausted.

### Configuration

A measure group to be configured is Inventory.

It has the following measure:

1. BOH

It takes numeric values. The number signifies the quantity of stock available at the specific material node on the plan current time bucket. If BOH at a material node is maintained at any other time bucket, SCS will ignore that stock in its calculations while planning demands.

Its Granularity comprises of 4 dimensions, namely

1. Material Node
  - a. Item.[Item]
  - b. Location.[Location]
2. [Lot].[Lot ID]
3. Time.[Fiscal Week]
4. Version.[Version Name]

There is flexibility regarding the naming convention of BOH. If beginning on hand measure is maintained as any other name other than the default name (BOH), same should be provided in the SCS config as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Inventory → BOH Measure Name → Select corresponding measure name from the drop-down menu.

BOH Measure Name ⓘ

BOH ▼

## Expected Receipts

### Business Need

Expected Receipts (ER) represent the amount of material producing into the material node in any time bucket. It is used to represent the production orders on shop floors, the distribution orders in the transportation process and the purchase orders placed on the supplier. It just specifies the material node and the time bucket in which this material will be available, but does not represent the source from which this is received.

### Functionality Description

SCS reads Expected Receipt from the Inventory measure group for all the time buckets including plan current. If Expected Receipts stock is not used in its corresponding time bucket then it gets carried forward to the subsequent time buckets until these are consumed by demand. While planning demands, SCS calculates the sum total of the BOH and Expected Receipts available in that time bucket to fulfill the demand.

### Configuration Steps



It is recommended that Expected receipts be maintained in the same Measure Group as BOH i.e Inventory measure group. Inventory measure group now has the following measures:

1. BOH
2. Expected Receipts  
It takes numeric values. The number signifies the quantity of stock available on a specific node at a specific time bucket. SCS honors Expected Receipts specified on the plan current bucket as well.

Its Granularity comprises of 4 dimensions, namely

1. Material Node
  - a. Item.[Item]
  - b. Location.[Location]
2. [Lot].[Lot ID]
3. Time.[Fiscal Week]
4. Version.[Version Name]

There is flexibility regarding the naming convention of Expected Receipts. If expected receipt measure is maintained as any other name other than the default name (Expected Receipts), it should be provided in the SCS config as follows

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Inventory → Expected Receipt Measure Name → Select corresponding measure name from the drop down menu

Expected Receipt Measure Name ⓘ

## WIP

### Business Need

Work In Progress (WIP) denotes plans that are already released/committed/firmed and should be honoured. These are usually in the near term where the plan needs to support execution realities (Intransit purchase order, distribution order and the production order on shop floor)

### Functionality Description

Planned WIP is passed as a measure to any of the graphs and solver tries to respect it with highest priority, before it solves for demand or safety stock. The solver solves for these as part of the infeasibilities run, producing flows along the supply chain. The output of the solver contains the WIP flows as well. If net new flow is required, one needs to write the computation to remove the WIP provided from the total flow.

### Configuration Steps

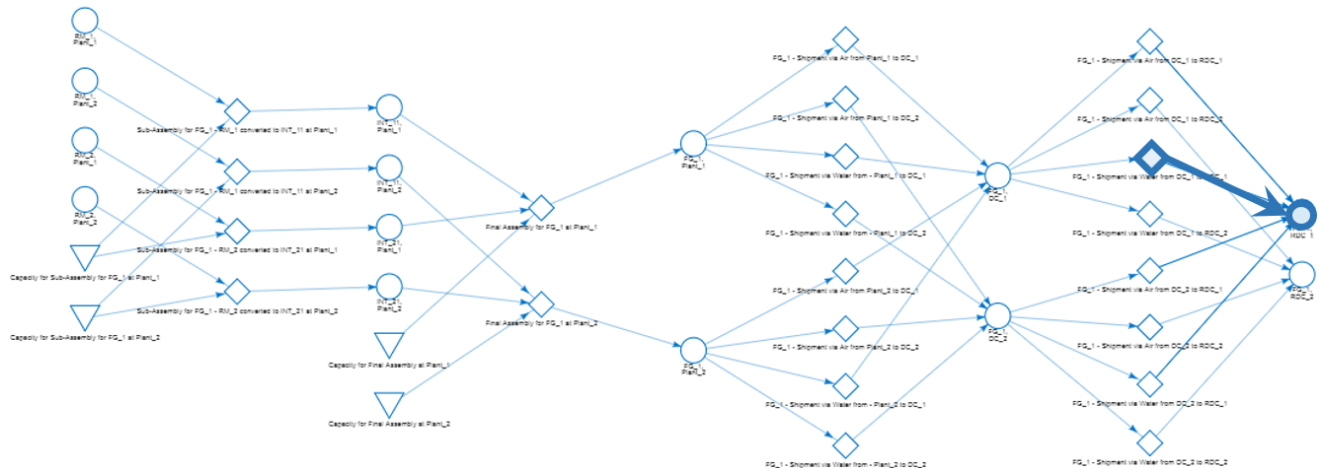
This concept is maintained in the Supply Chain Model through 3 measure groups:

1. Material Production Graph Planned WIP  
Measure Name: Material Production Planned WIP  
In this case the material quantity is pre-yield. Solver applies yield on the specified qty. In the case of co-product construct, it is sufficient to specify WIP on any one co-product.
2. Material Consumption Graph Planned WIP  
Measure Name: Material Consumption Planned WIP
3. Capacity Graph Planned WIP

## Measure Name: Capacity Consumption Planned WIP

All the above 3 graphs serve as input to the Solver. Each of the above 3 measures take numeric values. Consider the Activity: **FG\_1 - Shipment via Water from DC\_1 to RDC\_1** which is the primary activity used to ship goods from DC\_1 to RDC\_1.

Image: WIP\_001



This activity is maintained in FG Production Graph Planned WIP.

Image : WIP\_002

| Activity                                     | Item | Location | Lot ID | Data                            | W30-2015 |
|----------------------------------------------|------|----------|--------|---------------------------------|----------|
| FG_1 - Shipment via Water from DC_1 to RDC_1 | FG_1 | RDC_1    | WIP_0  | Material Production Planned WIP | 500      |

This specifies that irrespective of demand that is provided as an input, the solver also takes into consideration that this activity **FG\_1 - Shipment via Water from DC\_1 to RDC\_1** will be producing 500 units of FG\_1 at RDC\_1 in W30.

Granularity of each Planned Assembly WIP measure Group is as follows:

1. Material Node
  - a. Item.[Item name]
  - b. Location.[Location name]
2. Activity Node
  - a. Activity.[Activity name]
3. Lot.[Lot ID]
4. Version.[Version Name]

The Solver algorithm during planning in presence of WIP is as follows

1. First and foremost the whole network of the supply chain is loaded to the solver.
2. Secondly, all the data with respect to WIP (from all the three measures) are planned by consuming material from upstream. If it is not able to get material from upstream it is allocated to planner override producing into that buffer
3. Third, all the demands are loaded and planned taking into account the planned WIP.

4. If 'Respect Freeze Window For WIPs' is set to true in the config, solver will honour it. This means that the solver will not produce any flows in the frozen time periods, to satisfy the specified WIP.

*Note: WIP acts as an override by the planner and does not honour planning constructs like lot-size etc.*

## Lock

### Business Need

Planners would like to freeze certain plans in a bucket, so that no additional plans are created in that time bucket. For eg. after analyzing the plan the users might want to allocate certain supplies from a DC to a store and lock it, so that the next solver runs adheres to this planner input.

### Functional Description

This measure is at the same granularity as WIP. When specified it will lock the production in that bucket and will not allow any further production. It will consume material and capacity from upstream. When both Lock and WIP are specified in a time bucket, WIP is ignored (Lock is higher priority).

### Configuration Steps

This is modelled using four measure groups:

1. Material Production Graph Locked WIP  
Measure Name: Material Production Locked WIP
2. Material Consumption Graph Locked WIP  
Measure Name: Material Consumption Locked WIP
3. Capacity Consumption Graph Locked WIP  
Measure Name: Capacity Consumption Locked WIP

All the above 3 graphs serve as input to the Solver. Each of the above 3 measures take numeric values. Granularity of each Locked WIP measure Group is as follows:

1. Material Node
  - a. Item.[Item name]
  - b. Location.[Location name]
2. Activity Node
  - a. Activity.[Activity name]
3. Lot.[Lot ID]
4. Time.[Fiscal Week]
5. Version.[Version Name]

The solver algorithm in presence of Locked WIP is as follows:

1. First and foremost the whole network of the supply chain is loaded to the solver.
2. Secondly, all the data with respect to WIP (both planned and locked) are planned by consuming material from upstream. If it is not able to get material from upstream it is allocated to planner override producing into that buffer. If both WIPs are specified in the same time bucket, only Locked WIP is planned.

- Third all the demands are loaded and planned taking into account the locked and planned WIPs.

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Select corresponding measure name from the drop down menu

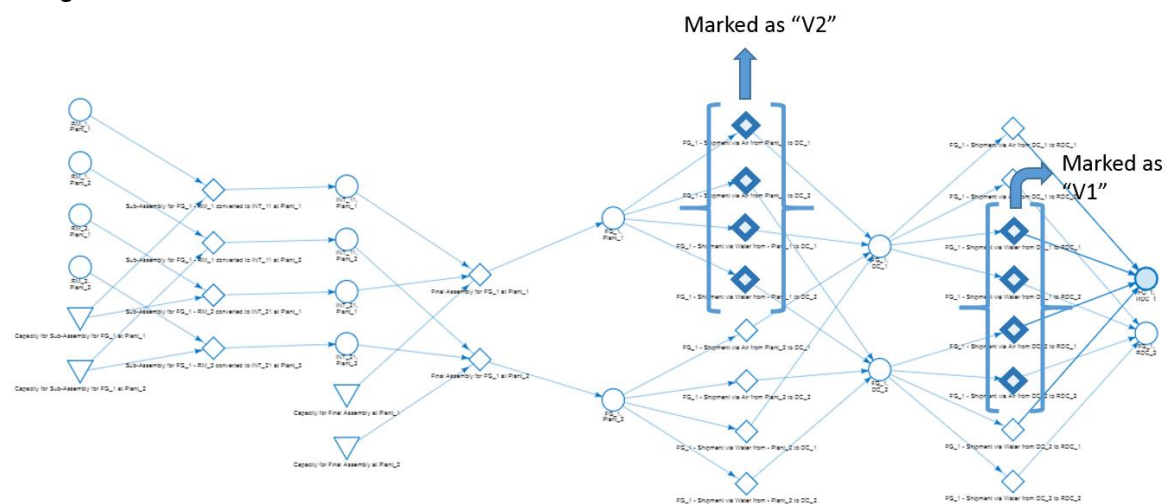
|                                              |                      |
|----------------------------------------------|----------------------|
| Prod Graph Planned WIP Measure Name ⓘ        | <input type="text"/> |
| Prod Graph Locked WIP Measure Name ⓘ         | <input type="text"/> |
| Consumption Graph Planned WIP Measure Name ⓘ | <input type="text"/> |
| Consumption Graph Locked WIP Measure Name ⓘ  | <input type="text"/> |

## Inventory Sweep (Activity Marker)

### Business Need

The business rule can be to use all the inventory in the system upto a certain level/stage before inducing any fresh start from upstream to meet the demand. This is also helpful when all new starts should happen on a new line combeen marked as V1 and V2 as shown in *INV\_SWEEP\_001*. This means when the solver is run, the activities marked as V1 and V2 and their related activities upstream are blocked. Only when no material is found via the exposed activities, the solver enables the activities marked as V1 and V2 and tries to solve demands through these paths.

Image: *INV\_SWEEP\_001*

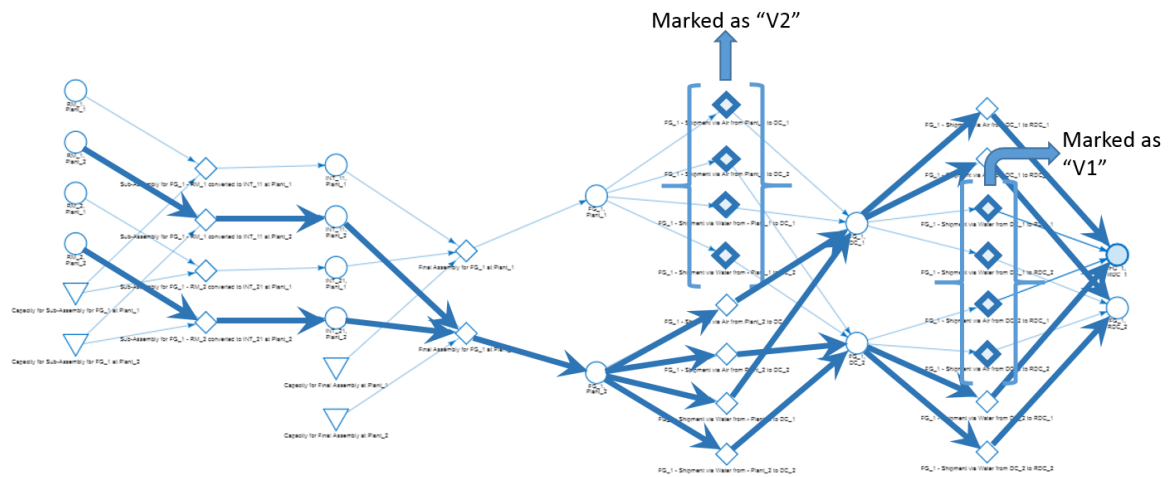


missioned on the shop floor, but inventory on older route need to consumed first without doing any fresh starts.

### Functional Description

Consider material node at RDC\_1. Order is incident on this node. A set of activities have So the paths initially exposed to the solver is shown below.

Image: *INV\_SWEEP\_002*



If no material is present in the network till the marked activities, the solver proceeds to solve demands as usual, i.e by relaxing the boundaries specified as marked.

Constraint Report for short orders is created along the primary path and not on the marked path. For example, in the above case, a demand incident on FG\_1, RDC\_1 will first try to procure from Plant\_2 and if material is not present in Plant\_2, SCS relaxes the boundaries and tries to procure from Plant\_1. If demand is still not fulfilled, constraint report for such an order indicates material to be made available at Plant\_1(which is the primary path for demand fulfillment by default)

## Configuration Steps

The measure Group to be configured is Activity Node Parameters.

- Measure Name
  - o Activity Marker
- Granularity
  - o Activity Node
    - [Activity].[Activity Name]
  - o [Version].[version Name]

The same can be specified in the solver config as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Activity Marker Exclusion List → Enter values in text

Activity Marker Exclusion List

V1 V2

**Activity marker exclusion list:** It is a comma separated list of activity markers, which tells the solver to exclude these mentioned activities in the order of priority given. It provides control on material search. It can be specified using wild card characters.

Image:INV\_SWEEP\_003

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Activity Marker Measure Name → Select corresponding measure from the drop down menu

Activity Marker Measure Name

## Limitations

Inventory sweep is at a demand level. Each demand first tries to find the supply till the activity marker, then relaxes this and goes to a higher level. It will then pick up the next demand.

## WIP Sweep [Iterative Deepening]

### Business Need

Business need is to use inventory (on hand, Expected Receipts, WIP's etc.) on all the available alternate paths, including cases where the inventory is lying deeper upstream along the alternate, before inducing any fresh start from upstream to meet the demand.

### Functional Description

WIP Sweep can be achieved using the following two ways:

- Setting up global WIP Sweep Parameter Default WIP Sweep Depth
- Setting up local parameter i.e., WIP Sweep Depth Measure Name

Apart from these two parameters there is WIP Sweep Priority setting.  
Let us understand each one of them

### Default WIP Sweep Depth

This parameter sets the default value of **WIP Sweep Depth** for all the material nodes having more than one producing activities.

|                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>In this modeling, there is no need to specify activity markers in the network as it can be counter productive, sometimes. So it is not recommended to use markers with WIP Sweep functionality.</p> <p>Also No Carry Node defined in between would be considered as one of the material node which will be used in Depth Sweep</p> <p>This Parameter only Takes Integer Values</p> <p>This Parameter can be set in the config or can be placed in arguments section of solver invocation</p> |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Configuration Steps

Go to Workspace ->Designer ->Edit model ->Rule ->Plugin ->Global ->o9.GraphCubePlugins ->SupplyChainSolver ->SupplyChainSolver ->Parameters  
→Default WIP Sweep Depth

SupplyChainSolver Successfully loaded instance 'SupplyChainSolver'.

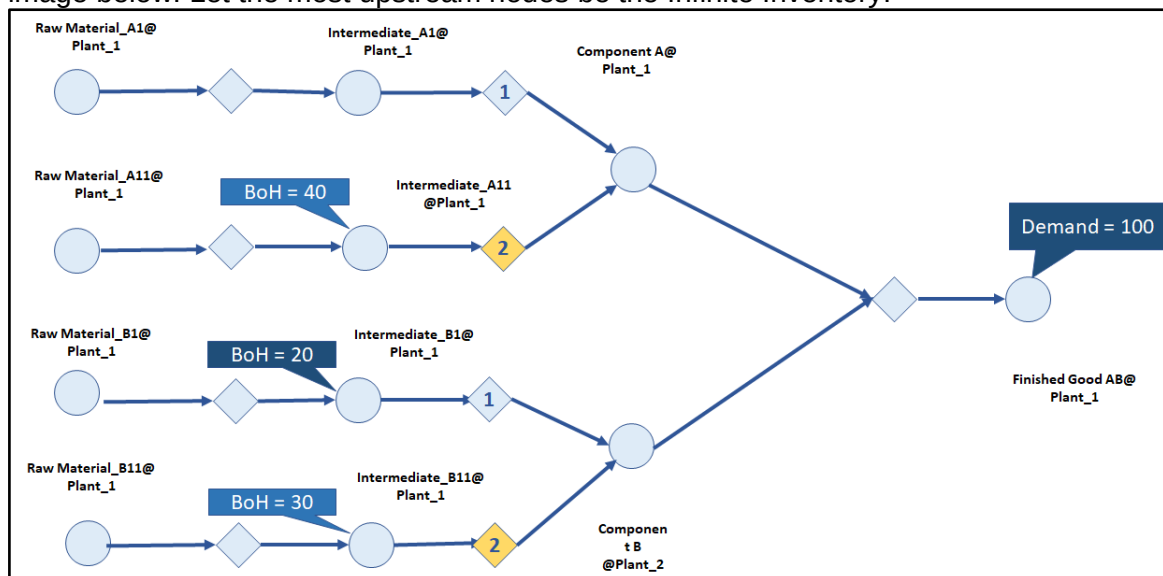
Instance Options
Parameters
Node
Demand
Inventory
Manufacturing
Distribution
Procurement

Default WIP Sweep Depth 0

| Default WIP Sweep Values | What does it Mean?                                                                            |
|--------------------------|-----------------------------------------------------------------------------------------------|
| 0 (Default Value)        | No Iterative Deepening search happens                                                         |
| n                        | Depth up to which iterative deepening search should continue is n level. Here n can be 1,2... |

## Use Case

Consider the below manufacturing operation. In this example, the Finished Good AB is manufactured using component A and B. Each of these components has alternates BOM. For example, A:A1 is Primary path and A11 is Secondary path. Let there be a demand of 100 on Finished Good AB, and on Hand as displayed in the image below. Let the most upstream nodes be the Infinite Inventory.



Let us see how the Inventories are being used by the Demand during the Demand Planning.

|             | Before or with Default WIP Sweep Depth as zero                                                                | Default WIP Sweep Depth as 1                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Component A | Planned using Fresh Production from A1<br>A11 inventory is left as is                                         | Planned using A11 Inventory and rest using Fresh Production from A1                                                          |
| Component B | Planned using existing inventory of B1 and rest using Fresh Production from B1<br>B11 inventory is left as is | Planned using existing inventory of B1,B11 and rest using Fresh Production from B1<br>Note: No Inventory is left at the end. |

## WIP Sweep Depth Measure Name

This is a local parameter which can override the value of WIP Sweep Depth at individual material nodes.

**Configuration Steps**

This is a measure that is maintained in the Material Node Parameters measure group. The Material Node parameters have the following granularity:

- 1. Material Node
  - a. Item.Item Name
  - b. Location.Location Name
- 2. Version.Version Name

This a Number Type measure which takes Integer values and controls the depth up to which iterative deepening search should continue.

**Example**

Edit Reports

Edit Model

Material Node Parameters

Editing Measure 'DepthWipSweep'.

DepthWipSweep

Name \*  
DepthWipSweep

Description \*  
DepthWipSweep

ToolTip  
DepthWipSweep

Data Type  
Number

Granularity

Item

Location

Version Name

Measures

constant min time

Constant Target Inventory

Constant WOS Target

DepthWipSweep

IL Demand Split

Infinite Inventory

Max Stock

MN\_segment

Rules

| Measure Values | What does it Mean?                                                                            |
|----------------|-----------------------------------------------------------------------------------------------|
| n              | Depth up to which iterative deepening search should continue is n level. Here n can be 1,2... |

|                                                                                     |                                                                                                                |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
|  | Material Nodes which do not have any alternate producing activities, need not set any values for this Measure. |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|

Associate this measure to **WIP Sweep Depth Measure Name\*** name in the SCS UI as displayed below:

Go to Workspace ->Designer ->Edit model ->Rule ->Plugin ->Global ->o9.GraphCubePlugins ->SupplyChainSolver ->SupplyChainSolver ->Inventory ->WIP Sweep Depth Measure Name

SupplyChainSolver

Successfully loaded Instance 'SupplyChainSolver'.

Instance Options

Parameters

Node

Demand

Inventory

Manufacturing

Distribution

Procurement

WIP Sweep Depth Measure Name

DepthWipSweep

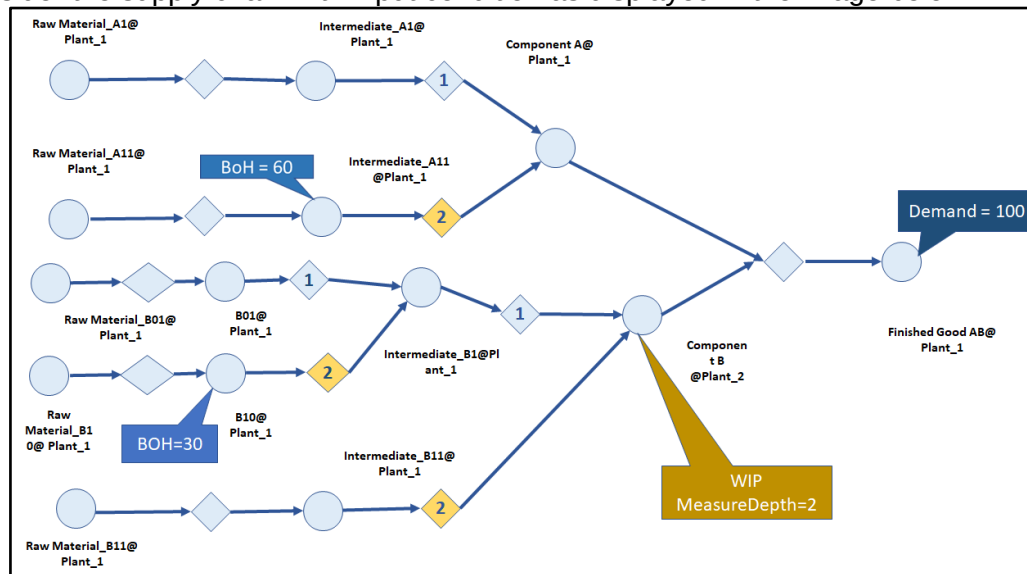


## WIP Sweep with Proportional sourcing

During WIP Sweep phase proportional sourcing will be ignored.  
Post WIP Sweep during Make New phase, proportional sourcing will be respected.

### Use Case

Consider the supply chain with input condition as displayed in the image below.



Let us see how the Inventories are being used by the Demand during the Demand Planning. The solver is invoked using Default WIP Sweep Depth as 1.

|             | Default WIP Sweep Depth as 1                                          | Default WIP Sweep Depth as 1 And WIP Sweep Depth Measure =2 for Component B |
|-------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Component A | Planned using Fresh Production from A1 and 60 from A11                | Planned using Fresh Production from A1 and 60 from A11                      |
| Component B | Planned using Fresh Production from B1<br>B10 inventory is left as is | Planned using 30 Units from B10 and for rest Fresh Production from B1       |

## Prod Graph WIP Sweep Priority

There is another feature which sets the priority in which activities are used during the WIP Sweep phase. This is controlled by the measure Prod Graph WIP Sweep Priority Measure Name.

### Configuration

This measure sets the priority in which the activities are used during the WIP Sweep phase. If there is no value specified, then the priority defaults Production Graph Edge Priority.

This is a measure which is maintained in the same measure group as Production Graph Edge or Production Graph Measure name.

Granularity:

1. Material Node
  - a. Item.Item Name
  - b. Location.Location Name
2. Activity Node
  - a. Activity.Activity
3. Version.Version Name

## Example

This measure should be associated with **Prod Graph WIP Sweep Priority** Measure name in the SCS UI.

Go to Workspace ->Designer ->Edit model ->Rule ->Plugin ->Global ->o9.GraphCubePlugins ->SupplyChainSolver ->SupplyChainSolver ->Manufacturing ->Prod Graph WIP Sweep Priority Measure name

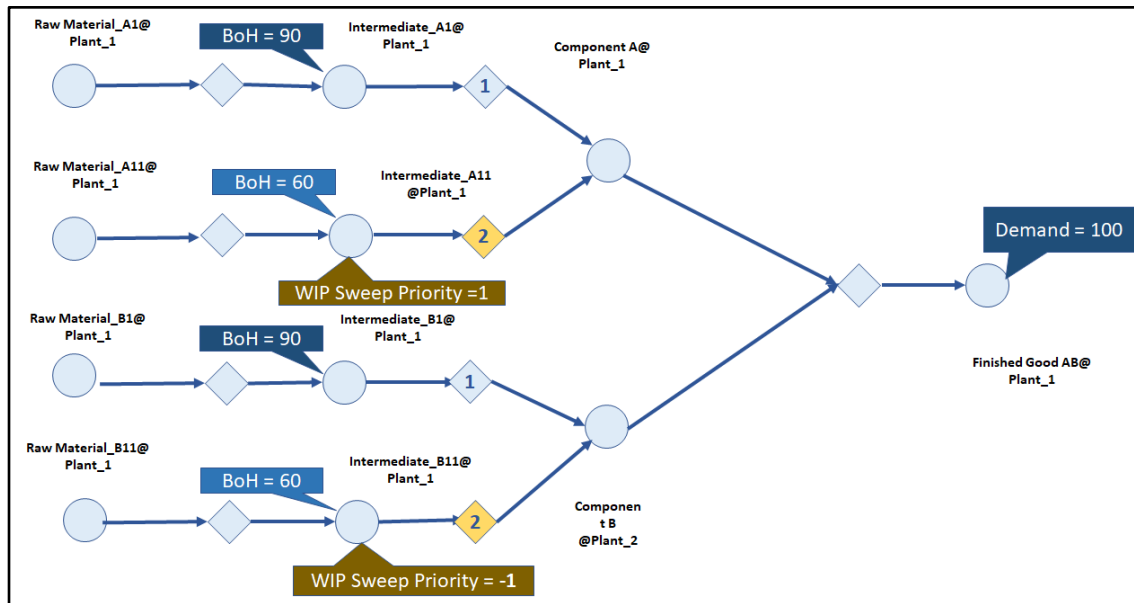
| Measure Values | What does it Mean?                                                                                                                                  |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| -n             | <p>If a -ve value is specified to an edge then that particular edge do not participate in the WIP Sweep phase.</p> <p>Here -n can be -1,-2 etc.</p> |
| n              | <p>Sets the priority in which activities are used during the WIP Sweep phase. Here n can be 1,2...</p>                                              |



Save and restart the tenant after all the changes.

## Use Case

Consider the supply chain with input condition as displayed in the image below.



Let us see how the Inventories are being used by the Demand during the Demand Planning.

The solver is invoked using the Default WIP Sweep Depth as 1.

| Component A with WIP Sweep Priority as 1 defined for Alternate A11                                              | Component B with WIP Sweep Priority as -1 defined for Alternate B11                                                 |
|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Planned using 60 from A11 Inventory and 40 from A1 Inventory.<br>50 units of A1 inventory is left               | Planned using existing inventory of B1 and rest using Fresh Production from B1<br>60 units of B11 inventory is left |
| Note: If a -ve value is specified to an edge then that particular edge does not participate in WIP Sweep phase. |                                                                                                                     |

## Limitations

While this feature improves the amount of inventory that can be picked up, it does not claim to be “globally optimal”. This is because it is still based on the existing heuristic solver framework.

## No Inventory Hold

### Business Need

In supply chain modeling there can be material nodes where one cannot stock any material. These are called as no inventory hold buffers or No carry material nodes. For example, in a mobile manufacturing unit where assembly, test and packaging are tightly integrated in a continuous process, inventory holding at intermediate stages is not allowed / possible. However planners require visibility at each stage of the network to plan other relevant items which are not modeled in the solver. Another example is where a large truck off loads all the goods to smaller trucks/vans at the outskirts of the city, where there is no physical location.

### Functional Description

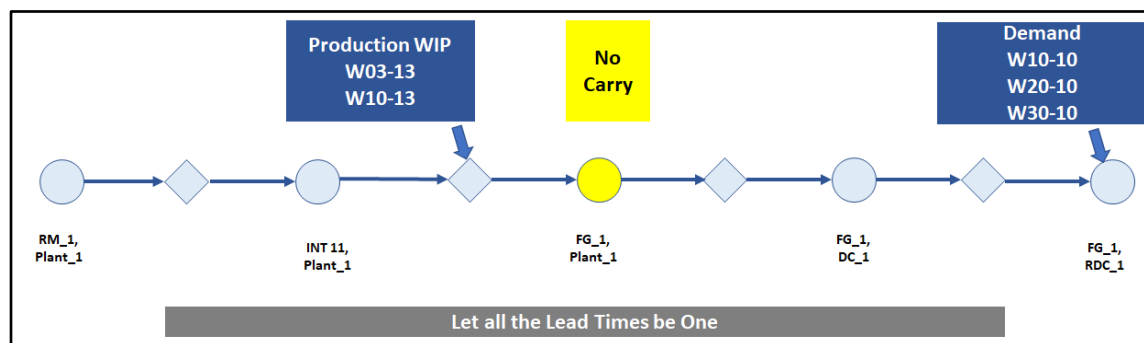
No Inventory hold is achieved such that any material arriving at a material node is pushed downstream in the same time bucket. This is achieved by internally using *local*

*build ahead limit as zero.* Solver algorithm works around the capacity and other constraints such that material passes through this material node only when it can be moved to the downstream nodes.

No Carry few points to note

- Solver will try to push the WIP quantity to downstream by consuming it earlier in time for a given demand violating its build ahead limit. It is extremely beneficial during such early consumption phases to consume earlier inventory first before trying to consume the later. Let us look at the simple use case here.

Consider the supply chain as follows where FG\_1@Plant\_1 is a no carry node . There are two production wip's and three demands.



Let us see what happens on the no carry node.

Demand 1 [W10] will use the WIP on Week 03 to fulfill the demand

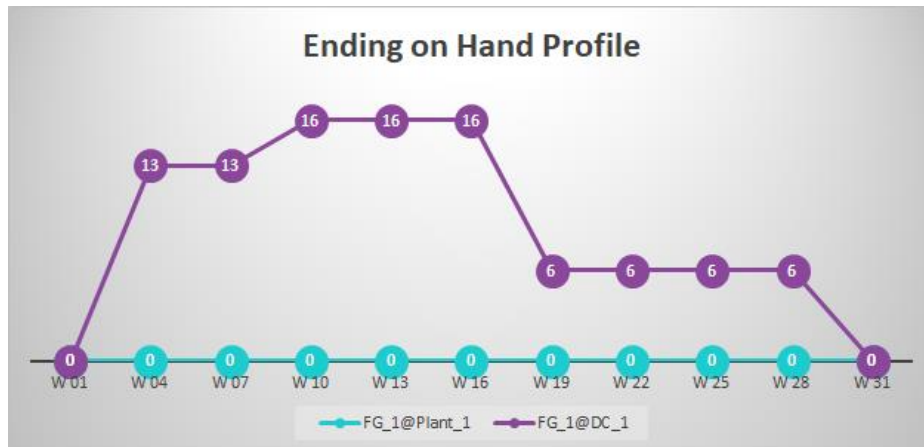
Demand 2 [W20] Demand will partially consume from W03 WIP, then from W10 WIP.

Demand 3 [W30] will only consumes from W10 WIP.



Post the Demand Planning Inventory Profile at FG\_1@Plant\_1 and FG\_1@DC\_1 is as follows:

As can be clearly seen all the Inventory at the No Carry Node have been pushed to the downstream node.

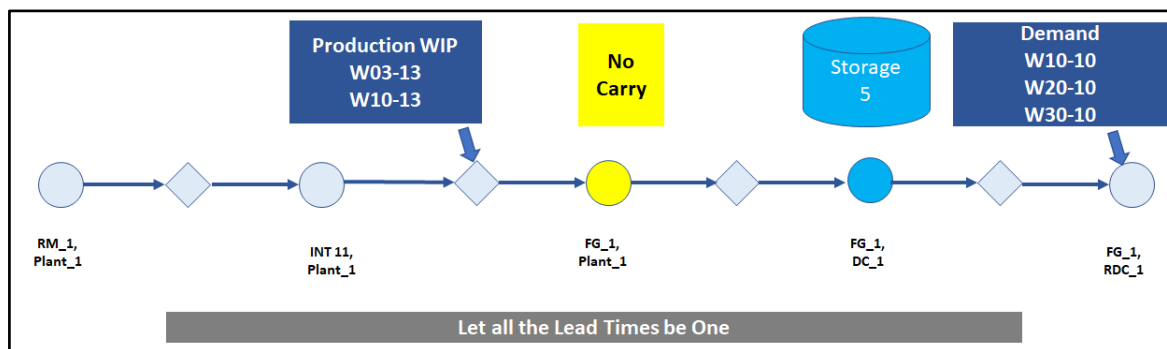


- In order to reduce storage violation the way No Carry works in combination with storage is push the demand further downstream of storage node.

Note : This only works when **Minimize NoCarry Violations flag is set to true**

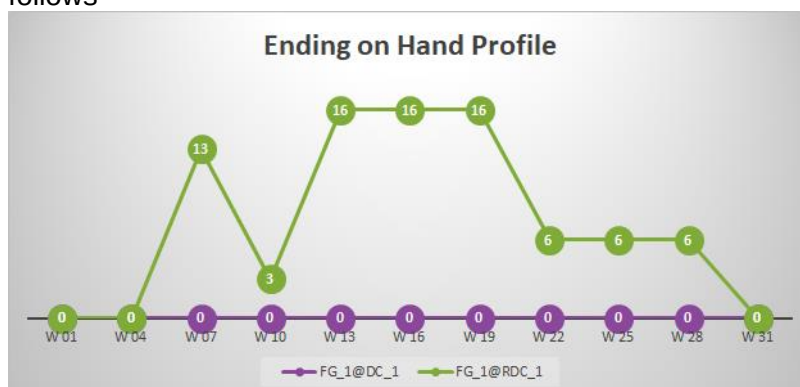
Let us visualize with simple example

Consider the modified supply chain, where storage defined on FG\_1@DC\_1.  
Storage availability is 5 units across the time horizon.



We know from the Previous example No Carry node pushes all the inventory to Next Node. Let us look at what happens when the next node is a storage node

Post the Demand Planning Inventory Profile at FG\_1@DC\_1, and FG\_1@RDC\_1 is as follows



As can be seen from the above graph, the demand is pushed downwards of storage node i.e. FG\_1@RDC\_1 to avoid storage constraints

- SCS has been enhanced to consider BOH and ER at no carry material nodes. The material will be pushed to downstream given downstream capacity availability. The demand build ahead is violated to respect no carry..
  - When No carry is set on Material Node having BOH/ER or Both  
[\[Functional Test #172\]](#), [\[Functional Test #173\]](#), [\[Functional Test #181\]](#)
- The material push happens during demand planning itself. Thus material can only be pushed downstream to respect no carry if there exist demands. Otherwise, material will not be pushed to downstream.
- Similarly any lot-size operation should be specified downstream of this no carry material node and not on activities producing into this no carry material node. Suppose, in case due to lotsize you produce more at no carry material node and downstream there are no further demands to plan, then the excess material will be lying at the no carry node thus violating no carry.
- For a node where planners define both Safety Stock and No Carry, it is no carry that should be honored .In this case, SCS should write a log pointing the ambiguity of data [\[Functional Test #170\]](#)

### Configuration Steps

A measure group Material Node Parameters is to be configured

1. Measure
  - a. No Carry: This measure takes boolean values.
    - i. TRUE: this means that as soon as goods reach the specified material node, it has to push it downstream along the requested path in the same time bucket. FALSE: this means that the material node can hold inventory (Default)
    - ii. NULL: null is synonymous with false
2. Granularity
  - a. Material Node
    - i. Item.Item
    - ii. Location.Location
  - b. Version.Version Name

The same should be specified in the **SCS config** as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →

o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Inventory → No Build Ahead Measure Name → Select corresponding measure from the drop down menu

No Build Ahead Measure Name ⓘ

No Carry ▼

## Infinite Inventory

### Business Need

Usually only critical components are modeled for planning. However one needs to account for the components like bolts and nuts, which are modeled as infinite material node to provide visibility into the material requirement in each planning bucket.

### Functional Description

The concept of a material node having infinite supply can be maintained in tenants by measure groups. If infinite inventory is specified at a particular material node, then SCS assumes that the mentioned material node has infinite supply of material , even during constrained planning. Therefore, the consuming activity at its corresponding time bucket

can draw any amount of material from the infinite specified material node. [\[Functional Test #40\]](#)

### Configuration Steps

Infinite Inventory or Infinite Material is a measure that is maintained in Material Node Parameters measure group.

Material Node parameters has the following granularity:

1. Material Node
  - a. Item.Item Name
  - b. Location.Location Name
2. Version.Version Name

Infinite material takes boolean values.

TRUE means the Material node has infinite supply of material and the consuming activity can draw any amount of material from the specified material node in any time bucket.

FALSE means the material node does not have an infinite supply of material.

NULL is equivalent to FALSE.

*Note : Ideally Infinite Inventory should be modelled only at the most upstream material node.*

## Inventory Plan

### Business Need

Planners need to understand the inventory profile at each of the material nodes. They need to understand the starting point – in the form of beginning on hand, what comes in – in the form of supply, what goes out – in the form of demand and what remains – in the form of ending on hand. This helps planners evaluate the profile, buckets of supply can help in inventory optimization

### Functional Description

As part of the SCS plan run, the Beginning on Hand and Ending on hand inventory for each material node that took part in the planning process within the planning horizon is considered. The data is output in the inventory plan tables as two measures of BOH and EOH when these measures are configured.

### Configuration

A measure Group Inventory Plan is maintained. It has 2 measures

1. Beginning On Hand Material Quantity
2. Ending On Hand Material Quantity

Inventory Plan has the following granularity

1. Material Node
  - a. [Item].[Item]
  - b. [Location].[Location]
2. [Time].[Fiscal Week] - Granularity of time bucket in which the plan is run
3. [Version].[Version Name]

In order to avoid SCS export overhead, Inventory planning measures are exported only when they are specified through the solver config. Both the measures stated below should be specified for SCS to export these Inventory Plan measures

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →

o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Inventory → Select corresponding measure names from the drop-down menu

Beginning On Hand Material Quantity  
Measure Name ⓘ

Beginning On Hand Material Quantity

Ending On Hand Material Quantity  
Measure Name ⓘ

Ending On Hand Material Quantity

*Note: Expected receipts generated on a node by SCS during unconstrained run are not seen in the inventory plan for that node.*

Consider the following supply chain



All activities have 1 week lead time

Material Node {FG\_1,VMI\_1} can procure its supply from {FG\_1, WH\_1} which in turn can procure its supply from {FG\_1, Factory}

Planning Horizon is between Week 1 and Week 10(both inclusive)

### Use Cases

Case:1

Let 100 units of demand be incident on {FG\_1, VMI\_1} . This demand is due on Week 31.

Let 1000 units of BOH be present on {FG\_1, Factory} in the form of BOH. This means this supply is available on material node {FG\_1, Factory} W01-2015 onwards.

Let min-lotsize on each of the activities are as follows:

- Min Lot for activity {Shipped from Factory to WH\_1} = 600.
- Min Lot for activity {Shipped form WH\_1 to VMI\_1} = 300.

When material constrained and capacity constrained plan is run on the above model, the inventory profile of the above 3 material nodes looks as follows:

| Item  | Location | Data                                | W27-2015 | W28-2015 | W29-2015 | W30-2015 | W31-2015 |
|-------|----------|-------------------------------------|----------|----------|----------|----------|----------|
| FG_1  | Factory  | Beginning On Hand Material Quantity | 1,000    | 1,000    | 1,000    | 400      | 400      |
|       |          | Material Quantity                   |          |          |          |          |          |
|       |          | Demand Quantity                     |          |          |          |          |          |
|       |          | Ending On Hand Material Quantity    | 1,000    | 1,000    | 400      | 400      | 400      |
| VMI_1 | VMI_1    | Beginning On Hand Material Quantity | 0        | 0        | 0        | 0        | 0        |
|       |          | Material Quantity                   |          |          |          |          | 300      |
|       |          | Demand Quantity                     |          |          |          |          | 100      |
|       |          | Ending On Hand Material Quantity    | 0        | 0        | 0        | 0        | 200      |
| WH_1  | WH_1     | Beginning On Hand Material Quantity | 0        | 0        | 0        | 0        | 300      |
|       |          | Material Quantity                   |          |          |          | 600      |          |
|       |          | Demand Quantity                     |          |          |          |          |          |
|       |          | Ending On Hand Material Quantity    | 0        | 0        | 0        | 300      | 300      |

The above report has been configured via Designer as follows:



The screenshot shows a software interface with filters at the top: Item, Location, Fiscal Week, and Version Name. Below these are more filters: Item, Location, and Measure. A 'Fiscal Week' filter is also visible with a settings icon. The main area is divided into a 'Measure' pane on the left and a 'Preview' pane on the right. The 'Measure' pane lists several measures: Beginning On Hand Material..., WOS For Total Demand, WOS For Independent Dema..., Material Quantity, Demand Quantity, and Ending On Hand Material Q... The 'Preview' pane shows a table with 12 rows of data. The table has columns for Item, Location, Data, and five fiscal weeks (W27-2015, W28-2015, W29-2015, W30-2015, W31-2015). The data rows show values for FG\_1 and VMI 1.

| Item  | Location | Data                                | W27-2015 | W28-2015 | W29-2015 | W30-2015 | W31-2015 |
|-------|----------|-------------------------------------|----------|----------|----------|----------|----------|
| FG_1  | Factory  | Beginning On Hand Material Quantity | 1,000    | 1,000    | 1,000    | 400      | 40       |
|       |          | Material Quantity                   |          |          |          |          |          |
|       |          | Demand Quantity                     |          |          |          |          |          |
|       |          | Ending On Hand Material Quantity    | 1,000    | 1,000    | 400      | 400      | 40       |
| VMI 1 |          | Beginning On Hand Material Quantity | 0        | 0        | 0        | 0        | 0        |

The measures (as selected in the bottom left pane of the above diagram) and their corresponding measure groups are as follows:

| Measure                             | Measure Groups            |
|-------------------------------------|---------------------------|
| Beginning On Hand Material Quantity | Inventory Plan            |
| Expected Receipts                   | Inventory                 |
| Material Quantity                   | Material Production Plan  |
| Material Consumption Quantity       | Material Consumption Plan |
| Demand Quantity                     | Demand                    |
| Ending On Hand Material Quantity    | Inventory Plan            |

Note that one can specify the display name or translation name of the measure as per requirement. Material quantity above specified the production flow into the material node

Inventory Plan is not written for material nodes

- that have no demands incident on it,
- that have no consumption flow incident on it,
- that have no production flow incident on it

Ending on hand Material Quantity for any material node M1 is calculated as follows:

**Ending on hand Material Quantity<sub>M1</sub>**

=

**Beginning on Hand<sub>M1</sub>**

**+ Supply<sub>M1</sub> (Expected Receipts<sub>M1</sub> + Material Production Quantity<sub>M1</sub>)**

**- Demand (Material Consumption Quantity<sub>M1</sub> + Demand<sub>M1</sub>)**

## Max Stock

### Business Need

Planner would like to specify the safety stock target and also the maximum level of stock that can be held at any material node. This could be specified based on quantity or time (Wos) and can be time varying. Usually demands should not be shorted/delayed because of max stock. This means flexibility needs to be provided to model it as soft or hard constraint during demand planning.

Max stock functionality is used mostly in metals and tire industry due to the product mix constraint by certain attributes (diameter, thickness, width, etc.). Also, many customers have the inventory policy of max stock by product line or product group due to the inventory budget.

### Functional Description

Max Stock when specified for a material node, honors the following condition:

$$\text{Ending-on-hand}(@ M_i \text{ on } T_j) \leq \text{Max Safety Stock}(@ M_i \text{ on } T_j)$$

where

$M_i$  : material node

$T_j$  :  $j^{\text{th}}$  time bucket

Thus, Max stock can be specified either constant or time varying as follows

- Constant max stock
  - Constant Max Stock for a material node remains constant across the planning horizon.
- Time varying max stock
  - Time Varying Max Stock for a material node may or may not vary across the time buckets in the planning horizon

Planners have the flexibility to set Max Stock as the soft constraint. This means SCS would try to honour the set max stock limit, but not at the cost of shorting a demand.

### Configuration

Max stock is implemented through measure groups and Flags.

#### Measure Groups

##### 1. Material Node Parameters

Constant Max Stock is maintained in this measure group.

- a. Measure:
  - i. Max Stock
- b. Granularity
  - i. Material Node
    1. [Item].[Item Name]
    2. [Location].[Location Name]
  - ii. [Version].[Version Name]

##### 2. Material Node Time Parameters

Time Varying Max Stock is maintained in this measure group.

- a. Measure
  - i. Max Stock in Bucket
- b. Granularity
  - i. Material Node
    1. [Item].[Item Name]
    2. [Location].[Location Name]
  - ii. [Time].[Fiscal Week]
  - iii. [Version].[Version Name]

Same should be specified in the SCS config as follows

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Inventory → Max Stock Measure Name → Select corresponding measure name from the drop down menu

Max Stock Measure Name  Max Stock ▼

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Inventory → Max Stock In Bucket Measure Name → Select corresponding measure name from the drop down menu

Max Stock In Bucket Measure Name  Max Stock in Bucket ▼

#### Configuration Parameters:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Respect Max Stock Constraint In JIT Planning → Set TRUE/FALSE

Respect Max Stock Constraint In JIT Planning  ☐

**TRUE:** when this flag is set to True, it means Demand Planning will respect Max Stock constraint. EOH at any time bucket for a material node will be limited by Max Stock

**FALSE** (default): when this flag is set to False, it means Demand Planning will not respect Max Safety Stock constraint. EOH at any time bucket for a material node will not be limited by Max Stock.

Goto Plans → Global Plugins → SupplyChainSolver → Edit → Parameters → Avoid Pegging Through Max Stock Node → Set TRUE/FALSE

Avoid Pegging Through Max Stock Node  ☐

**TRUE:** when this flag is set to True, it means that demand will not be pegged upstream of the Max Stock node.

**FALSE** (default): when this flag is set to False, it means that demand will be pegged upstream of Max Stock node. Thus flows upstream of max stock node, if converging to max stock node can get pegged twice(depends on the number of nodes converging) to the corresponding demand, thereby showing 2x quantity.

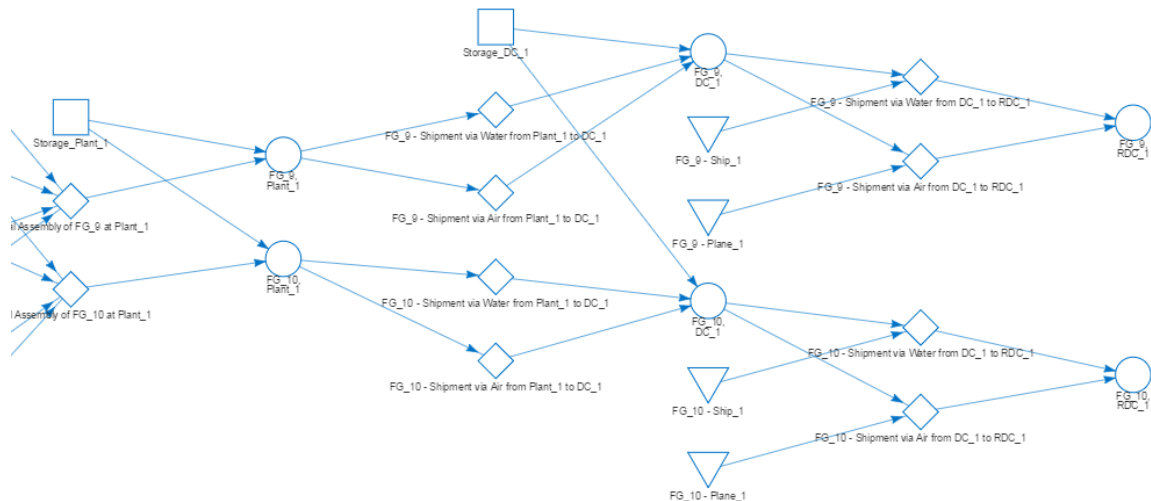
## Storage

### Business Need

There are physical constraints, which dictate the maximum amount of stock (inventory) that one can hold at a particular location. This is a hard constraint and the quantity of stock in any bucket should not be greater than the specified value. This is storage constraint.

### Functional Description

Consider the following supply chain segment

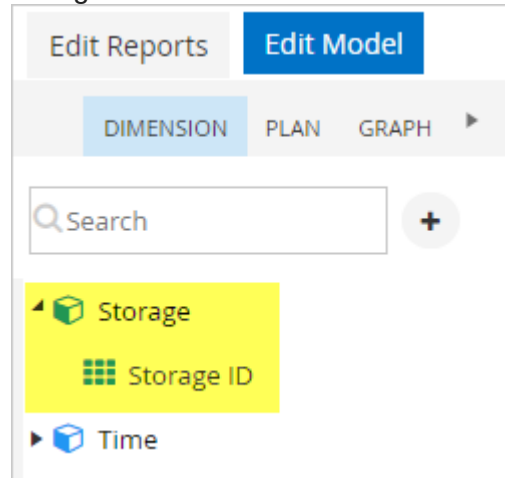


The above network diagram shows two Storage nodes in square shapes. Storage\_Plant\_1 is common to material nodes sharing the same location i.e Plant\_1. Storage\_DC\_1 is common to material nodes sharing the same location i.e DC\_1. From the above diagram one can infer that items arriving and stocking at location DC\_1 share a common space in the associated Storage. So 50 units of FG\_9 at DC\_1 and 10 units of FG\_10 at DC\_1 would acquire a total space of 60 units at Storage\_DC\_1, if they are not pushed downstream in the same bucket. Similarly, Storage\_Plant\_1 adheres to the same construct. Planners model storage availability as per the capacity availability in all time buckets. If demand is not satisfied in JIT time bucket, SCS looks for Storage space in earlier time buckets until build ahead limit specified is reached, and then in later time bucket until build late limit is reached. The corresponding storage constraint report can be seen in JIT time bucket only. This constraint report specifies that for the short/lated demand, how much storage quantity should have been held at that storage node, in order to fulfill the corresponding short/lated demand Just-In-Time.

## Configuration

Goto Workspace → Designer → Edit model → Dimensions → Click the  sign to create a dimension.

A dimension by the name Storage (shown as Storage in below image) needs to be configured. It can have a level attribute named Storage ID.



Maintaining hierarchies is optional. Hierarchies are advised only when a dimension has more than one level attributes.

Following measure groups have to be maintained

1. Storage Graph

- a. Measure
    - i. Storage Graph Association  
It takes boolean values
      - 1 means an Item, Location is constrained by corresponding Storage ID in the supply chain network
      - 0/NULL means an Item, Location is not constrained by corresponding Storage ID in the supply chain network
    - ii. Storage Qty Per(n)  
It is a multiplier and takes numeric values. It denotes the number of Storage Units(n) used by 1 unit of Item I at Location L.
  - b. Granularity
    - i. Material Node
      1. Item.Item
      2. Location.Location
    - ii. Storage Node
      1. Storage.Storage ID
    - iii. Version.Version Name
2. Storage Plan
    - a. Measure
      - i. Storage Plan(m)  
This measure is an output of the Solver. Storage Plan measure takes numeric values. It specifies that in time bucket T an Item, Location {I,L} will consume m units of Storage S.
    - b. Granularity
      - i. Material Node
        1. Item.Item
        2. Location.Location
      - ii. Storage Node
        1. Storage.Storage
      - iii. Time.Fiscal Week
      - iv. Version.Version Name
  3. Storage Availability
    - a. Measure
      - i. Storage Availability(s)  
This measure is an input to the solver. It takes numeric values. It specifies that at a time bucket T, the units of Storage s, available for demand planning.
    - b. Granularity
      - i. Storage Node
        1. Storage.Storage ID
      - ii. Time.Fiscal Week
      - iii. Version.Version Name

Following relationship types have to be maintained (Workspace → Models → Manage Graphs).

1. Storage\_Pegging
  - a. From node
    - i. Storage Node
      1. Storage.[Storage ID]
    - ii. Material Node
      1. Item.[Item]
      2. Location.[Location]
    - iii. Fiscal Week
  - b. To node
    - i. Demand Node

1. Order.[Order name]
2. Time.[Fiscal Week]
3. Customer.[Customer Name]
- ii. Material Node
  1. Item.[Item]
  2. Location.[Location]
- c. Edge Properties
  - i. Storage Pegged Quantity  
This is the output of the solver. It takes numeric values. The number of unit space of Storage used at a specific time bucket during demand planning is stored in this edge property
  - ii. Storage Constraint Type  
This is the output of the solver. If a demand is late or short, the storage type constraint in JIT bucket is reported
  - iii. Storage Constraint Quantity  
This is the output of the solver. It takes numeric values. The number of unit spaces of storage requested in JIT bucket using primary path is stored in this edge property, for the quantity of demand that is late or short. Note that adding this quantity to storage may not help demand to be satisfied in JIT as it is just a storage node.

The above mentioned node components, measure groups and relationship-types should be specified in SCS config as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Node → Storage Node Components → enter Storage Node components in the textbox.

Storage Node Components ⓘ

[Storage].[Storage ID]

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Storage Plan → Select corresponding measure from the drop down menu

Storage Plan Measure Name ⓘ

Storage Plan



Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Storage Graph Measure Name → Select corresponding measure from the drop down menu

Storage Graph Measure Name ⓘ

Storage Graph Association



Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Storage Graph Qty Per Measure Name → Select corresponding measure from the drop down menu

Storage Graph Qty Per Measure Name ⓘ

Storage Qty Per



Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Time Dep Available Storage Measure Name → Select corresponding measure from the drop down menu

Time Dep Available Storage Measure Name ⓘ Storage Availability ▼

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Storage Pegging Graph Name→Select corresponding measure from the drop down menu

Storage Pegging Graph Name ⓘ Storage\_Pegging ▼

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Storage Pegging Property→Select corresponding measure from the drop down menu

Storage Pegging Property ⓘ Storage Pegged Quantity ▼

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Storage Constraint Type Property→Select corresponding measure from the drop down menu

Storage Constraint Type Property ⓘ StorageConstraintType ▼

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Storage Constraint Qty Property→Select corresponding measure from the drop down menu

Storage Constraint Qty Property ⓘ StorageConstraintQty ▼

### Flags

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters→ Storage Constrained→Check if TRUE or Uncheck if FALSE [By default it is Unchecked i.e FALSE]

Storage Constrained ⓘ

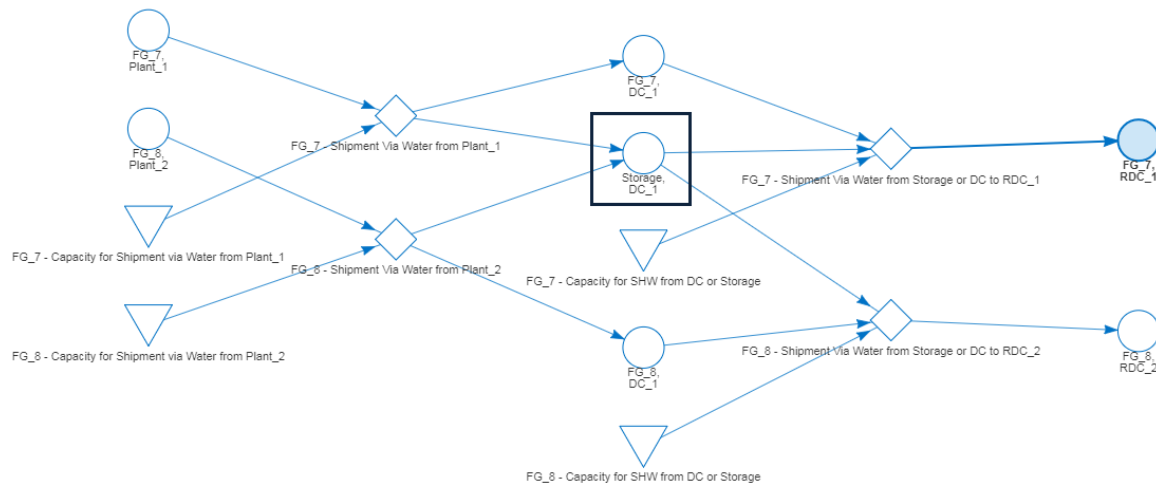


TRUE: If Storage Constrained is set as “True”, whatever Storage is associated with the material node will be considered for demand planning. Example if 100 units of Storage S1 is available for material nodes M1 and M2 at time T1, then this Storage S1 can satisfy not more than 100 units of demand collectively incident at M1 and/or M2 at time T1( assuming 0 build ahead and 0 build late). Subsequently, if the aggregated demands exceed 100 units then lower priority demands will get fully/partly shorted depending on the number of units by which these demands at T1 exceeds storage available at T1.

FALSE: If False, SCS assumes that the Storage S1 which is associated with material M1 and M2 at time T1 will be able to accommodate infinite units of demands incident on material node M1 and M2 at time T1.

### **Functional Description**

Consider the following supply chain



Storage is modelled using the co-product and Max Stock as the modelling constructs. Storage is just another material node. In the above diagram the material node enclosed within the red square represents the storage location for DC\_1. It is denoted as {Storage,DC\_1} which mirrors with {Item, Location} format. Modelling has been done such that all the activities producing at DC\_1 (there can be many other DCs) will also be producing at this node. Similarly all activities consuming from DC\_1 will also be consuming from this node i.e {Storage,DC\_1}.

Max Stock when specified for a material node, honors the following condition:

$$\text{Ending-on-hand}(@ M_i \text{ on } T_j) \leq \text{Max Safety Stock}(@ M_i \text{ on } T_j)$$

Where

$M_i$  :  $i^{\text{th}}$  material node

$T_j$  :  $j^{\text{th}}$  time bucket

Max Stock is specified on this common node to represent storage. Max stock is associated with material node and hence takes effect when material constrained or fully constrained plan is run. Max Stock is not honored when planners run an unconstrained plan.

### Configuration

Max stock is specified on the material node. Therefore it should be represented as {Item, Location}

Item Name : Storage

Location Name: DC\_1

This would mean all the activities producing into {Item\_n, DC\_1} would also go into {Storage, DC\_1}

Max Stock can be maintained in either (1 or 2) of the Measure Groups stated below

#### 1. Material Node Parameters (For value which is constant in time)

##### a. Measure

##### i. Storage :

It takes numeric values. The number denotes the value of storage in each bucket. The value remains the same for all time buckets in the planning horizon. Ending-on-hand on the specified material node should be less or equal to the Storage specified for that material node.

##### b. Granularity

##### i. Material Node

1. Item.[Item]
2. Location.[Location]

##### ii. Version.[Version]

#### 2. Material Node Time Parameters (For values which are time varying)



a. Measure

i. Storage in Bucket

It takes numeric values. The number denotes the value of Storage for a material node on a particular time bucket. Planners can have different value of storage for different time buckets in the planning horizon. Ending-on-hand for a material node in any time bucket should be less than or equal to the Storage specified for that material node in that time bucket.

b. Granularity

i. Material Node

1. Item.[Item]
2. Location.[Location]

ii. Time.[Fiscal Week]

iii. Version.[Version Name]

It should be configured in SCS config as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →

o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Inventory → Max Stock Measure Name → Select corresponding measure name from the drop down menu

Max Stock Measure Name  Max Stock ▼

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →

o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Inventory → Max Stock In Bucket Measure Name → Select corresponding measure name from the drop down menu

Max Stock In Bucket Measure Name  Max Stock in Bucket ▼

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →

o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters →

Respect Max Stock Constraint In JIT Planning → Set TRUE/FALSE

**TRUE:** when this flag is set to True, it means Demand Planning will respect Max Stock constraint. EOH at any time bucket for a material node will be limited by Max Stock

**FALSE (default):** when this flag is set to False, it means Demand Planning will not respect Max Safety Stock constraint. EOH at any time bucket for a material node will not be limited by Max Stock.

## Limitation

Root constraint of storage (max SS) is not reported.

## Storage Bands

### Business Need

Supply Planners need to make sure that they first use up all the available storage before looking for other options. There could be feasibility of having huge trucks in parking space in short term or leasing some storage at a nearby location. Planners need to understand the usage of this to make the cost vs benefit tradeoff. Planners can model these as storage bands and make sure that band1 is only used when all normal/regular storage is exhausted and then open up band2 when band1 is fully utilized.

### Functional Description

Overtime storages participate in demand planning only when regular storages in all time buckets within the planning horizon fail to fulfill demands in JIT, or within the build ahead and build late period specified. There are 3 types of storage

1. Regular
2. Band\_1

### 3. Band\_2

The order in which above are used in build late bucket can be toggled using a flag.

If the flag is False or NULL.

While demand planning {especially during capacity constraint} SCS uses regular Storages in all time buckets to fulfill demand until build late limit is reached. If regular Storages collectively fail to fulfill demands, then SCS looks for Band\_1 Storages. It tries to satisfy the remaining demands using Band\_1 Storages in all time buckets. If Band\_1 Storages in all time buckets are exhausted and demand is still unfulfilled, then SCS repeats the same by opening up the Band\_2 Storages. Thus, all the storages are used to fulfill demands in the following order

1. Regular Storages in all time buckets
2. Band\_1 Storages in all time buckets
3. Band\_2 Storages in all time buckets

If the flag is TRUE

While demand planning {especially during storage constraint} SCS uses regular + Band\_1 + Band\_2 storages in all time buckets to fulfill demand until Build late limit is reached.

To elaborate the above,

in order to satisfy an order JIT,

1st regular storages in JIT time bucket is utilised.

If regular Storages in JIT bucket is not sufficient enough to fulfill the demand JIT , then it tries to use regular storages in earlier buckets till build ahead limit is reached.

If still not satisfied, it opens Band\_1 storage in JIT bucket to fulfill the order. If it can't, it will try in earlier buckets using Band\_1 till build ahead limit is reached.

If still unsatisfied, it opens up the Band\_2 storage in JIT bucket and then tries with band\_2 storage in earlier buckets till BAL is reached.

If demand cannot be built JIT or within build ahead limit, then SCS tries to build the order late starting with utilizing

regular storages at JIT+1 bucket,

then Band\_1 Storages at JIT+1 bucket,

then Band\_2 Storages in JIT+1 bucket

and then goes to the JIT+2 bucket and so on until Build Late Limit is reached.

## Configuration Steps

### Flags:

- Use Bands To Minimize Lateness

The Flag that controls the behavior of regular and storage bands can be configured as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →

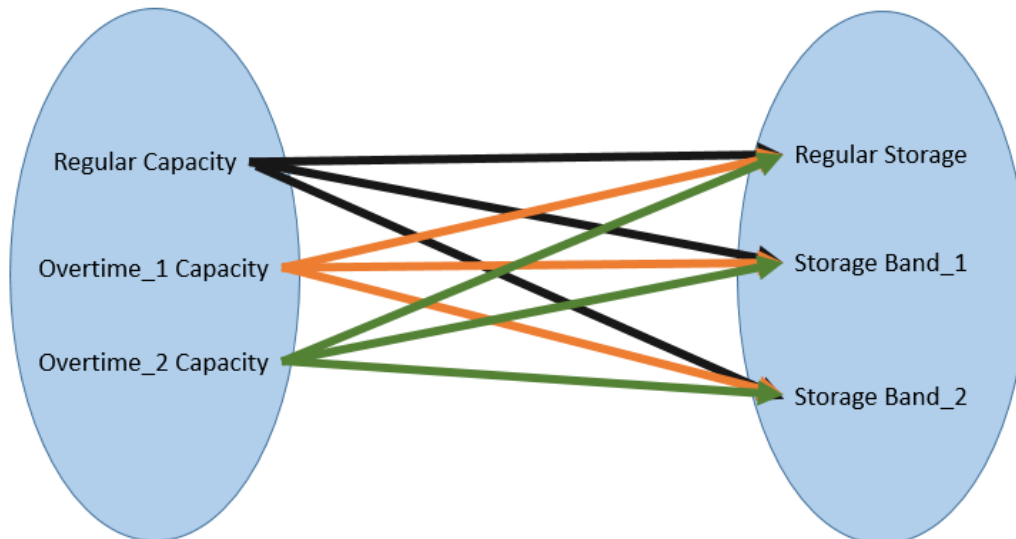
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Use Bands To Minimize Lateness → Check for TRUE and Uncheck for FALSE [By default it is Unchecked i.e FALSE]

Use Bands To Minimize Lateness 



- Band Priority

When both storage and capacity are modelled together, there can be the following combinations in which storage bands and capacity bands are used for demand planning.



Thus are total 9 combinations in which storage and capacity bands can be used for demand planning.

1. (Regular Capacity,Regular Storage)
2. (Regular Capacity,Band\_1 Storage)
3. (Regular Capacity,Band\_2 Storage)
4. (Overtime\_1 Capacity,Regular Storage)
5. (Overtime\_1 Capacity,Band\_1 Storage)
6. (Overtime\_1 Capacity,Band\_2 Storage)
7. (Overtime\_2 Capacity,Regular Storage)
8. (Overtime\_2 Capacity,Band\_1 Storage)
9. (Overtime\_2 Capacity,Band\_2 Storage)

However, the order in which the above combinations are searched can be controlled through a flag setting

If Band Priority is set to “Capacity”

This would mean that SCS uses Regular capacity first with all bands of storage in the following order for demand planning

1. (Regular Capacity,Regular Storage)
2. (Regular Capacity,Band\_1 Storage)
3. (Regular Capacity,Band\_2 Storage)

If demands are still unfulfilled, then Overtime\_1 capacities are used with all the bands of storage in the following order for demand planning

1. (Overtime\_1 Capacity,Regular Storage)
2. (Overtime\_1 Capacity,Band\_1 Storage)
3. (Overtime\_1 Capacity,Band\_2 Storage)

If demands are still unfulfilled, then Overtime\_2 capacities are used with all the bands of storage in the following order for demand planning

1. (Overtime\_2 Capacity,Regular Storage)
2. (Overtime\_2 Capacity,Band\_1 Storage)
3. (Overtime\_2 Capacity,Band\_2 Storage)

If Band Priority is set to “Storage”

Storage Bands are given priority over capacity bands. This means for demand planning SCS uses Regular Storage first with all bands of capacities in the following order

1. (Regular Capacity,Regular Storage)
2. (Overtime\_1 Capacity,Regular Storage)
3. (Overtime\_2 Capacity,Regular Storage)

If demands are still unfulfilled then Band\_1 Storage is used with all bands of capacities in the following order for demand planning

1. (Regular Capacity,Band\_1 Storage)
2. (Overtime\_1 Capacity,Band\_1 Storage)
3. (Overtime\_2 Capacity,Band\_1 Storage)

If demands are still unfulfilled then Band\_2 Storage is used with all bands of capacities in the following order for demand planning

1. (Regular Capacity,Band\_2 Storage)
2. (Overtime\_1 Capacity,Band\_2 Storage)
3. (Overtime\_2 Capacity,Band\_2 Storage)

If Band Priority is set to “Min\_Max\_Band”. This implies a search scheme where “minimum of maximum bands” gets higher priority. The combination in which the storage bands and capacity bands are picked up for demand planning is as follows:

1. (Regular Capacity,Regular Storage)
2. (Regular Capacity,Band\_1 Storage)
3. (Overtime\_1 Capacity,Regular Storage)
4. (Overtime\_1 Capacity,Band\_1 Storage)
5. (Regular Capacity,Band\_2 Storage)
6. (Overtime\_2 Capacity,Regular Storage)
7. (Overtime\_1 Capacity,Band\_2 Storage)
8. (Overtime\_2 Capacity,Band\_1 Storage)
9. (Overtime\_2 Capacity,Band\_2 Storage)

The above order ensures that there is optimal usage of all bands of capacities and storage

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Band Priority → enter values in the text box

Band Priority 

MIN\_MAX\_BAND  

The above flag can take the following values:

1. Capacity
2. Storage
3. MIN\_MAX\_BAND(default).

### Measure Groups and Relationship Types

For the implementation of Storage Bands, following measures/relationship types have to be maintained. These measures and edge properties go hand in hand with respect to Storage Band configuration. This means if measures are configured and corresponding pegging properties are not configured(and vice versa) then SCS throws an exception.

1. Storage Availability Measure Group
  - a. Measures
    - i. Band\_1 Storage : It takes numeric values. The number specifies overtime storage that will be available at a specific time bucket
    - ii. Band\_2 Storage :It takes numeric values.The number specifies overtime storage that will be available at a specific time bucket after regular and Overtime\_1 storage
  - b. Granularity
    - i. Storage Node
      1. Storage.Storage ID
    - ii. Time.Fiscal Week
    - iii. Version.Version Name

## 2. Storage Pegging

### a. Edge Properties

- i. Band\_1 Storage Pegged Quantity: It takes numeric values. SCS writes the demands that are pegged to this storage Band\_1.
- ii. Band\_2 Storage Pegged Quantity: It takes numeric values. SCS writes demands that are pegged to this storage band\_2

Same should be specified in the SCS config as follows

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Time Dep Band1 Available Storage Measure Name → Select corresponding measure from the drop down menu

Time Dep Band1 Available Storage Measure Name ⓘ

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Time Dep Band2 Available Storage Measure Name → Select corresponding measure from the drop down menu

Time Dep Band2 Available Storage Measure Name ⓘ

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Band1 Storage Pegging Property → Select corresponding measure from the drop down menu

Band1 Storage Pegging Property ⓘ

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Band1 Storage Pegging Property → Select corresponding measure from the drop down menu

Band2 Storage Pegging Property ⓘ

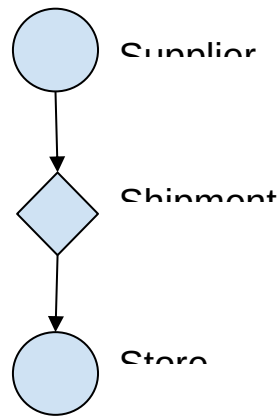
## Expedite WIP

### Business Need

There are cases where work-in-progress (WIP) operations can be expedited. For example, a usual job that takes two days can be completed in a day with extra personnel or by working extra hours when there is urgency. A user can specify both consumption and production buckets for a given work-in-progress (WIP) inventory. This overrides the lead time of the activity and honours the set WIP definition.

### Functional Description

Previously, a WIP was defined uniquely by bucket \* transaction ID. However, from now onwards WIP will be uniquely defined by only transaction ID (also known as LOT ID). A user should define the unique LOT ID for the WIPs defined along an activity. If a user has defined both consumption and production WIP for an activity with the same LOT ID, the solver considers it to a WIP expedite case and expedite the activity lead time as displayed in the examples below.

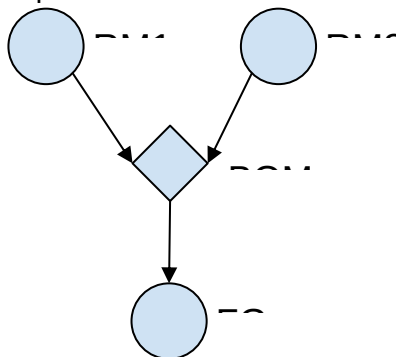


Activity Lead Time (shipment in the above schematic) is two Weeks.  
Input data looks like the following:

| Production Graph Time Dependent Properties |               |      |            |                |
|--------------------------------------------|---------------|------|------------|----------------|
| Activity                                   | Material Node | Time | SCS Doc ID | Production WIP |
| Shipment                                   | Store         | W10  | WIP_00     | 10             |

| Consumption Graph Time Dependent Properties |               |      |            |                 |
|---------------------------------------------|---------------|------|------------|-----------------|
| Activity                                    | Material Node | Time | SCS Doc ID | Consumption WIP |
| Shipment                                    | Supplier      | W09  | WIP_00     | 10              |

Ideally, the consumption WIP should have been on W08 because the lead time is two weeks. So this a case of expedite WIP with reduced lead time.



BOM lead time is two weeks.

Input data looks like the following:

| Production Graph Time Dependent Properties |               |      |            |                |
|--------------------------------------------|---------------|------|------------|----------------|
| Activity                                   | Material Node | Time | SCS Doc Id | Production WIP |
| BOM                                        | FG            | W10  | WIP_00     | 10             |

| Consumption Graph Time Dependent Properties |               |      |            |                 |
|---------------------------------------------|---------------|------|------------|-----------------|
| Activity                                    | Material Node | Time | SCS Doc Id | Consumption WIP |
| BOM                                         | RM1           | W06  | WIP_00     | 10              |
| BOM                                         | RM2           | W07  | WIP_00     | 10              |

Consumptions on both components can take place in different buckets. Here, the consumption takes place for RM1 and RM2 in Week 06 and Week 07, respectively. As per the BOM activity lead time, the consumption should have been in Week 08. In this case, the lead time of BOM activity is stretched due to expedite WIP. The expedite WIP support is added to FULL PLAN only for now. In the future, the support is expected to be added for incremental planning.

## Configuration

To maintain backward compatibility until all the existing tenants move to LOT ID based WIPs, this behaviour is protected with the following solver plugin parameter.

[Expedite WIP], true)

This is only available in dynamic arguments and not in the SCS Config UI because this will be removed (deprecated) after other tenants migrate to Lot ID-based WIPs.

[\[TestCase: ExpediteWIP-stretchLeadTime\]](#)

[\[TestCase: ExpediteWIP-squeezeLeadTime\]](#)

## Allow WIP with Activity Holiday Bucket

### Business Case

After write-back of purchase schedules/stock transfer orders from o9 platform to customer, their planners/analysts can modify those in their legacy systems owing to the (internal - within DCs/external - with suppliers) collaborations. These orders are then firmed and sent back to o9 platform as input data for planning. But, the legacy systems do not have the holiday calendar integrated in them hence creating the possibility of a schedule/order producing on a holiday. o9 platform should not change the delivery dates while planning with the firmed orders, neither should such orders be ignored while planning.

### Functionality Description

In o9 platform Supply Chain Solver (SCS) , currently the Work-In-Progress (WIP) inventory is discarded if it is falling on an activity holiday bucket. In SCS one can define WIP along any three edges of an activity i.e. production WIP, consumption WIP and capacity consumption WIP. The solver ignores the WIP if it falls on the activity holiday bucket by throwing an error message in the log file like below

*ERROR LiveServer (19)      Bucket 14 is holiday. Ignoring WIP record.*

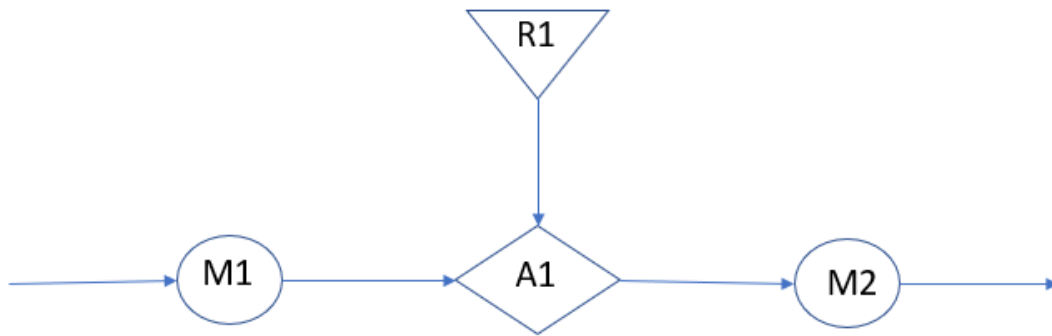
With this enhancement SCS now creates the WIP on activity holiday bucket.

This feature works with the tenant having LOTID or TransactionID modeled in the required SCS input and output tables.

### Example

Let's consider the following example.





Current bucket = W1

Lead Time for activity A1 = 7

Holiday buckets for A1 = W8, W9, W14, W15

- Production WIP on W15 for {A1, M2}

This WIP will be discarded in the default case. As part of this enhancement the WIP flow plan will be created on W15. Thus the material will be available at M2 on W15. The consumption bucket (both material and capacity) for the flow plan would be W4 (lead time is 7 and there are 4 holiday buckets, so it gets stretched).

- Consumption WIP on W8 for {A1, M1}

Like above this WIP would be created as part of this enhancement. The material consumption will happen on W8 at M1 so as capacity consumption from R1. The production bucket will be W19 (lead time is 7 and there are 4 holiday buckets, so it gets stretched) when material will be available at M2.

- Capacity consumption WIP on W9 for {A1, R1}

The material consumption will happen on W9 at M1 so as capacity consumption from R1. The production bucket will be W19 (lead time is 7 and there are 3 holiday buckets, so it gets stretched) when material will be available at M2.

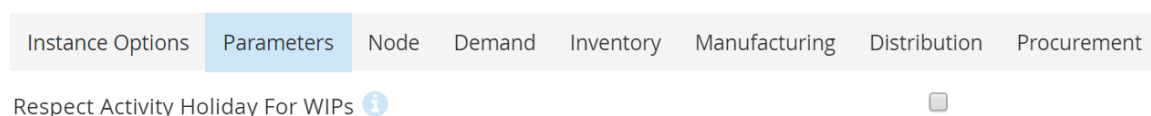
## Configuration

User needs to set a global parameter “Respect Activity Holiday For WIPs” to “false” to get this feature. The default value for the parameter is “true” i.e. WIPs are not created in activity holiday bucket. This ensures backward compatibility.

User can set this in command line argument or in SCS config UI as shown below

([Respect Activity Holiday For WIPs], false)

Or



By default, above checkbox will be ticked. User needs to untick it to get this current feature of creating WIP on activity holiday bucket.



Along with the above parameter user needs to set the following global parameter as command line argument.

([Expedite WIP], true)

## Safety Stock

### Safety Stock Planning

#### Business Need

Safety stock is the extra material that is maintained to mitigate the risk of stockouts due to the uncertainties in demand and supply. For specific items, planners like to hold safety stock. They can specify this as time based cover or quantity based.

#### Functional Description

Safety Stock can be constant throughout the planning horizon or can be time varying across the planning horizon. Safety Stock can be given in terms of Target Inventory and Weeks of Supply. Target Inventory is quantity based safety stock and represent the desired number of units to be held at the material node. Weeks of Supply or WOS the time based safety stock and represents a week's worth of supply desirable to held at the material mode. Planners have the flexibility to keep WOS and Target Inventory as time-varying or independent of time.

1. Constant Safety Stock:
  - a. Target Inventory: When planners want a constant quantity of inventory to be maintained at material nodes, they can specify a constant number indicating the same. [\[Functional Test #74\]](#)
  - b. Weeks of Supply: When planners want specific week's worth of inventory to be maintained throughout a material nodes, they can specify a constant number.
2. Time Varying Safety Stock
  - a. Target Inventory: Target inventory at material nodes can vary with time. Planners can specify the amount of inventory needed at a material node which can be different in different time buckets.
  - b. Weeks of Supply: Weeks worth of supply for material nodes can vary with time. For example, a material node at W01-2015 can have 4 weeks worth of supply, but at W02-2015 the same material node can have 5 weeks of supply and so on. In this case planners can specify the week's worth of inventory needed at a material node which can be different in different time buckets.

*Note: A material node can never have a combination of the above mentioned Safety Stocks i.e. Time Varying and Constant.*

#### Configuration

Constant Safety Stock is specified in Material Node Parameters measure group. It has the following measures:

1. Constant Target Inventory  
It takes numeric values. The number specifies the number of units of inventory desired at material nodes.
2. Constant WOS Target  
It takes numeric values. The number specifies the number of buckets worth of supply that is desired at material nodes.

Time varying safety stock is specified in a Measure Group named Material Node Time Parameters. It has the following measures:

1. IL Target Inventory

It takes numeric values. The number specifies time varying target inventory. A material node at W01-2015 can request 400 units of target inventory, but at W02-2015 the same material node can request 300 units of target inventory and so on.

## 2. IL WOS Target

It takes numeric values. The number specifies time varying weeks of supply. A material node at W01-2015 needs 4 weeks worth of supply, but at W02-2015 the same material node needs 5 weeks of supply and so on.

The granularity for safety stock(Constant) is

1. Material Node
  - a. [Item].[Item]
  - b. [Location].[Location]
2. [Version].[Version Name]

The granularity for safety stock(Time-Varying) is

1. Material Node
  - a. [Item].[Item]
  - b. [Location].[Location]
2. [Time].[Fiscal Week]
3. [Version].[Version Name]

Safety Stock can be supplied in the Plugin Configuration as follows

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Inventory → <Select measures from the drop down menu for the corresponding Safety Stock as shown below>

|                                          |                             |
|------------------------------------------|-----------------------------|
| Constant WOS Target Measure Name ⓘ       | Constant WOS Target ▼       |
| WOS Target Measure Name ⓘ                | IL WOS Target ▼             |
| WOS Reorder Point Measure Name ⓘ         | IL WOS Reorder Point ▼      |
| Constant Target Inventory Measure Name ⓘ | Constant Target Inventory ▼ |
| Target Inventory Measure Name ⓘ          | IL Target Inventory ▼       |
| Reorder Point Measure Name ⓘ             | IL Reorder Point ▼          |

## Limitations

- Currently safety stock is only planned for material nodes which are loaded. If SCS command to plan contains only specific orders, then it will load upstream the material nodes only on paths which are connected to these demands downstream. If there are other material nodes, on which safety stock is specified, then SCS will plan safety stock on these material nodes using the network that is loaded< otherwise safety stock on that material node will not be planned>.
- When the Inventory Plan Type is WoS, the quantity-based target inventory (if modeled) is ignored during the safety stock planning. Similarly, when the Inventory Plan Type is quantity, the WoS-based target inventory (if modeled) is ignored during the safety stock planning.

## Safety Stock Priority at Material Nodes

### Business Need

Safety stock acts as a buffer stock when either the actual demand is greater than forecasted or when suppliers are unable to deliver as per the standard lead time.

However demand and supply uncertainties can differ from one node in the network to another. For example,

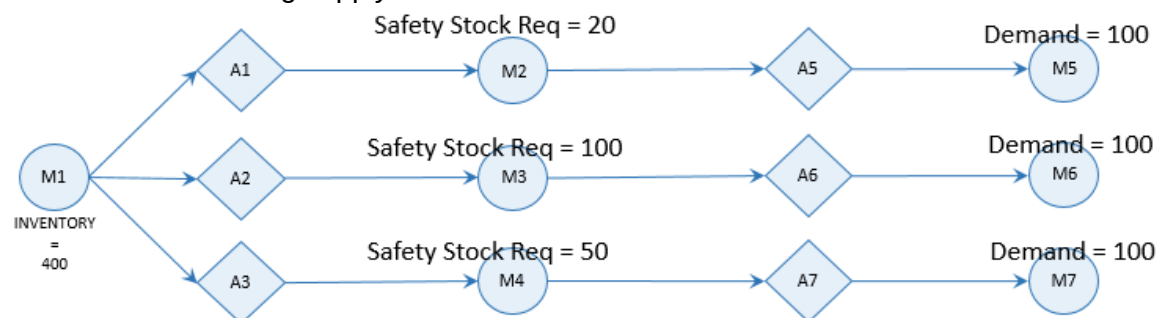
- there is limited supply from upstream and
- multiple locations in the network require safety stock that is higher than the supply.

Planners may decide to hold safety stock at locations that have higher probabilities of participating in the demand fulfilment process or where the uncertainties are higher.

### Functional Description

In constrained planning, when sufficient inventory is not available, planners can set priorities on material nodes such that safety stock allocation at higher priority material node takes precedence over safety stock allocation of lower priority material node. If priorities are not set, SCS picks material node from downstream to upstream. At the same level, the nodes are chosen based on the item dimension keys

Consider the following supply chain



Demand nodes M5, M6, M7 respectively have their own unique and independent paths all leading to the most upstream node M1. Inventory {BOH} is maintained at 400 . This inventory is sufficient to satisfy the 3 demands(total 300). It is observed that the total safety stock requirement is more than the actual inventory available for the same. SCS can pick any material node at random and do safety stock allocation and not meet safety stock at other locations. However, planner can prioritize the safety stock allocation such that SCS picks up material node of higher priority first, does the safety stock allocation on that node and then move on to the material node of the next priority. [[Functional Test #160](#)]

### Configuration Steps

A measure group Material Node Parameters is to be maintained

1. Measure
  - a. SS Priority  
This measure takes numeric values. Lower number means higher priority. Null is the least priority. [[Functional Test #161](#)]
2. Granularity
  - a. Material Node
    - i. Item.Item
    - ii. Location.Location
  - b. Version.Version Name

Same should be specified in the SCS config as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Inventory → SS Priority Measure Name → Select corresponding measure from the drop down menu

SS Priority Measure Name

SS Priority

Note: In general, the SS planning happens downstream to upstream but when the SS priority is defined, the safety stock planning is done based on the material node priority. In case of WoS-based or Quantity and WoS based safety stock planning, the SS priority is ignored i.e., the solver plans safety stock in the default sequence i.e., from downstream to upstream.

## Re-order Point & Order-Up-to-Level

### Business Need

Depending on the business, different inventory policies are set to trigger replenishment orders. One such policy is the reorder point (s), Order upto level policy (S). This is a continuous review system where orders are placed to bring the inventory back to the order up to level S when the inventory position drops to or below the reorder point s. In a continuous review system it will always order (S-s) as the replenishment quantity. However, SCS does planning at a bucket level and the EOH in the bucket can be below s, depending on the consumption from that bucket. Hence the replenished quantity will be greater than (S-s). Re-order point determines the safety stock needed to be held to meet uncertainties in demand and supply to avoid stockout. Hence this policy is named as MIN\_MAX policy.

### Functional Description

Reorder Point and Order-up-to-level (s,S) are quantities(numbers) that can vary over time. After JIT planning, SCS checks the EOH, the Reorder Point s, and order-up-to-level S, in every time bucket as mentioned below.

| Inventory Level |                                   |                                  |                                      |
|-----------------|-----------------------------------|----------------------------------|--------------------------------------|
| Time Buckets    | Calculated EOH for T <sub>i</sub> | ReOrder Point for T <sub>i</sub> | Order-up-to-level for T <sub>i</sub> |
| T1              | EOH_1                             | s <sub>1</sub>                   | S <sub>1</sub>                       |
| T2              | EOH_2                             | s <sub>2</sub>                   | S <sub>2</sub>                       |
| T3              | EOH_3                             | s <sub>3</sub>                   | S <sub>3</sub>                       |
| T4              | EOH_4                             | s <sub>4</sub>                   | S <sub>4</sub>                       |

Consider time bucket T<sub>1</sub>. If after demand JIT planning, corresponding EOH<sub>1</sub> is less than Re-order Point s<sub>1</sub>, then SCS triggers replenishment to bring the inventory level back to S<sub>1</sub>. If replenishment is triggered, then quantity to be ordered is calculated as {S<sub>1</sub> – EOH<sub>1</sub>}, so that the inventory is brought up to that level i.e S<sub>1</sub>. Similar calculations are initiated when Reorder point and order-up-to-level is given in terms of WOS. WOS is first converted to corresponding quantity by evaluating the demands in WOS time buckets. This means if Reorder Point is 2 weeks of supply and Order-up-to-level is 4 weeks of supply then

- Reorder Point = sum(Flows in the next 2 weeks)
- Order-up-to-level = sum(Flows in the next 4 weeks)
- Flows = Dependent and independent demands in a time bucket

### Configuration Steps

Order-up-to-level and ReOrder Points can be given in terms of Quantities or WOS. They are specified in a Measure Group named Material Node Time Parameters.

Order-up-to-level(nothing but target inventory at that material node) is configured as

- IL Target Inventory
  - It takes numeric values. The number specifies time varying target inventory. A material node at W01-2015 can request 400 units of

target inventory, but at W02-2015 the same material node can request 300 units of target inventory and so on.

- IL WOS Target
  - It takes numeric values. The number specifies time varying weeks of supply. A material node at W01-2015 needs 4 weeks worth of supply, but at W02-2015 the same material node needs 5 weeks of supply and so on.

ReOrder Point can be configured as

- IL Reorder Point
  - It takes numeric values. The number specifies target inventory at that material node in that time bucket.
- IL WOS Reorder Point
  - It takes numeric values. The number specifies time varying weeks of supply.

The granularity for Material Node Time Parameters is

1. Material Node
  - a. [Item].[Item]
  - b. [Location].[Location]
2. [Time].[Fiscal Week]
3. [Version].[Version Name]

The same can be supplied in the Plugin Configuration as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Inventory → Select corresponding measure from the drop down menu

|                                                                                                                    |                        |
|--------------------------------------------------------------------------------------------------------------------|------------------------|
| WOS Target Measure Name         | IL WOS Target ▼        |
| WOS Reorder Point Measure Name  | IL WOS Reorder Point ▼ |
| Target Inventory Measure Name   | IL Target Inventory ▼  |
| Reorder Point Measure Name      | IL Reorder Point ▼     |

Inventory Plan Policy can be set as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Inventory Plan Policy → Select one of the two values

|                                                                                                           |                                                                                                      |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Inventory Plan Policy  | TARGET_LEVEL  ▼ |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|

Inventory Plan Policy can take one of the below mentioned 2 values:

1. TARGET\_LEVEL - Policy is order upto level. <By default>
2. MIN\_MAX - Policy is reorder point, Order-up-to-level

## SS Bands

### Business Need

When there is limited supply to meet the specified safety stock(SS) requirements, there might be a need to fair share. This is similar to the demand slice factor, where demands are proportionally fulfilled (fairshare) when there is limited supply except that fairshare of

safety stock requirements is done across all material nodes irrespective of priorities. Consider the following scenario where the supply under constrained situation after demand fulfillment is less than the Safety Stock Requirement. If the material nodes are of the same priority, planners may want to fair share the remaining supply in the proportion of safety stock requirements to the extent possible.

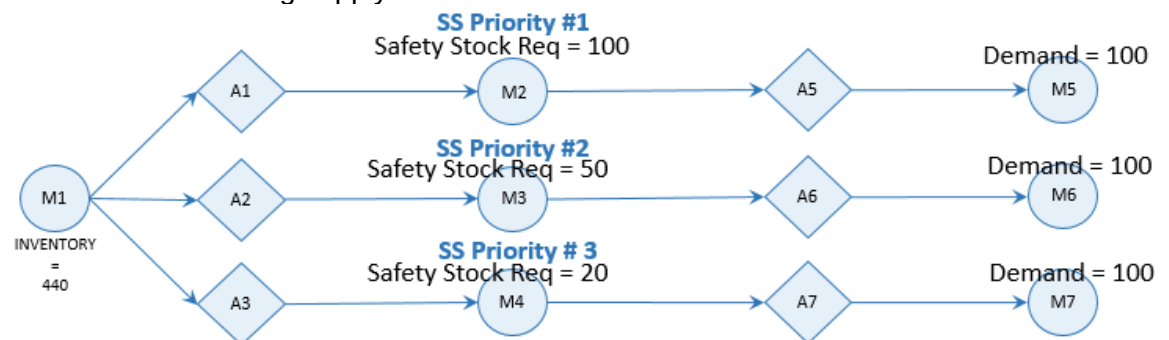
### Functional Description

SS Bands work in sync with Safety Stock Priorities at Material Node and the Safety Stock planning policy selected by planners.

Planners can specify the following safety stock policies:

1. Safety Stock Planning policy = TARGET LEVEL

Consider the following supply chain:



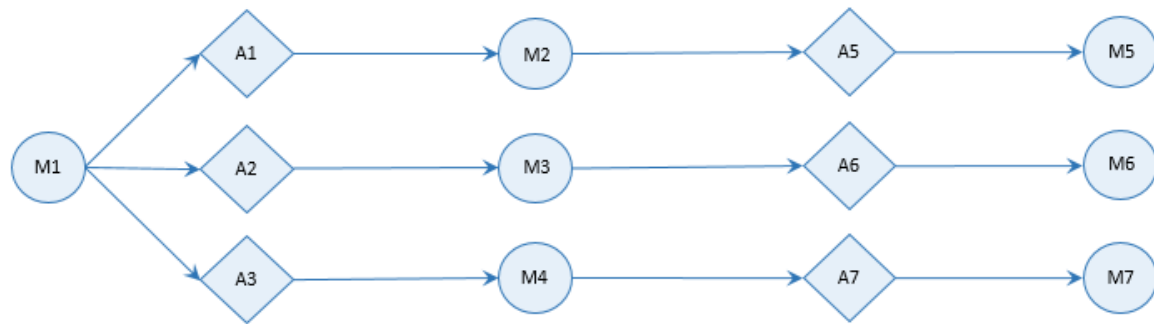
After the demand fulfillment, the inventory supply availability for safety stock planning is 140.

- SS Band = 0 or 1  
SCS will solve each Safety Stock requirement by picking up material nodes in the order of priority. Within the same priority, the downstream nodes are picked first. At the same level, the key determines the picking order of the node. In the above example, safety stock requirement is met first at M2(fully) and then at M3(partially). It shorts at M4, due to supply constraints.
- SS band = 2  
SCS divides each of the Safety Stock Requirements by 2 and solves them as follows:

| Material Node | SS requirement after MIN is fulfilled | SS Band = 2 | Order in which SCS plans SS |
|---------------|---------------------------------------|-------------|-----------------------------|
| M2            | 100                                   | 50          | 1st                         |
|               |                                       | 50          | 4th                         |
| M3            | 50                                    | 25          | 2nd                         |
|               |                                       | 25          | 5th(partially met)          |
| M4            | 20                                    | 10          | 3rd                         |
|               |                                       | 10          | 6th(shorted)                |

2. Safety Stock Planning policy = MIN\_MAX

Consider the same supply chain with different reorder point (s) and order upto levels (S)



| Material Node | Min            | Max            | Inventory Level  |  | SS Node Priority |
|---------------|----------------|----------------|------------------|--|------------------|
|               |                |                | EOH              |  |                  |
| M1            | s <sub>1</sub> | S <sub>1</sub> | EOH <sub>1</sub> |  | 1                |
| M2            | s <sub>2</sub> | S <sub>2</sub> | EOH <sub>2</sub> |  | 2                |
| M3            | s <sub>3</sub> | S <sub>3</sub> | EOH <sub>3</sub> |  | 3                |

EOH represents the inventory level after JIT planning, if it is greater than reorder point/min then SCS will not trigger replenishment to bring it to Max. This applicable even when SS Band is greater than 0.

- **SS Band = 0**  
At material node M2, after demand JIT planning, if corresponding EOH<sub>2</sub> is less than Re-order Point s<sub>2</sub> in any of the time buckets, then SCS triggers replenishment (by move in or produce new) to bring the inventory level back to S<sub>2</sub>. If replenishment is triggered, then quantity to be ordered is calculated as {S<sub>2</sub> – EOH<sub>2</sub>}, so that the inventory is brought up to that level i.e S<sub>2</sub>. SCS repeats the same process for material node M3 and M4 as per the SS node priority.
- **SS band = 1**  
SCS fulfills safety stock in 2 passes. In the 1st pass, it tries to satisfy s<sub>i</sub> of all material nodes M<sub>i</sub>, where min SS requirement is specified. Material node is chosen based on SS node priority and for that node it plans from first bucket to last bucket so as to make sure that inventory at least comes up to s<sub>i</sub> level. After minimum stock fulfilment at each of the above mentioned material nodes, SCS computes the EOH at that point. If supply is available, SCS tries to trigger replenishment. The quantity requested is calculated as {S<sub>i</sub> - EOH<sub>i</sub>} thereby bringing the inventory upto safety stock S<sub>i</sub> for each of material nodes M<sub>i</sub>..
- **SS band = 2**  
SCS fulfills safety stock in 3 passes. In the 1st pass, it tries to satisfy s<sub>i</sub> of all material nodes M<sub>i</sub> in the order of SS priority. After minimum stock fulfilment at each of the above mentioned material nodes, SCS computes the EOH. Then from the available supply, SCS triggers replenishment. The quantity that is requested is calculated as {(S<sub>i</sub> - s<sub>i</sub>)/ SS Band} thereby supplying safety stock to material nodes in a round robin fashion until it meets the SS requirements or meets the supply constraint. SCS also takes into consideration the priority in which the safety stock will be solved as follows:

| Material Node | SS requirement after MIN is fulfilled | Requested Quantity SS Band = 2           | Order in which SCS plans SS |
|---------------|---------------------------------------|------------------------------------------|-----------------------------|
| M2            | O2 = S <sub>2</sub> - s <sub>2</sub>  | s <sub>2</sub> + O2/2 - EOH <sub>2</sub> | 1st                         |



|    |                  |                      |     |
|----|------------------|----------------------|-----|
|    |                  | $S_2 - EOH_2$        | 4th |
| M3 | $O3 = S_3 - s_3$ | $s_3 + O3/2 - EOH_3$ | 2nd |
|    |                  | $S_3 - EOH_3$        | 5th |
| M4 | $O4 = S_4 - s_4$ | $s_4 + O4/2 - EOH_4$ | 3rd |
|    |                  | $S_4 - O4/2$         | 6th |

$EOH_i$  is the inventory level at the safety stock material node “i” at that point of time during planning

### Configuration Steps

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Inventory Plan Bands → Set value<any positive integer>

Inventory Plan Bands 

*Inventory Plan Bands* takes positive numeric values. The default value is NULL. this means Safety Stock requirements will not be divided into subparts and be treated as target inventory. The number denotes the total number of subparts in which a stand alone safety stock requirement will be divided into thereby achieving fair share. It is applied globally across all specified SS material nodes.

## SS quantity visibility after JIT plan

### Business Need

Safety stock is held to overcome uncertainties. Depending on the business, planners specify time varying SS numbers in terms of time or quantity. Planners specify that for this item at this location, need to hold SS equivalent to 2 weeks of supply. Planners would like to also know in quantity terms the independent demand (directly incident on that material node) and dependent demand (propagated to this node through consumptions) requests on this node to meet the specified time based SS. This helps planners in plan analysis and better the demand supportability by holding an appropriate amount of inventory at material stocking points.

### Functional Description

SCS accepts safety stock requirements based on time or quantity. The time requirement of safety stock is specified in buckets of supply, usually referred to as weeks of supply (WoS). SCS has the ability to write the actual WoS requirement in terms of quantity at the end of JIT planning, based on the WoS input provided. Planners now will be able to interpret the WoS quantity at a material node in specific time buckets when SCS is run with [inventory plan type] as WoS. SCS calculates WoS quantity for dependent and independent demands independently and when the measure is configured in the plugin, exports it out after the plan to these measures.

Consider the following supply chain



Demands incident on the material nodes are as follows:



| Order Name | Customer Name | Item | Location | Fiscal Week | Demand Quantity |
|------------|---------------|------|----------|-------------|-----------------|
| Order_1    | Cust_1        | FG_1 | VMI_1    | W06-2015    | 50              |
| Order_2    | Cust_1        | FG_1 | WH_1     | W06-2015    | 25              |

### **Order 1**

Demand at material node {FG\_1, VMI\_1} creates a subsequent requirement of 50 units of dependent demand at material node {FG\_1, WH\_1}, which in turn creates a subsequent demand of 50 units at material node {FG\_1, Factory}.

### **Order 2**

Demand at material node {FG\_1, WH\_1} creates a subsequent demand of 25 units at material node {FG\_1, Factory}.

Thus dependant and independent demands at each material node in the above mentioned supply chain would look as follows:

| Item | Location | Independent Demand | Dependant Demand |
|------|----------|--------------------|------------------|
| FG_1 | VMI_1    | 50                 |                  |
| FG_1 | WH_1     | 25                 | 75               |
| FG_1 | Factory  |                    | 75               |

When WOS { 2 for example } is set on a material node {FG\_1, WH\_1}

| Location | Item | Constant WOS Target |
|----------|------|---------------------|
| WH_1     | FG_1 | 2                   |

The safety stock that needs to be held at the material node {FG\_1, WH\_1} is pulled in by 2 time buckets as follows:

| Location | Item | Fiscal Week | Beginning On Hand Material Quantity | Ending On Hand Material Quantity | WOS For Total Demand | WOS For Independent |
|----------|------|-------------|-------------------------------------|----------------------------------|----------------------|---------------------|
| WH_1     | FG_1 | W01-2015    | 0                                   | 0                                |                      |                     |
|          |      | W02-2015    | 0                                   | 0                                |                      |                     |
|          |      | W03-2015    | 0                                   | 50                               | 50                   |                     |
|          |      | W04-2015    | 50                                  | 75                               | 75                   | 25                  |
|          |      | W05-2015    | 75                                  | 25                               | 25                   | 25                  |
|          |      | W06-2015    | 25                                  | 0                                |                      |                     |
|          |      | W07-2015    | 0                                   | 0                                |                      |                     |

SCS is capable of exporting WOS numbers when inventory plan type is set to WOS or QUANTITY\_WOS.

### **Configuration**

Following measures have to be configured

1. Inventory Plan Measure group
  - a. Measure
    - i. WOS For Total Demand : It takes numeric values. This measure stores the Total demand incident on a material node (both dependent and independent)
    - ii. WOS For Independent Demand : It takes numeric values. This measure stores the independent demand directly incident on this material node.
  - b. Granularity

- i. Material Node
  - [Item].[Item]
  - [Location].[Location]
- ii. [Time].[Fiscal Week] - Granularity of time bucket in which the plan is run
- iii. [Version].[Version Name]

Same should be specified in the SCS config as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Inventory → Select corresponding measure names from the drop-down menu

|                                         |                            |
|-----------------------------------------|----------------------------|
| WOS For Total Demand Measure Name       | WOS For Total Demand       |
| WOS For Independent Demand Measure Name | WOS For Independent Demand |

## Safety Stock Policy for Quantity and Time

### Business Need

For fast moving items, usually the safety stock is specified in terms of time. For slow moving items, it is specified in terms of quantity. The supply chain planner might need to define both the quantity and time (WoS type) based safety stock on material nodes in their network. Safety stock planning should adhere to all the given requirements.

### Functionality Description

The Inventory Plan Type is updated to take additional parameter i.e., QUANTITY\_WOS. This allows the planner to specify both the WoS-based and the quantity-based safety stock. During the safety stock planning if both are defined in the same node, the solver takes the maximum of the two and plan for the same. Check the example in the next page.

### Configuration Steps

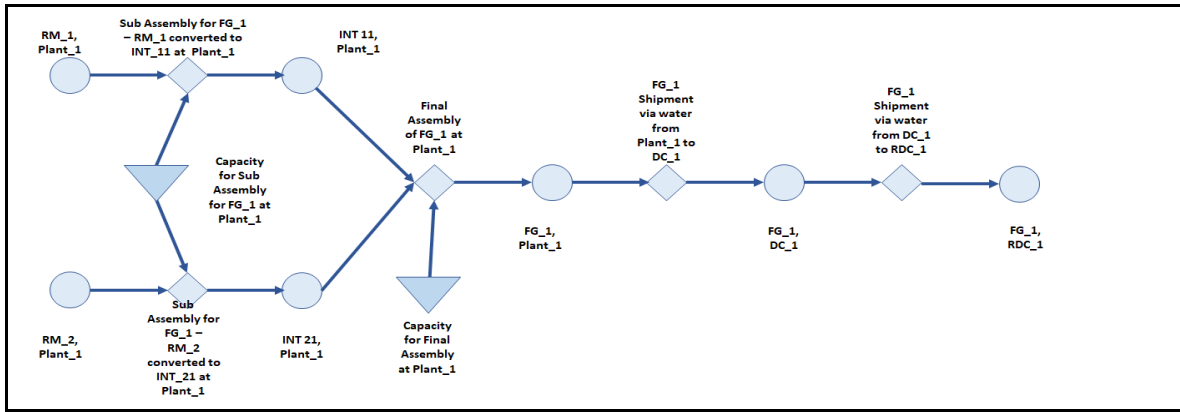
Navigate to Parameter tab of Solver

Designer → Edit Model → Rule → Plugins → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters

| Instance Options    | Parameters                                                                                                                                                                                                                                                                  | Node | Demand | Inventory | Manufacturing | Distribution |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------|-----------|---------------|--------------|
| Inventory Plan Type | This specifies if safety stock planning is required. NONE indicates no SS planning is done. QUANTITY indicates quantity based SS, WOS indicates cover based safety stock planning and QUANTITY_WOS indicates SS planning based on both Quantity and WOS Inventory planning. |      |        |           |               |              |

### Example:

Consider the Supply Chain presented below:



| Item | Location | Version Name       | Constant Target Inventory | Constant WOS Target |
|------|----------|--------------------|---------------------------|---------------------|
| FG_1 | RDC_1    | CurrentWorkingView | 20                        | 2                   |

Consider that there is a demand of 100 units at FG1,RDC\_1 on week 30.  
 Also, consider that there is a sufficient amount of raw material and capacity.  
 Run Constrained Plan with the Inventory Plan Type as “QUANTITY\_WOS”.  
 Let us see the impact of the quantity and the WoS planning at FG\_1,RDC\_1:

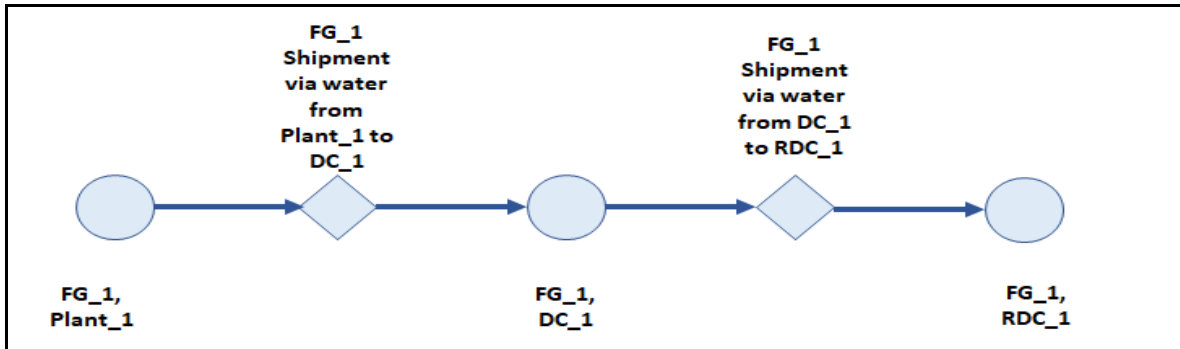
| Weeks                                                    | Constant Target Inventory | Constant WoS Target | Max of WoS vs Target Inventory |
|----------------------------------------------------------|---------------------------|---------------------|--------------------------------|
| Plan Start/When the material can be first produced - W27 | 20                        | 0                   | 20                             |
| W28 - W29                                                | 20                        | 100                 | 100                            |
| W30 - End of Horizon                                     | 20                        | 0                   | 20                             |

## Sequential WOS

In WoS-based safety stock planning, the solver computes the SS target just before the SS planning for the given node. If WoS is defined at multiple nodes (i.e., sequential WoS), it first plans for downstream nodes and then computes the target on the next upstream node where WoS is defined and plans for the same. The process is repeated until the safety stock planning is completed for all the nodes, where WoS is defined. After this, it exports WoS measures at the end (if configured in solver instance) i.e., WoS for total demand and WoS for independent demands. Check the below example for more details.

### Example:

Consider the Supply Chain presented below:



Consider that there is a demand of FG\_1@RDC\_1 and Constant WoS Targets are defined on FG\_1@RDC\_1 and FG\_1@DC\_1.

Please find below the sequence of steps in which safety stock planning will happen in this case

1. Safety stock planning starts at the downstream most node i.e. FG\_1@RDC\_1.
2. Once this safety stock planning is completed at FG\_1@RDC\_1 , it computes the targets at FG\_1@DC\_1.
3. Solver then does the safety stock planning at FG\_1@DC\_1.
4. Once Safety stock planning is done on all the nodes it finally exports the WoS related measures.

## Capacity

Capacity or resource is one of the basic building blocks of the supply chain network. It is used to represent the constraint that restricts the throughput of the system. The fundamental difference with material constraint is that material can be carried over from one planning bucket to another, but capacity expires at the end of the bucket.

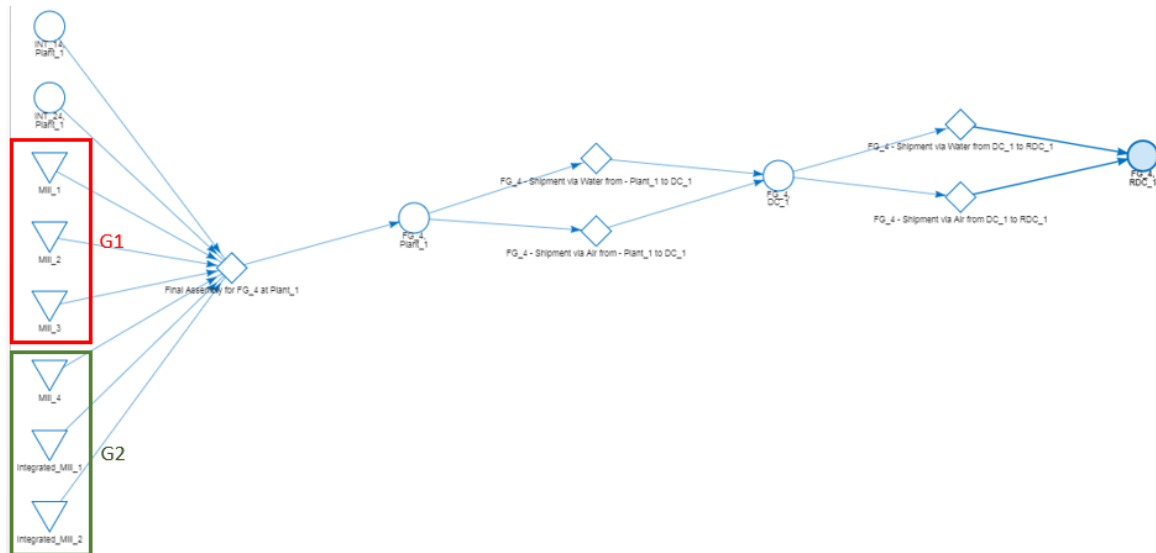
## Simultaneous

### Business Need

There can be cases, where more than one resource is needed to complete the task. These are called as simultaneous resources. For example, a manufacturing process might need both machinery and workforce to complete it. The non-availability of any one in a planning bucket restricts the process from running in that bucket.

### Functionality Description

Consider simultaneous resources R1 and R2. This means both R1 and R2 should be used together. Therefore R1 and R2 are assigned to a group G1. When specified in SCS config, that simultaneous capacity groups are to be used, SCS will simultaneously use both the capacities, R1 and R2, since they belong to same Group G1. Any activity can be linked to a Resource Group via association measures.



## Configuration Steps

A dimension by the name Group (shown as GroupComponents in below image) needs to be configured. It can have a level attribute named Group ID.

Designer

Edit Reports GroupComponents Editing Dimension 'GroupComponents'.

GroupComponents

Name \*

GroupComponents

Description \*

Sets of Simultaneous Capacities can

Attributes

Group ID

Hierarchies

Designer

Edit Reports GroupComponents Editing Attribute 'Group ID'.

Group ID

Name \*

Group ID

Description \*

Group ID can be assigned to Simulta

Order By

Name

☒ Key

Key Column Data Type

Integer

Key Generation

Auto

Attributes

Group ID

Hierarchies

Maintaining hierarchies is optional. Hierarchies are advised only when a dimension has more than one level attributes.

Following Measure Groups have to be maintained

1. Simultaneous Capacity Consumption Graph
2. Resource Group Priority

1. Simultaneous Capacity Consumption Graph

- Measure
  - a) Simultaneous Cap Cons Association  
This measure takes Boolean values 1/0.
    - 1 means that the corresponding Group ID and its Resources are active for simultaneous resources
    - 0 means that the corresponding Group ID and its Resources are inactive for simultaneous resources.
  - b) Simultaneous Distribution Capacity Consumption Graph Association
    - 1 means that the corresponding Group ID and its Resources are active for simultaneous resources in Distribution supply chain only
    - 0 means that the corresponding Group ID and its Resources are inactive for simultaneous resources in Distribution supply chain only
  - c) Simultaneous Supplier Capacity Consumption Graph Association
    - 1 means that the corresponding Group ID and its Resources are active for simultaneous resources in Procurement Supply chain only.
    - 0 means that the corresponding Group ID and its Resources are inactive for simultaneous resources in Procurement Supply chain only.
- Granularity
  - Activity Node
    - [Activity].[Activity]
  - [GroupComponents].[Group ID]
  - Capacity Node
    - [Resource].[Resource]
  - [Version].[Version Name]

2. Resource Group Priority

- Measure
  - Group Priority  
This measure takes numeric values. Lower value means higher priority.
- Granularity
  - [GroupComponents].[Group ID]
  - [Version].[Version Name]

*Note: For every data row inserted in Simultaneous Capacity Consumption Graph, similar entries have to be made in Capacity Consumption Graph (excluding the Group ID).*

The same can be specified in the solver config as follows:

1. Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Node → Group Components → Enter values in text

Group Components 

[GroupComponents].[Group ID]

2. Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Group Priority Measure Name → Select corresponding measure from drop-down menu

Group Priority Measure Name  Group Priority ▼

3. Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Simultaneous Capacity Consumption Graph Association Measure Name → Select corresponding measure from drop-down menu

Simultaneous Capacity Consumption Graph Association Measure Name  Simultaneous Cap Cons Association ▼

4. Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Distribution → Simultaneous Distribution Capacity Consumption Graph Association Measure Name → Select corresponding measure from drop-down menu

Simultaneous Distribution Capacity Consumption Graph Association Measure Name  ▼

5. Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Procurement → Simultaneous Supplier Capacity Consumption Graph Association Measure Name → Select corresponding measure from drop-down menu

Simultaneous Supplier Capacity Consumption Graph Association Measure Name  ▼

If an activity A requires resource R1 and R2 or resource R2 and R3, then one can create resource group with Group ID R12 and R23 in Group. Have the priority as 1 (primary) and 2 (secondary) respectively. This will help achieve both primary and alternate behaviour using groups. [\[Functional Test #66\]](#)

*Note: Specifying Group Priority Measure Name is optional and is needed only when the tenant has Simultaneous Resources and more than one resource groups are configured.*

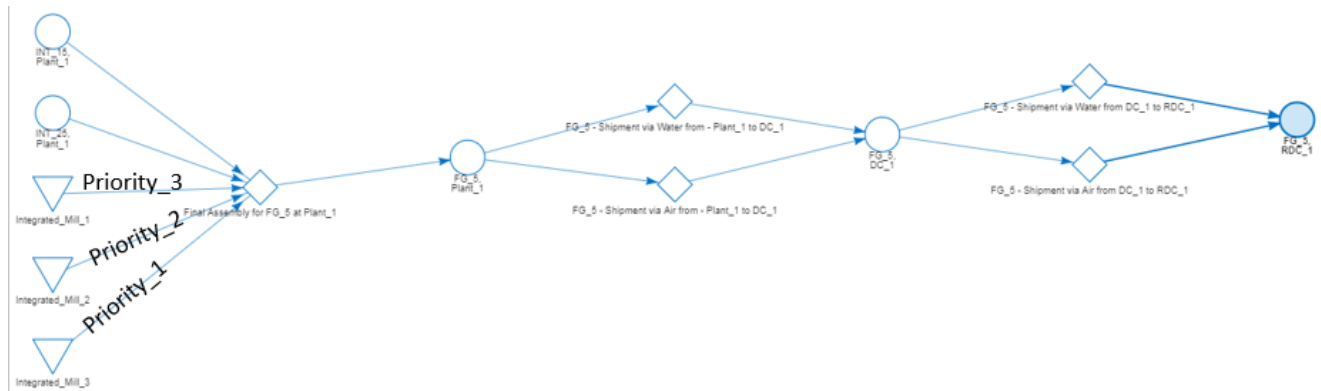
## Alternate

### Business Need

There can be cases, where one resource can be used instead of another. These are called as alternate resources. For example, a manufacturing process can use either resource1 or resource2 to complete the task. One can specify which resource is primary and which is secondary.

### Functionality Description

An activity can be linked to multiple resources. If simultaneous capacities are not specified in measure groups, then SCS assumes that the resources linked to that activity are alternate to each other. This is the default behaviour. Activity A1 can be linked to resources R1, R2 and R3. If R1, R2, R3 are alternate resources, it means activity A1 is processed when any one of these resources have available capacity in that bucket. However, resources can be picked up by an activity on a priority basis, which in turn can be set by the user.



## Configuration Steps

By default, if many resources load any activity, they are treated as alternate. Priority for resources corresponding to an activity can be set in Capacity Consumption Graph. The measure that needs to be configured is Capacity Graph Priority.

Capacity Graph Priority can be specified in the solver config as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →

o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Capacity Graph Priority Measure Name → Select corresponding measure name from the drop-down menu.

Capacity Graph Priority Measure Name ⓘ

Capacity Graph Priority ▼

## Limitations

SCS looks for primary resources and if not available loads the secondary resource in the same planning bucket. It will not load primary resource in earlier buckets, before offloading to secondary resource. Basically, given an activity, and a time bucket, all available capacities for that activity are considered. Thus, within a bucket, primary capacity is used first, then alternates in order of priority. Here it will use the regular first and then band1 and then band2.

One of the ways to overcome this limitation is to model two activities and provide priority of producing activities as primary and secondary. Set the parameter “Minimize Inventory Build Ahead” to TRUE if one wants to load primary in bucket T and then alternate in bucket T. Default is FALSE, which will load primary till build ahead limit and then offloads to secondary.

**Minimize Inventory Build Ahead:** SCS plans in JIT bucket and then by build ahead on primary path before going into secondary paths. However if one wants to minimize the inventory holding, one can set this flag to true, in which case, it will first use primary in bucket T and then secondary in bucket T. It will do build ahead on primary in bucket T-1 and then on secondary in bucket T-1 and so on. Default is False.

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →

o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Minimize Inventory Build Ahead → check if “TRUE”, uncheck if FALSE

Minimize Inventory Build Ahead ⓘ





## Varying Capacity Buckets

### Business Need

Business know the capacity availability at a monthly/quarterly level and would like to model it at that level. The actuals can be handled in execution system. For this, Planners need ability to model capacities with coarser buckets (integral multiple of planning buckets). For example, regular planning buckets could be weekly, and capacity buckets could be monthly (say, 4 weekly buckets or 5 weekly buckets, if fiscal calendar it is usually 4-4-5). Secondly, different capacities can have buckets of different length. Length of each bucket for a capacity can also vary.

### Functional Description

Capacities (supported for both regular and overtime) can have buckets of varying length. There is flexibility to interpret capacity availability of a resource in the following two ways. These 2 interpretations can be controlled by a flag.

1. Capacity availability for a resource specified in a time bucket is valid only in that time bucket. Other time buckets where value is not specified or null is interpreted as zero capacity. Planners have to explicitly provide capacity availability for a resource for all/some time buckets.
2. Capacity availability for a resource specified in a time bucket is valid till the next specified time bucket. If there is no other specification it is valid till end of the planning horizon. For example, if the buckets required are 4-4-5 as per fiscal months, then capacity will be specified in W1, W5, W9, W14 and so on. This means that the capacity specified in W1 hold across W1, W2, W3, W4. All these 4 weeks are interpreted as just one capacity bucket

Therefore capacities having varying time buckets can be incorporated in the 2nd interpretation using the existing constructs. The plans and pegging reports generated after SCS run remain the same for both interpretations.

### Configuration Steps

A measure group Capacity Availability is to be maintained

1. Measure
  - a. Capacity Availability : This measure takes numeric values. The number specifies the regular capacity that will be available at a specific time bucket.
2. Granularity
  - a. Capacity Node
    - i. Resource.[Resource]
  - b. Time.[Fiscal].[Week]
  - c. Version.[Version Name]

Same has to be specified in the SCS config as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → Time Dep Aval Cap Measure Name → choose measure from the drop down menu

Time Dep Aval Cap Measure Name ⓘ Capacity Availability ▼

Flags:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Varying Capacity Buckets → Check if TRUE or Uncheck if FALSE [By default it is Unchecked i.e FALSE]

Varying Capacity Buckets ⓘ



If FALSE(default)

When the flag is set to FALSE, the value of the Capacity Availability for a resource, if specified, remains as-is only for that time bucket. SCS is limited to use this capacity only in this bucket. For example,

| Resource | Time     | Capacity Availability |
|----------|----------|-----------------------|
| R_1      | W01-2015 | 30                    |
| R_1      | W05-2015 | 40                    |
| R_1      | W09-2015 | 10                    |
| R_1      | W14-2015 | 5                     |

From the above table SCS infers that capacity for Resource R\_1 is available only for 4 time buckets . For all other time buckets SCS assumes that the capacity for resource R\_1 is 0. SCS is limited to consume these resources only on the above specified time buckets. The consumption of these resources is agnostic to when the orders get fulfilled (JIT, early, or late).

If TRUE

When the flag is set to TRUE, capacity availability for a resource, if specified, is extended to all the subsequent time buckets in the planning horizon for that resource (until modified).

Thus if the Planning horizon is between W01-2015 to W52-2015, the above table is interpreted as follows:

Aggregated Capacity availability of resource R\_1 is 30 from W01 to W04.

This capacity of 30 units can be consumed by demand coming in any of buckets W01, W02, W03 or W04.

- W01 to W04 (treated as single bucket with 30 units of capacity availability)
- W05 to W08 (treated as next bucket with 40 units of capacity availability)
- W09 to W13 (treated as subsequent bucket with 10 units of capacity availability)
- W14 to W52 (treated as last bucket with 5 units of capacity availability)

## Capacity Bands

### Business Need

Shop floor managers need to effectively utilize the resources and optimize the throughput. Regular capacity could be one shift of 8 hours and there are options to run 2 additional shifts of 8 hours each. One needs to understand the trade off. Planners can model this two overtime capacities and make sure that overtime is only used when all normal/regular capacity is exhausted and by opening up one band at a time.

### Functional Description

Overtime capacities participate in demand planning only when regular capacities in all time buckets within the planning horizon fail to fulfill demands in JIT, or within the build ahead and build late period specified. Every resource in the tenant can have 3 types of capacities

1. Regular
2. 1st Overtime Band
3. 2nd Overtime Band

The order in which the above capacities are used in a bucket can be toggled using a flag.

If the flag is False or NULL.

While demand planning{especially during capacity constraint} SCS uses regular capacities in all time buckets to fulfill demand until Build late limit is reached. If regular capacities collectively fail to fulfill demands, then SCS looks for overtime capacities. It tries to satisfy the remaining demands using 1st Overtime Band capacities in all time buckets. If 1st Overtime Band capacities in all time buckets are exhausted and demands still remain unfulfilled, then SCS repeats the same by opening up the 2nd Overtime Band. Thus, all the capacities are used to fulfill demands in the following order

1. Regular Capacities in all time buckets
2. 1st Overtime Band Capacities in all time buckets
3. 2nd Overtime Band Capacities in all time buckets

If capacity re allocation takes place then, capacity is released in the reverse order i.e

1. 2nd Overtime Band Capacities
2. 1st Overtime Band Capacities
3. Regular Capacities

It should be noted that Capacity Flow plans show the overall capacity utilisation during demand planning. For example If a demand consumes 100 units of regular capacity and 100 units of overtime capacity then the Capacity Flow plan looks as follows:

| Fiscal Week | Resource                               | Activity                           | Capacity Consumption Plan |
|-------------|----------------------------------------|------------------------------------|---------------------------|
| W02-2015    | Capacity for Final Assembly at Plant_1 | Final Assembly for FG_1 at Plant_1 | 200                       |

If a demand is built Just-In-Time using Overtime capacities, then no constraint report is seen for the same[[#Testcase 263](#)]. However, if a demand is shorted even after using all the overtime capacities, then constraint report of Capacity Short is reported [[#Testcase 268](#)].The capacity constraint quantity in the report would suggest the amount of regular capacity needed at JIT bucket to build the order along primary path.

If the flag is TRUE

While demand planning{especially during capacity constraint} SCS uses regular + overtime (both 1st and 2nd bands) capacities in all time buckets to fulfill demand until Build late limit is reached.

To elaborate the above,

in order to satisfy an order JIT,

1st regular capacities in JIT time bucket is utilised.

If regular capacities in JIT bucket is not sufficient enough to fulfill the demand JIT , then it tries to use regular capacities in earlier buckets till build ahead limit is reached.

If still not satisfied, it open us 1st Overtime Band capacity in JIT bucket to fulfill the order. If it can't, it will try in earlier buckets using overtime band1 till build ahead limit is reached.

If still unsatisfied, it opens up the 2nd Overtime Band capacity in JIT bucket and then tries with band2 capacities in earlier buckets till BAL is reached.

If demand cannot be build JIT or ahead, then SCS tries to build the order late starting with utilizing  
regular capacities at JIT+1 bucket,  
then 1st Overtime Capacities at JIT+1 bucket,  
then 2nd Overtime band capacities in JIT+1 bucket  
and then goes to the JIT+2 bucket and so on until Build Late Limit is reached.

## Configuration Steps

### Flags:

The Flag that controls the behavior of regular and overtime capacity bands can be configured as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Use Bands To Minimize Lateness → Check for TRUE and Uncheck for FALSE [By default it is Unchecked i.e FALSE]

Use Bands To Minimize Lateness 



### Measure Groups

For the implementation of Capacity Bands, following measures/relationship types have to be maintained. These measures and edge properties go hand in hand with respect to Capacity Band configuration. This means if measures are configured and corresponding pegging properties are not configured (and vice versa) then SCS throws an exception.

1. Capacity Availability Measure Group
  - a. Measures
    - i. Overtime\_1: It takes numeric values. The number specifies overtime capacity that will be available at a specific time bucket
    - ii. Overtime\_2: It takes numeric values. The number specifies overtime capacity that will be available at a specific time bucket after regular and Overtime\_1 capacity
  - b. Granularity
    - i. Capacity Node
      1. Resource.Resource Name
    - ii. Time.Fiscal Week
    - iii. Version.Version Name
2. Capacity Pegging
  - a. Edge Properties
    - i. Overtime\_1 Pegged Quantity: It takes numeric values. SCS writes the demands that are pegged to this Capacity Band
    - ii. Overtime\_2 Pegged Quantity: It takes numeric values. SCS writes demands that are pegged to this capacity band.

*Note: Exceptions appear as follows:*

*For Manufacturing Supply Chain*

*Overtime1 capacity consumption pegging not defined for Assembly Activity*

*This would mean that 1st Overtime Capacity band is modelled as a measure but corresponding Pegging is not modelled in relationship graphs*

*For Distribution Supply Chain*

*Overtime1 capacity consumption pegging not defined for Distribution Activity*

*For Supplier Supply chain*

*Overtime1 capacity consumption pegging not defined for Supplier Activity*

Same should be specified in the SCS config as follows

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing→  
Overtime1 Time Dep Aval Cap Measure Name → choose measure from the drop down  
menu

Overtime1 Time Dep Aval Cap Measure Name 

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing→  
Overtime2 Time Dep Aval Cap Measure Name → choose measure from the drop down  
menu

Overtime2 Time Dep Aval Cap Measure Name 

## Manufacturing:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing→  
Overtime1 Capacity Consumption Pegging Property → choose property from the drop  
down menu

Overtime1 Capacity Consumption Pegging Property 

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing→  
Overtime2 Capacity Consumption Pegging Property → choose property from the drop  
down menu

Overtime2 Capacity Consumption Pegging Property 

## Distribution

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Distribution→  
Overtime2 Capacity Consumption Pegging Property → choose property from the drop  
down menu

Distribution Overtime1 Capacity Consumption Pegging Property 

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Distribution→  
Overtime2 Capacity Consumption Pegging Property → choose property from the drop  
down menu

Distribution Overtime2 Capacity Consumption Pegging Property 

## Procurement

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Procurement→  
Overtime1 Capacity Consumption Pegging Property → choose property from the drop  
down menu

Supplier Overtime1 Capacity Consumption Pegging Property 

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Procurement→

Overtime2 Capacity Consumption Pegging Property → choose property from the drop down menu

Supplier Overtime2 Capacity  
Consumption Pegging Property 

## Handling (Inbound & Outbound) Capacities

### Business Need

Planners would like to model certain capacities such that one of them is consumed at the start of the bucket and the other at the end of the bucket. It could be inbound (which comes into the material node) or outbound (which goes out of the material node). Apart from this, there could be regular capacities directly associated with the activity. Resource categories can hence be classified as inbound, outbound or regular (default). When these capacities are attached to a particular activity, it implies that all are required, meaning they act as simultaneous resources (even though not defined as resource grouping).

### Functional Description

Note that inbound is with respect to the activity from the modeling perspective. So whatever is inbound to the activity is actually going out from the material node. So please model it accordingly.

Consider the following supply chain

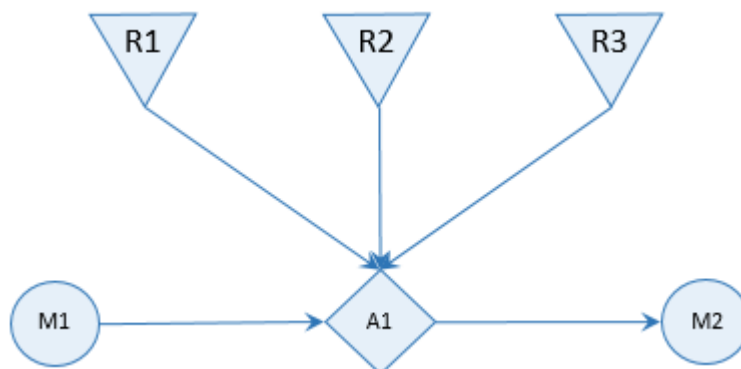


Image : HC001

Material node M2 can get its supply from material node M1 via Activity A1. Let's assume there are 3 resources R1, R2 and R3 that are consumed by Activity A1 upon its usage. At the end of the operation, material becomes available at M2 in the produced bucket. Capacities need to be available at the start of the bucket on all the resources for the production to take place.

However, Planners need flexibility to model resources that consume capacity in the produced bucket. Resources whose capacities are consumed by an activity at the production bucket are known as Outbound Resources. There could be alternate outbound resources, simultaneous outbound resources etc. Capacity from inbound and regular resources is consumed at the (Start) consumption bucket. The category is specified at the resource-activity level and hence gives ability to specify the same resource as inbound resource on one activity but as outbound resource on another activity as follows:

| Resource                    | Activity                              | Capacity Category |
|-----------------------------|---------------------------------------|-------------------|
| FG_12 - Inbound capacity_1  | FG_12 - Shipment from Factory to DC_1 | I                 |
| FG_12 - Inbound capacity_2  | FG_12 - Shipment from DC_1 to RDC_1   | I                 |
| FG_12 - Outbound capacity_1 | FG_12 - Shipment from Factory to DC_1 | O                 |
| FG_12 - Outbound capacity_2 | FG_12 - Shipment from DC_1 to RDC_1   | O                 |
| FG_12 - Regular capacity_1  | FG_12 - Shipment from Factory to DC_1 | R                 |
| FG_12 - Regular capacity_2  | FG_12 - Shipment from DC_1 to RDC_1   | R                 |

Image: HC002

### Configuration

Following measure has to be maintained

#### 1. Measure

- a. Capacity Category: This measure takes string values. Capacity category value for a specific Resource Activity {R1, A1} combination should be unique. This Resource Activity {R1, A1} combination cannot have multiple values of Capacity Category. There may be use cases where planners want a resource to be used an inbound for one activity and outbound for another activity. In such a case the measure values looks as shown in the above image (Image: HC002).
  - i. I : This would mean that the resource is an inbound resource and hence can only be consumed at the consumption time bucket (Start) of its corresponding activity
  - ii. O: This would mean that the resource is an outbound resource and hence can only be consumed at the production time bucket (End) of its corresponding activity
  - iii. R: This would mean that the resource is a Regular Resource and hence can only be consumed at the consumption time bucket (Start) of its corresponding activity
  - iv. Null: If no value or NULL is assigned to the resource category for a resource on an activity, then that resource is a regular resource and hence can only be consumed at the consumption time bucket (Start) of its corresponding activity.

If there are resources of different category loading the same activity, all of them will load simultaneously. For the same activity resource, if there are multiple capacities which need to be used simultaneously, then they need to be defined using resource grouping.

#### 2. Granularity

- a. Capacity Node
  - i. Resource.[Resource]
- b. Activity Node
  - i. Activity.[Activity Node]
- c. Version.[Version Name]



Capacity Consumption Graph ⚙️ 🔄 ✔️ Successfully Loaded Rules

Edit Reports

Capacity Category

Name \*

Capacity Category

Description \*

Resource Capacity measure name

Data Type

String

Type

Regular

☒ Editable

☐ Association Graph Measure

Format

#

Aggregation

Null

☐ Apply Conversion

Granularity **Resource** Activity Version Name

Measures

Capacity Category

Capacity Consumption Graph Association

Capacity Consumption Qty Per

Capacity Graph Priority

Rules

Same should be specified in the SCS plugin as follows:

Goto Plans → Global Plugins → SupplyChainSolver → Edit → Manufacturing → Resource Category Measure Name → choose measure from the drop down menu

Resource Category Measure Name ℹ️ Capacity Category

### Limitations

- A capacity cannot be part of more than one category when connected to the same activity.
- Simultaneous capacities in a Resource group must belong to the same Category.

## WIP consumes outbound capacity Only

### Business Need

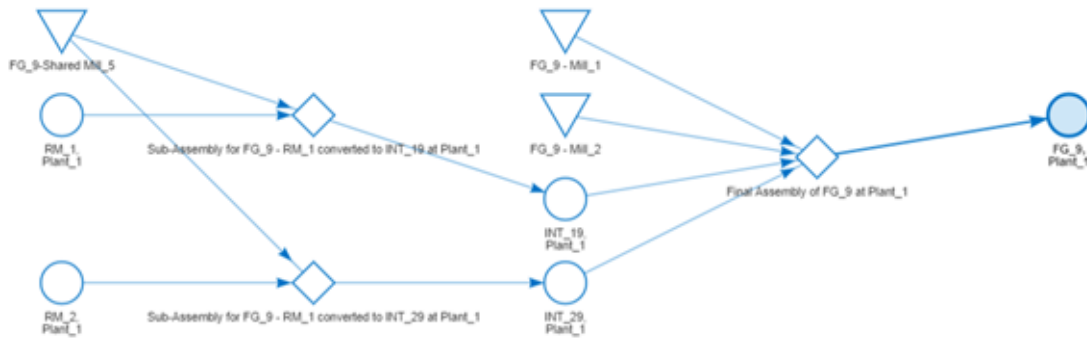
In many industries, transaction records represents what comes into the DC, RDC or Store. The materials would have left the source destination already and yet to reach the target destination. There are capacity restrictions throughout the supply chain. Specifically there can be handling resources to off load the incoming material at the destination. The planning need to cater this WIP (purchase order or distribution order which is arriving at a material node) is to consume only the outbound capacity resources. This WIP should not be propagated upstream and hence should not consume any other capacity and material upstream.

### Functionality Description

mPower Supply Chain Solver (SCS) does the WIP planning by consuming the material and capacity all the way upstream. In the current release SCS has been enhanced to support WIP to consume only outbound capacity.

Example,





In the above example FG\_9-Mill\_2 is outbound capacity.  
 The WIP is of 500 producing at FG\_9@Plant\_1 on W30.  
 Demand is 100 on W30 at FG\_9@Plant\_1.  
 The demand is satisfied using WIP qty.  
 With this enhancement, users would see only FG\_9-Mill\_2 to be consumed on W30 (WIP producing bucket). There will be no upstream traversal for WIP plan i.e. no upstream capacity or material consumption for the WIP.

## Configuration

Following Boolean measure needs to be created in the model. It should be of the same grain as the WIP measure. Ideally, one should define this in the same measure group. The default value of the measure is False. This needs to be set to True (or 1) for WIP records which should only consume outbound capacity and not consume any material and capacity upstream of it in the supply network.

Next, the measure should be added to SCS config (Manufacturing Tab)

Designer → Edit Models → Rules → Plugins → Global → o9.GraphCube.Plugins → SupplyChainSolver → <SCS instance> → Manufacturing → “Prod Consume Outbound Capacity for WIP Measure Name”.

When the value of the new measure is set to true, the scheduled WIP on that activity will consume only the outbound capacity and plan the WIP. Any material, inbound capacity and regular capacity will not be consumed.

*Note: This functionality is not supported alongwith Expedite WIP, which means it does not behave as per expected when ([Expedite WIP], “true”)*

## Minimize build ahead for capacity bands

### Business Need

Certain types of business (like retail) have capacity bands and would prefer to use the JIT buckets first before doing build ahead. The other flavor is to minimize overtime, where all the regular capacities are used first by doing build ahead and then only the band (overtime) capacities are opened up for consumption. This use case is more relevant in demand attribute based planning, where you would like to exactly do a plan and peg and know if a particular demand is utilizing overtime or not. Also when there are alternate activities, it needs to check all the alternates for regular capacity first before utilizing the overtime bands.

### Functionality Description

A resource can have regular as well as overtime bands of capacity. By default, solver chooses regular capacities from JIT, JIT-1, JIT-2 buckets and so on till build ahead is reached. If still demand is not satisfied, it consumes band1 overtime capacity sequentially as regular capacity for all the buckets (JIT through build ahead period). Similarly, it does for band2 overtime capacity. In this way, the demand is built much earlier.

On the contrary, if all the capacities (regular and overtime) are available on a resource in every bucket, this earliness can be reduced. Current functionality provides this flavor to minimize the build ahead on demands([#Testcase 389](#))

In case you have alternate capacities modeled through alternate activities and those resources have regular as well as overtime bands defined, in such a case in a given bucket first all the regular capacity will be used (across alternate paths) and then overtime bands will be used.

Example:

Let's assume activity A is loading the resource R.

R has capacity as follows

|           | Regular | Band1 | Band2 |
|-----------|---------|-------|-------|
| W5 Bucket | 50      | 30    | 20    |
| W4 Bucket | 50      | 30    | 20    |
| W3 Bucket | 50      | 30    | 20    |

Let's say demand is 180 on W5 (JIT bucket).

Assume lead time is 0 for activity A.

When this flag is not specified (default behaviour), all the 3 buckets regular capacities will be used along with Band1 capacity on JIT bucket. Thus, demand is built on W3 bucket.

On the contrary, through the current functionality, based on following configuration, solver can choose all the available capacities in W5 (i.e. JIT) bucket first which is 100.

Then it will move to W4 (i.e. JIT-1) bucket and consume regular and Band1 capacities to fulfil the demand. In this case, one has saved one bucket of pre-build (W3 bucket)

### Configuration

Configure SCS plugin as follows:

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters →  
Use Bands To Minimize Build Ahead → check if “TRUE”, uncheck if FALSE

Enable the flag by clicking the checkbox as shown below.

Use Bands To Minimize Build Ahead  ☐

When this flag is enabled, one must enable “Minimize Inventory Build Ahead”

Goto Workspace → Designer → Edit model → Rule → Plugin → Global →  
o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters →  
Minimize Inventory Build Ahead → check if “TRUE”, uncheck if FALSE

Minimize Inventory Build Ahead  ☐

## Capacity Re-entrant flow

### Business Need

The same machine or resource can be used across multiple operation. This leads to capacity re-entrant flows where we now need to fair share the shared capacity (in case of constrained scenario). This ultimately leads to improved demand satisfaction in some cases.

### Functionality Description

The planner can set the activity level option i.e., Activity Resource Re-entrant Measure Name which is provided to indicate that the re-entrant algorithm must kick that particular activity [ please set on the downstream activity].

The algorithm works in the following manner:

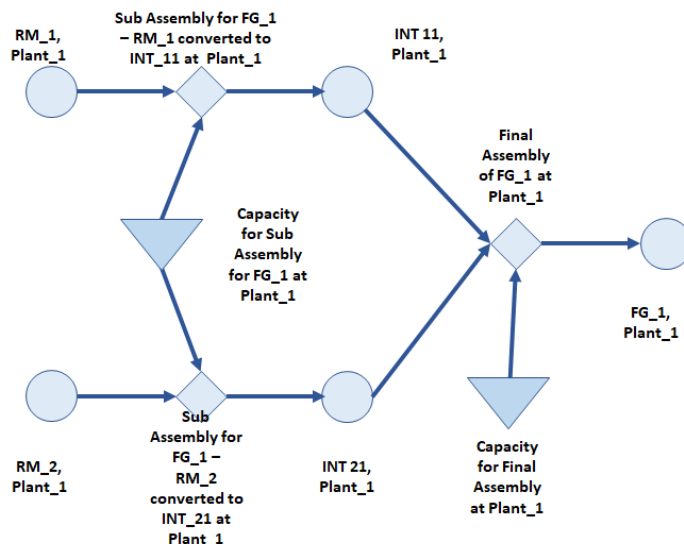
- At first, the solver tries to produce the requested units at an activity. If we are able to meet the demand, we continue with the normal flow and re-entrant algo does not get executed.
- In case, the solver is unable to produce the requested units and if the re-entrant flag is enabled for that activity, we need to reduce the requested quantity by half and retry with the reduced value. For example, if the demand is 100, the solver will divide by two and try to see if 50 can be achieved.
- If the solver is not able to satisfy the demand, then it further divides the remaining half of the demand request (i.e., out of the 50 remaining from the original demand request of 100, the solver will divide by 2 i.e., 25) and then the solver will retry with the updated demand request.
- If the solver is not able to satisfy the reduced demand request, then it further divides the demand request and continue the procedure until retry count is exceeded or the demand request is met.



In case a resource is 0 and re-entrant is associated with it, we need to ensure that the activity is not producing any items. In order to avoid this, retry counter has been added with the retry count set to 5 (there is no option for the user to change retry count) and then exit re-entrant algo after 5 trials.

## Example

Consider the below supply chain here sub assembly capacity is shared between two activities



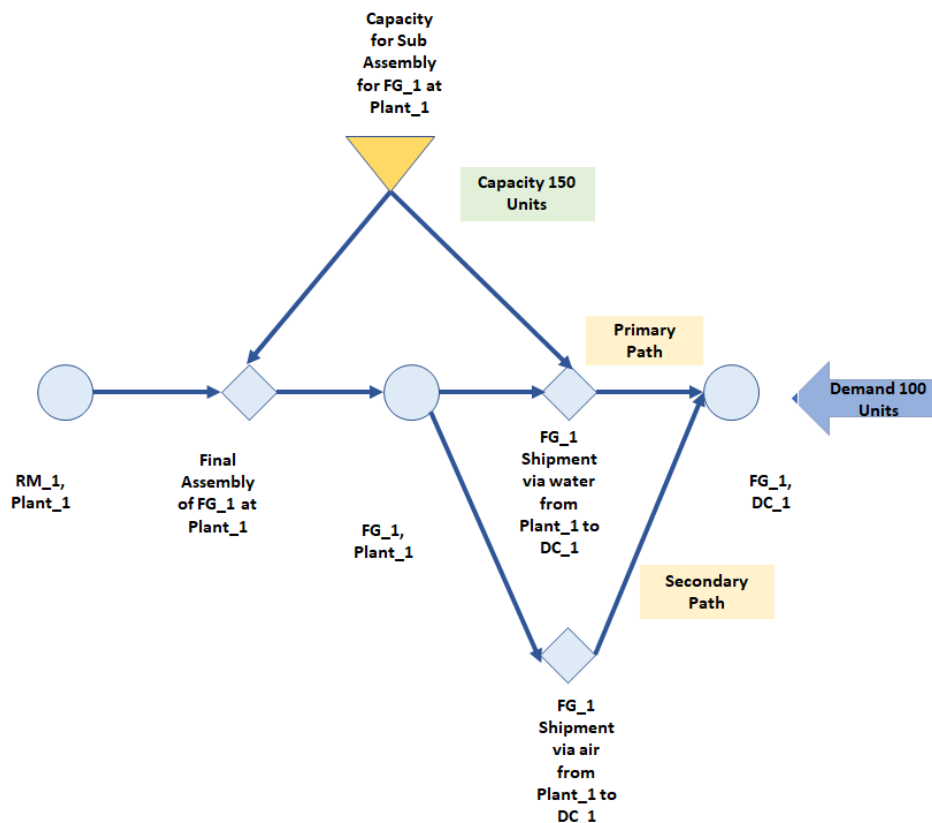
Consider there is enough material and Capacity at Final Assembly to Satisfy demand  
Consider there is demand of 100 units at FG\_1@Plant\_1  
Capacity at Sub Assembly let it be 120.

In case of non re-entrant flow demand fulfilment is only 20.

Let there be **Re-entrant flow enabled at Final Assembly of FG\_1 at Plant\_1**, with this demand fulfilment is improved i.e. 60, let us see what happens additionally here

1. After Solver is able to meet 20, Capacity left at Subassembly is 80
2. Next due to re-entrant flow it divides the rest of the demand by 2 i.e.  $(100 - 20) / 2$  i.e. 40
3. It asks for 40 at INT\_11, which is produced because of below actions  
When Solver asks at Sub Assembly for FG\_1 – RM\_1 converted to INT\_11 at Plant\_1 for 40 units of INT\_11 it internally sees there is capacity of 80 left, hence it can be used to produce 40 units of INT\_11
4. It asks for 40 at INT\_21, which is produced because of below actions  
When Solver asks at Sub Assembly for FG\_1 – RM\_2 converted to INT\_21 at Plant\_1 for 40 units of INT\_21 it internally sees there is capacity of 40 left, hence it can be used to produce 40 units of INT\_21
5. Post this demand satisfaction increases to 60 from 20
6. Since Capacity at Sub assembly becomes 0 now, solver will not go any further

Consider another case where we will see demand satisfaction reduces due to reentrant flow, let us take a look at the supply chain below



In Case of Non Reentrant Flow demand will be fully satisfied, 50 from primary path and 50 from secondary path.

Let there be **Re-entrant flow enabled at FG\_1 - Shipment via Water from DC\_1 to RDC\_1**, with this demand will be partially met i.e. only 75

- Pass 1 : From primary path demand will be partially fulfilled i.e. 50  
Capacity units remaining is 50 units ( i.e. 150 - 100 (since capacity is shared between two activities))
- Pass 2 : Next due to re-entrant flow it divides the rest of the demand by 2  
i.e.  $(100-50)/2 = 25$ . This can be satisfied since there is 50 capacity left.  
This also can be satisfied from the primary path

Post this since the capacity profile becomes zero, hence no further demand can be satisfied. As we can see in this scenario demand is partially fulfilled in case of reentrant flow.

### Configuration

To enable the capacity re-entrant flow, one measure needs to be configured in the same grain as Activity and type Boolean.

**Measure:** ActivityReEntrant

**Granularity:** <Activity>, <Version>

**Type:** Boolean



The default value is false. To improve the demand satisfaction, set to true for the most downstream activity which is connected to a capacity re-entrant flow.

Link the measure in the SCS config. Go to:

Designer → Edit Model → Rule → Plugins → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Manufacturing → **Activity Resource**

**ReEntrant Measure Name**

Select the corresponding measure name in the list.

### Telescopic Buckets

#### Business Need

Organizations want to have the visibility of supply planning for a longer horizon and at the same time, they also want to have the overall runtime of the enterprise application within some tolerable limit. By selecting a larger planning time bucket in the future, the user can eliminate unnecessary plan precision and detail for the entire plan and thus, it gives a benefit on runtime.

In this case as the planning horizon increases, the time bucket increases from a finer grain to a coarser grain.

Eg: Weeks to Months to Quarter. 1st 4 weeks planning can be at a week level, Next 8 weeks planning can be at a Monthly level, Next 12 weeks can be at Quarterly level

|                      |               |      |      |      |                |      |      |      |      |      |      |      |                  |      |      |      |      |      |      |      |      |      |      |      |
|----------------------|---------------|------|------|------|----------------|------|------|------|------|------|------|------|------------------|------|------|------|------|------|------|------|------|------|------|------|
| Finer grain buckets: | W 01          | W 02 | W 03 | W 04 | W 05           | W 06 | W 07 | W 08 | W 09 | W 10 | W 11 | W 12 | W 13             | W 14 | W 15 | W 16 | W 17 | W 18 | W 19 | W 20 | W 21 | W 22 | W 23 | W 24 |
| Telescopic buckets:  | TB 1          | TB 2 | TB 3 | TB 4 | TB 5           | TB 5 | TB 5 | TB 5 | TB 6 | TB 6 | TB 6 | TB 6 | TB 7             | TB 7 | TB 7 | TB 7 | TB 7 | TB 7 | TB 7 | TB 7 | TB 7 | TB 7 | TB 7 | TB 7 |
|                      | Weekly bucket |      |      |      | Monthly bucket |      |      |      |      |      |      |      | Quarterly bucket |      |      |      |      |      |      |      |      |      |      |      |

### Functionality Description

This is one of the performance optimization techniques followed in the o9 Platform Supply Chain Solver (SCS).

The following are the assumptions with telescopic buckets w.r.t. SCS:

1. The capacity balance equation holds good at the Telescopic Bucket (TB) grain. The capacity is aggregated and is available at the Telescopic Bucket Start boundary. (It is available for the TB)
2. It avoids intra-bucket negative on hand. Thus, the material feasibility is w.r.t. the finer grain bucket.

For example:

The material available is in the middle of TB and the demand is in the same TB but earlier to the material availability. In this case, the demand checks the Build Ahead Limit (BAL) if able to get planned. If not, it will short the demand. If Build Late Limit (BLL) is allowed, it will try at the start of the next TB. There it will find the material and get planned.

3. BAL search skips to the end of the previous TB.
4. BLL search skips to the beginning of the next TB.

### Example

|                   |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |
|-------------------|-----|-----|-----|-----|-----|-----|-----|-----|----|-----|-----|-----|-----|-----|-----|
| Regular Bucket    | W1  | W2  | W3  | W4  | W5  | W6  | W7  | W8  | W9 | W10 | W11 | W12 | W13 | W14 | W15 |
| Telescopic Bucket | TB1 | TB2 | TB3 | TB4 | TB5 | TB6 | TB7 | TB8 |    |     |     | TB9 |     |     |     |

### Case1 - Capacity Aggregation

- Demand of 100 on W13
- Supply chain lead time from demand node to Resource node is 0
- Resource capacity is 50 on W13, W15, respectively
- Both W13 and W15 falls in telescopic bucket TB9
- Internally, total 100 units capacity is available on TB9 start i.e., on W12. This 100 units is available for consumption in TB9 bucket, anywhere. Hence, the above demand will be planned on time.
- Please note the behavior is same if Capacity input is in Telescopic grain or in finer grain. For eg: in the above case, the result would have been the same if instead of inputting 50 units each on W13 and W15, the user would have input 100 units on TB9

### Case2 - Demand Build Ahead Limit

- Demand due on W14
- Once solver plans on W14 and is not able to meet demand completely, it will do build ahead and ask material on W11, the boundary of previous telescopic bucket TB8. In normal case (when telescopic buckets are not used), the solver does the sequential search i.e., W13, then W12 and so on.
- Here once it checks on W11, next build ahead search happens on W7 i.e., TB7 and so on.
- BAL defined is as per finer grain bucket, not Telescopic Bucket.
  - If BAL = 1, no BAL search is done
  - If BAL = 2, no BAL search is done
  - If BAL = 3, BAL search is done only at W11
  - If BAL = 4, BAL search is done only at W11
  - If BAL = 5, BAL search is done only at W11
  - If BAL = 6, BAL search is done only at W11
  - If BAL = 7, BAL search is done at W11 and W7
  - If BAL = 8, BAL search is done at W11 and W7 and W6
  - and so on...

### Case3 - Demand Build Late Limit

- Demand due on W7
- If it is not able to meet in due bucket, delay demand planning occurs.
- In this case with the above telescopic buckets, solver asks for the next telescopic buckets boundary. Here it will ask on W8 (TB8 start), then W12 (TB9 start) and so on.
- BLL defined is as per finer grain bucket, not Telescopic Bucket.
  - If BLL = 1, BLL search is done only at W8
  - If BLL = 2, BLL search is done only at W8

- If BLL = 3, BLL search is done only at W8
- If BLL = 4, BLL search is done only at W8
- If BLL = 5, BLL search is done at W8 and at W12
- If BLL = 6, BLL search is done at W8 and at W12
- If BLL = 7, BLL search is done at W8 and at W12
- If BLL = 8, BLL search is done at W8 and at W12
- and so on...

#### **Case4 - WOS**

- Demand due on W10 100 units, W11 200 units, W12 400 units
- WOS is calculated proportional to the no. of buckets in that Telescopic bucket
- At W8, WOS = 1 would mean 75 units (  $(100+200)*1/4$  units)
- At W8, WOS = 2 would mean 150 units (  $(100+200)*2/4$  units)
- At W8, WOS = 3 would mean 225 units (  $(100+200)*3/4$  units)
- At W8, WOS = 4 would mean 325 units (  $(100+200)*3/4 + (400)*1/4$  units)

#### **Case5 - Material**

- Demand of 100 on W13
- Supply chain lead time from Material node to demand node is 0
- Material is 50 on W13, W15, respectively
- Although both W13 and W15 falls in telescopic bucket TB9, ONLY W13 50 units is available for consumption because material is considered at finer grain
- Internally, ONLY 50 units material is available on W13. Hence, the above demand will NOT be planned on time (50 units either short or delayed depending on BLL)

#### **Case6 - Impact on Supportability**

- Since it is a Performance Optimization functionality, it is expected to plan faster and increase demand supportability in future coarser buckets
- But in certain exceptional cases, supportability might decrease because of activity lead times. For instance,
  - Demand of 100 due on W11
  - Supply chain lead time from demand node to Resource node is 1
  - Resource capacity is 100 on W7
  - Once solver plans on W11 and is not able to meet demand completely, it will do build ahead and ask material on W7, the boundary of previous telescopic bucket TB7.
  - Because the activity lead time is 1, it asks for capacity on W6 and not W7, so the demand gets shorted.
  - In normal case (when telescopic buckets are not used), the solver does the sequential search i.e., W10, then W9, then W8.
  - Once it checks on W8, it would have consumed the capacity available at W7.
  - So, there are some borderline scenarios where supportability goes down, in those scenarios, it is a tradeoff between speed and fulfillment.

#### **Note:**

- Expected Receipts (ER), SS, WIP, etc. are wrt finer grain bucket
- Activity Lead times, Activity Effectivity, Activity Holiday, Activity Marker, etc. are wrt finer grain bucket
- Plan and pegging output tables are available in finer as well as telescopic grain
- Since it is a performance optimization functionality, the planning time is expected to be shorter as compared to search methodologies like Binary Search and



Uniform Interval, etc.

- Currently, Varying Capacity Buckets is NOT supported with Telescopic Buckets. If a user has changed configuration according to Telescopic and changed Varying Capacity Buckets = TRUE, SCS will assume Varying Capacity Buckets = FALSE

## Configuration

- Create a Level Attribute (for example, Telescopic) as an alternate parent of the Leaf Timelike dimension's Level Attribute.
- **Regular Hierarchy:** Level Attr: [Year]->[Quarter]->[Month]->[Week]->[Day]
- **Alt Hierarchy:** Level Attr: [Telescopic]->[Day]
- The Alternate Hierarchy can have the following members:  
Alt parent-Child  
// assume 7 Daily Buckets  
Bucket1-Day1  
Bucket2-Day2  
...  
Bucket7-Day7  
//assume Bucket 8 is a Weekly Bucket, so it will have 7 children at Day level  
Bucket8-Day8  
Bucket8-Day9  
.....  
Bucket8-Day14  
//assume Bucket 9 is a monthly bucket, so it will have 30/31 children at day level  
Bucket9-Day15  
Bucket9-Day16  
.....  
Bucket9-Day45
- Telescopic attribute needs to be configured in SCS config UI:  
Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Telescopic Attribute Name → select relevant Time dimension and Telescopic LA from the drop-down menu

Telescopic Attribute Name ⓘ

|               |            |
|---------------|------------|
| Telescopic    |            |
| DimensionName | Time       |
| AttributeName | Telescopic |

## Others

### Negative On Hand

#### Business Need

In certain businesses (for eg: Logistic service providers), expeditors / progress chasers are assigned to certain parts and logistics centres. Their role is to identify shortages and attempt to resolve by collaborating with the supplier or by attempting to have a stock transfer from an alternate location. These expedited actions are purely manual and not expected to be an output of the plan. For this they need to run an unconstrained plan so as to meet the demands on time, but leave demand supply imbalance (no

supply/production, but there is demand/consumption) causing negative on hand (NOH) at intermediate nodes.

Other use case is to generate an unconstrained plan that calculates the unconstrained dependent demand at every level of the supply chain because sometimes the result you get with [Material Constrained] = false is skewed because demand is still constrained by lead-time.

This can also be used to generate an unconstrained "forecast" of build requirements at multiple levels of the supply chain.

### Functionality Description

Solver accounts for all the static constructs like lead time, lot size and plans the demand. These are hard constraints and certain demands are met late or shorted when they are incident within the lead time. This feature provides the planners the ability to have NOH at intermediate nodes and the upstream production can come at a later time after adhering to lead time constraints, but supply happens immediately.

### Configuration Steps

In unconstrained (capacity unconstrained, material unconstrained) mode of plan, 'Allow NOH' flag can be set to True to achieve this behaviour.

Goto Workspace → Designer → Edit model → Rule → Plugin → Global → o9.GraphCubePlugins → SupplyChainSolver → SupplyChainSolver → Parameters → Allow NOH → Check if TRUE or Uncheck if FALSE

Allow NOH 



For ideal flow of goods across material nodes, production at a material node should occur before consumption at that same material node. Negative on hand only occurs when consumption at a material node occurs before the production. It gives visibility to the planners so that they can make some exceptions (cross transfer between DC's which is not allowed in the base plan) to satisfy some of the demands on time.

### Limitations

'Allow NOH' flag should be used only when running an Unconstrained Plan.

## Graph Widget

### Business Need

Planners need the ability to view the supply chain network in a visual manner, so that they can comprehend the network connections and the changes required for the static network. This also helps them to figure out issues related to data (broken BOD, BOM etc). Planners also need the ability to drill down from a node after visual inspection to know additional details about the same.

### Functional Description

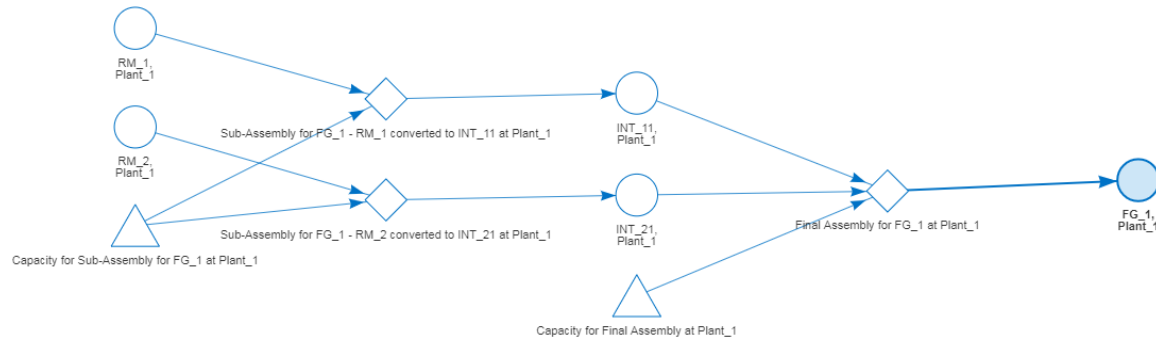
Consider a simple manufacturing process. The process is as follows:

Sub-assembly activities:

1. {RM\_1, Plant\_1} can be transformed to {INT\_11, Plant\_1} via activity {Sub-Assembly for FG\_1 - RM\_1 converted to INT\_11 at Plant\_1}
2. {RM\_2, Plant\_1} can be transformed to {INT\_21, Plant\_1} via activity {Sub-Assembly for FG\_1 - RM\_1 converted to INT\_21 at Plant\_1}
3. Activities {Sub-Assembly for FG\_1 - RM\_1 converted to INT\_11 at Plant\_1} and {Sub-Assembly for FG\_1 - RM\_1 converted to INT\_21 at Plant\_1} have a shared resource {Capacity for Sub-Assembly for FG\_1 at Plant\_1}

Final assembly activities:

1. {INT\_11,Plant\_1} and {INT\_21, Plant\_1} can be converted to {FG\_1,Plant\_1} via activity {Final Assembly for FG\_1 at Plant\_1}.
2. Activity {Final Assembly for FG\_1 at Plant\_1} utilize capacity of resource {Capacity for Final Assembly at Plant\_1}



The arrows converging to a material node is represented through Material Production Graph

The arrows converging to an activity node and leaving from a material node is represented through Material Consumption Graph

The arrows converging to an activity node and leaving from a capacity node is represented through Capacity Consumption Graph

### Configuration

In order to bring the above layout, following relationship types have to be maintained

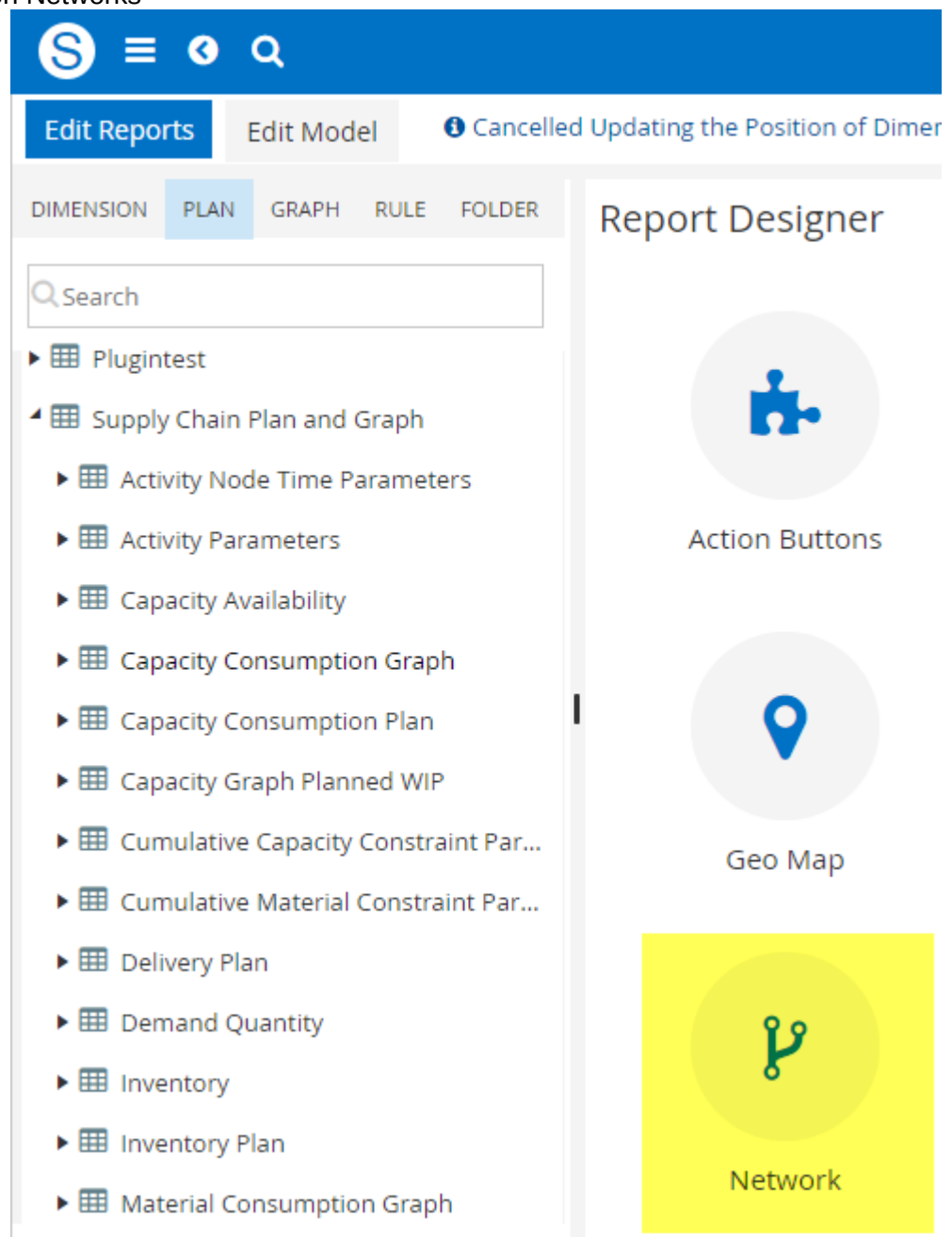
1. Material Consumption Graph
  - a. From Members :
    - i. Material Node
      1. [Item].[Item]
      2. [Location].[Location]
  - b. To Members
    - i. Activity Node
      1. [Activity].[Activity Name]
  - c. Properties
    - i. Material Consumption Graph Association
      1. Data Type: number
      2. Aggregation: Sum
2. Material Production Graph
  - a. From Members:
    - i. Activity Node
      1. [Activity].[Activity Name]
  - b. To Members
    - i. Material Node
      1. [Item].[Item]
      2. [Location].[Location]
  - c. Properties
    - i. Material Production Graph Association
      1. Data Type: number
      2. Aggregation: Sum
3. Capacity Consumption Graph
  - a. From Members:
    - i. Capacity Node
      1. [Resource].[Resource]
  - b. To Members

- i. Activity Node
  - 1. [Activity].[Activity Name]
- c. Properties
  - i. Capacity Consumption Graph Association
    - 1. Data Type: number
    - 2. Aggregation: Sum

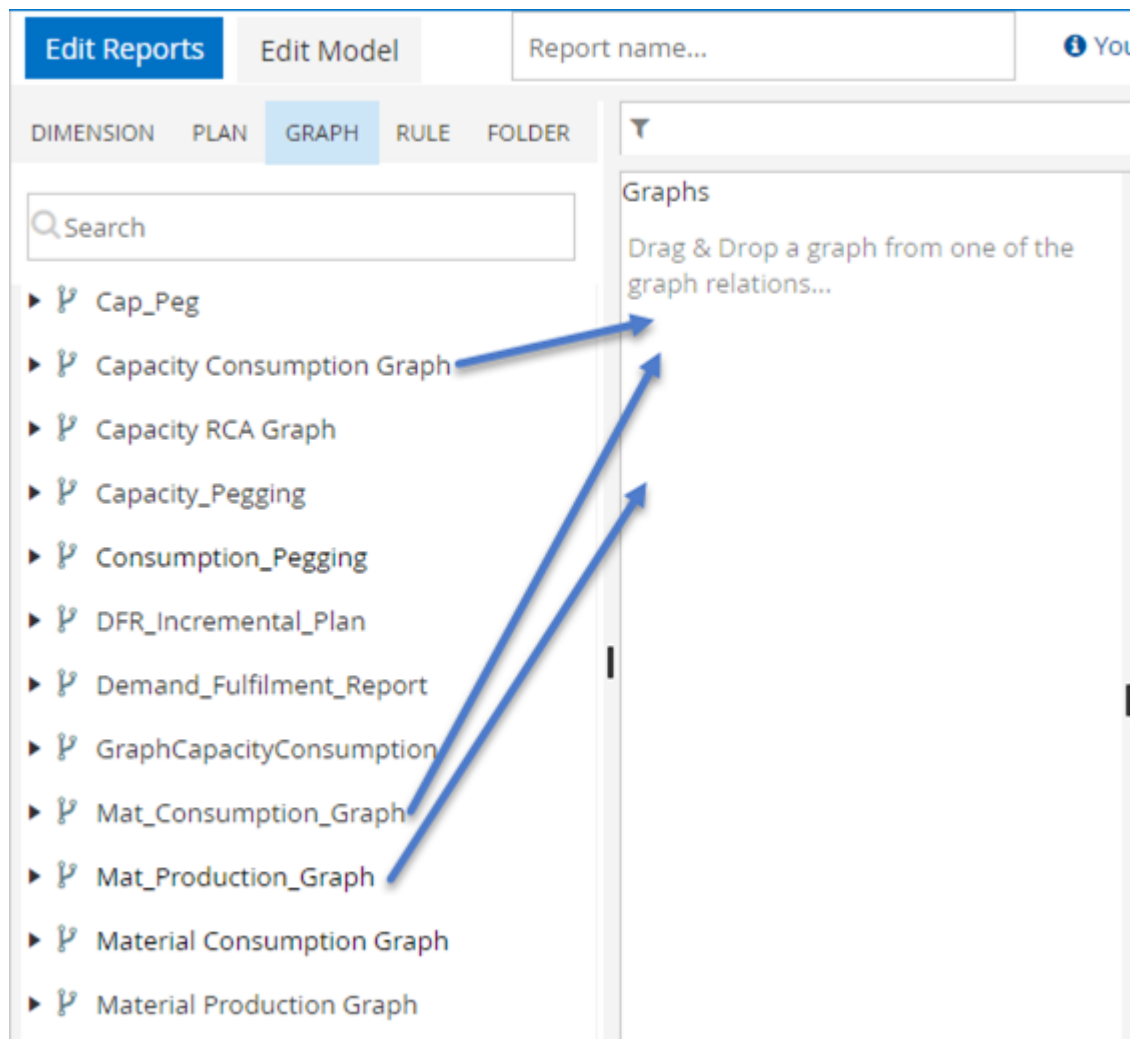
After all the 3 graphs are configured, user can proceed towards configuring the Graph Widget through designer as follows:

*Note: The following steps are in accordance with the multi-graph widget configured in tenant SCS-o9-hm*

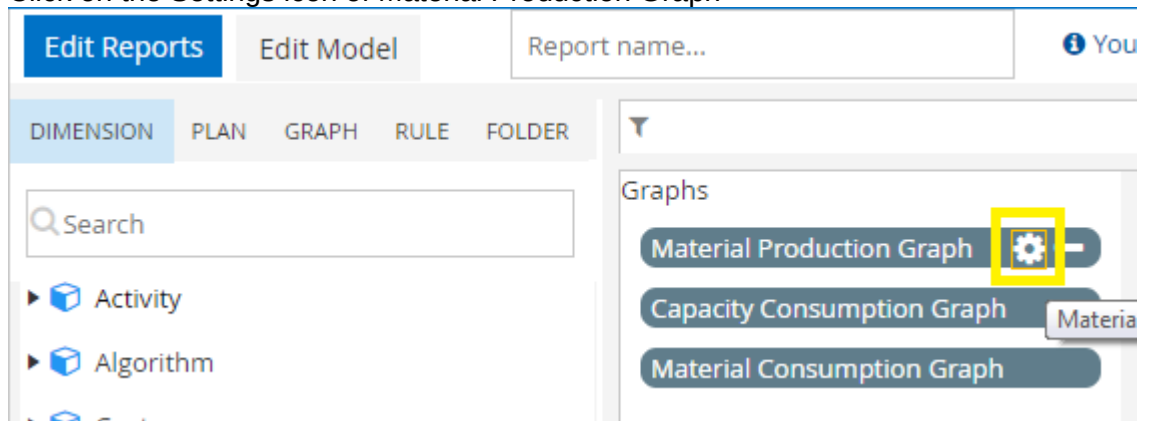
1. Goto Workspace → Designer → Edit Reports
2. Click on Networks



- a.
  3. Drag and Drop the graphs as follows



- a.
- Click on the Settings icon of Material Production Graph



5. Check the box for “Is Start Graph” and set the Start Node as TO from the drop down menu

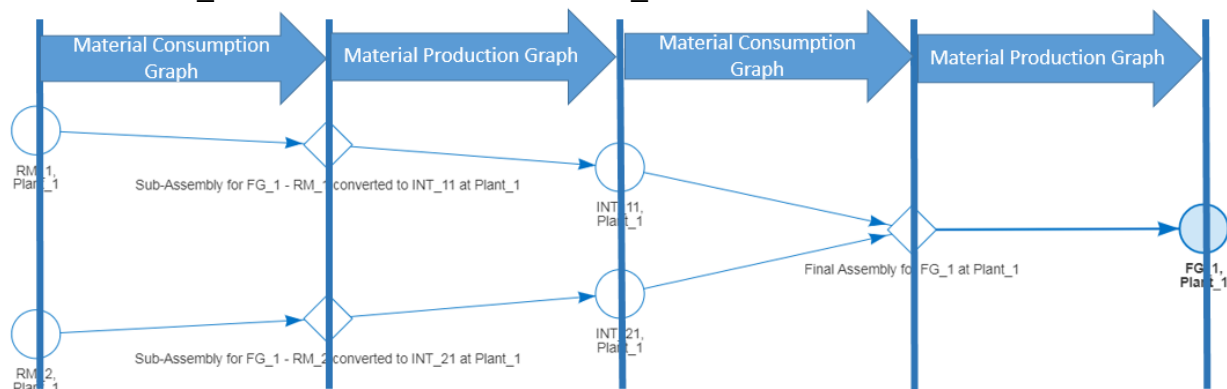
As evident from the supply chain, {FG\_1,Plant\_1} is the start node and present in the Material\_Production\_Graph. Therefore, Material\_Production\_Graph RelationshipType becomes the start graph.

Start node of a relationship Type can be a from node or a to node. This can be specified from the dropdown menu against Start Node. If the selected node is a FROM node then it is assumed that graph traversal will happen downstream. If it is a TO node then that graph traversal will happen upstream.

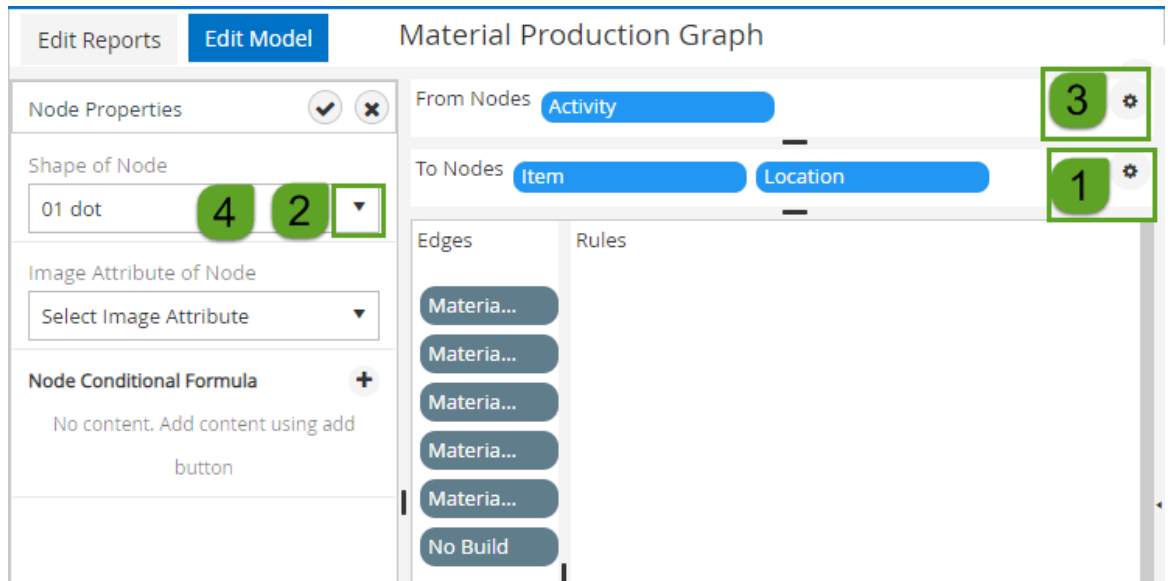
In this case , graph traversal has to happen upstream, so TO is selected from the drop down menu for Start Node

*Note: Filters are tied only to the start node*

6. Each of the above mentioned graphs have a from\_node and a to\_node. In order to successfully bring up the layout, make sure that to\_node of one graph becomes the from\_node of the consequent graph.
  - a. For example, Material\_Consumption\_Graph has material node as its from\_node and activity node as its to\_node
  - b. Similarly, in continuation, Material\_Production\_Graph has activity node as its from\_node and material node as its to\_node



7. Shape and colors of From\_node and To\_Node of each of the graphs can be configured as follows:  
Material Production Graph



Step 1: Click on To Nodes properties  as shown above.

Step 2: Select the shape of the node form the drop down menu

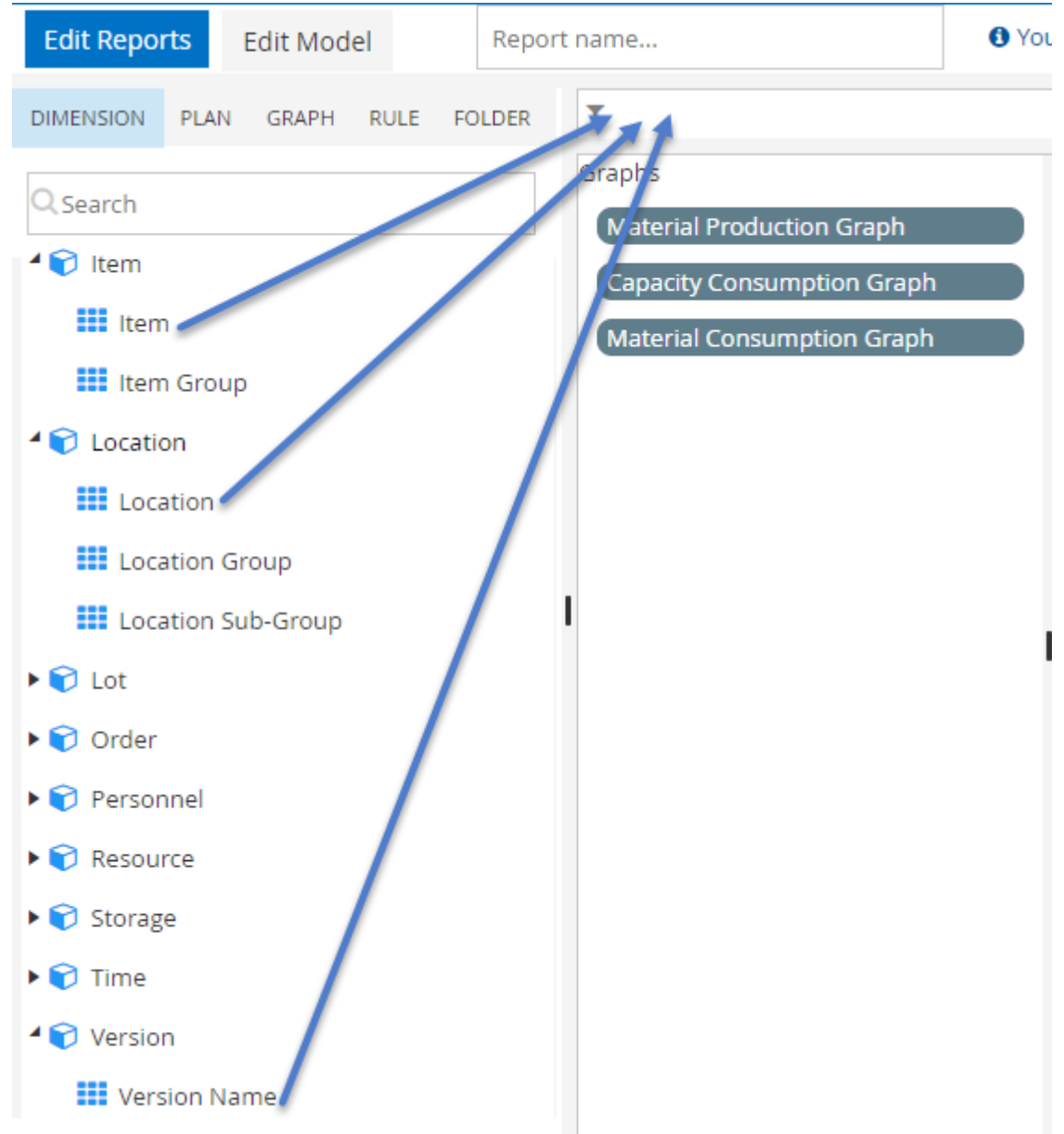
Step 3: Click on  to update the shape for the To Node

Step 4: Repeat Step 1 to Step 3 to select the shape of From Nodes

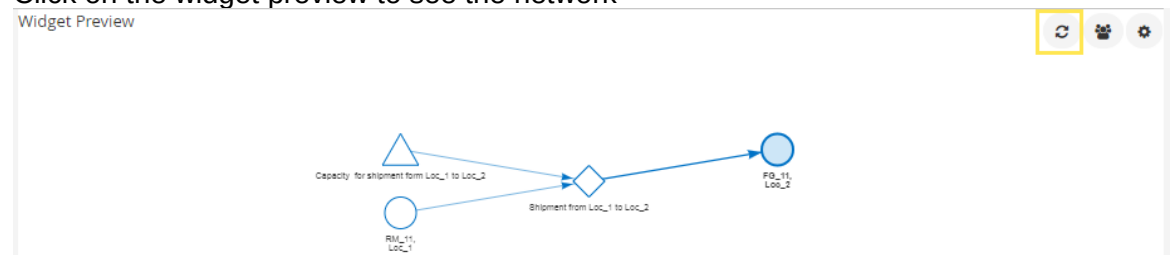
Repeat Step 1 to Step 4 for Material Consumption Graph and Capacity Consumption Graph

8. Add dimension "Item", "Location" and "Version" in the filters as follows. This is because in step 5, Material Production Graph has been set as the start graph, the

to nodes of which are item and location



9. Click on the widget preview to see the network

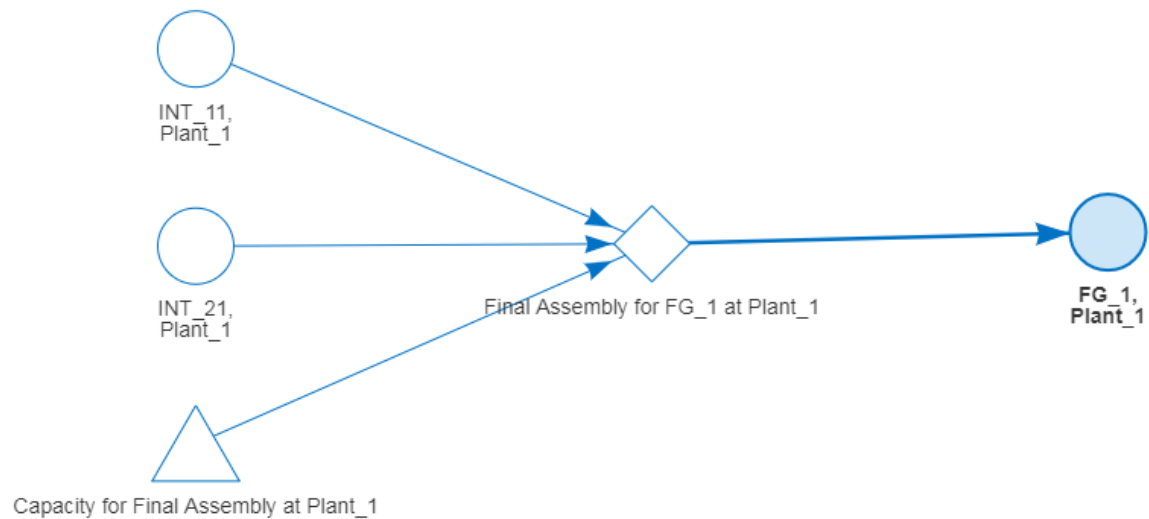




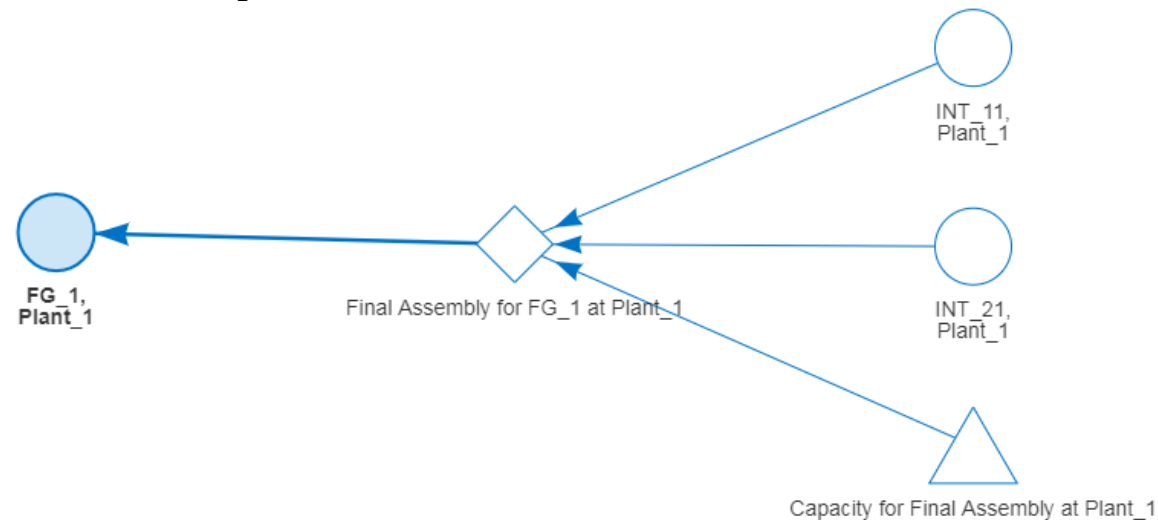
10. Click on widget settings to view the following menu

The screenshot shows a settings menu for a widget. At the top, there are three tabs: 'Edit Reports' (highlighted in blue), 'Edit Model', and a circular icon with a double arrow. Below the tabs, the menu is titled 'Widget' with a checkmark and a close button. The settings are organized into sections: 'Name' with a text input field containing 'o9Graph'; 'Title' with a text input field containing 'Graph Title'; 'Layout Direction' with a dropdown menu set to 'LR'; 'Number of Traversal Steps' with a text input field containing '0'; a group of three checkboxes: 'Show Animations' (checked), 'Show Pegging Values' (unchecked), and 'Show Image' (checked); and a section titled 'Show Node Labels' containing three checkboxes: 'Activity' (unchecked), 'Item,Location' (checked), and 'Resource' (unchecked).

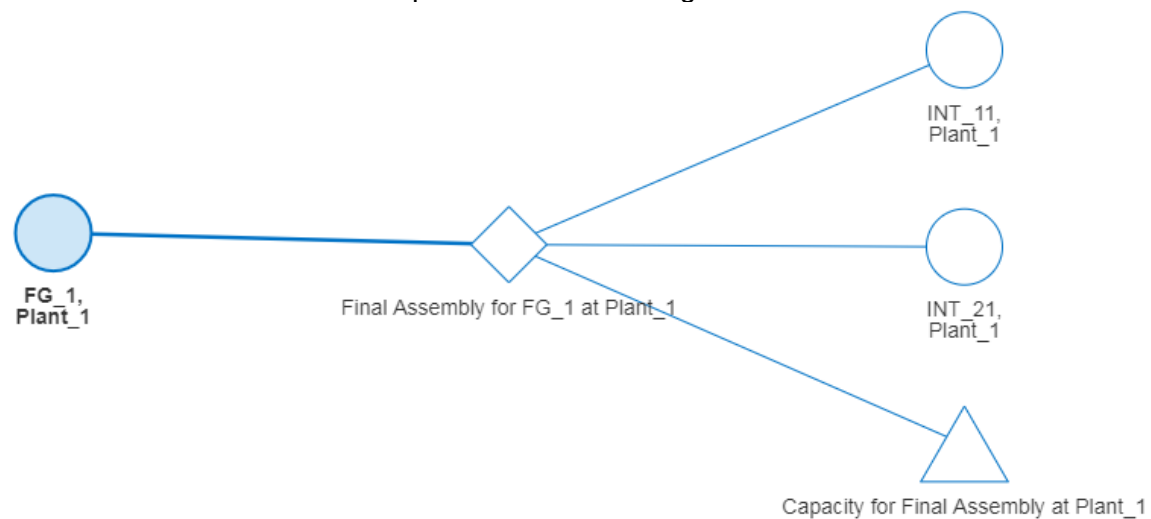
- a. Name: Name of the widget
- b. Title: Title of the widget
- c. Layout Direction: Layout direction can be set as
  - i. Left to Right (LR) : Demand node appears on the extreme right and the supply nodes appear relatively towards the left of the demand node. The direction of the arrows represent the flow of materials, i.e from Left to Right



- ii. Right to Left (RL) : Demand node appears on the extreme left and the supply nodes appear relatively towards the right of the demand node. The direction of the arrows represent the flow of materials, i.e from Right to Left



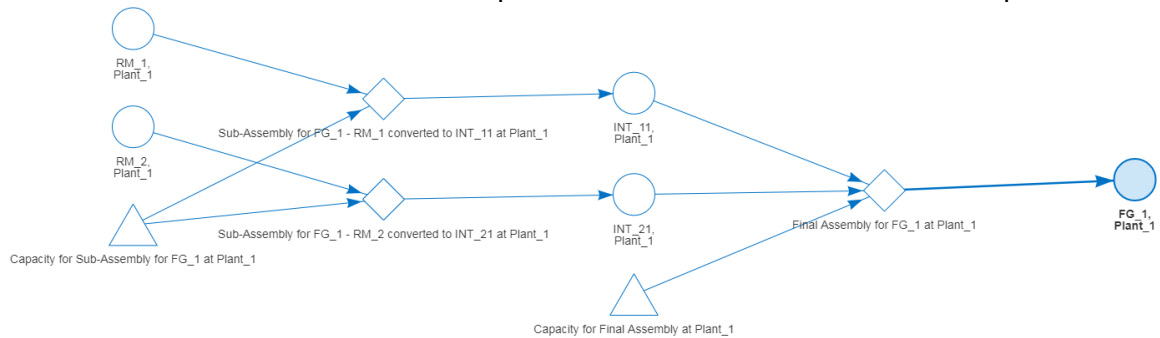
- iii. NONE : Demand node appears on the extreme left and the supply nodes appear relatively towards the right of the demand node. Arrow directions are not provided in this configuration



d. Number of Traversal steps:

- i. Its takes numeric values. The number denotes the number of levels of the supply chain that will be displayed in the network widget.

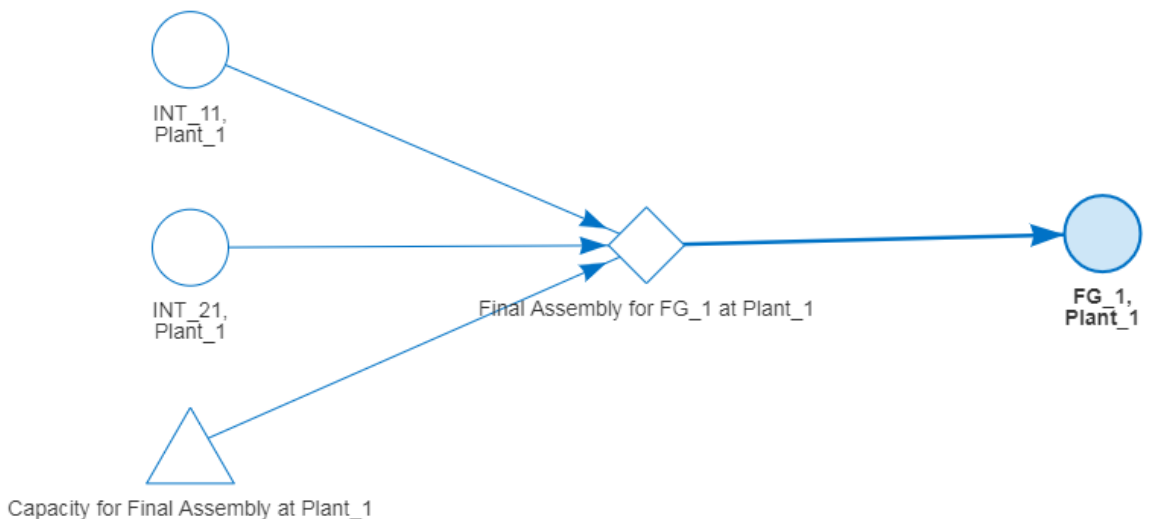
1. If number of steps = 0 then the entire network comes up



2. If traversal steps = 1 then we see the following



3. If the number of steps = 2 then we see the following



- e. Show animations : Network Widget has the ability to rearrange itself.
  - i. Checked : The data gets rendered onto the screen in systematic and clean way. If number of records are more, then user has to zoom in to check the specific network.
  - ii. Unchecked : If the data to be rendered is more, then rendering of the network onto the screen takes time. In such a case, if Show Animation is false , then rendering of the network is skipped. Data gets loaded onto the screen faster but is not rendered.
- f. Show pegging values
  - i. Checked : when checked , demand fulfillment path is highlighted in the network and pegging values are shown as follows:
  - ii. Unchecked : The static supply chain as configured by the user id displayed.
- g. Show Image : Users can also show an image of a node participating in the network by checking and unchecking this flag
  - i. Checked : The image of the node appears on the report as follows:

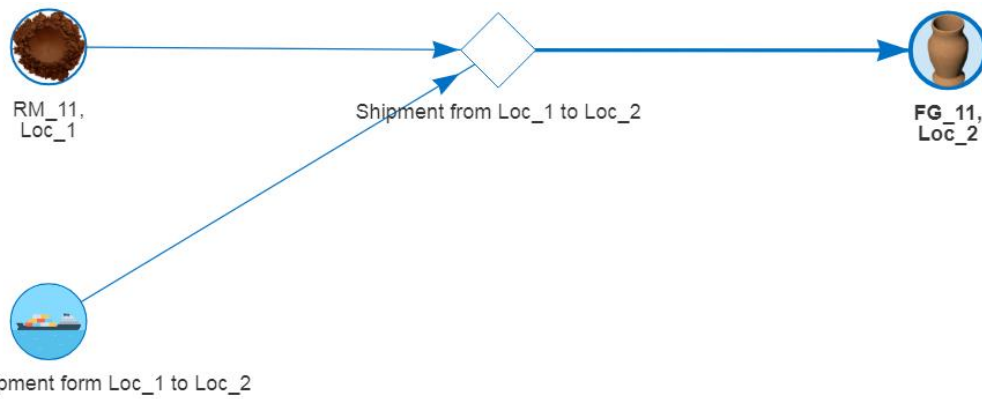


Image of a dimension can be uploaded in the form of member data. User has to give the url of the image. As of now, image cannot be uploaded to the GraphCube server from the local.

The material node is a combination of 2 dimension i.e Item, Location. User can select image of which dimension (whether item or location) gets to represent the material node. This can be done as follows:

Goto Workspace → Designer → Edit model → Material Consumption

Graph → From Nodes → Properties → Image Attribute of the node →

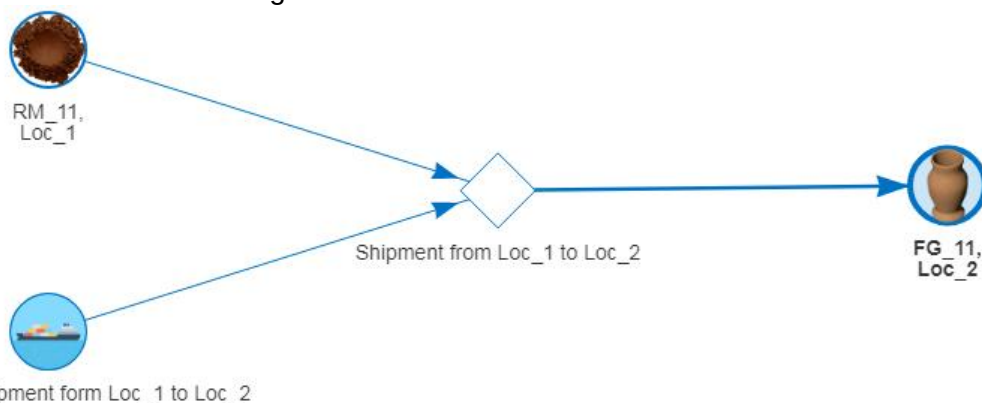
Select corresponding Attribute from the drop down menu → Click on when done

Edit Reports
Edit Model
Material Consumption Graph
Loaded 0 Rules.

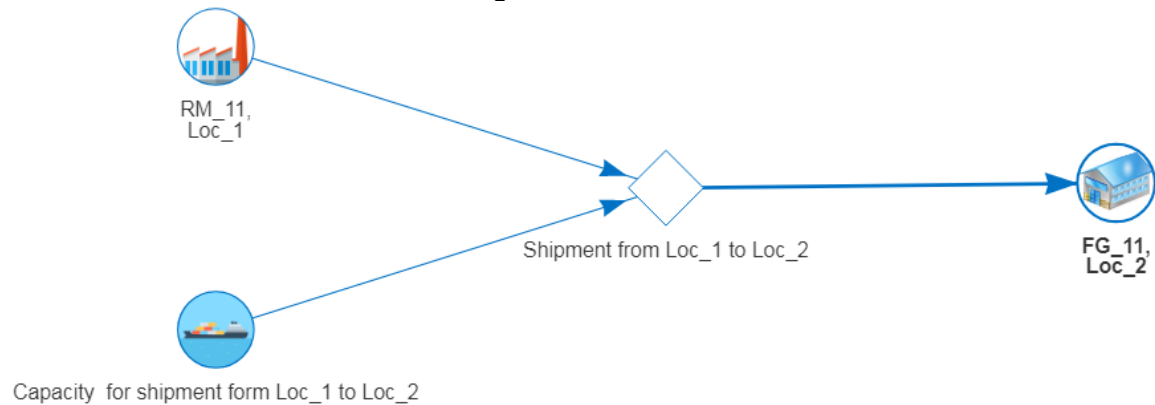
Node Properties
Shape of Node
01 dot
Image Attribute of Node
Item
Node Conditional Formula
No content. Add content using add button

From Nodes
Location
Item
To Nodes
Activity
Edges
Material Consumption Graph Association
Material Consumption Qty Per
Rules

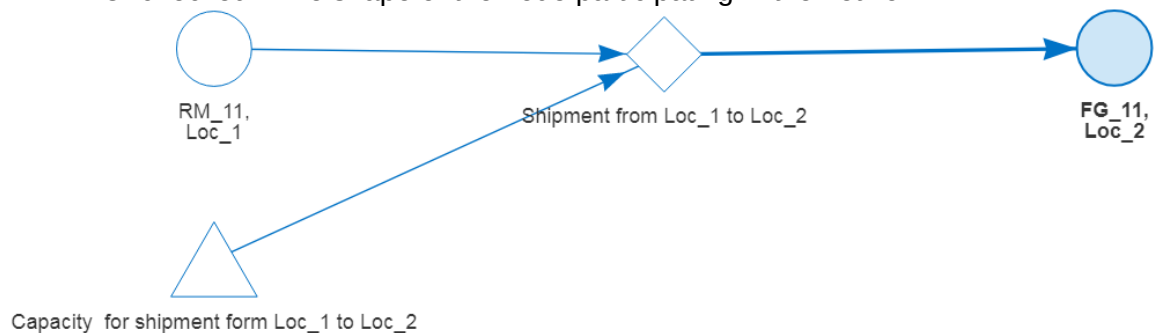
When Item is selected as the image attribute of the node



When Location is selected as the image attribute of the same node



ii. Unchecked : The shape of the node participating in the network



Shape of the node can be selected as follows:

Goto Workspace → Designer → Edit model → Capacity Consumption Graph → To Nodes → Properties → Shape of the node → Select

corresponding Shape from the drop down menu → Click on  when done

Node Properties

Shape of Node

03 diamond

Image Attribute of Node

Activity

Node Conditional Formula

No content. Add content using add button

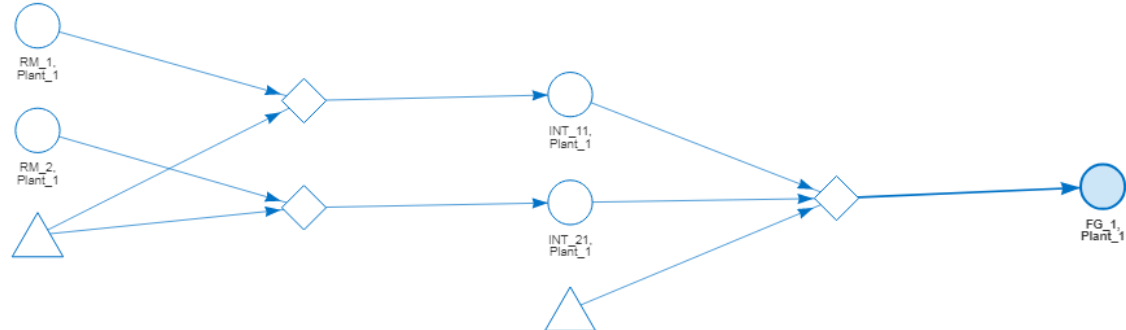
User can select the shape of From Node and To Node of the following graphs (or all graphs that are part of the network widget)

- Capacity Consumption Graph
- Material Production Graph
- Material Consumption graph
- Storage Graph

h. Show Node labels

- i. Checked(by default): When checked the names of the nodes appear on the widget.
- ii. Unchecked: when Unchecked, the names of the nodes don't appear on the widget

For example , if {Item, Location} box is checked and {Activity} and {Resource} box is unchecked, then one sees the following output:



11. Click on Save to save the report

## Export Enhancements

Following two SCS export features are available.

### Distribution Plan

With this enhancement SCS users will be able to have consumed and produced distribution plan quantity by a certain distribution activity in the same output table.

### Configuration

Need to define a graph table for this in Graph Cube Service with following nodes and edges.

- From Node → <Material Node Attr>, <Time Attr>
- To Node → <Material Node Attr>, <Activity node attr>, <Time Attr>
- Edge Properties → Consumed Qty, Produced Qty

This works only when “Distribution Activity Node Components” is defined in SCS config. This means when users have defined distribution activity granularity differently.

Distribution Activity Node Components [i](#)

[BOD].[BOD ID], [TransportMode].[Trans ID]

Need to define the graph table as below

Dist\_Plan\_Graph
⚙️
🔄
Loaded 0 Rules.

From Nodes
Item
Location
Fiscal Week

To Nodes
BOD ID
Trans ID
Item
Location
Fiscal Week

Edges
+
Rules
ConsumeShipQty
ProduceShipQty

In SCS config need to add the graph and edges as shown below.

|                                                                                                                                        |                 |
|----------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Distribution Plan Graph Name                          | Dist_Plan_Graph |
| Distribution Plan Material Production Property Name   | ProduceShipQty  |
| Distribution Plan Material Consumption Property Name  | ConsumeShipQty  |

Before this enhancement users need to check two measure groups to get the consumption and production w.r.t. a distribution activity as below

| Fiscal Week | BOD ID | Item | Location | Trans ID | Version Name       | Distribution Consumption Quantity |
|-------------|--------|------|----------|----------|--------------------|-----------------------------------|
| W27-2015    | BOD_1  | FG_1 | Plant_1  | Land     | CurrentWorkingView | 456                               |

| Fiscal Week | BOD ID | Item | Location | Trans ID | Version Name       | Distribution Production Quantity |
|-------------|--------|------|----------|----------|--------------------|----------------------------------|
| W30-2015    | BOD_1  | FG_1 | DC_1     | Land     | CurrentWorkingView | 456                              |

Now in one graph table users can get the relevant data as below

| Item | Location | Fiscal Week | BOD ID | Trans ID | Item | Location | Fiscal Week | ConsumeShipQty | ProduceShipQty |
|------|----------|-------------|--------|----------|------|----------|-------------|----------------|----------------|
| FG_1 | Plant_1  | W27-2015    | BOD_1  | Land     | FG_1 | DC_1     | W30-2015    | 456            | 456            |

## Load Plan by Produced Material

With this enhancement SCS users will be able to get the consumed capacity by an activity along with the corresponding produced material quantity. The existing capacity consumption plan SCS output provides the capacity consumption for a resource by an activity. It does not indicate how much of the material is produced with the given capacity consumption.

### Configuration

Need to define a graph table for this in Graph Cube Service with following nodes and edges.

- From Node → <Resource Node Attr>, <Time Attr>
- To Node → <Material Node Attr>, <Activity node attr>, <Time Attr>
- Edge Properties → Consumed Res Qty, Produced Material Qty

This is supported for all the three types of activity – Manufacturing, Distribution and Supplier.

Need to define the load graph table as below. Here it is shown for Manufacturing load graph.

Mfg Load Graph


✔ Loaded 0 Rules.

From Nodes
Resource
Fiscal Week

To Nodes
Activity
Item
Location
Fiscal Week

Edges
+
Rules
ConsumedResQty
ProducedMaterialQty

Need to add the graph and edges to SCS config as below

|                                              |                     |
|----------------------------------------------|---------------------|
| Load Plan Graph Name                         | Mfg Load Graph      |
| Load Plan Capacity Consumption Property Name | ConsumedResQty      |
| Load Plan Material Production Property Name  | ProducedMaterialQty |

After planning, the load graph output will look like below

Here the assembly capacity and sub-assembly capacity have quantity per as 1.5 and 0.5 respectively.

| Resource                                      | Fiscal Week | Activity                                                    | Item   | Location | Fiscal Week | ConsumedResQty | ProducedMaterialQty |
|-----------------------------------------------|-------------|-------------------------------------------------------------|--------|----------|-------------|----------------|---------------------|
| Capacity for Final Assembly at Plant_1        | W02-2015    | Final Assembly for FG_1 at Plant_1                          | FG_1   | Plant_1  | W03-2015    | 150            | 100                 |
| Capacity for Sub-Assembly for FG_1 at Plant_1 | W01-2015    | Sub-Assembly for FG_1 - RM_1 converted to INT_11 at Plant_1 | INT_11 | Plant_1  | W02-2015    | 50             | 100                 |
| Capacity for Sub-Assembly for FG_1 at Plant_1 | W01-2015    | Sub-Assembly for FG_1 - RM_2 converted to INT_21 at Plant_1 | INT_21 | Plant_1  | W02-2015    | 50             | 100                 |

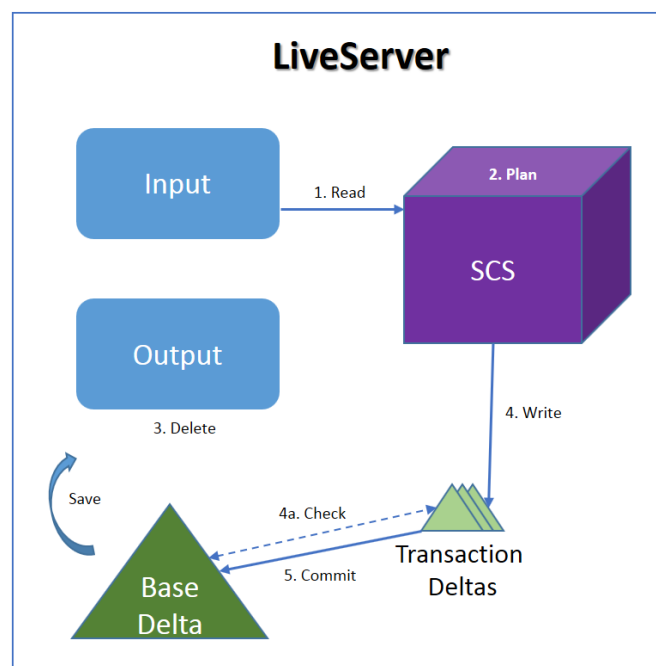
In case of co-products, this export will only report data by primary produced node. The LOTID needs to be specified in TO nodes of Load Plan Graph.

## Technical Architecture

### Technical Architecture

#### SCS Algorithm

1. Reads data from input tables and builds SCS models
2. Plan run does a demand supply match
3. Deletes output data  
Deletion is local till transaction is committed
4. Write output to a transaction delta  
4a. Checks for data in base delta when it writes so that it can hold it appropriately
5. Commit moves data from transaction to base delta
  - Save clears the base delta and moves it to the Tables
  - Each SCS run is a transaction and writes to a specific transaction delta



- LiveServer contains inputs and output Tables
- It also contains Base and Transaction Delta
- SCS is a plugin that runs within the LiveServer process

## Workflow

o9 SCM workflows support the order promising functionality by leveraging SCS's incremental planning feature.

Commonly, the modeling employed is to create a Master Planning sub-model, which models the full and deep supply network, and a shallower Order promising sub-model. Based on the business requirements for order promising, the Order promising sub-model could be single level, or multi-level, with or without alternates, with or without capacity/allocation constraints. Both of these sub-model could exist within the same



tenant in order to facilitate in-memory integration between the Master Planning function and the order promising function.

The supply Plan of the Master Planning function could be projected onto the Order Promising model. The pegging function could be utilized to project only the supply that is pegged to forecast/stock if the workflow is that of net ATP. Customer allocations could be modeled either directly in the MP model, or for additional re-allocation flexibility, customer allocations could be modeled in the order promising model by modeling constraints that control maximum allocation to a given customer.

o9 Order promising model will receive a set of incoming orders and SCS will be triggered on this model in incremental mode to plan and promise these orders. This can be done in a commit mode (eg in CWV) or in a what-if mode by performing the order drop in a what-if scenario.

At the time of the next Master Planning run, these orders and their commit dates are reflected back into the MP model.